

Supporting Information

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SI Methods

Metabolic rates of 3,006 species across life's major domains were analyzed in the article, as listed in the accompanying [Datasets S1–S11](#). Here, we report additional information on data conversions used in our analyses.

Mass Units Conversions. To make the data reported on either wet or dry mass bases across different groups studied comparable, we chose the ratio of DM/WM = 0.3 for converting dry-mass based q_{DM} to wet-mass-based metabolic rates q , $q = q_{DM} \times 0.3$. The value 0.3 was chosen as a crude mean for the DM/WM ratio of the nongelatinous groups where metabolic rates were reported on wet mass basis (letter W in the U column of Table 1).

Note that applying a single DM/WM = 0.3 ratio is conservative with respect to the main conclusion of the article about the narrow range of the observed mass-specific metabolic rates between the studied groups. Using a lower DM/WM (e.g., DM/WM = 0.15–0.20) for heterotrophic unicells or a DM/WM > 0.3 ratio for vascular plants would have left the observed range of mean taxonomic mass-specific metabolic rates (Table 1) essentially unchanged. Given the scarce information on DM/WM ratio across the groups, it seems unjustified to be very specific here, so a universal DM/WM = 0.3 ratio was applied.

Further Details on Temperature Conversions: Information on Q_{10} Determination in Macroalgae seen in Dataset S9. All nonendothermic data were adjusted to 25°C before analyses. No temperature adjustments of metabolic rates were performed for endothermic vertebrates. Endotherms do not live at body temperatures of 25°C, and very few ectotherms live at ambient temperatures in the vicinity of 40°C. Therefore, expression of endothermic and ectothermic metabolic rates at one and the same temperature (White *et al.*, 2006), which is fully relevant for testing recent models of metabolic rate dependence on body size and temperature, does not conform to the goal of the present study, which is to explore and describe the realistic range of metabolic rates. Noteworthy, the temperature of 25°C is representative of tropical forest habitats where the diversity of life forms is the greatest.

On the Question of Minimal Metabolic Rates in Aquatic Organisms. Because the buoyancy of the living matter is not precisely zero, to sustain their position in space aquatic organisms (unless they are bottom-dwellers) need to swim, i.e., they make periodical mechanical movements to adjust the position of their bodies in the water column. Terrestrial animals, helped by gravity, can remain in a given point of the Earth surface without locomotive energy expenditures.

The theoretical issue on what is the true “standard metabolism” in aquatic animals and how it compares with that of terrestrial animals is a big and important one. Metabolic rate measurements are characterized by some duration, i.e., the period during which metabolic rate is measured. For example, during a many hours’ measurement the completely motionless state can be found as being unnatural or even health-detrimental for many terrestrial animals, especially the metabolically active ones like shrews. For such animals, too, locomotion can be thought of as an essential part of maintenance expenditures. At the other extreme, during short-term measurements in the order of several minutes many aquatic animals (otherwise periodically performing swimming movements) can be found motionless,

similar to terrestrial animals during standard metabolic rate measurements.

On a practical basis, with the advent of measurement facilities that allow for a long-term, high-resolution, real-time monitoring of metabolic rates, it became possible to discriminate such periods of minimal activity in aquatic animals; accordingly, these were sometimes thought of as representing the true standard metabolic rate or “minimal” metabolic rate (e.g., Steffensen 2002). For example, Kawall *et al.* (2001) studied Antarctic copepods making dozens of sequential 30-min runs of metabolic rate measurements for each individual. Mean minimum 30-min values were found to be approximately one-third as high as the mean—i.e., “routine”—metabolic rate in the studied species.

However, a great deal of data that are available in the literature (and which we make use of in our article) was obtained by standard techniques. Such data pertain to routine metabolic rate, i.e., the one that accounts for some swimming. Here, in the following paragraph we compare the existing data on “minimal” metabolic rate from the direct long-term, high-resolution measurements described above with the averages we obtained for the same aquatic taxa (see Table 1).

It is important to note that in our dataset we took the minimal (after temperature correction) value available in the literature for each species. When the directly measured minimum data of Kawall *et al.* (2001) were corrected for temperature and body size, these values appeared on average to be 50% lower than the copepod mean values we used in our study (see [Table S1](#)). Similarly, using data of Steffensen (2002) as “etalons” for minimal metabolic rate in fish, we found even better agreement with the established taxonomic mean from Table 1 ($\approx 10\%$ lower on average). These data indicate that the elevation of the reported taxonomic means of metabolic rate in aquatic animals (Table 1) above the “minimal” metabolic rate is within several dozens of percent, i.e., it is significant, but is significantly smaller than the severalfold range of mean values among taxa and taxon groups discussed in the article.

Comparing terrestrial versus aquatic vertebrates, e.g., reptiles versus fish, as suggested by an anonymous referee, is a very interesting idea. For reptiles the mean taxonomic q (AMR) from Table 1 is 0.30 W kg^{-1} at mean body mass of 700 g, whereas for fish it is 0.38 W kg^{-1} at 400 g (data for 25°C). Using the established mass scaling coefficient $\beta = -0.22$ for reptiles (Table 1), we calculate that at body mass of $M = 400 \text{ g}$ the average reptile would have 0.34 W kg^{-1} , which is statistically indistinguishable from the fish mean.

As follows from [Table S1](#), our data for fish are very close to minimal rather than routine metabolic rates. This coincidence between reptile and fish average values hints that namely the minimal (rather than routine) metabolic rates of aquatic ectothermic vertebrates might be the relevant metabolic analogy of standard metabolic rate measured in motionless terrestrial ectothermic vertebrates. But clearly much more analysis is needed to reach a definitive conclusion here.

Miscellaneous. Out of 245 measurements of endogenous metabolic rates in heterotrophic prokaryotes, that were analyzed in this study, only 14 (5.7%) were accompanied by some information on cell size. In all other cases this information had to be found elsewhere.

Data on basal metabolic rates of mammals were taken from appendix 1 of the work of Savage *et al.* (2004). Minimal mass-specific values for each species were taken. Note that using,

for each species, mean species values given by Savage *et al.* (2004) in their appendix 1 instead of minimum values changes the geometric mean of the sample by 7%, from 4.4 W kg⁻¹ (Table 1) to 4.7 W kg⁻¹.

All other information can be found in the corresponding files for taxonomic groups. As specified in *Methods*, we used the conversion factor of 20 J per 1 ml of O₂ consumed to convert oxygen consumption rates to energy consumption rates. While in plants, bacteria and fungi O₂ uptake may or may not be coupled to ATP synthesis (where the so-called cyanide-resistant respiration can indeed constitute a substantial portion of total respiration), this does not affect the energetic conversion factor, which only depends on the products of chemical reaction.

That is, if a carbohydrate molecule (CH₂O)_n is oxidized to produce water and carbon dioxide, (CH₂O)_n + nO₂ = nCO₂ + nH₂O, the energy released will not depend on the chemical pathway, i.e., whether ATP synthesis was involved or not. For this reason, for example, the caloric equivalent of organic food (i.e., how much energy is released after its oxidation) can be measured (and originally was in the first food measurements made for military troops, fodder for cattle, etc.) by simply burning the food in the oven, where apparently no ATP synthesis occurs.

To summarize, the particular biochemical pathway provided the reaction products are the same does not influence the energetic equivalent of oxygen. The value of 20 J per 1 ml of O₂ that we use is representative of the main biochemical substrates oxidized by aerobic life, proteins (≈19 J per ml of O₂), lipids (≈20 J per ml of O₂) and carbohydrates (≈21 J per ml of O₂), to the accuracy of 5%.

References for SI Methods and Tables S1–S3

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Table S1. Comparison of minimal metabolic rates with taxonomic means from Table 1 in fish and copepods

Species	Body mass, g	Temp, °C	MMR, W kg ⁻¹	MMR adjusted to 25°C and to the mean taxonomic body mass from Table 1	AMR, W kg ⁻¹	MMR/AMR
Fish						
<i>Onchorhynchus mykiss</i>	392	10	0.19	0.40	0.38	1.1
<i>Trematomus hansonii</i>	100	−1	0.11	0.32	0.38	0.8
<i>Pagothenia borchgrevinkii</i>	100	−1	0.11	0.32	0.38	0.8
Mean for fish						0.9
Copepods						
<i>Calanoides acutus</i>	0.0033	0			3.0	0.4
<i>Calanus propinquus</i>	0.0047	0			3.0	0.7
<i>Metridia gerlachei</i>	0.0014	0			3.0	0.7
<i>Gaetanus tenuispinus</i>	0.0036	0			3.0	0.1
<i>Rhincalanus gigas</i>	0.0117	0		1.	3.0	0.3
<i>Paraeuchaeta antarctica</i>	0.0196	0			3.0	0.5
<i>Heterohabdus farrani</i>	0.0032	0			3.0	0.8
Mean for copepods						0.5

MMR, minimal metabolic rate; AMR, average taxonomic metabolic rate from Table 1. Data for fish are from Steffensen (2002), and for copepods are from Kawall et al. (2001). Note: for temperature and body mass adjustments of MMR Q_{10} values of 1.65 and 2.0 (*Methods*) and scaling exponents $\beta = -0.15$ and -0.30 (Table 1) were used for fish and copepods, respectively.

Table S2. Nitrogen content in the taxonomic groups studied

Taxonomic group	N/DM, %	Reference	Comment
Heterotrophs			
Prokaryotes	6.5–8.9	<i>Arthrobacter globiformis</i>	van Veen and Paul 1979
	9.1–11.1	<i>Enterobacter aerogenes</i>	van Veen and Paul 1979
	10.0–14.1	<i>Bacillus cereus</i>	Dataset S1
	5.6–8.9	<i>Mycobacterium phlei</i>	Tepper 1968
	10	<i>Streptococcus agalactiae</i>	
Protozoa	9.5	MEAN	
	10.6	Used as a mean value for analysis of an extensive metabolic dataset	Vladimirova and Zotin 1983
	12	Used as a mean value for analysis of an extensive metabolic dataset	Fenchel and Finlay 1983
Insects	9.7 ± 0.2	119 herbivores insect species, mean ± 1 SE	Fagan <i>et al.</i> 2002
	11 ± 0.2	33 predator species, mean ± 1 SE	Fagan <i>et al.</i> 2002
Aquatic invertebrates	9–11	Crustacean zooplankton in freshwater lake	Andersen and Hessen 1991
Crustacea: copepods and krill	9–11	Sargasso Sea	Beers 1966
	9.8 ± 0.06	n = 10 species (mean ± 1 SD)	Dataset S4
Crustacea: peracarids	6.4 ± 1.9	n = 13 species (mean ± 1 SD)	Dataset S4
Crustacea: decapods	8.3 ± 1.8	n = 12 species (mean ± 1 SD)	Dataset S4
Mollusca: cephalopods	n.d.		
Gelatinous invertebrates	10	chaetognath <i>Sagitta elegans</i>	Ikeda and Skjoldal 1989
	4.3	medusa <i>Aequorea victoria</i>	Ikeda and Skjoldal 1989
Ectothermic vertebrates			
Amphibians	n.d.		
Fish	11.5	<i>Ictalurus punctatus</i>	Brugger 1993
	14.2	<i>Dorosoma cepedianum</i>	Brugger 1993
	13.7	<i>Lepomis cepedianum</i>	Brugger 1993
Reptiles	n.d.		
Endothermic vertebrates			
Birds	12–14	<i>Zonotrichia leucophrys</i> , lean dry mass	Chilgren 1985
Mammals	11.2–13.2	Various tissues of <i>Dugong dugong</i>	Yamamuro <i>et al.</i> 2002
	9.7	Rat, whole body	Truszkowski 1927
	12	Guinea-pig, whole body	Truszkowski 1927
	14.5	Horse, skeletal muscle	Truszkowski 1927
	15.2	Mean for 24 tropical bats, whole body	Studier <i>et al.</i> 1994
Photoautotrophs			
Cyanobacteria	4–9		Fogg <i>et al.</i> 1973
	4.2–13.1	<i>Spirulina platensis</i>	Gordillo <i>et al.</i> 1999
	9.4	<i>Anacystis nidulans</i>	Kratz and Myers 1955
Eukaryotic microalgae	6.9–8.6	<i>Chatocerus furcellatus</i> *	Dataset S8
	5.1–6.3	<i>Coscinodiscus</i> sp. C38B*	Dataset S8
	6.0–7.5	<i>Coscinodiscus</i> sp. CoA*	Dataset S8
	6.8–8.5	<i>Ditylum brightwellii</i> *	Dataset S8
	9.1–11.4	<i>Dunaliella tertiolecta</i> *	Dataset S8
	8.3–10.4	<i>Emiliania huxleyi</i> *	Dataset S8
	4.8–6.0	<i>Gonyaulax tamarensis</i> *	Dataset S8
	7.3–9.1	<i>Isochrysis galbana</i> *	Dataset S8
	6.9–8.6	<i>Leptocylindrus danicus</i> *	Dataset S8
	3.1–3.9	<i>Monochrysis lutheri</i> *	Dataset S8
	5.6–6.9	<i>Olisthodiscus luteus</i> *	Dataset S8
	8.2–10.2	<i>Phaeodactylum tricornutum</i> *	Dataset S8
	10.8–13.5	<i>Prorocentrum micans</i> *	Dataset S8
	5.9–7.4	<i>Skeletonema costatum</i> *	Dataset S8
	8.7–10.9	<i>Stephanodiscus neoastraea</i> *	Dataset S8
	7.5–9.4	<i>Thalassiosira nordenskiöldii</i> *	Dataset S8
	5.1–6.3	<i>Thalassiosira pseudonana</i> *	Dataset S8
	6.3–7.9	<i>Thalassiosira weissflogii</i> *	Dataset S8
	6.3–7.9	MEAN	
Phytoplankton	5.5 ± 2.5	Mean ± 1 SD for 112 measurements	Duarte 1992
Eukaryotic macroalgae	1.9 ± 0.8	Mean ± 1 SD for 298 measurements	Duarte 1992
Vascular plants: green leaves	1.7	Geometric mean for 2,061 measurements; 95% C.I. 0.6–5	Data of Wright <i>et al.</i> 2004
Vascular plants: tree saplings	0.54	Geometric mean for 118 measurements; 95% C.I. 0.3–1	Data of Reich <i>et al.</i> 2006
Vascular plants: seedlings	2.8	Geometric mean for 198 measurements; 95% C.I. 1.6–5.0	Data of Reich <i>et al.</i> 2006

N/DM, nitrogen mass to dry mass ratio.

*N/DM calculated from the known C/N mass ratio assuming either 40% or 50% carbon in dry mass (lower and upper value of the range, respectively). The two species with known carbon/dry mass ratios were *Navicula pelliculosa* (C/DM = 0.412) and *Stephanodiscus neoastraea* (C/ODM = 0.46; ODM, organic dry matter). For each species, the N/DM ratio shown corresponds to the measurement with the lowest metabolic rate.

Table S3. Dry matter content in the taxonomic groups studied

Taxonomic group	U	DM/WM	Reference	Comment
Heterotrophs				
Prokaryotes	D	0.16–0.40	Posch <i>et al.</i> 2001	Depends on cell size and measurement technique
Protozoa	D	0.15	Fenchel and Finlay 1983	
		0.135	Vladimirova and Zotin 1983	
Insects	W	0.2–0.6	Hadley 1994	
Aquatic invertebrates	W			
Crustacea: copepods and krill	W	0.19 ± 0.05	Dataset S4	<i>n</i> = 55 species (mean ± 1 SD)
Crustacea: peracarids	W	0.22 ± 0.15	Dataset S4	<i>n</i> = 38 species (mean ± 1 SD)
Crustacea: decapods	W	0.23 ± 0.08	Dataset S4	<i>n</i> = 18 species (mean ± 1 SD)
		0.29	Ileva 1980	208 observations for eight species of tropical and temperate decapods
Mollusca: cephalopods	W	n.d.		
Gelatinous invertebrates	W	0.035–0.05	Hirst and Lucas 1998	medusae
		0.035–0.30	Hirst and Lucas 1998	chaetognaths
		0.1	Ikedo and Skjoldal 1989	chaetognaths
Ectothermic vertebrates	W			
Amphibians	W	n.d.		
Fish	W	0.26	Tierney <i>et al.</i> 2002	Mesopelagic Antarctic fishes
		(0.18–0.30)		
		0.29	Torres and Somero 1988	Mesopelagic Antarctic fishes
		(0.13–0.36)		
		0.24–0.29	Brugger 1993	3 North American fishes
Reptiles	W	n.d.		
Endothermic vertebrates	W			
Birds	W	0.30–0.38	Skadhauge 1981	
Mammals	W	0.38	Balonov and Zhesko 1989	Rats, mice, dogs, humans
Photoautotrophs				
Cyanobacteria	D	0.42*	Fietz and Nicklisch 2002	<i>Planktothrix agardhii</i>
		0.067, 0.172	Scherer <i>et al.</i> 1984	Two fully hydrated Nostoc spp.
		0.035	Li and Gao 2004	<i>Nostoc sphaeroides</i>
Eukaryotic microalgae	C	0.18–0.26*	Myers and Graham 1971	<i>Chlorella pyrenoidosa</i>
		0.27*	Fietz and Nicklisch 2002	<i>Stephanodiscus neoastraea</i>
Eukaryotic macroalgae	D	0.23	Weykam <i>et al.</i> 1996	35 Antarctic species
		(0.12–0.54)		
Vascular plants: green leaves	D	0.16–0.41	Vile <i>et al.</i> 2005	DM/WM increases from short-lived forbs to trees
Vascular plants: tree saplings	D	n.d.		
Vascular plants: seedlings	D	n.d.		

DM/WM, dry mass to wet mass ratio; U, dominant mass units in the original data sources: metabolic rates reported mostly per dry (D), wet (W), or carbon (C) mass basis.

*Calculated from DM/volume ratio assuming cell density of 1 g ml⁻¹.

Other Supporting Information Files

[Dataset S1](#)

[Dataset S2](#)

[Dataset S3](#)

[Dataset S4](#)

[Dataset S5](#)

[Dataset S6](#)

[Dataset S7](#)

[Dataset S8](#)

[Dataset S9](#)

[Dataset S10](#)

[Dataset S11](#)

[SI Appendix](#)

SI Appendix: Physical Limitations on Metabolic Rates Attainable via Breathing

Mechanical oxygen pumping involves certain energy expenditures (breathing cost), which depend on body size and ambient oxygen concentration. Beyond some critical body size and in oxygen-poor environments, maintenance of a high, size-independent metabolic rate appears to be prohibited by energy conservation law due to an overly high breathing cost. We will now show that the larger ectotherms, with their mean metabolic rates deviating consistently from the proposed metabolic optimum ($3\text{--}9 \text{ W kg}^{-1}$), exist precisely in such prohibitive conditions. This provides further support for the proposed causal coupling between metabolic optimality and animal breathing.

In oxygen balance, when energy consumption is balanced by oxygen delivery,

$$Q = qV\rho = \omega V_T \rho_{O_2} KE, \quad [1]$$

where Q (W) is whole-body metabolic rate, q (W kg^{-1}) is mass-specific metabolic rate, V (m^3) is body volume, $\rho = 10^3 \text{ kg m}^{-3}$ is live body density, E is oxygen extraction coefficient (proportion of oxygen that is extracted from the medium during breathing), ω (s^{-1}) is breathing frequency, V_T (m^3) is tidal volume (water or air volume delivered into the body per breath), ρ_{O_2} (kg m^{-3}) is oxygen density in the medium (water or air), and $K = 1.4 \times 10^7 \text{ J (kg of } O_2)^{-1}$ is energy conversion coefficient for aerobic metabolism.

Breathing involves periodical movements of some parts of the body that occupy volume V_p (e.g., chest volume in mammals, intraopercular volume in fish, mantle volume in cephalopods) and move there and back along a linear scale $l_T \sim V_T^{1/3}$ at a frequency ω . This corresponds to mean linear velocity $u = 2\omega l_T$, acceleration $a = 4\omega^2 l_T$, force $F = 8\rho V_p u \omega$ and mechanical power

$$W = Fu = 8\rho V_p l_T^2 \omega^3. \quad [2]$$

This is a lower estimate of the mechanical power spent by the organism to make its breathing pump move, because it does not take into account either the work against friction forces, or the work on moving the medium where breathing occurs.

Expressing ω in terms of q using Eq. 1, one obtains from Eq. 2 that at a size-independent q the mechanical cost $\kappa \equiv W/(\varepsilon Q)$ of oxygen pumping (cost of breathing) grows as squared body length $l = V^{1/3}$:

$$\kappa = \left(\frac{\rho}{\rho_{O_2} EK} \right)^3 \frac{8\beta}{\varepsilon \alpha^{7/3}} q^2 l^2, \quad [3]$$

where $\alpha \equiv V_T/V$, $\beta \equiv V_p/V$, and ε is muscle efficiency (the ratio of mechanical work performed by the muscles to the metabolic power of the muscle tissue consumed during this work). In cuttlefish *Sepia officinalis*, for which all parameters entering Eq. 3 were recently measured in detail (1), q and α increased, while E decreased, by 2.9, 3, and 2.3 times, respectively, between 11 and 23°C. The value of κ was measured to increase by 8.3 times (1) compared with 7.9 times predicted from Eq. 3. At $\beta \sim 1$, Eq. 3 also accurately predicts the absolute magnitude of κ (2.7% vs. the observed 2.9% at $\varepsilon = 0.03$).

After reaching a critical body size for which κ becomes large, further maintenance of a size-independent optimum q is impossible. Putting maximum $q_{\max}$ attainable by organisms living

at the basal $q \sim 10 \text{ W kg}^{-1}$ as $q_{\max} \sim 10^2 \text{ W kg}^{-1}$, and assuming $E_{\max} \sim 0.8$, $\alpha \sim 10^{-2}$, $\beta \sim 1$, $\varepsilon_{\max} \sim 0.1$ and $\kappa_{\max} \sim 0.1$, the upper estimate of critical body length for aquatic animals, where $\rho_{\text{O}_2} \approx 0.007 \text{ (kg of O}_2\text{) m}^{-3}$ at 25 °C, is

$$l_{\text{cr}} = \left(\frac{\kappa_{\max} \varepsilon_{\max}}{8\beta} \right)^{1/2} \frac{\alpha^{7/6}}{q_{\max}} \left(\frac{\rho_{\text{O}_2} E_{\max} K}{\rho} \right)^{3/2} \sim 10^{-3} \text{ m.} \quad [4]$$

The obtained value of critical body size of the order of 1 mm for aquatic media indicates that the maintenance of optimal value of mass-specific metabolic rate q_{opt} is physically permitted up to body masses of $M_{\text{cr}} \sim 1 \text{ mg}$, as is the case with copepods still featuring optimal q (Table 1). All aquatic organisms with $M > M_{\text{cr}}$ do not have the option of maintaining a constant q_{opt} . In such organisms mass-specific metabolic rates should decrease with increasing body size. In agreement with this prediction, the larger aquatic organisms (decapods, cephalopods, fish, as well as the evolutionarily and biochemically closely related amphibians and reptiles, all with mean mass $M \gg M_{\text{cr}}$) have mean q several times lower than all the other groups studied (Table 1).

Increase of ambient oxygen concentration significantly reduces the energetic costs of pumping a unit oxygen volume into the body (Eq. 3). Transition from aquatic medium with $\rho_{\text{O}_2,1} \approx 0.007 \text{ (kg of O}_2\text{) m}^{-3}$ to atmospheric air with $\rho_{\text{O}_2,2} = 0.3 \text{ (kg of O}_2\text{) m}^{-3}$ allows for an increase in metabolic rate q_2 in the air as compared to q_1 in water by as much as $q_2 / q_1 = (\rho_{\text{O}_2,2} / \rho_{\text{O}_2,1})^{3/2} \sim 280$ -fold (see Eq. 4), all other parameters remaining the same. This explains why, despite their large body size, air-breathing endotherms were able to evolve metabolic rates back to the vicinity of the metabolic optimum, similar to much smaller species both on land and in the sea (Fig. 3). Fig. 3 shows mass-specific metabolic rates of endotherms and of all heterotroph taxa with mean mass M less than, or of the order of, M_{cr} (Table 1). Fig. 3 emphasizes that in all body size intervals occupied by life, from the smallest bacteria to the largest animals, there are taxonomic groups that fit into one and the same optimal range of mass-specific metabolic rates, in terms of similar mean values and confidence intervals (Table 1 and Fig. 2).

The limitation on maximum body size l_{cr} , Eq. 4, compatible with optimal mass-specific metabolic rate holds independently of the properties of the internal networks distributing oxygen within the body. It does not depend on the type of cardiovascular system, or oxygen carrying capacity of the blood, or any other parameters that determine the process of oxygen distribution within the body after oxygen has been consumed. The estimate of l_{cr} is based, instead, on the physics of the primary process of oxygen intake from the environment. The transition from aquatic to aerial milieu is a necessary, but insufficient, condition for elevation of metabolic rate up to the optimum in large animals. To benefit from the energetically cheap oxygen delivery, the air-breathing animals had first to modify their internal distributive networks and blood and tissue biochemistry. There is no use in high oxygen intake if one cannot properly distribute it within the body. This explains why reptiles and amphibians, whose blood biochemistry is close to that of fish, retain low metabolic rates despite air breathing. And only in endotherms, with their radical evolutionary changes in the cardiovascular system and biochemistry (2), was the elevation of metabolic rates back to the optimum (Figs. 2 and 3) made possible. The analyzed body of evidence is thus consistent with the statement that natural selection favors the optimal metabolic rate in all taxa where this rate is physically achievable.

The proposed theoretical approach, Eq. 4, opens a wide field for the quantitative analysis, in different groups of organisms, of the numerous organismal parameters that affect breathing costs. The fact that realistic values of these parameters yield an estimate of l_{cr} much smaller than the linear size of species with mean $q \ll q_{\text{opt}}$, makes the breathing cost limitation a novel and

numerically competitive explanation of the relatively low metabolic rates observed in the larger ectotherms. This approach unambiguously predicts that for taxa with $l > l_{cr}$ (high breathing cost), metabolic rates *must* decline with growing body size, $q \propto l^{-1}$, if the parameters of Eqs. 3 and 4 are evolutionarily conserved. These fundamental physical considerations show that for such organisms there is no other option than metabolic allometry. In contrast, because breathing cost decline very rapidly with diminishing body length (see Eq. 3), organisms smaller than l_{cr} have negligible breathing costs and can, at physiological rest, thrive without metabolic scaling. These physical limitations can explain the apparently increasing conspicuousness of scaling patterns with increasing mean body mass of the taxa studied (Table 1 and Fig. 3).

1. Melzner F, Bock C, Pörtner HO (2006) Temperature-dependent oxygen extraction from the ventilatory current and the costs of ventilation in the cephalopod *Sepia officinalis*. *J Comp Physiol* 176B:607-621.
2. Else PL, Turner N, Hulbert AJ (2004) The evolution of endothermy: Role for membranes and molecular activity. *Phys Biochem Zool* 77:950-958.

Dataset S1. Endogenous respiration rates in heterotrophic prokaryotes

Notes to Table **S1a**:

On data collection:

Prokaryote data were compiled by searching the www.pubmedcentral.nih.gov full-text library for "bacterium" and "endogenous respiration" and subsequent analysis of the returned 570 documents (mostly papers in the Journal of Bacteriology and Journal of Applied and Environmental Microbiology, time period 1940-2006) and references therein.

On cell size and taxonomy:

Data on endogenous respiration rates (i.e. respiration rates of non-growing cells in nutrient-deprived media) in heterotrophic eukaryotes are presented. Studies of bacterial respiration **very rarely** report information on cell size, which had therefore to be retrieved from different sources. To do so, an attempt was made to assign the bacterial strains described in the metabolic sources to accepted species names, to further estimate the cell size for these species in the relevant literature. This was done using strain designations and information in the official bacterial culture collections, like ATCC (American Type Culture Collection), NCTC (National Type Culture Collection) and others.

Column "**Species (Strain)**" gives the strain designation **as given by the authors** of the respiration data paper. Column "**Valid Name**" gives the relevant valid species name for this strain, as determined from culture collections' information and/or other literature sources. Valid names follow Euzéby (1997). Cell size in the "**Mpg**" column correspond to species indicated in the "Valid Name" column. Note that this information is of approximate nature, because many respiration data come from quite old publications and it was sometimes difficult to find out the valid name of the strain used with great precision. "**Class:Order**" column contains the relevant taxonomic information for the species listed in the "Valid Name" column as given by Euzéby (1997) (<http://www.bacterio.net>).

For example, Hareland et al. (1975) reported respiration rate for *Pseudomonas acidovorans* (ATCC strain number 17455). ATCC web site (www.atcc.org) says that this strain is *Delftia acidovorans* originally deposited as *Pseudomonas acidovorans*. Cell size for *Pseudomonas acidovorans* was therefore determined from the species description of *Delftia acidovorans* given by Wen et al. (1999). "**Class:Order**" for these data was determined as given at <http://www.bacterio.net> for *Delftia acidovorans*.

Note that the taxonomic uncertainty exclusively relates to the cell size determination. Cell size information participates in the paper's results only as a crude mean for all the 173 species studied, which is unlikely biased in any significant way. Taxonomic uncertainties, if any, do not influence any of the conclusions regarding the range, mean and frequency distribution of the prokaryotic respiration rates analysed in the paper.

Abbreviations and universal conversions: DM – dry mass; WM – wet mass; N – nitrogen mass; C – carbon mass; Pr – protein mass; X/Y – X by Y mass ratio in the cell, e.g. DM/WM is the ratio of dry to wet cell mass; 1 W = 1 J s⁻¹; 1 mol O₂ = 32 g O₂.

Original units are the units of endogenous respiration rate measurements as given in the original publication (**Source**); **qou** is the numeric value of endogenous respiration rate in the original units. E.g., if it is “ $\mu\text{l O}_2$ (5 mg DM) $^{-1}$ (2 hr) $^{-1}$ ” in the column “**Original units**” and “200” in the column “**qou**”, this means that cells amounting to 5 mg dry mass consumed 200 microliters oxygen in two hours.

qWkg is the original endogenous respiration rate **qou** converted to W (kg WM) $^{-1}$ (Watts per kg wet mass) using the following conversion factors: C/DM = 0.5 (Kratz & Myers 1955; Bratbak & Dundas 1984; Nagata 1986), Pr/DM = 0.5 (Gronlund & Campbell 1961; Sobek et al. 1966; Smith & Hoare 1968 (see Table S1a); Zubkov et al. 1999), N/DM = 0.1 (SI Methods, Table S12b) if not indicated otherwise, and DM/WM = 0.3 as a crude mean for all taxa applied in the analysis (SI Methods, Table S12a). Energy conversion: 1 ml O₂ = 20 J. The respiratory quotient of unity was used (1 mol CO₂ released per 1 mol O₂ consumed).

TC is ambient temperature during measurements, degrees Celsius.

q25Wkg is endogenous respiration rate converted to 25 °C using $Q_{10} = 2$, **q25Wkg** = **qWkg** $\times 2^{(25 - TC)/10}$, dimension W (kg WM) $^{-1}$. For each species rows are arranged in the order of increasing **q25Wkg**.

Mpg: estimated cell mass, pg (1 pg = 10⁻¹² g). In most cases it is estimated from linear dimensions (using geometric mean of the available linear size range) assuming spherical cell shape for cocci and cylindrical shape for rods. Square brackets around the **Mpg** value indicate that the cell size information was obtained from a different source than the source of endogenous respiration rate data. When converting cell volume to cell mass, cell density of 1 g ml⁻¹ was assumed.

Source: the first, unbracketed reference in this column is where the value of **qou** is taken from; references and data in square brackets refer to cell size determination. Cell size reference "BM" in brackets corresponds to Bergey's Manual of Systematic Bacteriology, 1st Edition (Holt, 1984, 1986, 1989); BM9 is Bergey's Manual of Determinative Bacteriology, 9th Edition (Holt et al. 1994). Word "genus" in brackets indicates that cell size is determined as mean for the genus. This was done for those genera where the range of minimum to maximum cell masses did not exceed a factor of ten. E.g. for an unknown Chromatium sp. (BM9 genus: rods 1-6×1.5-15 μm , which corresponds to cell mass range from 1.2 to 420 pg) cell mass was left undetermined (empty "Mpg" column).

Culture age: Information on culture age and the duration of respiration measurements, if available.

Comments: this column provides relevant information on culture conditions and cellular composition of the studied species, often including additional data on respiration rates that were obtained for the same strain (species) by the same group of authors.

Log₁₀-transformed values of **q25Wkg** (W (kg WM) $^{-1}$), minimum for each species, were used in the analyses shown in Figures 1-3 and Table 1 in the paper (a total of 173 values for n = 173 species). The corresponding rows are highlighted in [blue](#).

References within Table S1a to Tables, Figures etc. refer to the corresponding items in the original literature indicated in the **Source** column.

Table **S1a**. Endogenous respiration rates in heterotrophic prokaryotes.

Species (strain)		Valid name	Class: Order	Original units	MIN	qou	qWkg	TC	q25Wkg	Mpg	Source	Culture age	Comments
Acetobacter (Ch 31)	aceti	1.Acetobacter aceti	Clostridia: Clostridiales	$\mu\text{mol O}_2 (1.8 \times 45 \text{ mg WM})^{-1} (5 \text{ hr})^{-1}$	MIN	2	0.6	30	0.42	[0.75]	De Ley & Schell 1959 [BM, ellipsoid or rod-shaped 0.6-0.8×1.0-4.0 μm]	Cells incubated for 4-5 days on gelatin slants at 20 °C; respiration measured for 5 hr	Cells additionally incubated for 2-3 days at 30 °C in a shaking apparatus "occasionally displayed higher endogenous respiration, 13.5 $\mu\text{mol O}_2 (1.8 \times 45 \text{ mg WM})^{-1} (2.5 \text{ hr})^{-1}$ = 8.3 W/kg; this respiration increased exponentially during incubation
Acholeplasma laidlawii (NCTC 10116)	laidlawii	2.Acholeplasma laidlawii	Mollicutes: Acholeplasmatales	$\text{nmol O}_2 (\text{mg protein})^{-1} \text{ min}^{-1}$	MIN	1.2	1.4	37	0.61	[0.04]	Abu-Amero et al. 1996 [Wieslander et al. 1987, sphere diam 0.3-0.6 μm]	Cells harvested after 24-72 hr incubation	
Hydrogenomonas ruhlandii		3.Achromobacter ruhlandii	Betaproteobacteria: Burkholderiales	$\mu\text{l O}_2 (0.5 \text{ mg DM})^{-1} (2 \text{ hr})^{-1}$	MIN	9	15	30	10.61	0.2	Packer & Vishniac 1955 [rods 0.4-0.75×0.75-2.0 μm , mean 0.5×1.1 μm]	Bacteria harvested after 4-5 days' incubation on agar plates; respiration of resting cells measured for 2 h	Hydrogen-oxidizing bacterium isolated from soil
Achromobacter sp.		4.Achromobacter sp.	Betaproteobacteria: Burkholderiales	$\mu\text{mol O}_2 (100 \text{ mg DM})^{-1} (2 \text{ hr})^{-1}$	MIN	45.1	8.4	30	5.94	[0.6]	Gronlund & Campbell 1961 [Chester & Cooper 1979, 0.5-0.8×1.5-2.5 μm]	cells harvested after 20 hr growth; respiration measured for 2 hr	Classification and size determination made for the Achromobacter genus as described at www.bacterio.cict.fr . There is no such species at www.bacterio.cict.fr
Achromobacter (B8)	sp.	5.Achromobacter sp.	Betaproteobacteria: Burkholderiales	$\mu\text{l O}_2 (5 \text{ mg DM})^{-1} (2 \text{ hr})^{-1}$		200	35	30	24.75	[0.6]	Tomlinson & Campbell 1963 [Chester & Cooper 1979, 0.5-0.8×1.5-2.5 μm]	cells harvested after 20 hr growth; respiration measured for 2 hr	No drop of respiration during the first two hours Classification and size determination made for the Achromobacter genus as described at www.bacterio.cict.fr . There is no such species at www.bacterio.cict.fr
Achromobacter viscosus (ATCC 12448)		6.Achromobacter viscosus	Betaproteobacteria: Burkholderiales	$\mu\text{l O}_2 (5 \text{ mg DM})^{-1} (2 \text{ hr})^{-1}$	MIN	200	35	30	24.75	[0.6]	Tomlinson & Campbell 1963 [Chester & Cooper 1979, 0.5-0.8×1.5-2.5 μm]	cells harvested after 20 hr growth; respiration measured for 2 hr	No drop of respiration during the first two hours Classification and size determination made for the Achromobacter genus as described at www.bacterio.cict.fr . There is no such species at www.bacterio.cict.fr
Achromobacter xerosis (ATCC 14780)		7.Achromobacter xerosis	Betaproteobacteria: Burkholderiales	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	MIN	14	23	30	16.26	[0.5]	Jurtshuk & McQuitty 1976 [Groupé et al. 1954, 0.5×2-3 μm]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	Classification and size determination made for the Achromobacter genus as described at www.bacterio.cict.fr . There is no such species at www.bacterio.cict.fr
Hydrogenomonas facilis		8.Acidovorax facilis	Betaproteobacteria: Burkholderiales	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	MIN	4-11	7	30	4.95	0.3	Schatz & Bovell 1952 [rods, 0.4×2.5 μm in	Heterotrophic cultures: cells grown for 48, 21, 72, and 48 hr on lac-	Synonym Pseudomonas facilis

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Table 1. Physiological characteristics of the strains used in the present study											References		
Strain	Strain	Strain	Strain	Strain	Strain	Strain	Strain	Strain	Strain	Strain	Strain	Strain	Strain
tumefaciens (ATCC 15955)	tumefaciens	Rhizobiales	DM) ⁻¹ hr ⁻¹								McQuitty 1976 [BM9, genus, rods 0.6-1.0×1.5-3.0 μm]	late-logarithmic growth phase (two thirds of the maximal growth concentration)	
Hydrogenomonas eutropha (ATCC 17697)	18.Alcaligenes eutrophus	Betaproteobacteria: Burkholderiales	μmol O ₂ (3.6 mg DM) ⁻¹ hr ⁻¹	MIN	8	83	33	47.67	[0.8]	Bongers 1970 [BM, rods, 0.7×1.8-2.6 μm]	Respiration of cells withdrawn from turbidostat (steady-state growth)	Hydrogen-oxidizing bacterium, max. respiration (in the presence of H ₂) is ten times the endogenous rate	
Alcaligenes faecalis (ATCC 8750)	19.Alcaligenes faecalis	Betaproteobacteria: Burkholderiales	μl O ₂ (mg DM) ⁻¹ hr ⁻¹	MIN	16	27	30	19.09		Jurtshuk & McQuitty 1976	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)		
Alcaligenes sp. (strain 5)	20.Alcaligenes sp.	Betaproteobacteria: Burkholderiales	μmol O ₂ (14 mg DM) ⁻¹ (5 hr) ⁻¹	MIN	4	2.1	30	1.48		Subba-Rao & Alexander 1985	Bacteria grown for 2 days; washed; incubated in buffer for 6 to 12 hr on a rotary shaker at 30 C to reduce endogenous respiration; washed again and added to respirometer flasks (Subba-Rao & Alexander 1977)	Bacteria isolated from enrichments with benzhydrol as the sole carbon source.	
Unnamed methylo-troph (CC495)	21.Aminobacter lissarensis	Alphaproteobacteria: Rhizobiales	nmol O ₂ (mg WM) ⁻¹ min ⁻¹	MIN	0.22	1.6	25	1.60	[0.6]	Coulter et al. 1999 [BM9, genus, rods, 0.6-1.0×1.0-3.0 μm]	Cells harvested in the late exponential phase	Bacterium isolated from the top 5 cm of soil in a beech wood in Northern Ireland	
Amoebobacter purpureus (ML1)	22.Amoebobacter purpureus	Gammaproteobacteria: Chromatiales	nmol O ₂ (mg protein) ⁻¹ min ⁻¹	MIN	9.8	11	30	7.78	[36]	Overmann & Pfennig 1992 [Eichler & Pfennig 1988, 3.3-3.8×3.5-4.5 μm]	Respiration of cells without microscopically visible sulfure globules at oxygen concentrations of 11-67 μM; respiration rates of phototrophically (anaerobically) and chemotrophically (microaerobically) grown cells do not differ; the species displays poor if any growth in the dark.	Purple sulfur bacteria isolated from the chemocline of meromictic Mahoney Lake (British Columbia, Canada) Endogenous respiration of cells with visible sulfur globules is lower, 5.7 nmol O ₂ (mg protein) ⁻¹ min ⁻¹	
Amoebobacter roseus (6611)	23.Amoebobacter roseus	Gammaproteobacteria: Chromatiales	nmol O ₂ (mg protein) ⁻¹ min ⁻¹	MIN	4.8	5.4	30	3.82	[5]	Overmann & Pfennig 1992 [BM9, genus, spherical cells 1.5-3 μm diam]	Respiration of cells without microscopically visible sulfure globules at oxygen concentrations of 11-67 μM; respiration rates of phototrophically (anaerobically) and chemotrophically	Purple sulfur bacteria Endogenous respiration of cells with visible sulfur globules is higher, up to 22 nmol O ₂ (mg protein) ⁻¹ min ⁻¹	

Strain		Taxonomy		Respiration rate	Q ₁₀	Q ₁₀ (20-30 °C)	Q ₁₀ (30-40 °C)	Q ₁₀ (40-50 °C)	Q ₁₀ (50-60 °C)	Q ₁₀ (60-70 °C)	Q ₁₀ (70-80 °C)	Q ₁₀ (80-90 °C)	Q ₁₀ (90-100 °C)	Other	Notes
Amoebobaeter pendens (5813)	24.Amoebobaeter pendens	Gammaproteobacteria: Chromatiales		nmol O ₂ (mg protein) ⁻¹ min ⁻¹	MIN	7.6	8.5	30	6.01	[5]	Overmann & Pfennig 1992[BM9, genus, spherical cells 1.5-3 µm diam]	(microaerobically) grown cells do not differ; the species is capable of chemotrophic growth in the dark.	Purple sulfur bacteria		
															Endogenous respiration of cells with visible sulfur globules is higher, up to 35 nmol O ₂ (mg protein) ⁻¹ min ⁻¹
Spirillum (Aquaspirillum) itersonii (ATCC 12639)	25.Aquaspirillum itersonii	Betaproteobacteria: Neisseriales		µl O ₂ (mg DM) ⁻¹ hr ⁻¹	MIN	6	10	30	7.07	[0.9]	Jurtshuk & McQuitty 1976 [Krieg 1976, helical shape, Fig. 1E, Table 3, diam 0.3-0.4 µm, full length ~10 µm] Ensign 1970	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)			
Arthrobacter tallopoietes	26.Arthrobacter crystallopoietes	Actinobacteria: Actinomycetales		µl O ₂ (mg DM) ⁻¹ hr ⁻¹	MIN	0.1	0.2	30	0.14	1.7		Cells harvested during the exponential phase of growth (48 hr for spherical cells, 4-8 hr for rods); stable endogenous respiration during 24 days of starvation at 100% viability	cell mass estimated from the dry mass data for spherical cells (0.5 mg dry mass per 10 ⁹ cells)		
															Endogenous respiration at harvest was about 8-9 µl O ₂ (mg DM) ⁻¹ hr ⁻¹ and decreased 80-fold during the first two days of starvation
															Growing spherical cells contain about 40% (dry mass) of a glycogen-like polysaccharide; rods — 10% (Boylen & Ensign 1970)
															Boylen 1973: Bacteria of this species survived 6 months of extreme desiccation at 50% viability converting 0.0005% of their carbon per hour to carbon dioxide (≈ 10 ⁻² W/kg)
Arthrobacter globiformis (ATCC 8010)	27.Arthrobacter globiformis	Actinobacteria: Actinomycetales		µl O ₂ (mg DM) ⁻¹ hr ⁻¹	MIN	5	8.3	30	5.87	[0.5]	Jurtshuk & McQuitty 1976 [Conn & Dimmick 1947, rods 0.6-0.8×1-1.5 µm]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)			
Arthrobacter globiformis (NCIB)	28.Arthrobacter sp.	Actinobacteria: Actinomycetales		µl O ₂ (mg DM) ⁻¹ hr ⁻¹	MIN	0.45	0.75	25	0.75	0.2	Luscombe & Gray 1974	Cells harvested from continuous cultures	Cocci survive better than rods; initial endogenous respiration was 1.74 and		

10683)													and starved for more than 2 days, steady-state viability approximately 80%.	7.33 μl O ₂ (mg DM) ⁻¹ hr ⁻¹ for cocci and rods, respectively, but “after 2 days these had both fallen to a relatively stable level of 0.45”, which was monitored for 7 days. Cell volume corresponds to the minimum dilution rate (0.01 h ⁻¹); it grows up to 0.96 μm ³ at 0.3 h ⁻¹ . Data from Fig. 4a (induced cells), last 20 min of endogenous respiration; bacteria living in soil of Citrus plantations; rods 2 μm in length Respiration decreases with time (mean 2.3 W/kg) Isolated from soil Bacteria isolated from field soil contaminated by aviation fuel, UK Cell mass estimated from dry mass data, Table 1, 3.8 pg DM/cell
Arthrobacter sp.	29.Arthrobacter sp.	Actinobacteria: Actinomycetales	μl O ₂ (15 mg DM) ⁻¹ min ⁻¹	MIN	3.6	1.2	30	0.85	[1.5]	Devi et al. 1975 [BM9, genus, rods 0.8-1.2×1-8 μm; Devi et al. 1975 indicate their strain is 2 μm in length]	Bacteria grown for 48 hr; endogenous respiration of resting cells measured for 80 min; data taken for the last 20 min			
Arthrobacter (TMP)	sp. 30.Arthrobacter sp.	Actinobacteria: Actinomycetales	μl O ₂ (50 mg WM) ⁻¹ min ⁻¹	MIN	0.9	6	30	4.24		Donnelly et al. 1981	Cells harvested during exponential growth			
Arthrobacter (CA1)	sp. 31.Arthrobacter sp.	Actinobacteria: Actinomycetales	μmol O ₂ (5.3 mg DM) ⁻¹ hr ⁻¹	MIN	2	14	30	9.90		Ougham & Trudgill 1982	Bacteria harvested at late logarithmic phase; respiration measured for 2 hr			
Azotobacter agile	32.Azomonas agilis?	Gammaproteobacteria: Pseudomonadales	μl O ₂ (mg DM) ⁻¹ hr ⁻¹	MIN	12.6	21	26	19.59	13	Gunter & Kohn 1956	Cells harvested from 16 to 18-hr yeast agar plates			
Azorhizobium caulinodans (ORS571)	33.Azorhizobium caulinodans	Alphaproteobacteria: Rhizobiales	nmol O ₂ (mg protein) ⁻¹ min ⁻¹	MIN	34	38	30	26.87	[0.5]	Allen et al. 1991 [BM9, rods 0.5-0.6×1.5-2.5 μm]	Endogenous respiration of cells taken from continuous culture; measured for 10 min			
Azospirillum brasiliense (ATCC 29145)	34.Azospirillum brasiliense	Alphaproteobacteria: Rhodospirillales	μmol O ₂ (mg protein) ⁻¹ min ⁻¹	MIN	0.02	27	37	11.75	[1]	Loh et al. 1984 [BM, species image]	cells harvested during mid log-phase, starved for 4 hr at 4 C; constant respiration rate throughout the experiment (≈4 hr)			
Azospirillum lipoferum (ATCC 29707)	35.Azospirillum lipoferum	Alphaproteobacteria: Rhodospirillales	μmol O ₂ (mg protein) ⁻¹ min ⁻¹	MIN	0.03	39	37	16.98	[4]	Loh et al. 1984 [BM, species image]	cells harvested during mid log-phase, washed and starved for 4 hr at 4 C	Synonym Spirillum lipoferum (BM)		
Azotobacter chroococcum (NCIB 8003)	36.Azotobacter chroococcum	Gammaproteobacteria: Pseudomonadales	μl O ₂ (mg DM) ⁻¹ hr ⁻¹	MIN	15	25	30	17.68	[14]	Bishop et al. 1962 [Bisset & Hale 1958, Figs. 1-3, 7, 13, diam approx. 3 μm]	Respiration measured immediately after harvesting the cells (aerobic culture) at the end of the logarithmic growth phase	Max. resp. (in the presence of glucose) was 130 μl O ₂ (mg DM) ⁻¹ hr ⁻¹		
Azotobacter vinelandii (O)	37.Azotobacter vinelandii	Gammaproteobacteria: Pseudomonadales	μl O ₂ (mg DM) ⁻¹ hr ⁻¹	MIN	0.9	1.5	30	1.06	[0.5]	Sobek et al. 1966 [Tsai et al. 1979, Fig. 3, diam 1 μm, ATCC 12837, ≈0.06 pgDM/cell, Fig. 1]	Respiration of glucose-grown cells harvested at 20 hr and starved for 48 hr (Table 1); viability >95% (Fig. 2)	During starvation respiration diminishes from 4.6-5.8 μl O ₂ (mg DM) ⁻¹ hr ⁻¹ (depending on growth substrate) during the first four hours and to 0.9-1.4 μl O ₂ (mg DM) ⁻¹ hr ⁻¹ between 48th and 52th hours		

													Johnson et al. 1958 report that "well-washed cells of this organism possess no significant endogenous respiration"	Viability and respiration depend on the remaining store of PHB (poly-β-hydroxybutyric acid). 16-hr grown cultures were deprived of significant PHB stores and rapidly lost viability during starvation
Azotobacter vinelandii (O, ATCC 12518)	38.Azotobacter vinelandii	Gamma	μl O ₂ (mg DM) ⁻¹ hr ⁻¹	19	30	30	21.21	[0.5]	Jurtshuk & McQuitty 1976 [Tsai et al. 1979, Fig. 3, diam 1 μm, ATCC 12837, ≈0.06 pgDM/cell, Fig. 1]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	Jurtshuk et al. 1975: Strain O cells grown to the late logarithmic phase, washed and "allowed to sit overnight at 4 C to reduce the intracellular endogenous reserve by starvation"; respiration at 30 C was 9-12 μl O ₂ (mg DM) ⁻¹ hr ⁻¹ = 15-20 W/kg			
Bacillus cereus (C5-25)	39.Bacillus cereus	"Bacilli": Bacillales	μl O ₂ (mg DM) ⁻¹ hr ⁻¹	MIN	8.5	14	37	6.09	[3.7]	Nickerson & Sherman 1952 [BM, species]	Normal (not filamentous) cells grown for 18 hr; respiration measured for about 1.5 hr; data of Table 3	Q ₁₀ for this species in Ingram 1940		
Bacillus (USDA)	cereus	40.Bacillus cereus	μl O ₂ (mg N) ⁻¹ hr ⁻¹	188	31	30	21.92	[3.7]	Dietrich & Burris 1967 [BM, species]	Cells cultured for 2 weeks; respiration measured for 1 hr after 10 min equilibration				
Bacillus cereus	41.Bacillus cereus	"Bacilli": Bacillales	μl O ₂ (mg DM) ⁻¹ hr ⁻¹	56	97	30	68.59	[3.7]	Jurtshuk & McQuitty 1976 [BM, species]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)				
Bacillus firmus (ATCC 14575)	42.Bacillus firmus	"Bacilli": Bacillales	μl O ₂ (mg DM) ⁻¹ hr ⁻¹	MIN	8	13	30	9.19	[0.9]	Jurtshuk & McQuitty 1976 [Kanso et al. 2002, rods 0.6-0.9×1.2-4 μm]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)			
Bacillus megaterium	43.Bacillus megate-	"Bacilli": Bacillales	μl O ₂ (mg MIN	2	3.3	30	2.33	[7]	Jurtshuk &	Cells harvested at the				

Strain	Strain ID	Genus	Family	Respiration rate				Growth rate				References	Notes
				$\mu\text{l O}_2$ (mg DM) ⁻¹ hr ⁻¹	MIN	MAX	MEAN	$\mu\text{l O}_2$ (mg DM) ⁻¹ hr ⁻¹	MIN	MAX	MEAN		
Bacillus megaterium (KM)	44.Bacillus megaterium	Bacillus	"Bacilli": Bacillales	$\mu\text{l O}_2$ (mg DM) ⁻¹ hr ⁻¹		7.7	13	30	9.19	[7]	McQuitty 1976 [Bisset 1953, Fig. 17, rods 1.4×4.3 μm] [[Kubitschek 1969, Coulter counter]] Frederick et al. 1974 [Bisset 1953, Fig. 17, rods 1.4×4.3 μm]	late-logarithmic growth phase (two thirds of the maximal growth concentration)	Obligately aerobic asporogenous strain
Bacillus megaterium (ATCC 19213)	45.Bacillus megaterium	Bacillus	"Bacilli": Bacillales	natoms O (mg protein) ⁻¹ min ⁻¹		<40	25	30	17.68	[7]	Decker & Lang 1977 [Bisset 1953, Fig. 17, rods 1.4×4.3 μm]	Log-phase harvested cells (strain ATCC 19213) respired at a rate of less than 10% of 400 natoms O (mg protein) ⁻¹ min ⁻¹ = 25 W/kg; respiration measured for 1 min	Respiration of unstarved cells was 11 μl O ₂ (mg DM) ⁻¹ hr ⁻¹
Bacillus megaterium (NCTC 9848)	46.Bacillus megaterium	Bacillus	"Bacilli": Bacillales	$\mu\text{l O}_2$ (mg DM) ⁻¹ hr ⁻¹		28	47	37	20.46	[7]	Bishop et al. 1962 [Bisset 1953, Fig. 17, rods 1.4×4.3 μm]	Cells harvested at the end of the logarithmic growth phase	
Bacillus megaterium (KM)	47.Bacillus megaterium	Bacillus	"Bacilli": Bacillales	$\mu\text{l O}_2$ (mg DM) ⁻¹ hr ⁻¹		35	58	25	58.00	[7]	Marquis 1965 [Bisset 1953, Fig. 17, rods 1.4×4.3 μm]	Cells harvested in the phase of declining growth rate; respiration measured for 90 min	
Bacillus popilliae (NRRL B-2309-P)	48.Bacillus popilliae	Bacillus	"Bacilli": Bacillales	$\mu\text{l O}_2$ (mg DM) ⁻¹ hr ⁻¹	MIN	0.5	0.8	30	0.57	[0.8]	Pepper & Costilow 1964 [Mitraka et al. 1967, Fig. 1, rods 0.6×2-4 μm]	Cells harvested in the stationary phase after 24 hr incubation	Bacterium causing "milky disease" of the Japanese beetle (Popilla japonica) larvae Similar respiration rates were measured for B. lentimorbus
Bacillus pumilus (ATCC 70)	49.Bacillus pumilus	Bacillus	"Bacilli": Bacillales	$\mu\text{l O}_2$ (mg DM) ⁻¹ hr ⁻¹	MIN	3	5	30	3.54	[0.7]	Jurtshuk & McQuitty 1976 [Hayase et al. 2004, rods 0.5-0.7×2.0-3.0 μm]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	
Bacillus stearothermophilus (PH24)	50.Bacillus stearothermophilus	Bacillus	"Bacilli": Bacillales	$\mu\text{l O}_2$ (14.2 mg DM) ⁻¹ min ⁻¹ (30)	MIN	35	8.2	50	1.45	[0.7]	Buswell 1975 [Montesinos et al. 1983, Coulter counter]	Cells harvested in late exponential phase (14-16 hr of growth on phenol) at 55 C and either used immediately or stored at -20 C; respiration measured for 30 min	The organism, obligate thermophile, was isolated from industrial sediment at Ravenscraig Steel Works near Motherwell, Scotland.
Bacillus subtilis (W-)	51.Bacillus subtilis	Bacillus	"Bacilli": Bacillales	$\mu\text{l O}_2$ (mg DM) ⁻¹ hr ⁻¹	MIN	2	3.3	30	2.33	[1.4]	Jurtshuk &	Cells harvested at the	Crook 1952 reports 4-5 μl O ₂ (mg DM) ⁻¹ hr ⁻¹

23)				DM) ⁻¹ hr ⁻¹											McQuitty 1976 [Kubitschek 1969, Coulter counter]	late-logarithmic growth phase (two thirds of the maximal growth concentration); respiration was 2 µl O ₂ (mg DM) ⁻¹ hr ⁻¹ for B. subtilis W-23 and B. (vulgatus) subtilis	DM) ⁻¹ hr ⁻¹ for spores after heat shock and 0.3 for untreated spores at 25 °C; Nitrogen content B. subtilis N/DM=0.111; B. cereus N:DM=0.128 Bohin et al. 1976: sporulating bacteria: 90 µl O ₂ (0.9 mg DM) ⁻¹ (40 min) ⁻¹ = 250 W/kg at 37 C
Bacillus subtilis (Marburg strain C4)	52.	Bacillus subtilis	“Bacilli”: Bacillales	µl O ₂ (5 mg DM) ⁻¹ (700 min) ⁻¹		225	6.5	37	2.83	[1.4]					Gary & Bard 1952 [Kubitschek 1969, Coulter counter]	"Resting cell suspensions were prepared by harvesting cells from flask cultures after 6 to 8 hr incubation at 37 C, the periods of maximum respiratory and fermentative activity." Respiration of cells grown on "complex medium" (Fig. 7) (C-cells) was 225 µl O ₂ (5 mg DM) ⁻¹ (700 min) ⁻¹ = 6.5 W/kg; cells grown on "simple medium" (S-cells) respired at 600 µl O ₂ (5 mg DM) ⁻¹ (700 min) ⁻¹ = 17 W/kg	
Bacillus subtilis (ATCC 6633)	53.	Bacillus subtilis	“Bacilli”: Bacillales	µmol O ₂ (100 mg DM) ⁻¹ (2 hr) ⁻¹		29.8	5.6	30	3.96	[1.4]					Gronlund & Campbell 1961 [Kubitschek 1969, Coulter counter]	Cells harvested after 20 hr growth; respiration measured for 2 hr	
Bacillus subtilis (D76)	54.	Bacillus subtilis	“Bacilli”: Bacillales	µl O ₂ (5.7 mg DM) ⁻¹ (10 min) ⁻¹		20	32	30	22.63	[1.4]					Clifton & Cherry 1966 [Kubitschek 1969, Coulter counter]	Cells harvested after 16 hr growth at 30 C respired initially at a rate of 60 µl O ₂ (5.7 mg DM) ⁻¹ (10 min) ⁻¹ = 105 W/kg; this rate decreased approximately threefold by the end of 160 min' measurements (Fig. 1)	
Bacillus subtilis (NCTC 9848)	55.	Bacillus subtilis	“Bacilli”: Bacillales	µl O ₂ (mg DM) ⁻¹ hr ⁻¹		24	75	37	32.65	[1.4]					Bishop et al. 1962 [Kubitschek 1969, Coulter counter]	Respiration measured immediately after harvesting the cells at the end of the logarithmic growth phase; respiration of spores was below detection limit.	
Bdellovibrio bacte-	56.	Bdellovibrio	Deltaproteobacteria:	µl O ₂ (mg MIN		14.8-	25	30	17.68	0.3					Hespell et al.	Cells incubated for 18-	Bacterium-predator attacking E. coli

Table 1. Growth characteristics of the strains used in the study											
Strain	Host	Strain	Phylum	Medium	Temperature (°C)	Incubation time (h)	Optical density (OD ₆₀₀)	Yield (g/L)	Yield (g/kg)	Yield (g/kg)	References
riovorus (109J)	bacteriovorus	Bdellovibrionales	DM) ⁻¹ hr ⁻¹	17.3							1973 [DM=10 ⁻¹³ pg/cell]
											24 hr in a medium containing E. coli cells that were all lyzed by the time of harvest; respiration measured for 2-4 hr
											Rittenberg & Shilo 1970 report 0.42 mg protein per 10 ¹⁰ cells of strain 109
											Straley et al. (1979) characterize this respiration rate as "unusually high"; Friedberg & Friedberg (1976) as "extremely high"
											Rittenberg & Shilo 1970: 13-27 nmol O ₂ (0.42 mg protein) ⁻¹ min ⁻¹ = 35-72 W/kg at 30 C
Pseudomonas triegens	na-	57.Beneckea triegens	na-	Gammaproteobacteria: "Vibrionales"	μl O ₂ (1.07 mg DM) ⁻¹ min ⁻¹	MIN	85	265	30	187.38	[1.5]
											Cho & Eagon 1967 [Baumann et al. 1971, Figs. 12, 16, rods 0.6-1.2×1.9-3.6 μm depending on culture conditions?]
											Cells harvested at the end of the logarithmic phase; respiration of resting cells measured for 45 min
											Oxygen uptake with all substrates is characterized as "low" (the lowest with glucose, 268 μl O ₂ (mg DM) ⁻¹ hr ⁻¹ = 450 W/kg is within the upper range of maximum specific metabolic rates in bacteria
											Marine bacterium with shortest known generation time (9.8 min) (Eagon 1962)
											Synonym Vibrio natriegens
Bradyrhizobium japonicum (I-110)		58.Bradyrhizobium japonicum	Alphaproteobacteria: Rhizobiales	nmol O ₂ (mg protein) ⁻¹ hr ⁻¹	MIN	53	1.0	29	0.76	[0.7]	Frustaci et al. 1991 [BM]
Neisseria catarrhalis		59.Branhamella catarrhalis	Gammaproteobacteria: Pseudomonadales	μl O ₂ (mg DM) ⁻¹ hr ⁻¹	MIN	1	1.7	37	0.74	[1.3]	Bishop et al. 1962 [Baumann et al. 1968, diam 1.0-1.7 μm, coccoid]
											Respiration of late-log phase cells
											Respiration measured immediately after harvesting the cells at the end of the logarithmic growth phase
											Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)
											Cells grown for 24 hr; respiration varies among equally prepared suspensions
Branhamella catarrhalis (Gp4)	catarrhalis	60.Branhamella catarrhalis	Gammaproteobacteria: Pseudomonadales	μl O ₂ (mg DM) ⁻¹ hr ⁻¹		14	23	30	16.26	[1.3]	Jurtshuk & McQuitty 1976 [Baumann et al. 1968, diam 1.0-1.7 μm, coccoid]
											Cells grown for 15-18 hr
											Bacterium isolated from Miami soil
Brucella abortus (19)	abortus	61.Brucella melitensis	Alphaproteobacteria: Rhizobiales	μl O ₂ (mg N) ⁻¹ hr ⁻¹	MIN	40-82	6.5	34	3.48	[0.3]	Gerhard et al. 1950 [BM9, genus, cocci or short rods 0.5-0.7×0.6-1.5 μm]
											Cells grown for 15-18 hr
											Bacterium isolated from Miami soil
Burkholderia sp. (JT 1500)		62.Burkholderia sp.	Betaproteobacteria: Burkholderiales	nmol O ₂ (mg DM) ⁻¹ min ⁻¹	MIN	19	21	30	14.85	0.7	Morawski et al. 1997 [rods 0.5-0.8×1.5-3.0]
											Cells grown for 15-18 hr
											Bacterium isolated from Miami soil
Cellvibrio gilvus		63.Cellvibrio gilvus	Gammaproteobacteria: Pseudomonadales	μl O ₂ (3 mg wet packed cells) ⁻¹ (30 min) ⁻¹	MIN	10	37	30	26.16	2	Hulcher & King 1958a [Hulcher & King 1958b, rods 0.75-1.5×1.5-3.75 μm]
											Cells grown on cellobiose for 18 hr at room temperature
											Cells in young cultures are very large (3×10 μm), 24-hr cells grown on cellulose were 0.3-0.5×0.8-1.5 μm (Hulcher & King 1958b)
											There is no such species at http://www.bacterio.cict.fr ; higher taxon as for Cellvibrio genus.

Chromatium sp. (Miami PBS1071)	64.Chromatium sp.		Gamma	Chromatiales	nmol O ₂ (mg DM) ⁻¹ min ⁻¹	MIN	4.8	11	25	11.00		Kumazawa et al. 1983 [BM9 genus: rods 1-6×1.5-15 µm]	2-3 days' old cultures grown anaerobically in the light; respiration increased severalfold upon addition of H ₂ and fell to the endogenous level after H ₂ was exhausted	Marine purple sulfur bacterium
Chromatium vinosum (2811)	65.Chromatium vinosum		Gamma	Chromatiales	nmol O ₂ (mg protein) ⁻¹ min ⁻¹	MIN	2.0	2.2	30	1.56	[1.5] [[1.21]]	Overmann & Pfennig 1992 [Montesinos et al. 1983, Coulter counter] [[Mas et al. 1985]]	Respiration of cells without microscopically visible sulfure globules at oxygen concentrations of 11-67 µM; respiration rates of phototrophically (anaerobically) and chemotrophically (microaerobically) grown cells do not differ; the species is capable of chemotrophic growth in the dark.	Purple sulfur bacteria Respiration rate increases with oxygen concentration reaching a plateau at approx. 6 µM. Endogenous respiration of cells with visible sulfur globules is higher, up to 24 nmol O ₂ (mg protein) ⁻¹ min ⁻¹
Corynebacterium diphtheriae (ATCC 11913)	66.Corynebacterium diphtheriae		Actinobacteria:	Actinomycetales	µl O ₂ (mg DM) ⁻¹ hr ⁻¹	MIN	4	6.7	30	4.74		Jurtshuk & McQuitty 1976 [	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	
Corynebacterium sp.	67.Corynebacterium sp.		Actinobacteria:	Actinomycetales	µl O ₂ (100 mg WM) ⁻¹ (120 min) ⁻¹	MIN	180	10	30	7.07		Levine & Kram-pitz 1952	Cells harvested after 2-4 days growth; respiration measured for 2 hr	A soil-isolated bacterium sometimes capable of acetone degradation
Pseudomonas acidovorans (ATCC 17455=NCIB 10013)	68.Delftia acidovorans		Betaproteobacteria:	Burkholderiales	µl O ₂ (4 mg DM) ⁻¹ min ⁻¹	MIN	0.5	13	30	9.19	[0.8]	Hareland et al. 1975 [Wen et al. 1999, rods, 0.4-0.8×2.5-4.1 µm]	Bacteria grown for 20 hr (end of logarithmic growth); respiration measured for about 10 min	Bacteria originally isolated from poultry house deep-litter
Pseudomonas B ₂ aba	69.Delftia acidovorans		Betaproteobacteria:	Burkholderiales	µl O ₂ (mg DM) ⁻¹ hr ⁻¹		8	13	30	9.19	[0.8]	Kornberg & Gotto 1961 [Wen et al. 1999, rods, 0.4-0.8×2.5-4.1 µm]	Cells grown on glycollate harvested during the logarithmic phase; for succinate-grown cells 10 µl O ₂ (mg DM) ⁻¹ hr ⁻¹ (17 W/kg). Cooper & Kornberg 1964: cells grown on itaconate harvested during the logarithmic phase, 39 µl O ₂ (mg DM) ⁻¹ hr ⁻¹ = 65 W/kg; for succinate-grown	

Pseudomonas desmolytica (S449B1)	70.Delftia vorans	acidovorans	Betaproteobacteria: Burkholderiales	$\mu\text{l O}_2$ (0.6 mg DM) ⁻¹ (160 min) ⁻¹		14	15	30	10.61	[0.8]	Jigami et al. 1979 [Wen et al. 1999, rods, 0.4-0.8×2.5-4.1 μm]	cells 40 $\mu\text{l O}_2$ (mg DM) ⁻¹ hr ⁻¹ Cells harvested after 3 days of growth	
Pseudomonas acidovorans (ATCC 15688)	71.Delftia vorans	acidovorans	Betaproteobacteria: Burkholderiales	$\mu\text{l O}_2$ (mg DM) ⁻¹ hr ⁻¹		18	30	30	21.21	[0.8]	Jurtshuk & McQuitty 1976 [Wen et al. 1999, rods, 0.4-0.8×2.5-4.1 μm]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	
Desulfovibrio salexigens (Mast1)	72.Desulfovibrio salexigens		Deltaproteobacteria: Desulfovibrionales	nmol O ₂ (mg protein) ⁻¹ min ⁻¹	MIN	12	13	30	9.19	[1.5]	van Niel & Gottschal 1998 [BM]	Cells harvested at the end of the exponential growth phase	Organism isolated from the oxic-anoxic (top) layer of a marine microbial mat, Greece
Enterobacter aerogenes	73.Enterobacter aerogenes		Gammaproteobacteria: “Enterobacteriales”	$\mu\text{l O}_2$ (mg DM) ⁻¹ hr ⁻¹	MIN	6	10	30	7.07	[0.3]	Jurtshuk & McQuitty 1976 [Fu et al. 2003, rods 0.6×1.2 μm]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	
Enterobacter cloacae (17/97)	74.Enterobacter cloacae		Gammaproteobacteria: “Enterobacteriales”	nmol O ₂ (mg DM) ⁻¹ min ⁻¹	MIN	13.9	31	30	21.92	[0.9]	Majtán & Majtánová 1999 [BM9, genus, rods 0.6-1.0×1.2-3.0 μm]	Cells harvested from the exponential phase; respiration measured for 10 min	Bacterium isolated from a patient suffering from nosocomial infection.
Enterococcus cecorum (DSM 20682 ¹)	75.Enterococcus cecorum		“Bacilli”: “Lactobacillales”	nmol O ₂ (mg DM) ⁻¹ min ⁻¹	MIN	1.7	3.8	30	2.69	[0.4]	Bauer et al. 2000 [BM]	Respiration of cells from aerobically glucose-grown cultures; respiration of anaerobically grown cells <1 nmol O ₂ (mg DM) ⁻¹ min ⁻¹	
Streptococcus faecalis (NCDO 581)	76.Enterococcus faecalis		“Bacilli”: “Lactobacillales”	$\mu\text{l O}_2$ (mg DM) ⁻¹ hr ⁻¹	MIN	0.16	0.3	30	0.21	[0.2]	Bryan-Jones & Whittenbury 1969 [BM]	respiration of resting suspensions of cells grown aerobically with glucose	
Streptococcus faecalis (ATCC 8043)	77.Enterococcus hirae		“Bacilli”: “Lactobacillales”	$\mu\text{mol O}_2$ (100 mg DM) ⁻¹ (2 hr) ⁻¹	MIN	8.0	1.5	30	1.06		Gronlund & Campbell 1961	Cells harvested after 17 hr growth; respiration measured for 2 hr	
Enterococcus (ATCC 8043, ATCC 9790)	78.Enterococcus hirae		“Bacilli”: “Lactobacillales”	$\mu\text{l O}_2$ (mg DM) ⁻¹ hr ⁻¹		1	1.7	30	1.20		Jurtshuk & McQuitty 1976	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	
Enterococcus (RfL6)	79.Enterococcus sp.		“Bacilli”: “Lactobacillales”	nmol O ₂ (mg DM) ⁻¹ min ⁻¹	MIN	15.1	33	30	23.33	0.8	Bauer et al. 2000 [rods, 0.6-1×1.1-3 μm]	Respiration of cells from aerobically glucose-grown cultures; respiration of anaerobically grown cells	

Strain		Reference		Medium		Temperature		pH		Growth rate		Respiration rate		Notes	
Escherichia coli (EMG-2 K12 Ymel)	80.	Escherichia coli	Gammaproteobacteria: "Enterobacteriales"	ngatom O (mg protein) ⁻¹ min ⁻¹	MIN	58-86	32	37	14	[0.7]	Lawford & Had-dock 1973 [Kubitschek 1969, Coulter counter, 0.33-1.46 μm ³ depending on growth rate]	15.1 nmol O ₂ (mg DM) ⁻¹ min ⁻¹	Cells were grown to the early exponential phase and starved for 2 hr by vigorous shaking at 37 C; respiration varied (depending on carbon source for growth		
Escherichia coli (W-1485)	81.	Escherichia coli	Gammaproteobacteria: "Enterobacteriales"	μl O ₂ (mg DM) ⁻¹ hr ⁻¹		2.7	4.5	26	4.20	[0.7]	Gunter & Kohn 1956 [Kubitschek 1969, Coulter counter, 0.33-1.46 μm ³ depending on growth rate]		Cells harvested from 16 to 18-hr yeast agar plates		
Escherichia coli (O11a ₁)	82.	Escherichia coli	Gammaproteobacteria: "Enterobacteriales"	μl O ₂ (mg DM) ⁻¹ hr ⁻¹		4	6.7	30	4.74	[0.7]	Jurtshuk & McQuitty 1976 [Kubitschek 1969, Coulter counter, 0.33-1.46 μm ³ depending on growth rate]		Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	Respiration was 13, 7, 9, 10, 6, 8, 7, 6, and 5 μl O ₂ (mg DM) ⁻¹ hr ⁻¹ for strains B, B/r, C, Crookes, K12, K12S UTH 2593, K12S UTH 3672, and F , respectively (8.3-22 W/kg).	
Escherichia coli	83.	Escherichia coli	Gammaproteobacteria: "Enterobacteriales"	μl O ₂ (mg DM) ⁻¹ hr ⁻¹		7	12	37	5.22	[0.7]	Dawes & Ribbons 1965 [Kubitschek 1969, Coulter counter, 0.33-1.46 μm ³ depending on growth rate]		Stable respiration of aerobically grown cells harvested during the exponential phase and starved aerobically for 150-180 min; no loss of viability during 12 hr of starvation	"Stationary-phase cells respire endogenously at higher rates and contain larger reserves of glycogen, which is the initial substrate oxidized [than exponential-phase cells]. Carbohydrate content of exponential-phase cells drops rapidly together with endogenous respiration during the first 100 min of starvation	
Escherichia coli (B)	84.	Escherichia coli	Gammaproteobacteria: "Enterobacteriales"	μl O ₂ (10 mg DM) ⁻¹ (2 hr) ⁻¹		180	15	35	7.50	[0.7]	Sobek & Talburt 1968 [Kubitschek 1969, Coulter counter, 0.33-1.46 μm ³ depending on growth rate]		Cells grown for 20 hr at 35 C; respiration measured for 2 hr		
Escherichia coli (ATCC 4157)	85.	Escherichia coli	Gammaproteobacteria: "Enterobacteriales"	μl O ₂ (mg DM) ⁻¹ hr ⁻¹		12	20	37	8.71	[0.7]	Davis & Bateman 1960 [Kubitschek 1969, Coulter counter, 0.33-1.46 μm ³ depending on growth rate]		Cells harvested after 16.5 hr growth at 37 C; respiration measured for 2 hr		
Escherichia coli (ATCC 6894)	86.	Escherichia coli	Gammaproteobacteria: "Enterobacteriales"	μmol O ₂ (100 mg DM) ⁻¹ (2 hr) ⁻¹		79	15	30	10.61	[0.7]	Gronlund & Campbell 1961 [Kubitschek 1969, Coulter counter, 0.33-1.46 μm ³ depending on		Cells harvested after 20 hr growth; respiration measured for 2 hr		

Table 1. Respiration of various bacteria											References		Notes	
Strain	Species	Strain	Phylum	Class	Units	MIN	MAX	MIN	MAX	Notes	References	Notes	Notes	Notes
Escherichia (Gratia)	coli	87.Escherichia coli	Gammaproteobacteria: "Enterobacteriales"	$\mu\text{l O}_2 \text{ (mg N)}^{-1} \text{ hr}^{-1}$		105	17.5	30	12.37	[0.7]	growth rate] Dietrich & Burris 1967 [Kubitschek 1969, Coulter counter, 0.33-1.46 μm^3 depending on growth rate]	Cells cultured for 2 weeks; respiration measured for 1 hr after 10 min equilibration	Bacteria "had a lower endogenous [respiration] ... if harvested during exponential growth than after reaching the stationary phase"	
Escherichia coli		88.Escherichia coli	Gammaproteobacteria: "Enterobacteriales"	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$		37	62	37	26.99	[0.7]	Bishop et al. 1962 [Kubitschek 1969, Coulter counter, 0.33-1.46 μm^3 depending on growth rate]	Respiration measured immediately after harvesting the cells (aerobic culture) at the end of the logarithmic growth phase	Anaerobically grown culture respired at 2 $\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$ = 3.3 W/kg	
Flavobacterium capsulatum (ATCC 14666)		89.Flavobacterium capsulatum	Flavobacteria: Flavobacteriales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	38	63	30	44.55	[0.3]	Jurtshuk & McQuitty 1976 [BM9, genus, rods 0.5x1.0-3.0 μm]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	Listed in ATCC as Novosphingobium capsulatum	
Pasteurella tularensis (SCHU S4, 38 A)		90.Francisella tularensis	Gammaproteobacteria: Thiotrichales	$\mu\text{l O}_2 \text{ (mg N)}^{-1} \text{ hr}^{-1}$	MIN	22	3.7	37	1.61	[0.01]	Weinstein et al. 1962 [BM9, rods 0.2x0.2-0.7 μm]	Cells grown for 16 to 18 hr; respiration of strains 503, CHUR, JAP, LVS, and MAX was 60, 38, 53, 105, and 94 $\mu\text{l O}_2 \text{ (mg N)}^{-1} \text{ hr}^{-1}$, respectively (6.3-17.5 W/kg)		
Frankia (EAN1 _{pec})	sp.	91.Frankia sp.	Actinobacteria: Actinomycetales	$\text{nmol O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	242	9	25	9.00		Tisa & Ensign 1987 [BM, genus, hyphae 0.5-2.0 μm diam]	Bacteria grown for 21 days	Members of the genus <i>Frankia</i> are filamentous actinomycetes that infect roots and induce nodule formation in a variety of woody dicotyledonous plants.	
Haemophilus influenzae (641b)	influenzae	92.Haemophilus influenzae	Gammaproteobacteria: Pasteurellales	$\mu\text{l O}_2 \text{ (mg N)}^{-1} \text{ hr}^{-1}$	MIN	3	0.5	37	0.22	[0.14]	Biberstein & Spencer 1962 [BM, coccobacilli or small regular rods 0.3-0.5 - 0.5-3.0 μm]	Bacteria grown for 18-24 hr at 37 C; respiration of washed cells measured for 60 min	Respiration of strains b, c, d, 525, and K 75 was 14, 6, 7, 32, and 9 $\mu\text{l O}_2 \text{ (mg N)}^{-1} \text{ hr}^{-1}$, respectively (1-5.3 W/kg).	
Haemophilus parahaemolyticus (9796)		93.Haemophilus parahaemolyticus	Gammaproteobacteria: Pasteurellales	$\mu\text{l O}_2 \text{ (mg N)}^{-1} \text{ hr}^{-1}$	MIN	21	3.5	37	1.52		Biberstein & Spencer 1962	Bacteria grown for 18-24 hr at 37 C; respiration of washed cells measured for 60 min	Respiration of strain 7901 was 24 $\mu\text{l O}_2 \text{ (mg N)}^{-1} \text{ hr}^{-1}$ (4 W/kg).	
Haemophilus parainfluenzae (K 98)	parainfluenzae	94.Haemophilus parainfluenzae	Gammaproteobacteria: Pasteurellales	$\mu\text{l O}_2 \text{ (mg N)}^{-1} \text{ hr}^{-1}$	MIN	21	3.5	37	1.52	[0.4]	Biberstein & Spencer 1962 [Kahn 1981, Fig. 1, 0.6x1.3-1.7 μm ; Kowalski et al. 1991, diam 0.75-1.25 μm]	Bacteria grown for 18-24 hr at 37 C; respiration of washed cells measured for 60 min	Respiration of strains K 8, K 17, and K 45 was 24, 32, and 41 $\mu\text{l O}_2 \text{ (mg N)}^{-1} \text{ hr}^{-1}$, respectively (4-6.8 W/kg).	
Halobacterium salinarum (1)		95.Halobacterium salinarum	Archaea: Halobacteria: Halobacteriales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	10	17	30	12.02	[3.9]	Stevenson 1966 [Mescher & Strominger 1976,	Cells grown for about 70 hr to the end of the exponential phase	White 1962: Stationary-phase cells (>15 hr old) have an insignificant endogenous respiration. Log-phase cells have a small endogenous respiratory rate. An extremely halophilic bacterium	

Micrococcus denitrificans	halo-	96.Halomonas halodenitrificans	Gammaproteobacteria: Oceanospirillales	$\mu\text{l O}_2 \text{ (2 mg DM)}^{-1} \text{ (30 min)}^{-1}$	MIN	40	67	25	67.00	[0.4]	rods 0.5×5 μm] Sierra & Gibbons 1962 [Ventosa et al. 1998, rods, 0.5-0.9×0.9-1.2 μm]	Cells harvested towards the end of the logarithmic phase (about 40 hr)	Endogenous respiration and viability of starved cells remained constant at around 40 $\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$ and 100%, respectively, for 3 hr, while the amount of intracellular polyester rapidly declined. After 3 hr both endogenous respiration and viability started to decline rapidly. In polyester-poor cells this process was initiated immediately after the beginning of starvation.
Aerobacter aerogenes	aerogenes	97.Klebsiella pneumoniae? Enterobacter aerogenes?	Gammaproteobacteria: "Enterobacteriales"	$\mu\text{l O}_2 \text{ (mg N)}^{-1} \text{ hr}^{-1}$		67	11	30	7.78		Dietrich & Burris 1967	Bacteria cultured for 2 weeks in an extract from wheat plants; respiration measured for 1 hr after 10 min equilibration	Bacteria "had a lower endogenous [respiration] ... if harvested during exponential growth than after reaching the stationary phase"
Aerobacter aerogenes	aerogenes	98.Klebsiella pneumoniae? Enterobacter aerogenes?	Gammaproteobacteria: "Enterobacteriales"	$\mu\text{mol O}_2 \text{ (100 mg DM)}^{-1} \text{ (2 hr)}^{-1}$		69.3	13	30	9.19		Gronlund & Campbell 1961	Cells harvested after 20 hr growth	
Klebsiella pneumoniae (M5a1)		99.Klebsiella pneumoniae	Gammaproteobacteria: "Enterobacteriales"	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	2	3.3	30	2.33	[0.4]	Jurtshuk & McQuitty [BM] 1976	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	
Klebsiella pneumoniae (ATCC 13882)		100.Klebsiella pneumoniae	Gammaproteobacteria: "Enterobacteriales"	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$		4	6.7	30	4.74	[0.4]	Jurtshuk & McQuitty [BM] 1976	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	
Aerobacter aerogenes (NCTC 418)		101.Klebsiella pneumoniae	Gammaproteobacteria: "Enterobacteriales"	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$		12	20	37	8.71	[0.4]	Bishop et al. 1962 [BM]	Respiration measured immediately after harvesting the cells at the end of the logarithmic growth phase	
Lactobacillus brevis (12)		102.Lactobacillus brevis	"Bacilli": "Lactobacillales"	$\mu\text{l O}_2 \text{ (12 mg DM)}^{-1} \text{ (2 hr)}^{-1}$	MIN	2.9	0.2	30	0.14	[1.6]	Walker [BM, rods 0.7-1.0×2-4] 1959	Cells (strain 1.2) were grown for 48 hr; endogenous respiration measured for 3 hr (Fig. 3); mean respiration rate for the second and third hour was taken; during the first hour, respiration was 4 times higher	Bacterium originally isolated from New Zealand cheddar cheese Respiration decreases with starvation time
Lactobacillus casei subsp. rhamnosus (ATCC 7469)		103.Lactobacillus casei	"Bacilli": "Lactobacillales"	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	1	1.7	30	1.20		Jurtshuk & McQuitty 1976	Cells harvested at the late-logarithmic growth phase (two	

Streptococcus lactis (strains 8, 9, 32)	104.Lactococcus lactis	“Bacilli”: “Lactobacillales”	μl O ₂ (115 mg DM) ⁻¹ (4 hr) ⁻¹	MIN	170	0.6	30	0.42	[0.2]	Spendlove et al. 1957 [BM]	thirds of the maximal growth concentration) Cells grown for 11 hr; for 7- and 9-hr grown cells respiration was around 400 μl O ₂ (115 mg DM) ⁻¹ (4 hr) ⁻¹	In 11-hr cells respiration decreases in the first 30 min of the 4-hr’ measurement, then remains relatively constant; in 7-hr and 9-hr cells respiration first decreases, then starts to increase again after 2-3 hr.
Lactococcus ssp. lactis (DSM 20481 ^T)	105.Lactococcus lactis	“Bacilli”: “Lactobacillales”	nmol O ₂ (mg DM) ⁻¹ min ⁻¹		<1	2.3	30	1.63	[0.2]	Bauer et al. 2000 [BM]	Respiration of cells from aerobically and anaerobically glucose-grown cultures	
Lactococcus sp. (TmLO5)	106.Lactococcus sp.	“Bacilli”: “Lactobacillales”	nmol O ₂ (mg DM) ⁻¹ min ⁻¹	MIN	<1	2.3	30	1.63	0.8	Bauer et al. 2000 [rods, 0.6-1×1.1-3 μm]	Respiration of cells from aerobically and anaerobically glucose-grown cultures	
Legionella pneumophila (Knoxville-1, serotype 1)	107.Legionella pneumophila	Gammaproteobacteria: Legionellales	μl O ₂ (mg DM) ⁻¹ min ⁻¹	MIN	0.2	20	37	8.71	[0.3]	Tesh et al. 1983 [Faulkner & Garduño 2002, rods 0.3-0.5×1.5-3.0 μm, prereplicative phase; Kowalski et al. 1999, 0.3-0.9×2 μm]	Bacteria harvested at mid- to late exponential growth phase (15-18 hr growth time) when the mass-specific oxygen consumption is highest (~70 W/kg); washed; “held at 37 C until used”; calibrated for 3 min; respiration measured when “a steady rate of endogenous respiration was established”.	
Legionella pneumophila (serogroup 1)	108.Legionella pneumophila	Gammaproteobacteria: Legionellales	μl O ₂ (400 μg protein) ⁻¹ (40 min) ⁻¹		25	77	37	33.52	[0.3]	Bonach & Snyder 1983 [Faulkner & Garduño 2002, rods 0.3-0.5×1.5-3.0 μm, prereplicative phase; Kowalski et al. 1999, 0.3-0.9×2 μm]]	Cells harvested at mid-log phase, shaken for 30 min, respiration measured for 40 min	
Pseudomonas AM1 (NCIB 9133)	109.Methylobacterium extorquens	Alphaproteobacteria: Rhizobiales	nmol O ₂ (mg protein) ⁻¹ min ⁻¹	MIN	5	6	30	4.24	[1]	Keevil & Anthony 1979 [Peel & Quayle 1961, rods 0.8×2.0 μm]	Cells harvested at the end of logarithmic phase; depending on growth conditions, respiration ranged from 5 to 16 orig. units; the same result obtained by O’Keeffe & Anthony 1978	
Methylomicrobium sp. (AMO 1)	110.Methylomicrobium sp.	Gammaproteobacteria: Methylococcales	nmol O ₂ (mg protein) ⁻¹ min ⁻¹	MIN	10	11	30	7.78	3	Sorokin et al. 2000 and personal communication	Cells grown with methane at pH 10 Endogenous respira-	

											with Dr. D.Yu. Sorokin (15 Nov 2006) (ovoid rods, 1-1.5×2-3 μm)	tion is usually below 10 nmol O ₂ (mg protein) ⁻¹ min ⁻¹ , except in cells grown on acetate and at high pH (10.8-11.5), when it can be 20-50 nmol O ₂ (mg protein) ⁻¹ min ⁻¹	
Methylophilus methylotrophus (NCIB 10515)	111.Methylophilus methylotrophus	Betaproteobacteria: Methylophiliales	ng-atoms O (mg DM) ⁻¹ min ⁻¹	MIN	1.4	1.6	40	0.57	[0.15]	Dawson & Jones 1981 [Jenkins et al. 1987, rods 0.3-0.6×0.8-1.5 μm]	Cells harvested from continuous culture and used within 3 hr	Obligate methylotroph using methanol as the sole source of carbon and energy	
Methylosinus trichosporium OB3b (ATCC 35070) Sarcina lutea	112.Methylosinus trichosporium	Alphaproteobacteria: Rhizobiales	nmol O ₂ (mg protein) ⁻¹ min ⁻¹	MIN	38	42	30	29.70	[1]	Lontoh et al. 1999 [Reed & Dugan 1978, Fig. 1]	Cells harvested in the exponential phase	Cell mass estimated from cell linear dimensions as shown in Fig. 1 of Reed & Dugan 1978	
	113.Micrococcus luteus	Actinobacteria: Actinomycetales	μl O ₂ (mg DM) ⁻¹ hr ⁻¹	MIN	0.7	1.2	37	0.52	[1.1]	Burleigh & Dawes 1967 [BM]	Cells were harvested after 24 hr growth on peptone; starved for 29 hr; viability 96%; respiration fell to "barely measurable" (0.3 μl O ₂ (mg DM) ⁻¹ hr ⁻¹) when starvation was prolonged to 72 hr with viability falling to 25%	Endogenous respiration decreased during starvation most rapidly in the first 5 hr of starvation (from 21.1 to 0.7 orig. units)	
											Bishop et al. 1962: respiration measured immediately after harvesting the cells at the end of the logarithmic growth phase was below detection limit (0 orig. units) (148 μl O ₂ (mg DM) ⁻¹ hr ⁻¹ in the presence of lactate)	Initial endogenous respiration depended on the time of harvesting: mid-exponential (21 hr) — 30 orig. units; onset of the stationary phase (34 hr) — 15.9; 60 hr — 6.3 orig. units; the decline of respiration is correlated with the decline of intracellular contents of free amino acids, carbohydrate etc.	
Sarcina lutea	114.Micrococcus luteus	Actinobacteria: Actinomycetales	μl O ₂ (mg DM) ⁻¹ hr ⁻¹		3.5	6	35	3.00	[1.1]	Dawes & Holms 1958 [BM]	Cells harvested after 24 hr growth at 35 C and aerated for 5 hr; respiration 3.5 μl O ₂ (mg DM) ⁻¹ hr ⁻¹ = 6 W/kg	Mathews & Siström 1959: endogenous respiration is reduced by 75% by shaking for 3 hr at 34 C (cells harvested in the exponential phase after 10-12 hr incubation at 34 C)	
Micrococcus lysodeikticus	115.Micrococcus luteus	Actinobacteria: Actinomycetales	μl O ₂ (mg DM) ⁻¹ hr ⁻¹		11	18	37	7.83	[1.1]	Davis & Bateman 1960 [BM]	Cells harvested after 16.5 hr growth at 37 C; respiration measured for 2 hr		
Sarcina flava	116.Micrococcus luteus	Actinobacteria: Actinomycetales	μl O ₂ (mg DM) ⁻¹ hr ⁻¹		9	15	30	10.61	[1.1]	Jurtshuk & McQuitty [BM] 1976	Cells harvested at the late-logarithmic growth phase (two		

Table 1. Micrococci and other bacteria isolated from soil											
Strain	Isolation source	Genus	Family	Medium	Temperature	Time	pH	Salinity	Moisture	Respiration	Notes
Micrococcus (S9)	117. Micrococcus sp.	Actinobacteria: Actinomycetales	$\mu\text{l O}_2 \text{ (4 mg DM)}^{-1} \text{ min}^{-1}$	MIN	0.40	10	30	7.07			thirds of the maximal growth concentration); respiration was 16, 9 and 30 $\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$ for Micrococcus (Sarcina) luteus, M. (S. flava) luteus and M. (lyso-deikticus) luteus ATCC 4698, respectively (27, 15 and 50 W/kg)
Micrococcus sp.	118. Micrococcus sp.	Actinobacteria: Actinomycetales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	10	17	30	12.02			Cells harvested during the exponential phase
Moraxella osloensis	119. Moraxella osloensis	Gammaproteobacteria: Pseudomonadales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	8	13	30	9.19	[2.3]		Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)
Mycobacterium fortuitum	120. Mycobacterium fortuitum	Actinobacteria: Actinomycetales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	10	17	30	12.02			Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)
Mycobacterium leprae	121. Mycobacterium leprae	Actinobacteria: Actinomycetales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	2	3.3	37	1.44			Cells isolated from mice
Mycobacterium lepraemurium (Hawaiian strain)	122. Mycobacterium lepraemurium	Actinobacteria: Actinomycetales	$\mu\text{l O}_2 \text{ (mg N)}^{-1} \text{ hr}^{-1}$	MIN	20-70	3.3	37	1.44			Culture isolated from infected rats; occasionally, some suspensions respired less than 5 $\mu\text{l O}_2 \text{ (mg N)}^{-1} \text{ hr}^{-1}$
Mycobacterium phlei (72)	123. Mycobacterium phlei	Actinobacteria: Actinomycetales	$\mu\text{mol O}_2 \text{ (mg N)}^{-1} \text{ hr}^{-1}$	MIN	4.8	14	37	6.09	[0.4]		Glucose-grown cells (strain 72) harvested from 5-day culture; respiration calculated assuming N/DM ratio of 0.08 (Tepper 1968); respiration measured for a minimum of 90 min; glycerol-grown cells had 7.3 $\mu\text{mol O}_2 \text{ (mg N)}^{-1} \text{ hr}^{-1}$ = 16 W/kg

Mycobacterium phlei	124.Mycobacterium phlei	Actinobacteria: Actinomycetales	Acti-	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	(mg	16	27	30	19.09	[0.4]	Jurtshuk & McQuitty [BM]	1976	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	
Mycobacterium smegmatis (607)	125.Mycobacterium smegmatis	Actinobacteria: Actinomycetales	Acti-	$\mu\text{l O}_2 (\text{mg N})^{-1} \text{ hr}^{-1}$	MIN	146	24	37	10.45		Forbes et al. 1962 [BM, genus, rods 0.2-0.6x3.0-3.5 μm]		Cells harvested after 22 hr incubation.	
Mycobacterium stercoris (NCTC 3820)	126.Mycobacterium smegmatis	Actinobacteria: Actinomycetales	Acti-	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	(mg	11.9	20	30	14.14		Hunter 1953 [BM, genus, rods 0.2-0.6x3.0-3.5 μm]		Cells grown for 5-6 days; respiration of M. (butyricum) smegmatis (NCTC 337) and M. smegmatis (NCTC 523) was 13.4 and 15 $\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$, respectively	
Mycobacterium smegmatis	127.Mycobacterium smegmatis	Actinobacteria: Actinomycetales	Acti-	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	(mg	13	22	30	15.56		Jurtshuk & McQuitty [BM, genus, rods 0.2-0.6x3.0-3.5 μm]	1976	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration); respiration was 13 $\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$ = 22 W/kg at 30 C	
Mycobacterium tuberculosis var. hominis	128.Mycobacterium tuberculosis	Actinobacteria: Actinomycetales	Acti-	$\mu\text{l O}_2 (\text{mg DM})^{-1} (5 \text{ hr})^{-1}$	MIN	6.5	2.2	37	0.96	[0.2]	Engelhard et al. 1957 [BM, rods 0.2-0.5x2-4 μm]		Bacteria grown for 25 to 28 hr in batches; magnetically mixed in buffer solution for 5 hr; stored at -5 C for no more than 12 hr before analysis; respiration measured for 5 hr	Human tuberculosis bacteria
Mycobacterium tuberculosis var. hominis (H37Rv, LRv)	129.Mycobacterium tuberculosis	Actinobacteria: Actinomycetales	Acti-	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	(mg	4.2-5.8	7	37.8	2.9	[0.2]	Segal & Bloch 1955 [BM, rods 0.2-0.5x2-4 μm]		Various growth conditions	
Mycobacterium tuberculosis var. avium	130.Mycobacterium tuberculosis	Actinobacteria: Actinomycetales	Acti-	$\mu\text{l O}_2 (10 \text{ mg WM})^{-1} (90 \text{ min})^{-1}$	(mg	100	37	37	16.11	[0.2]	Minami 1957 [BM, rods 0.2-0.5x2-4 μm]		Cultures 3 days old	
Mycobacterium tuberculosis var. hominis (H37Rv)	131.Mycobacterium tuberculosis	Actinobacteria: Actinomycetales	Acti-	$\mu\text{l O}_2 (\text{mg WM})^{-1} (\text{hr})^{-1}$	(mg	0.46-1.10	2.6-6	37	23.33	[0.2]	Youmans et al. 1960 [BM, rods 0.2-0.5x2-4 μm]		Various growth conditions. Similar results were obtained later by the same team (Youmans & Youmans 1962a,b)	
Myxococcus xanthus (FB)	132.Myxococcus xanthus	Deltaproteobacteria: Myxococcales		$\mu\text{l O}_2 (15 \text{ mg DM})^{-1} (\text{min})^{-1}$	MIN	7	4.6	30	3.25	[1]	Dworkin & Niederpruem 1964 [McVittie et al. 1962, rods,		Vegetative cells harvested in the early stationary phase (30 hr); respiration meas-	No measurable respiration in microcysts

<i>Neisseria elongata</i> (ATCC 25295)	133.	<i>Neisseria elongata</i>	Betaproteobacteria: Neisseriales	$\mu\text{l DM}^{-1} \text{O}_2 \text{ hr}^{-1}$ (mg MIN	5	8.3	30	5.87	[0.2]	0.5×3-8 μm] Jurtshuk & McQuitty 1976 [BM, short rods 0.5 μm width]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	Cell mass estimated from linear dimensions given in BM
<i>Neisseria flava</i> (ATCC 14221)	134.	<i>Neisseria flava</i>	Betaproteobacteria: Neisseriales	$\mu\text{l DM}^{-1} \text{O}_2 \text{ hr}^{-1}$ (mg MIN	7	12	30	8.49	[0.2]	Jurtshuk & McQuitty 1976 [BM, genus, cocci, diam 0.6-1.0 μm]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	
<i>Neisseria gonorrhoeae</i>	135.	<i>Neisseria gonorrhoeae</i>	Betaproteobacteria: Neisseriales	$\text{nmol protein}^{-1} \text{O}_2 \text{ (mg MIN}^{-1}$	0.5	0.6	37	0.26	[0.2]	Kenimer & Lapp 1978 [BM, genus, cocci, diam 0.6-1.0 μm]	Cells harvested during early stationary phase after 18-20 hours of incubation; respiration measured for 2-3 min	
<i>Neisseria mucosa</i>	136.	<i>Neisseria mucosa</i>	Betaproteobacteria: Neisseriales	$\mu\text{l DM}^{-1} \text{O}_2 \text{ hr}^{-1}$ (mg MIN	9	15	30	10.61	[0.2]	Jurtshuk & McQuitty 1976 [BM, genus, cocci, diam 0.6-1.0 μm]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	
<i>Neisseria sicca</i>	137.	<i>Neisseria sicca</i>	Betaproteobacteria: Neisseriales	$\mu\text{l DM}^{-1} \text{O}_2 \text{ hr}^{-1}$ (mg MIN	8	13	30	9.19	[0.2]	Jurtshuk & McQuitty 1976 [BM, genus, cocci, diam 0.6-1.0 μm]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	
<i>Nitrobacter agilis</i>	138.	<i>Nitrobacter winogradskyi</i>	Alphaproteobacteria: Rhizobiales	$\mu\text{l O}_2 \text{ (mg protein}^{-1} \text{ hr}^{-1}$	12	10	30	7.07	[0.24]	Smith & Hoare 1968 [Tappe et al. 1999, 0.17-0.3 μm^3]	Incubation time 20 hr	Facultative heterotroph otherwise growing on nitrite, = <i>Nitrobacter winogradskyi</i> (Pan 1971)
<i>Nitrobacter agilis</i> (ATCC 14123)	139.	<i>Nitrobacter winogradskyi</i>	Alphaproteobacteria: Rhizobiales	$\text{ng-atoms O (mg protein}^{-1} \text{ min}^{-1}$	20	11	25	11.00	[0.24]	Hollocher et al. 1982 [Tappe et al. 1999, 0.17-0.3 μm^3]	Cells grown at 30 C to the late exponential phase, stored for no more than 3 days at 0 C before measurements	Protein to dry mass ratios (Protein/DM) are 0.55, 0.47 and 0.44 for autotrophic, autotrophic plus 1mM acetate, and autotrophic plus 5mM acetate, grown cells, respectively.
<i>Nitrosomonas europaea</i>	140.	<i>Nitrosomonas europaea</i>	Betaproteobacteria: Nitrosomonadales	$\text{ng-atoms O (mg protein}^{-1} \text{ min}^{-1}$	20	11	25	11.00	[0.6]	Hollocher et al. 1982 [Tappe et al. 1999, 0.5-0.7 μm^3]	Bacteria grown at 30 C to the late exponential phase, stored for no more than 3 days at 0 C	1 mg WM $\approx$ 0.1 mg protein
<i>Nocardia corallina</i>	141.	<i>Nocardia corallina</i>	Actinobacteria: Actinomycetales	$\mu\text{l DM}^{-1} \text{O}_2 \text{ (mg MIN}^{-1}$	1	1.7	30	1.20	2.1	Robertson & Batt 1973 (rods, 0.8-1.2×2-4 μm)	Cells harvested at the end of growth phase; respiration measured after 45 hours of starvation; viability more than 90%	Soil bacterium
												Dry mass data (Fig. 1): 4.3×10 ⁹ cells (90% viable) make up 1.6 mg dry mass: 0.4 × 10 ⁻¹² g dry mass cell ⁻¹

Nocardia (A-6)	corallina	142.Nocardia coral-lina	Actinobacteria: Acti-nomycetales	Acti-nomycetales	μl O ₂ (4.8 mg DM) ⁻¹ min ⁻¹	(mg DM) ⁻¹ hr ⁻¹	MIN	40	4	30.3	2.77	[2.1]	Raymond et al. 1967 [Robertson & Batt 1973, rods, 0.8-1.2×2-4 μm]	Respiration rate fell from 10 to 1 μl O ₂ (mg DM) ⁻¹ hr ⁻¹ during the first 45 hr of the total 450 hr' period of starvation	Cell mass estimated from linear dimensions given by Robertson & Batt (1973), 2-4 μm (length) × 0.8-1.2 μm (diameter), is greater, 2.4× 10 ⁻¹² g cell ⁻¹
Nocardia asteroides (ATCC 3308)		143.Nocardia far-cinica	Actinobacteria: Acti-nomycetales	Acti-nomycetales	μl O ₂ (mg DM) ⁻¹ hr ⁻¹	(mg DM) ⁻¹ hr ⁻¹	MIN	5	8.3	30	5.87	[0.1]	Jurtshuk & McQuitty 1976 [Takeo & Uesaka 1975, Fig. 1, hyphae diam 0.6 μm mycellium; Kowalski et al. 1999, 1-1.25×3-5 μm airborne]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	
Nocardia sp. (Z1)		144.Nocardia sp.	Actinobacteria: Acti-nomycetales	Acti-nomycetales	μl O ₂ (42 mg DM) ⁻¹ hr ⁻¹	(mg DM) ⁻¹ hr ⁻¹	MIN	56	2.2	30	1.56		Watson & Cain 1975 [BM9, genus, hyphae diam 0.5-1.2 μm]	Cells grown for 24 hr	Bacterium isolated from soil with by enrichment with pyridine
Nocardia erythropolis		145.Nocardia sp.	Actinobacteria: Acti-nomycetales	Acti-nomycetales	μl O ₂ (7.5 mg DM) ⁻¹ min ⁻¹	(mg DM) ⁻¹ min ⁻¹	MIN	40	3	30	2.12		Cartwright & Cain 1959 [BM9, genus, hyphae diam 0.5-1.2 μm]	Respiration measured for 200 min	No change of respiration with time
														Cain et al. 1968: Cells harvested after 24-36 hr incubation (Smith et al. 1968); respiration measured for 90 min was 185 μl O ₂ (14.3 mg DM) ⁻¹ hr ⁻¹ = 22 W/kg at 30 C	The organism was originally isolated from soil by enrichment with <i>p</i> -nitrobenzoate (Cain et al. 1958)
Nonomuraea (ATCC 39727)	sp.	146.Nonomuraea sp.	Actinobacteria: Acti-nomycetales	Acti-nomycetales	nmol O ₂ (mg protein) ⁻¹ min ⁻¹	(mg protein) ⁻¹ min ⁻¹	MIN	1	1	25	1.00		Palese et al. 2003	Respiration of cells in the lag phase (incubated for 15 hr) prior to exponential growth	Classification and size determination made for the Nocardia genus. There is no N. erythropolis at www.bacterio.cict.fr.
															Endogenous respiration increases from lag to exponential phase from 1 to 35 nmol O ₂ (mg protein) ⁻¹ min ⁻¹ , then starts to drop abruptly in the stationary phase (min. value measured after 160 hr incubation was 7 nmol O ₂ (mg protein) ⁻¹ min ⁻¹)

Micrococcus denitrificans	147.Paracoccus denitrificans	Alphaproteobacteria: Rhodobacterales	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	MIN	5	8.3	30	5.87	[0.16]	Kornberg & Morris 1965 [BM, spheres (0.5-0.9 μm in diam) or short rods (0.9-1.2 μm long)] Bishop et al. 1962	Cells harvested during the exponential phase	Endogenous reserves of cells were "depleted" by 4 hr aerobic shaking at 30 C in another experiment
Pasteurella pseudotuberculosis (NCTC 1101)	148.Pasteurella pseudotuberculosis	Gammaproteobacteria: Enterobacteriales	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	MIN	7	12	25	12.00			Respiration measured immediately after harvesting the cells at the end of the logarithmic growth phase	Max. resp. (in the presence of glucose) was 331 $\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$
Pediococcus cerevisiae (ATCC 8081)	149.Pediococcus acidilactici	"Bacilli": "Lactobacillales"	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	MIN	2	3.3	30	2.33	[2.2]	Jurtshuk & McQuitty 1976 [BM9, genus, spheres 1.0-2.0 μm diam]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	
Phaeospirillum fulvum (5K, KM MGU 325)	150.Phaeospirillum fulvum	Alphaproteobacteria: Rhodospirillales	$\text{nmol O}_2 (\text{mg protein})^{-1} \text{ min}^{-1}$	MIN	25.3	28	28	22.74	[2]	Berg et al. 2002 [BM]	Bacteria harvested from early exponential cultures grown photoheterotrophically	Purple bacteria
Picrophilus oshimae	151.Picrophilus oshimae	Thermoplasmata (Archaea): Thermoplasmatales	$\text{nmol O}_2 (\text{mg protein})^{-1} \text{ min}^{-1}$	MIN	22.7	25	60	2.21	[1]	Van de Vossenberg et al. 1998 [Schleper et al. 1995, cocci, diam 1-1.5 μm]	Starvation for several hours	Thermoacidophilic archaeon
Proteus morganii [=Morganella morganii]	152.Proteus morganii	Gammaproteobacteria: "Enterobacteriales"	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	MIN	3	5	30	3.54	[0.4]	Jurtshuk & McQuitty 1976 [BM, rods, 0.6-0.7x1-1.7 μm]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	
Proteus vulgaris	153.Proteus vulgaris	Gammaproteobacteria: "Enterobacteriales"	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	MIN	3	5	37	2.18	[0.4]	Bishop et al. 1962 [BM9, genus, rods 0.4-0.8x1-3 μm]	Respiration measured immediately after harvesting the cells (aerobic culture) at the end of the logarithmic growth phase; endogenous respiration of anaerobic culture was below detection limit (0 orig. units)	Max. resp. (in the presence of glucose) was 87 orig. units
Proteus vulgaris	154.Proteus vulgaris	Gammaproteobacteria: "Enterobacteriales"	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$		3	5	30	3.54	[0.4]	Jurtshuk & McQuitty 1976 [BM9, genus, rods 0.4-0.8x1-3 μm]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	
Pseudomonas aeruginosa	155.Pseudomonas aeruginosa	Gammaproteobacteria: Pseudomonadales	$\mu\text{l O}_2 (\text{mg N})^{-1} (75 \text{ min})^{-1}$	MIN	50	6.7	30	4.74	[0.5]	Rogoff 1962 [Montesinos et al. 1983, Coulter counter; see also datra of Hou et al. 1966, 0.2-0.8 pg]	Cells isolated from soil and closely resembling P. aeruginosa were grown for 18 to 48 hr; respiration was measured for 75 min	

Pseudomonas aeruginosa (120Na)	156.Pseudomonas aeruginosa	Gammaproteobacteria: Pseudomonadales	$\mu\text{mol O}_2 \text{ (100 mg DM)}^{-1} \text{ (2 hr)}^{-1}$	55.5	10	30	7.07	[0.5]	Gronlund & Campbell 1961 [Montesinos et al. 1983, Coulter counter; see also datra of Hou et al. 1966, 0.2-0.8 pg]	Cells (strain 120Na) harvested after 20 hr growth; respiration measured for 2 hr; for strain ATCC 9027 respiration was 80 orig. units	Hou et al. 1966: Cell size of P. aeruginosa: 4.5×10^8 viable cells = 29.6×10^{-6} g = 17.5×10^{-6} g protein, cells harvested after 8 hr growth: 1 cell = 0.06×10^{-12} g dry mass = 0.2×10^{-12} g wet mass
										Tomlinson & Campbell 1963: cells (strain 120Na) harvested after 20 hr growth at 30 C; respiration measured for 2 hr; $120 \mu\text{l O}_2 \text{ (5 mg DM)}^{-1} \text{ (2 hr)}^{-1} = 20 \text{ W/kg}$	For 24-hr phosphorus starved cells, they observed 4.18×10^8 viable cells = 44.4×10^{-6} g = 22.8×10^{-6} g protein: 1 cell = 0.35×10^{-12} g wet mass
										Gronlund & Campbell 1963: cells (strain ATCC 9027) harvested after 20 hr growth at 30 C; respiration measured for 3 hr; hourly rates decreased from 4.75 to 4.0 to $3.19 \mu\text{mol O}_2 \text{ (10 mg DM)}^{-1} \text{ hr}^{-1}$, with no loss of viability (min. rate 12 W/kg)	For refed cells at 30 hr: 7.68×10^8 viable cells = 182×10^{-6} g = 75.1×10^{-6} g protein: 1 cell = 0.8×10^{-12} g wet mass
										Gronlund & Campbell 1966: cells (strain ATCC 9027) harvested after 20 hr growth at 30 C; respiration measured for 2 hr; $95 \mu\text{l O}_2 \text{ (5 mg DM)}^{-1} \text{ (2 hr)}^{-1} = 16 \text{ W/kg}$	Protein to DM (Protein/DM) ratio = 0.59, 0.51 and 0.41, respectively.
										Warren et al. 1960 report practically the same rate, $2500 \mu\text{l O}_2 \text{ (100 mg DM)}^{-1} \text{ (2 hr)}^{-1} = 20 \text{ W/kg}$; the same age of culture (20 hr), strain ATCC 9027	
Pseudomonas aeruginosa (ATCC 15442)	157.Pseudomonas aeruginosa	Gammaproteobacteria: Pseudomonadales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	17	28	30	19.80	[0.5]	Jurtshuk & McQuitty 1976 [Montesinos et al.	Cells harvested at the late-logarithmic growth phase (two	

Pseudomonas (pyo- ceanea) aeruginosa	158.Pseudomonas aeruginosa	Gammaproteobacteria: Pseudomonadales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	28	47	37	20.46	[0.5]	1983, Coulter counter; see also datra of Hou et al. 1966, 0.2-0.8 pg]	thirds of the maximal growth concentration)
Pseudomonas aeru- ginosa (9/93 and 72/92)	159.Pseudomonas aeruginosa	Gammaproteobacteria: Pseudomonadales	$\text{nmol O}_2 \text{ (mg DM)}^{-1} \text{ min}^{-1}$	23.7	53	30	37.48	[0.5]	Bishop et al. 1962 [Montesinos et al. 1983, Coulter counter; see also datra of Hou et al. 1966, 0.2-0.8 pg]	Respiration measured immediately after harvesting the cells (aerobic culture) at the end of the logarithmic growth phase; anaero- bically grown culture respired at $15 \mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1} = 25 \text{ W/kg}$
Pseudomonas fluo- rescens (A.3.12)	160.Pseudomonas fluorescens	Gammaproteobacteria: Pseudomonadales	$\mu\text{l O}_2 \text{ (3 mg DM)}^{-1} \text{ hr}^{-1}$	8	4.4	28	3.57	[1]	Majtán et al. 1995 [Montesinos et al. 1983, Coulter counter; see also datra of Hou et al. 1966, 0.2-0.8 pg]	Cells harvested at the exponential phase; respiration measured for 10 min
Pseudomonas fluo- rescens (A.3.12)	161.Pseudomonas fluorescens	Gammaproteobacteria: Pseudomonadales	$\mu\text{l O}_2 \text{ (3 mg DM)}^{-1} \text{ hr}^{-1}$	5.9	10	26	9.33	[1]	Altekar & Rao 1963 [BM, 0.7- 0.8×2.0-3.0 μm ; Gunter & Kohn 1956, 0.24 pg DM/cell = 0.8 pg/cell at 70% water content]	Cells washed after incubation for 24 hr in a growth medium on a shaker; respiration of “resting” cells meas- ured for 4 hr; value for the last hour taken (Fig. 4)
Pseudomonas fluo- rescens (KBI)	162.Pseudomonas fluorescens	Gammaproteobacteria: Pseudomonadales	$\mu\text{mol (10 mg DM)}^{-1} \text{ hr}^{-1}$	4	15	30	10.61	[1]	Gunter & Kohn 1956 [BM, 0.7- 0.8×2.0-3.0 μm ; Gunter & Kohn 1956, 0.24 pg DM/cell = 0.8 pg/cell at 70% water content]	Respiration of washed cell suspensions har- vested from 16 to 18- hr yeast agar plates
Pseudomonas fluo- rescens (A.3.12)	163.Pseudomonas fluorescens	Gammaproteobacteria: Pseudomonadales	$\mu\text{mol O}_2 \text{ (100 mg DM)}^{-1} \text{ (2 hr)}^{-1}$	98	18	30	12.73	[1]	Hughes 1966 [BM, 0.7-0.8×2.0- 3.0 μm ; Gunter & Kohn 1956, 0.24 pg DM/cell = 0.8 pg/cell at 70% water content]	Cells grown for 22-24 hr at 25 C; respiration measured after equili- bration for 30 min at 30 C
									Gronlund & Campbell 1961 [BM, 0.7-0.8×2.0- 3.0 μm ; Gunter & Kohn 1956, 0.24 pg DM/cell = 0.8	Cells harvested after 20 hr growth; respira- tion measured for 2 hr at 30 C Tomlinson & Camp-

Pseudomonas fluorescens (A.3.12)	164.Pseudomonas fluorescens	Gamma	proteobacteria: Pseudomonadales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	20	33	30	23.33	[1]	Jacoby 1964 [BM, 0.7-0.8×2.0-3.0 μm ; Gunter & Kohn 1956, 0.24 pg DM/cell = 0.8 pg/cell at 70% water content]	bell 1963: 190 $\mu\text{l O}_2 \text{ (5 mg DM)}^{-1} \text{ (2 hr)}^{-1}$ = 32 W/kg for 20-hr cultures at 30 C		
Pseudomonas fluorescens (KBI)	165.Pseudomonas fluorescens	Gamma	proteobacteria: Pseudomonadales	$\mu\text{l O}_2 \text{ (2 mg DM)}^{-1} \text{ (160 min)}^{-1}$	110	34	30	24.04	[1]	Kornberg 1958 [BM, 0.7-0.8×2.0-3.0 μm ; Gunter & Kohn 1956, 0.24 pg DM/cell = 0.8 pg/cell at 70% water content]	Cells grown for 10-12 hr, harvested during logarithmic phase; respiration measured for 160 min		
Pseudomonas fluorescens (ATCC 13525)	166.Pseudomonas fluorescens	Gamma	proteobacteria: Pseudomonadales	$\text{nmol O}_2 \text{ (mg DM)}^{-1} \text{ (min)}^{-1}$	45.7	60	30	42.43	[1]	Eisenberg et al. 1973 [BM, 0.7-0.8×2.0-3.0 μm ; Gunter & Kohn 1956, 0.24 pg DM/cell = 0.8 pg/cell at 70% water content]	Respiration of washed cells harvested at midlog or early stationary phase		
Pseudomonas formicans	167.Pseudomonas formicans	Gamma	proteobacteria: Pseudomonadales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ min}^{-1}$	MIN	4	10	30	7.07	[7]	Sabina & Pivnick 1956 [Crawford 1954, rods 1-2×4-5 μm]	Bacteria grown for 5 days in a mixture of soluble oils; incubated for 24 hr in a nutrient broth; aerated vigorously in nutrient broth for 18 hr; harvested and washed; washed suspension "shaken for 2 to 8 hr at room temperature in an attempt to reduce the endogenous respiration"; respiration measured for 210 min (Fig. 2); value taken for the last 40 min	Bacteria isolated from used emulsifiers of industrial oil, Illinois, USA Endogenous respiration decreases with time during 2-4 hr of respiration (Figs. 2, 4)
Pseudomonas oleovorans	168.Pseudomonas oleovorans	Gamma	proteobacteria: Pseudomonadales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	2	3.3	30	2.33	[0.16]	Sabina & Pivnick 1956 [Lee & Chandler 1941, almost coccoid, 0.5×0.8 μm]	Bacteria grown for 5 days in a mixture of soluble oils; incubated for 24 hr in a nutrient broth; aerated vigorously in nutrient broth for 18 hr; harvested and washed; washed	Bacteria isolated from used emulsifiers of industrial oil, England Endogenous respiration decreases with time during 2-4 hr of respiration (Figs. 2, 4)

Pseudomonas oleovorans	169.Pseudomonas oleovorans	Gamma proteobacteria: Pseudomonadales	nmol O ₂ (mg DM) ⁻¹ min ⁻¹	16-19	40	30	28.28	[0.16]	Peterson 1970 [Lee & Chandler 1941, almost coccoid, 0.5×0.8 μm]	suspension "shaken for 2 to 8 hr at room temperature in an attempt to reduce the endogenous respiration"; respiration measured for 210 min (Fig. 3); value taken for the last hour Cells harvested in late-log-phase	
Pseudomonas putida (O1OC)	170.Pseudomonas putida	Gamma proteobacteria: Pseudomonadales	nmol O ₂ (25 mg DM) ⁻¹ min ⁻¹	MIN	11	1	30	0.71	[1.7]	Chapman & Ribbons 1976 [BM, 0.7-1.1×2.0-4.0 μm]	Succinate-grown cells harvested after 16-20 hr incubation in the late-logarithmic phase; respiration measured for 2 min; respiration of strains O1OC, ORC and O1 varied from 11 to 29 nmol O ₂ (25 mg DM) ⁻¹ min ⁻¹ (1-2.6 W/kg) depending on growth substrate. The organism was isolated from orcinol enrichments Respiration measured for 1 hr (strain ORC) was 43 μl (20 mg WM) ⁻¹ hr ⁻¹ (Fig. 3) = 12 W/kg
Pseudomonas putida (arvilla (PaM1))	171.Pseudomonas putida	Gamma proteobacteria: Pseudomonadales	μl O ₂ (20 mg WM) ⁻¹ hr ⁻¹		10	2.8	30	1.98	[1.7]	Kunz & Chapman 1981 [BM, 0.7-1.1×2.0-4.0 μm]	P. putida (arvilla) mt-2 cells grown for 24 hr on toluene or 48 hr on pseudocumene; respiration 10 μl O ₂ (20 mg WM) ⁻¹ hr ⁻¹ = 2.8 W/kg
Pseudomonas putida (biotype B)	172.Pseudomonas putida	Gamma proteobacteria: Pseudomonadales	μl O ₂ (mg DM) ⁻¹ min ⁻¹		0.27/6.1	4.4	30	3.11	[1.7]	Sebek & Barker 1968 [BM, 0.7-1.1×2.0-4.0 μm]	Cells harvested towards the end of the exponential phase, shaken for 18 hr "to reduce endogenous respiration"
Pseudomonas putida (TM)	173.Pseudomonas putida	Gamma proteobacteria: Pseudomonadales	μl O ₂ (10 mg WM) ⁻¹ min ⁻¹		0.25	8.3	30	5.87	[1.7]	Donnelly & Dagle 1980 [BM, 0.7-1.1×2.0-4.0 μm]	Cells harvested during exponential growth Strain isolated from rotting oak leaves
Pseudomonas fluorescens-putida	174.Pseudomonas putida	Gamma proteobacteria: Pseudomonadales	μl O ₂ (mg DM) ⁻¹ hr ⁻¹		11.2	19	37	8.27	[1.7]	Chakrabarty & Roy 1964 [BM, 0.7-1.1×2.0-4.0 μm]	Cells grown to the stationary phase (72 hr) Respiration decreases from mid-exponential to late-exponential to stationary phase (20.3 → 17.8 → 11.2 μl O ₂ (mg DM) ⁻¹ hr ⁻¹); in cells harvested at 20 hr (mid-exponential) and starved for 0-4 hr respiration decreases from 27 (0th) to 20 (2nd) to 16.5 (3rd) to 15.6

Pseudomonas putida (PRS1)	175.Pseudomonas putida	Gammaproteobacteria: Pseudomonadales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$		10	17	30	12.02	[1.7]	Meagher et al. 1972 [BM, 0.7-1.1×2.0-4.0 μm]	Cells harvested during exponential growth	(4th hr) $\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$
Pseudomonas putida	176.Pseudomonas putida	Gammaproteobacteria: Pseudomonadales	$\text{nmol O}_2 \text{ (mg DM)}^{-1} \text{ min}^{-1}$		15	34	30	24.04	[1.7]	Peterson 1970 [BM, 0.7-1.1×2.0-4.0 μm]	Cells harvested in late-log-phase	
Pseudomonas saccharophila	177.Pseudomonas saccharophila	Gammaproteobacteria: Pseudomonadales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	20-28	33	30	23.33		Doudoroff et al. 1956	Cells harvested from early stationary phase cultures	Bacteria isolated from activated sludge taken from a domestic sewage plant, The Netherlands
Pseudomonas (B1)	178.Pseudomonas sp.	Gammaproteobacteria: Pseudomonadales	$\text{nmol O}_2 \text{ (mg protein)}^{-1} \text{ min}^{-1}$	MIN	2	2.2	30	1.56		Van Ginkel et al. 1992 [BM, genus, rods 0.5-1.0×1.5-5.0 μm]	Yeast-glucose plate inoculates were incubated in a stationary fashion at 30 °C; washed; endogenous respiration measured for 5 min	
Pseudomonas sp.	179.Pseudomonas sp.	Gammaproteobacteria: Pseudomonadales	$\mu\text{mol O}_2 \text{ (50 mg)}^{-1} \text{ (4 hr)}^{-1}$	MIN	13	8	30	5.66		Shaw 1956 [BM, genus, rods 0.5-1.0×1.5-5.0 μm]	Bacteria grown for 24 hr at 30 °C; washed; respiration measured for 4 h	An airborne organism isolated in the laboratory; University of Otago; New Zealand; a fluorescent species of Pseudomonas
Pseudomonas sp.	180.Pseudomonas sp.	Gammaproteobacteria: Pseudomonadales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$		10	17	30	12.02		Sariaslani et al. 1982 [BM, genus, rods 0.5-1.0×1.5-5.0 μm]	Bacteria harvested during log phase; respiration measured for about 1 h	Bacteria isolated from California soil, USA.
Pseudomonas (OD1)	181.Pseudomonas sp.	Gammaproteobacteria: Pseudomonadales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	2	3.3	25	3.30		Jayasuriya 1956 [BM, genus, rods 0.5-1.0×1.5-5.0 μm]	Bacteria grown on oxalate for 40 hr at 25 °C	Synonym Acetomobacter sp. P6
Pseudomonas (PN-1)	182.Pseudomonas sp.	Gammaproteobacteria: Pseudomonadales	$\mu\text{mol O}_2 \text{ (6.05 mg protein)}^{-1} \text{ (145 min)}^{-1}$	MIN	6.5	8.3	30	5.87		Taylor 1983 [BM, genus, rods 0.5-1.0×1.5-5.0 μm]	Endogenous respiration of anaerobically grown cells	
Pseudomonas sp. P6 (NCIB 10431)	183.Pseudomonas sp.	Gammaproteobacteria: Pseudomonadales	$\mu\text{mol O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	0.4	15	30	10.61		Jones & Turner 1973 [BM, genus, rods 0.5-1.0×1.5-5.0 μm]	Respiration measured for not less than 90 min	
Azotomonas insolita (ATCC 12412)	184.Pseudomonas sp.	Gammaproteobacteria: Pseudomonadales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	11	18	30	12.73		Jurtshuk & McQuitty 1976 [BM, genus, rods 0.5-1.0×1.5-5.0 μm]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	Size determination is made for Pseudomonas genus, since ATCC 12412 corresponds to a Pseudomonas sp.
Pseudomonas (B13)	185.Pseudomonas sp.	Gammaproteobacteria: Pseudomonadales	$\mu\text{g O}_2 \text{ (mg DM)}^{-1} \text{ min}^{-1}$	MIN	0.5	35	20	49.50		Tros et al. 1996 [BM, genus, rods 0.5-1.0×1.5-5.0 μm]	Bacteria grown for about 210 hr in a recycling fermentor to reach the stationary phase; endogenous respiration measured for 5-10 min of a sample taken from the recycling fermentor	Bacteria isolated from activated sludge taken from a domestic sewage plant, The Netherlands
Streptomyces nitri-	186.Pseudonocardia	Actinobacteria: Acti-	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	1.5	2.5	30	1.77		Isenberg et al.	Mycellium grown for	5 mg DM= 25 mm ³ wet packed volume

ficans Rhizobium japonicum (61A76)	nitrificans 187.Rhizobium japonicum	nomycetes Alphaproteobacteria: Rhizobiales	$\mu\text{mol O}_2 \text{ (0.71 mg protein)}^{-1} \text{ (10 min)}^{-1}$	MIN	0.1	16	23	18.38	[0.7]	1954 Peterson & LaRue 1982 [BM9, genus, rods, 0.5-0.9×1.2-3.0 μm]	5-6 days Bacteria harvested at mid-log phase of growth	→ DM/WM=0.2 Free-living bacteria
Rhizobium leguminosarum (128 C 53)	188.Rhizobium leguminosarum	Alphaproteobacteria: Rhizobiales	$\mu\text{l O}_2 \text{ (mg N)}^{-1} \text{ hr}^{-1}$	MIN	57	9.5	30	6.72	[0.6]	Dietrich & Burris 1967 [BM]	Bacteria cultured for 2 weeks in an extract from wheat plants; respiration measured for 1 hr after 10 min equilibration	
Rhizobium meliloti (F-28)	189.Rhizobium meliloti	Alphaproteobacteria: Rhizobiales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	7	12	30	8.49	[0.7]	Jurtshuk & McQuitty 1976 [BM9, genus, rods, 0.5-0.9×1.2-3.0 μm]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration); respiration of strain 3DOa1 was 9 orig. units	
Rhodobacter sphaeroides (2R)	190.Rhodobacter sphaeroides	Alphaproteobacteria: Rhodobacterales	$\text{nmol O}_2 \text{ (mg protein)}^{-1} \text{ min}^{-1}$	MIN	14.9	17	28	13.81	[0.8]	Berg et al. 2002 [Gunter & Kohn 1956 0.23 pg DM/cell = 0.8 pg/cell at 70% water content]	Bacteria (strain 2R) harvested from early exponential cultures grown photoheterotrophically	Purple bacteria
Rhodopseudomonas spheroides	191.Rhodobacter sphaeroides	Alphaproteobacteria: Rhodobacterales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$		20.8	35	26	32.66	0.8	Gunter & Kohn 1956 [0.23 pg DM/cell = 0.8 pg/cell at 70% water content]	Cells harvested from 16 to 18-hr yeast agar plates	Cell mass estimated from dry mass data, Table 1, 0.23 pg DM/cell
Corynebacterium (7E1C, ATCC 19067)	192.Rhodococcus rhodochrous	Actinobacteria: Actinomycetales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	10	17	30	12.02		Jurtshuk & McQuitty 1976	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	
Rhodococcus (094)	193.Rhodococcus sp.	Actinobacteria: Actinomycetales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	1.3	2.2	25	2.20		Bruheim et al. 1999	Cells grown to the early stationary phase on oil	The isolate was obtained from enrichment cultures by using inocula from Norwegian coastal waters
Nocardia opaca	194.Rhodococcus sp.	Actinobacteria: Actinomycetales	$\mu\text{l O}_2 \text{ (14.8 mg DM)}^{-1} \text{ (60 min)}^{-1}$	MIN	60	7	30	4.95		Cartwright & Cain 1959	Respiration measured for 60 min	
Nocardia sp. (NCIB 11216)	195.Rhodococcus sp.	Actinobacteria: Actinomycetales	$\mu\text{l O}_2 \text{ (5 mg DM)}^{-1} \text{ min}^{-1}$	MIN	0.37	7.4	25	7.40		Harper 1977	Bacteria grown at 25 C to early exponential phase	A microorganism capable of using benzonitrile as sole carbon, nitrogen and energy source was isolated by elective culture from mud obtained from the bed of the River Lagan in Belfast
Rhodospirillum rubrum (2R KM MGU 301)	196.Rhodospirillum rubrum	Alphaproteobacteria: Rhodospirillales	$\text{nmol O}_2 \text{ (mg protein)}^{-1} \text{ min}^{-1}$	MIN	3.4	3.8	28	3.09	[9]	Berg et al. 2002 [BM]	Bacteria harvested from early exponential cultures grown photoheterotrophically;	Purple bacteria Breznak et al. 1978 found half-life survival time of 3-4 days in the dark for

Table 1. <i>Respiration of selected Gram-negative bacteria</i>													
Bacterial strain		Reference		Phylum		Medium		Temperature		pH		Respiration rate	

											sheathed filaments in young cultures, liberated flagellated cells in old cultures]	washed suspensions were aerated for 3-5 hr to reduce endogenous respiration, which is characterized as “rather high” perhaps “due to the large amount of fatty material stored in the cells”	Aeration for more than 5 hr “tended to destroy the oxidizing capacity of the cells”
Sporosarcina ureae	206.Sporosarcina ureae	“Bacilli”: Bacillales	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	MIN	7	12	30	8.49	[3.8]	Jurtshuk & McQuitty 1976 [BM9, genus, rods, 1-2×2-3 μm]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)		
Staphylococcus aureus	207.Staphylococcus aureus	“Bacilli”: Bacillales	$7 \mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	MIN	9.3-15.7	16	37	7	[0.04-0.17] [[0.27]]	Ramsey 1962 [Watson et al. 1998, diam 0.41 μm for long-starved cells, 0.69 μm for exponential phase cells] [[Montesinos et al. 1983, Coulter counter]]	Cells grown for 12, 24, 48 and 72 hr on agar respired at 14.8, 12.0, 10.0 and 0 $\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$, respectively, at 37 °C; depending on the growth medium, endogenous respiration ranged from 9.3 to 15.7 $\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$ = 16-26 W/kg		
Staphylococcus aureus (FDA 209P)	208.Staphylococcus aureus	“Bacilli”: Bacillales	$\mu\text{l O}_2 (14 \text{ mg DM})^{-1} (4 \text{ hr})^{-1}$		44	1.3	30	0.92	[0.04-0.17] [[0.27]]	Huber & Schuhardt 1970 [Watson et al. 1998, diam 0.41 μm for long-starved cells, 0.69 μm for exponential phase cells] [[Montesinos et al. 1983, Coulter counter]]	Cells grown for 18 hr at 37 °C with shake aeration; respiration measured for 4 hr.		
Staphylococcus aureus	209.Staphylococcus aureus	“Bacilli”: Bacillales	$\mu\text{l O}_2 (0.1-0.15 \text{ mg N}) \text{ hr}^{-1}$		2	2.2	37	0.96	[0.04-0.17] [[0.27]]	Yotis & Ekstedt 1959 [Watson et al. 1998, diam 0.41 μm for long-starved cells, 0.69 μm for exponential phase cells] [[Montesinos et al. 1983, Coulter counter]]	Cells incubated for 16 hr at 37 °C; respiration measured for 6 hr; it decreases with time	Respiration decreases with starvation time from 21 $\mu\text{l O}_2 (0.1-0.15 \text{ mg N}) \text{ hr}^{-1}$ in the first hour, 11.5 (second hour), 6.5 (third, fourth hour), 2 (fifth, sixth hour), min. rate = 2.2 W/kg	
Staphylococcus	210.Staphylococcus	“Bacilli”: Bacillales	$\mu\text{l O}_2 (6.02 \text{ mg}$		9.73	2.7	37	1.18	[0.04-	Bluhm & Ordal	Cells grown for 10 hr		

aureus (MF-31)	aureus		DM) ⁻¹ hr ⁻¹							0.17] [[0.27]]	1969 [Watson et al. 1998, diam 0.41 µm for long-starved cells, 0.69 µm for exponential phase cells] [[Montesinos et al. 1983, Coulter counter]]	at 37 C; respiration of heat injured cells is three times lower	
Staphylococcus aureus (31-r)	211.Staphylococcus aureus	“Bacilli”: Bacillales	µmol O ₂ (mg DM) ⁻¹ hr ⁻¹		0.19	7	37	3.05		[0.04-0.17] [[0.27]]	Krzemiński et al. 1972 [Watson et al. 1998, diam 0.41 µm for long-starved cells, 0.69 µm for exponential phase cells] [[Montesinos et al. 1983, Coulter counter]]	Cells harvested after 20 hr growth at 37 C and starved for 3 hr; respiration decreases from 0.68 to 0.43 to 0.19 µmol O ₂ (mg DM) ⁻¹ h ⁻¹ for the 1st, 2nd and 3rd hrs at 37 C, respectively.	Respiration decreases with starvation time; it also depends on growth phase: maximum at approx. 10 hr and then decreases towards the stationary phase
Staphylococcus (albus) aureus	212.Staphylococcus aureus	“Bacilli”: Bacillales	µl O ₂ (mg DM) ⁻¹ hr ⁻¹		3	5	30	3.54		[0.04-0.17] [[0.27]]	Jurtshuk & McQuitty 1976 [Watson et al. 1998, diam 0.41 µm for long-starved cells, 0.69 µm for exponential phase cells] [[Montesinos et al. 1983, Coulter counter]]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration); respiration was 3, 4 and 5 µl O ₂ (mg DM) ⁻¹ hr ⁻¹ for strains S. (albus) aureus, S. aureus (University of Houston) and S. aureus ATCC 6538, respectively (5-8.3 W/kg), at 30 C	
Staphylococcus epidermidis (AT2)	213.Staphylococcus epidermidis	“Bacilli”: Bacillales	µl O ₂ (mg DM) ⁻¹ hr ⁻¹	MIN	16	27	30	19.09		[0.5]	Jacobs & Conti 1965 [BM]	Cells grown for 8 hr at 37 C harvested at the end of the log phase	Cell mass estimated from linear dimensions given in BM
Staphylococcus albus (Micrococcus pyogenes var. albus)	214.Staphylococcus simulans	“Bacilli”: Bacillales	µl O ₂ (mg DM) ⁻¹ hr ⁻¹	MIN	6	10	37	4.35			Bishop et al. 1962	Respiration measured immediately after harvesting the cells (aerobic culture) at the end of the logarithmic growth phase; endogenous respiration of anaerobic culture was 4 orig. units	Max. resp. (in the presence of lactate) was 111 orig. units
Gaffkya tetragena (ATCC 10875)	215.Staphylococcus sp.	“Bacilli”: Bacillales	µl O ₂ (mg DM) ⁻¹ hr ⁻¹	MIN	10	17	30	12.02			Jurtshuk & McQuitty 1976	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration); strain ATCC 10875	

Table 1. Literature data on the respiration of Streptococcus and Streptomyces strains												
Strain	Reference	Medium	Incubation time (hr)	Temperature (°C)	pH	Respiration rate (μl O ₂ (mg DM) ⁻¹ hr ⁻¹)	Respiration rate (mg O ₂ (g DM) ⁻¹ hr ⁻¹)	Respiration rate (W/kg)	Respiration rate (g O ₂ (g DM) ⁻¹ hr ⁻¹)	Respiration rate (g O ₂ (g DM) ⁻¹ hr ⁻¹)	Respiration rate (g O ₂ (g DM) ⁻¹ hr ⁻¹)	Respiration rate (g O ₂ (g DM) ⁻¹ hr ⁻¹)
Streptococcus mastitidis (70b)	216. Streptococcus agalactiae	"Bacilli": "Lactobacillales"	10	1.7	37	0.74	[0.3]	Greisen & Gun-salus 1944 [BM]	respired at 14 μl O ₂ (mg DM) ⁻¹ hr ⁻¹ = 23 W/kg			
Streptococcus agalactiae	217. Streptococcus agalactiae	"Bacilli": "Lactobacillales"	5	8.3	37	3.61	[0.3]	Mickelson 1961 [BM]	Cells grown for 12 hr	N/DM≈0.1		
Streptococcus pneumoniae (ATCC 6360)	218. Streptococcus pneumoniae	"Bacilli": "Lactobacillales"	1	1.7	30	1.20	[0.4]	Jurtshuk & McQuitty 1976 [Kowalski et al. 1999, diam 0.8-1 μm]	Cells harvested after 10-15 hr incubation; respiration measured for 5 hr			
Streptococcus pyogenes (ATCC 10389)	219. Streptococcus pyogenes	"Bacilli": "Lactobacillales"	2	3.3	30	2.33	[0.2]	Jurtshuk & McQuitty 1976 [Kowalski et al. 1999, diam 0.6-1 μm]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)			
Streptomyces coelicolor	220. Streptomyces coelicolor	Actinobacteria: Actinomycetales	24	40	30	28.28		Niederpruem & Hackett 1961	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)			
Streptomyces fradiae (ATCC 11903)	221. Streptomyces fradiae	Actinobacteria: Actinomycetales	13	22	30	15.56		Niederpruem & Hackett 1961	Respiration of mycelium			
Streptomyces griseus (3475 Waksman)	222. Streptomyces griseus	Actinobacteria: Actinomycetales	105	18	30	12.73		Gilmour et al. 1955	Respiration of mycelium			
Streptomyces griseus (ATCC 10137)	223. Streptomyces griseus	Actinobacteria: Actinomycetales	12	20	30	14.14		Jurtshuk & McQuitty 1976	Cells grown for 24 hr			
Actinomyces gispororuber	224. Streptomyces longispororuber	Actinobacteria: Actinomycetales	2	3.3	30	2.33		Feofilova et al. 1966	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)			
									Niederpruem & Hackett 1961: Respiration of mycelium was 21 μl O ₂ (mg DM) ⁻¹ hr ⁻¹ = 35 W/kg at 30 C			
									Respiration of 24-hr-old mycelium			
									Endogenous respiration was minimal (around 2 μl O ₂ (mg DM) ⁻¹ hr ⁻¹) in the beginning of growth before the exponential phase (lag phase); at maximal growth rate it increased up to 14 μl O ₂ (mg DM) ⁻¹ hr ⁻¹ (48 hr) and then started to decline gradually to 7-11 μl O ₂ (mg DM) ⁻¹ hr ⁻¹ at 72 hr and 4-8 μl O ₂ (mg DM) ⁻¹ hr ⁻¹ at 96 hr			
Streptomyces olivaceus (NRRL B-1125)	225. Streptomyces olivaceus	Actinobacteria: Actinomycetales	54	9	37	3.92		Maitra & Roy 1961	Cells harvested at the end of growth at 24 hr; respiration measured for 60 min; pH 5.5; at pH 7.2 endogenous			

Sulfolobus acidocalcadareus	226.Sulfolobus acidocalcadareus	Archaea: Thermoprotei (Crenarchaeota): Sulfolobales	nmol O ₂ (mg protein) ⁻¹ min ⁻¹	MIN	13.5	15	60	1.33	[0.4]	Schäfer 1996 [Takayanagi et al. 1996, lobed cells, diam 0.8-1.0 µm]	respiration was 201.1 orig. units Steady-state respiration on endogenous substrate	
Contagious equine metritis bacterium (E-CMO)	227.Taylorella equigenitalis	Betaproteobacteria: Burkholderiales	pmol O ₂ (mg protein) ⁻¹ min ⁻¹	MIN	82	0.1	30	0.07	[0.4]	Lindmark et al. 1982 [BM9, genus, 0.7×0.7-1.8 µm]	Cells (English E-CMO strain) harvested from late log phase after 24 hr	Contagious equine metritis bacterium
Ferrobacillus ferro-oxidans (Thiobacillus ferrooxidans)	228.Thiobacillus ferrooxidans	Betaproteobacteria: Hydrogenophilales	µmol O ₂ (5.6 mg protein) ⁻¹ hr ⁻¹	MIN	0.12	0.5	25	0.50	[0.25]	Silver 1970 [Kelly & Wood 2000, rods 0.4×2.0 µm]	respiration of cells during 60 min of substrate deprivation	
Thiobacillus intermedius	229.Thiobacillus intermedius	Betaproteobacteria: Hydrogenophilales	µmol O ₂ (mg protein) ⁻¹ hr ⁻¹	MIN	0.6	11	30	7.78	[0.4]	London & Rittenberg 1966 [BM9, genus, 0.5-1.0-4.0 µm]	Cells were grown in the presence of glucose to the stationary phase; respiration of cells grown without glucose was “nil”	Facultative autotroph oxidizing thiosulfate
Thiobacillus thio-oxidans	230.Thiobacillus thiooxidans	Betaproteobacteria: Hydrogenophilales	µl O ₂ (mg N) ⁻¹ hr ⁻¹	MIN	4-10	1.5	28	1.22	[0.25]	Vogler 1942b [Kelly & Wood 2000, rods 0.4×2.0 µm]	young cultures (as cited by Newburgh 1954) Vogler 1942a: late cultures respire at 10-40 µl O ₂ (mg N) ⁻¹ hr ⁻¹	Autotroph growing on sulfur; in the presence of sulfur respiration increases by 20-100 times
Thiocapsa roseopersicina (M1)	231.Thiocapsa roseopersicina	Gammaproteobacteria: Chromatiales	nmol O ₂ (mg protein) ⁻¹ min ⁻¹	MIN	5.0	5.6	30	3.96	[1]	Overmann & Pfennig 1992 nmol O ₂ [Montesinos et al. 1983, Coulter counter]	Respiration of cells without microscopically visible sulfure globules at oxygen concentrations of 11-67 µM; respiration rates of phototrophically (anaerobically) and chemotrophically (microaerobically) grown cells do not differ; the species is capable of chemotrophic growth in the dark.	Purple sulfur bacteria Endogenous respiration of cells with visible sulfur globules is higher, up to 15 nmol O ₂ (mg protein) ⁻¹ min ⁻¹
Thiocystis violacea (2711)	232.Thiocystis violacea	Gammaproteobacteria: Chromatiales	nmol O ₂ (mg protein) ⁻¹ min ⁻¹	MIN	2.2	2.5	30	1.77	[11]	Overmann & Pfennig 1992 [BM9, genus, spherical or ovoid, 2.5-3.0 µm diam]	Respiration of cells without microscopically visible sulfure globules at oxygen concentrations of 11-67 µM; respiration rates of phototrophically (anaerobically) and chemotrophically (microaerobically) grown cells do not differ; the species is capable of chemotrophic growth in the dark.	Purple sulfur bacteria Endogenous respiration of cells with visible sulfur globules is higher, up to 30 orig. units

Table 1. Physiological characteristics of the isolates													
Isolate	Strain	Phylum	Unit	Temperature (°C)	Time (min)	Rate (mg O ₂ (mg protein) ⁻¹ min ⁻¹)	Rate (µl O ₂ (mg DM) ⁻¹ min ⁻¹)	Rate (mg O ₂ (mg protein) ⁻¹ min ⁻¹)	Rate (µl O ₂ (mg DM) ⁻¹ min ⁻¹)	Rate (mg O ₂ (mg protein) ⁻¹ min ⁻¹)	Rate (µl O ₂ (mg DM) ⁻¹ min ⁻¹)	Reference	Notes
Thiorhodovibrio winogradskyi (SSP1)	233.Thiorhodovibrio winogradskyi	Gamma	nmol O ₂ (mg protein) ⁻¹ min ⁻¹	5.9	6.6	30	4.67					Overmann & Pfennig 1992	differ; the species is capable of chemotrophic growth in the dark. Respiration of cells without microscopically visible sulfure globules at oxygen concentrations of 11-67 µM; respiration rates of phototrophically (anaerobically) and chemotrophically (microaerobically) grown cells do not differ; the species is capable of chemotrophic growth in the dark. Purple sulfur bacteria isolated from the littoral sediment of meromictic Mahoney Lake (British Columbia, Canada) Maximum respiration in the presence of substrate (H ₂ S) is 264 orig. units
Pseudomonas butanovora (ATCC 43655)	234.Unidentified bacterium	Gamma	nmol O ₂ (mg protein) ⁻¹ min ⁻¹	10-25	11	30	7.78		[0.6]			Vangnai et al. 2002 and personal communication with Dr. Luis A. Sayavedra-Soto (7 Nov 2006) [BM, species, rods, 0.6-0.8×1.1-2.4 µm]	Bacteria grown to the stationary phase (35-40 hr); kept at 25 C for at least 1 hr to lower endogenous respiration. Classification made for the Pseudomonas genus as described at www.bacterio.cict.fr. There is no such species at www.bacterio.cict.fr; strain ATCC 43655 is an “unidentified bacterium”
Unknown sp. (A-50)	235.Unknown	Unknown	µl O ₂ (16 mg DM) ⁻¹ min ⁻¹	32	6.7	30	4.74					Trudinger 1967	With a cell suspension kept at room temperature, the cells would show less endogenous respiration as time passes, down to 1-5 nmol O ₂ (mg protein) ⁻¹ min ⁻¹ Cells grown for 16 hr at 28 C; respiration measured for 30 min "Where necessary to reduce endogenous respiration, the bacteria were shaken at 30 C for 1 to 2 hr"
Unknown sp. (C-3)	236.Unknown	Unknown	µl O ₂ (3.1 mg DM) ⁻¹ min ⁻¹	28	30	30	21.21					Trudinger 1967	Cells grown for 16 hr at 28 C; respiration measured for 30 min "Where necessary to reduce endogenous respiration, the bacteria were shaken at 30 C for 1 to 2 hr"
Achromobacter hartlebii (NCIB 8129)	237.Unnamed rhizobiaceae	Alpha	µl O ₂ (mg DM) ⁻¹ hr ⁻¹	25	42	25	42.00					Bishop et al. 1962	Respiration measured immediately after harvesting the cells at The organism was isolated from percolation units containing garden soil and elemental sulfur. Max. resp. (in the presence of lactate) was 220 µl O ₂ (mg DM) ⁻¹ hr ⁻¹

Vibrio parahaemolyticus (biotype alginolyticus 156-70)	238.Vibrio alginolyticus	Gammaproteobacteria: "Vibrionales"	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	2	3.3	30	2.33	[0.6]	Jurtshuk & McQuitty 1976[BM9, genus, rods 0.5-0..8×1.4-2.6 μm]	the end of the logarithmic growth phase Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	
Vibrio fischeri (MAC 401)	239.Vibrio fischeri	Gammaproteobacteria: "Vibrionales"	$\text{n atoms O (mg DM)}^{-1} \text{ min}^{-1}$	MIN	20	22	20	31.11	[0.6]	Droniuk et al. 1987 [Allen & Baumann 1971, Fig. 16, 0.7×1.5 μm]	Respiration at 100 mM Na^+	Marine bacterium
Vibrio (metschnikovii) cholerae biotype proteus (ATCC 7708)	240.Vibrio metschnikovii	Gammaproteobacteria: "Vibrionales"	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	2	3.3	30	2.33	[0.6]	Jurtshuk & McQuitty 1976[BM9, genus, rods 0.5-0..8×1.4-2.6 μm]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration); respiration 1 orig. unit both studied strains, FC1011 and SAK3	
Vibrio parahaemolyticus (FC1011, SAK3)	241.Vibrio parahaemolyticus	Gammaproteobacteria: "Vibrionales"	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	1	1.7	30	1.20	[0.6]	Jurtshuk & McQuitty 1976 [BM9, genus, rods 0.5-0..8×1.4-2.6 μm]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration); respiration 1 orig. unit both studied strains, FC1011 and SAK3	
Vibrio sp. (Ant-300)	242.Vibrio sp.	Gammaproteobacteria: "Vibrionales"	% cellular carbon hr^{-1}	MIN	0.00 71	0.11	5	0.44	0.14	Novitsky & Morita 1977 [Novitsky & Morita 1976, Figs. 3b,c, cells starved for several weeks, cocci, diam 0.6-0.7 μm ; unstarved cells, rods, 1×2-4 $\mu\text{m} = 2 \mu\text{m}^3$]	Stable respiration of cells starved for several weeks; viability 50-100% judged by plate counts; during the first week of starvation, respiration is reduced by over 99%	Cell size estimated from linear dimensions of starved cells as shown in Fig. 3b of Novitsky & Morita 1976 During starvation, cells first increase in numbers at the expense of internal genetic material (nuclear bodies); as testified by Amy and Morita 1983 for 16 other marine bacterial isolates, this property of Ant-300 is not unique
Vitreoscilla stercoraria (ATCC 15218)	243.Vitreoscilla stercoraria	Betaproteobacteria: Neisseriales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	21	35	30	24.75		Jurtshuk & McQuitty 1976 [BM9, genus, filaments 1-3 μm diam]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	
Xanthomonas phaseoli (ATCC 9563)	244.Xanthomonas axonopodis	Gammaproteobacteria: Xanthomonadales	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	MIN	22	37	30	26.16	[0.25]	Jurtshuk & McQuitty 1976 [BM9, genus, rods 0.4-0.7×0.7-1.8 μm]	Cells harvested at the late-logarithmic growth phase (two thirds of the maximal growth concentration)	
Pasteurella pestis (virulent Alexander strain)	245.Yersinia pestis	Gammaproteobacteria: Enterobacteriales	$\mu\text{l O}_2 \text{ (mg N)}^{-1} \text{ hr}^{-1}$	MIN	29	4.8	28	3.90	[0.55]	Wessman & Miller 1966 [Kowalski et al.	Cells (virulent Alexander strain) harvested after 24 to 30 hr	Respiration decreased by approx. 1.5-fold during the first 1-2 hr of starvation and then stabilized for 72 hr

1999, 0.5-1×1-2 growth; stable respira-
μm] tion after 1.5 hr of
starvation

Notes on additional data not included into Table **S1a**:

- 1) *Listeria monocytogenes* (Friedman & Alm 1962) resting cells from cultures grown for 16 hr had no measurable endogenous respiration. The lowest reported values were 17.7 and 8.1 $\mu\text{L O}_2 (\text{mg N})^{-1} \text{ hr}^{-1}$ for growth in the presence of pyruvate. The same result was obtained by Welch et al. 1979.
- 2) Data needing verification (can be unrealistic): Gaudy et al. 1963, *E. coli* 500 mg DM/l consumed 10 mg O_2 in 7 hr (Fig. 3) = 0.003 W/kg at 25 C.
- 3) Goldshmidt & Wiss 1966: "Since the high endogenous respiration of 24-hr *Azotobacter* cultures can be reduced markedly by aerating in saline, washed vegetative cells were shaken for 4 hr before exposure to the EDTA-Tris system."
- 4) *Neisseria meningitidis* (Yu & deVoe 1980): washed whole cells were devoid of detectable endogenous respiration; Mallavia & Weiss 1970 obtained the same result (typical rates in the presence of substrates were about 5 $\mu\text{mol O}_2 (\text{mg protein})^{-1} \text{ hr}^{-1}$ = 93 W/kg)
- 5) Data needing verification (can be unrealistic): Mårdén et al. 1985 studied the decline of endogenous respiration of marine bacteria during several days' starvation and observed values of the order of $0.5 \times 10^{-10} \text{ mg O}_2 (\mu\text{m}^3)^{-1} \text{ hr}^{-1} \sim 200 \text{ W/kg}$. Since this value is in the upper range of respiration values in the presences of substrates (!) (Makarieva et al. 2005), there should be some error in the reported respiration units. Moreover, in the inlet to Fig. 2 showing respiration rate per biosurface the units are again similar (μm^{-3}) instead of (μm^{-2}) again indicating an inconsistency. Finally, Morton et al. 1994 characterize these rates as "low but detectable", which could hardly be plausible if they were indeed in the vicinity of several hundred W/kg. In the related work by Kjelleberg et al. 1982 it is stated that after five days of starvation a marine *Vibrio* respired at a rate of not less than 9 ng atoms $\text{O}_2 (10^9 \text{ viable cells})^{-1} \text{ min}^{-1}$ and cell volume was 0.4 μm^3 . This gives a mass-specific rate of 90 W/kg. Even taking a conservative estimate of energy content of the living matter of $4 \times 10^6 \text{ J/kg}$, we conclude that the bacterium should have eaten itself about ten times in five days, had it possessed such a high respiration rate. The data of Mårdén et al. 1985 and Kjelleberg et al. 1982 were not included into the present analysis.
- 6) Acidophilic bacterium PW2 (Goulbourne et al. 1986) starved at pH 3 or 4 progressively lost both respiration and viability, with no stabilization. In the presence of Mg^+ ions half-life of cells was 72 hr and respiration dropped from 84 nmol $\text{O}_2 (\text{mg protein})^{-1} \text{ min}^{-1}$ = 94 W/kg to virtually zero.
- 7) Data needing verification (can be unrealistic): *Azospirillum brasiliense* Sp7 and *Azospirillum lipoferum* Sp59b respired endogeneously at 0.73 and 0.98 $\mu\text{mol O}_2 (\text{mg protein})^{-1} \text{ min}^{-1}$, respectively (820 and 1100 W/kg) (Martinez-Drets et al. 1984)

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Table S1b. Numeric values used in the analyses presented in Figures 1-3 and Table 1 in the paper (after Table S1a). Log is decimal logarithm of the corresponding variable.

	Valid name	Class: Order	MIN	qWkg	LogqWkg	TC	q25Wkg	Logq25Wkg	Mpg	LogMpg
1.	Acetobacter aceti	Clostridia: Clostridiales	MIN	0.6	-0.222	30	0.42	-0.377	0.75	-0.125
2.	Acholeplasma laidlawii	Mollicutes: Acholeplasmatales	MIN	1.4	0.146	37	0.61	-0.215	0.04	-1.398
3.	Achromobacter ruhlandii	Betaproteobacteria: Burkholderiales	MIN	15	1.176	30	10.61	1.026	0.2	-0.699
4.	Achromobacter sp.	Betaproteobacteria: Burkholderiales	MIN	8.4	0.924	30	5.94	0.774	0.6	-0.222
5.	Achromobacter viscosus	Betaproteobacteria: Burkholderiales	MIN	35	1.544	30	24.75	1.394	0.6	-0.222
6.	Achromobacter xerosis	Betaproteobacteria: Burkholderiales	MIN	23	1.362	30	16.26	1.211	0.5	-0.301
7.	Acidovorax facilis	Betaproteobacteria: Burkholderiales	MIN	7	0.845	30	4.95	0.695	0.3	-0.523
8.	Acinetobacter baumannii	Gammaproteobacteria: Pseudomonadales	MIN	12	1.079	30	8.49	0.929	2	0.301
9.	Acinetobacter calcoaceticus	Gammaproteobacteria: Pseudomonadales	MIN	6.7	0.826	30	4.74	0.676	2	0.301
10.	Acinetobacter johnsonii	Gammaproteobacteria: Pseudomonadales	MIN	46	1.663	30	32.53	1.512	2	0.301
11.	Acinetobacter sp.	Gammaproteobacteria: Pseudomonadales	MIN	3.3	0.519	25	3.30	0.519	2	0.301
12.	Acinetobacter sp.	Gammaproteobacteria: Pseudomonadales	MIN	6	0.778	30	4.24	0.627	2	0.301
13.	Acinetobacter sp.	Gammaproteobacteria: Pseudomonadales	MIN	8	0.903	30	5.66	0.753	2	0.301
14.	Aeromonas hydrophila	Gammaproteobacteria: Aeromonadales	MIN	38	1.580	30	26.87	1.429		
15.	Aeromonas veronii	Gammaproteobacteria: Aeromonadales	MIN	20	1.301	30	14.14	1.150		
16.	Agrobacterium tumefaciens	Alphaproteobacteria: Rhizobiales	MIN	20	1.301	30	14.14	1.150	1.5	0.176
17.	Alcaligenes eutrophus	Betaproteobacteria: Burkholderiales	MIN	83	1.919	33	47.67	1.678	0.8	-0.097
18.	Alcaligenes faecalis	Betaproteobacteria: Burkholderiales	MIN	27	1.431	30	19.09	1.281		
19.	Alcaligenes sp.	Betaproteobacteria: Burkholderiales	MIN	2.1	0.322	30	1.48	0.170		
20.	Aminobacter lissarensis	Alphaproteobacteria: Rhizobiales	MIN	1.6	0.204	25	1.60	0.204	0.6	-0.222
21.	Amoebobacter purpureus	Gammaproteobacteria: Chromatiales	MIN	11	1.041	30	7.78	0.891	36	1.556
22.	Amoebobacter roseus	Gammaproteobacteria: Chromatiales	MIN	5.4	0.732	30	3.82	0.582	5	0.699
23.	Amoebobaeter pendens	Gammaproteobacteria: Chromatiales	MIN	8.5	0.929	30	6.01	0.779	5	0.699
24.	Aquaspirillum itersonii	Betaproteobacteria: Neisseriales	MIN	10	1.000	30	7.07	0.849	0.9	-0.046
25.	Arthrobacter crystallopoietes	Actinobacteria: Actinomycetales	MIN	0.2	-0.699	30	0.14	-0.854	1.7	0.230
26.	Arthrobacter globiformis	Actinobacteria: Actinomycetales	MIN	8.3	0.919	30	5.87	0.769	0.5	-0.301
27.	Arthrobacter sp.	Actinobacteria: Actinomycetales	MIN	1.2	0.079	30	0.85	-0.071	1.5	0.176
28.	Arthrobacter sp.	Actinobacteria: Actinomycetales	MIN	0.75	-0.125	25	0.75	-0.125	0.2	-0.699
29.	Arthrobacter sp.	Actinobacteria: Actinomycetales	MIN	6	0.778	30	4.24	0.627		
30.	Arthrobacter sp.	Actinobacteria: Actinomycetales	MIN	14	1.146	30	9.90	0.996		

31. <i>Azomonas agilis</i> ?	Gammaproteobacteria: Pseudomonadales	MIN	21	1.322	26	19.59	1.292	13	1.114
32. <i>Azorhizobium caulinodans</i>	Alphaproteobacteria: Rhizobiales	MIN	38	1.580	30	26.87	1.429	0.5	-0.301
33. <i>Azospirillum brasiliense</i>	Alphaproteobacteria: Rhodospirillales	MIN	27	1.431	37	11.75	1.070	1	0.000
34. <i>Azospirillum lipoferum</i>	Alphaproteobacteria: Rhodospirillales	MIN	39	1.591	37	16.98	1.230	4	0.602
35. <i>Azotobacter chroococcum</i>	Gammaproteobacteria: Pseudomonadales	MIN	25	1.398	30	17.68	1.247	14	1.146
36. <i>Azotobacter vinelandii</i>	Gammaproteobacteria: Pseudomonadales	MIN	1.5	0.176	30	1.06	0.025	0.5	-0.301
37. <i>Bacillus cereus</i>	"Bacilli": Bacillales	MIN	14	1.146	37	6.09	0.785	3.7	0.568
38. <i>Bacillus firmus</i>	"Bacilli": Bacillales	MIN	13	1.114	30	9.19	0.963	0.9	-0.046
39. <i>Bacillus megaterium</i>	"Bacilli": Bacillales	MIN	3.3	0.519	30	2.33	0.367	7	0.845
40. <i>Bacillus popilliae</i>	"Bacilli": Bacillales	MIN	0.8	-0.097	30	0.57	-0.244	0.8	-0.097
41. <i>Bacillus pumilus</i>	"Bacilli": Bacillales	MIN	5	0.699	30	3.54	0.549	0.7	-0.155
42. <i>Bacillus stearothermophilus</i>	"Bacilli": Bacillales	MIN	8.2	0.914	50	1.45	0.161	0.7	-0.155
43. <i>Bacillus subtilis</i>	"Bacilli": Bacillales	MIN	3.3	0.519	30	2.33	0.367	1.4	0.146
44. <i>Bdellovibrio bacteriovorus</i>	Deltaproteobacteria: Bdellovibrionales	MIN	25	1.398	30	17.68	1.247	0.3	-0.523
45. <i>Beneckea natriegens</i>	Gammaproteobacteria: "Vibrionales"	MIN	265	2.423	30	187.38	2.273	1.5	0.176
46. <i>Bradyrhizobium japonicum</i>	Alphaproteobacteria: Rhizobiales	MIN	1.0	0.000	29	0.76	-0.119	0.7	-0.155
47. <i>Branhamella catarrhalis</i>	Gammaproteobacteria: Pseudomonadales	MIN	1.7	0.230	37	0.74	-0.131	1.3	0.114
48. <i>Brucella melitensis</i>	Alphaproteobacteria: Rhizobiales	MIN	6.5	0.813	34	3.48	0.542	0.3	-0.523
49. <i>Burkholderia</i> sp.	Betaproteobacteria: Burkholderiales	MIN	21	1.322	30	14.85	1.172	0.7	-0.155
50. <i>Cellvibrio gilvus</i>	Gammaproteobacteria: Pseudomonadales	MIN	37	1.568	30	26.16	1.418	2	0.301
51. <i>Chromatium</i> sp.	Gammaproteobacteria: Chromatiales	MIN	11	1.041	25	11.00	1.041		
52. <i>Chromatium vinosum</i>	Gammaproteobacteria: Chromatiales	MIN	2.2	0.342	30	1.56	0.193	1.5	0.176
53. <i>Corynebacterium diphtheriae</i>	Actinobacteria: Actinomycetales	MIN	6.7	0.826	30	4.74	0.676		
54. <i>Corynebacterium</i> sp.	Actinobacteria: Actinomycetales	MIN	10	1.000	30	7.07	0.849		
55. <i>Delftia acidovorans</i>	Betaproteobacteria: Burkholderiales	MIN	13	1.114	30	9.19	0.963	0.8	-0.097
56. <i>Desulfovibrio salexigens</i>	Deltaproteobacteria: Desulfovibrionales	MIN	13	1.114	30	9.19	0.963	1.5	0.176
57. <i>Enterobacter aerogenes</i>	Gammaproteobacteria: "Enterobacteriales"	MIN	10	1.000	30	7.07	0.849	0.3	-0.523
58. <i>Enterobacter cloacae</i>	Gammaproteobacteria: "Enterobacteriales"	MIN	31	1.491	30	21.92	1.341	0.9	-0.046
59. <i>Enterococcus cecorum</i>	"Bacilli": "Lactobacillales"	MIN	3.8	0.580	30	2.69	0.430	0.4	-0.398
60. <i>Enterococcus faecalis</i>	"Bacilli": "Lactobacillales"	MIN	0.3	-0.523	30	0.21	-0.678	0.2	-0.699
61. <i>Enterococcus hirae</i>	"Bacilli": "Lactobacillales"	MIN	1.5	0.176	30	1.06	0.025		
62. <i>Enterococcus</i> sp.	"Bacilli": "Lactobacillales"	MIN	33	1.519	30	23.33	1.368	0.8	-0.097
63. <i>Escherichia coli</i>	Gammaproteobacteria: "Enterobacteriales"	MIN	32	1.505	37	14	1.146	0.7	-0.155
64. <i>Flavobacterium capsulatum</i>	Flavobacteria: Flavobacteriales	MIN	63	1.799	30	44.55	1.649	0.3	-0.523
65. <i>Francisella tularensis</i>	Gammaproteobacteria: Thiotrichales	MIN	3.7	0.568	37	1.61	0.207	0.01	-2.000
66. <i>Frankia</i> sp.	Actinobacteria: Actinomycetales	MIN	9	0.954	25	9.00	0.954		
67. <i>Haemophilus influenzae</i>	Gammaproteobacteria: Pasteurellales	MIN	0.5	-0.301	37	0.22	-0.658	0.14	-0.854
68. <i>Haemophilus parahaemolyticus</i>	Gammaproteobacteria: Pasteurellales	MIN	3.5	0.544	37	1.52	0.182		
69. <i>Haemophilus parainfluenzae</i>	Gammaproteobacteria: Pasteurellales	MIN	3.5	0.544	37	1.52	0.182	0.4	-0.398
70. <i>Halobacterium salinarum</i>	Archaea: Halobacteria: Halobacteriales	MIN	17	1.230	30	12.02	1.080	3.9	0.591
71. <i>Halomonas halodenitrificans</i>	Gammaproteobacteria: Oceanospirillales	MIN	67	1.826	25	67.00	1.826	0.4	-0.398
72. <i>Klebsiella pneumoniae</i>	Gammaproteobacteria: "Enterobacteriales"	MIN	3.3	0.519	30	2.33	0.367	0.4	-0.398
73. <i>Lactobacillus brevis</i>	"Bacilli": "Lactobacillales"	MIN	0.2	-0.699	30	0.14	-0.854	1.6	0.204
74. <i>Lactobacillus casei</i>	"Bacilli": "Lactobacillales"	MIN	1.7	0.230	30	1.20	0.079		
75. <i>Lactococcus lactis</i>	"Bacilli": "Lactobacillales"	MIN	0.6	-0.222	30	0.42	-0.377	0.2	-0.699
76. <i>Lactococcus</i> sp.	"Bacilli": "Lactobacillales"	MIN	2.3	0.362	30	1.63	0.212	0.8	-0.097

77. Legionella pneumophila	Gammaproteobacteria: Legionellales	MIN	20	1.301	37	8.71	0.940	0.3	-0.523
78. Methylobacterium extorquens	Alphaproteobacteria: Rhizobiales	MIN	6	0.778	30	4.24	0.627	1	0.000
79. Methylobacterium sp.	Gammaproteobacteria: Methylococcales	MIN	11	1.041	30	7.78	0.891	3	0.477
80. Methylophilus methylotrophus	Betaproteobacteria: Methylophilales	MIN	1.6	0.204	40	0.57	-0.244	0.15	-0.824
81. Methylosinus trichosporium	Alphaproteobacteria: Rhizobiales	MIN	42	1.623	30	29.70	1.473	1	0.000
82. Micrococcus luteus	Actinobacteria: Actinomycetales	MIN	1.2	0.079	37	0.52	-0.284	1.1	0.041
83. Micrococcus sp.	Actinobacteria: Actinomycetales	MIN	10	1.000	30	7.07	0.849		
84. Micrococcus sp.	Actinobacteria: Actinomycetales	MIN	17	1.230	30	12.02	1.080		
85. Moraxella osloensis	Gammaproteobacteria: Pseudomonadales	MIN	13	1.114	30	9.19	0.963	2.3	0.362
86. Mycobacterium fortuitum	Actinobacteria: Actinomycetales	MIN	17	1.230	30	12.02	1.080		
87. Mycobacterium leprae	Actinobacteria: Actinomycetales	MIN	3.3	0.519	37	1.44	0.158		
88. Mycobacterium lepraemurium	Actinobacteria: Actinomycetales	MIN	3.3	0.519	37	1.44	0.158		
89. Mycobacterium phlei	Actinobacteria: Actinomycetales	MIN	14	1.146	37	6.09	0.785	0.4	-0.398
90. Mycobacterium smegmatis	Actinobacteria: Actinomycetales	MIN	24	1.380	37	10.45	1.019		
91. Mycobacterium tuberculosis	Actinobacteria: Actinomycetales	MIN	2.2	0.342	37	0.96	-0.018	0.2	-0.699
92. Myxococcus xanthus	Deltaproteobacteria: Myxococcales	MIN	4.6	0.663	30	3.25	0.512	1	0.000
93. Neisseria elongata	Betaproteobacteria: Neisseriales	MIN	8.3	0.919	30	5.87	0.769	0.2	-0.699
94. Neisseria flava	Betaproteobacteria: Neisseriales	MIN	12	1.079	30	8.49	0.929	0.2	-0.699
95. Neisseria gonorrhoeae	Betaproteobacteria: Neisseriales	MIN	0.6	-0.222	37	0.26	-0.585	0.2	-0.699
96. Neisseria mucosa	Betaproteobacteria: Neisseriales	MIN	15	1.176	30	10.61	1.026	0.2	-0.699
97. Neisseria sicca	Betaproteobacteria: Neisseriales	MIN	13	1.114	30	9.19	0.963	0.2	-0.699
98. Nitrobacter winogradskyi	Alphaproteobacteria: Rhizobiales	MIN	10	1.000	30	7.07	0.849	0.24	-0.620
99. Nitrosomonas europaea	Betaproteobacteria: Nitrosomonadales	MIN	11	1.041	25	11.00	1.041	0.6	-0.222
100. Nocardia corallina	Actinobacteria: Actinomycetales	MIN	1.7	0.230	30	1.20	0.079	2.1	0.322
101. Nocardia farcinica	Actinobacteria: Actinomycetales	MIN	8.3	0.919	30	5.87	0.769	0.1	-1.000
102. Nocardia sp.	Actinobacteria: Actinomycetales	MIN	2.2	0.342	30	1.56	0.193		
103. Nocardia sp.	Actinobacteria: Actinomycetales	MIN	3	0.477	30	2.12	0.326		
104. Nonomuraea sp.	Actinobacteria: Actinomycetales	MIN	1	0.000	25	1.00	0.000		
105. Paracoccus denitrificans	Alphaproteobacteria: Rhodobacterales	MIN	8.3	0.919	30	5.87	0.769	0.16	-0.796
106. Pasteurella pseudotuberculosis	Gammaproteobacteria: Enterobacteriales	MIN	12	1.079	25	12.00	1.079		
107. Pediococcus acidilactici	"Bacilli": "Lactobacillales"	MIN	3.3	0.519	30	2.33	0.367	2.2	0.342
108. Phaeosporillum fulvum	Alphaproteobacteria: Rhodospirillales	MIN	28	1.447	28	22.74	1.357	2	0.301
109. Picrophilus oshimae	Thermoplasmata (Archaea): Thermoplasmatales	MIN	25	1.398	60	2.21	0.344	1	0.000
110. Proteus morganii	Gammaproteobacteria: "Enterobacteriales"	MIN	5	0.699	30	3.54	0.549	0.4	-0.398
111. Proteus vulgaris	Gammaproteobacteria: "Enterobacteriales"	MIN	5	0.699	37	2.18	0.338	0.4	-0.398
112. Pseudomonas aeruginosa	Gammaproteobacteria: Pseudomonadales	MIN	6.7	0.826	30	4.74	0.676	0.5	-0.301
113. Pseudomonas fluorescens	Gammaproteobacteria: Pseudomonadales	MIN	4.4	0.643	28	3.57	0.553	1	0.000
114. Pseudomonas formicans	Gammaproteobacteria: Pseudomonadales	MIN	10	1.000	30	7.07	0.849	7	0.845
115. Pseudomonas oleovorans	Gammaproteobacteria: Pseudomonadales	MIN	3.3	0.519	30	2.33	0.367	0.16	-0.796
116. Pseudomonas putida	Gammaproteobacteria: Pseudomonadales	MIN	1	0.000	30	0.71	-0.149	1.7	0.230
117. Pseudomonas saccharophila	Gammaproteobacteria: Pseudomonadales	MIN	33	1.519	30	23.33	1.368		
118. Pseudomonas sp.	Gammaproteobacteria: Pseudomonadales	MIN	2.2	0.342	30	1.56	0.193		
119. Pseudomonas sp.	Gammaproteobacteria: Pseudomonadales	MIN	8	0.903	30	5.66	0.753		
120. Pseudomonas sp.	Gammaproteobacteria: Pseudomonadales	MIN	3.3	0.519	25	3.30	0.519		
121. Pseudomonas sp.	Gammaproteobacteria: Pseudomonadales	MIN	8.3	0.919	30	5.87	0.769		
122. Pseudomonas sp.	Gammaproteobacteria: Pseudomonadales	MIN	15	1.176	30	10.61	1.026		

123. <i>Pseudomonas</i> sp.	Gammaproteobacteria: Pseudomonadales	MIN	18	1.255	30	12.73	1.105		
124. <i>Pseudomonas</i> sp.	Gammaproteobacteria: Pseudomonadales	MIN	35	1.544	20	49.50	1.695		
125. <i>Pseudomonas</i> sp.	Gammaproteobacteria: Pseudomonadales	MIN	17	1.230	30	12.02	1.080		
126. <i>Pseudonocardia nitrificans</i>	Actinobacteria: Actinomycetes	MIN	2.5	0.398	30	1.77	0.248		
127. <i>Rhizobium japonicum</i>	Alphaproteobacteria: Rhizobiales	MIN	16	1.204	23	18.38	1.264	0.7	-0.155
128. <i>Rhizobium leguminosarum</i>	Alphaproteobacteria: Rhizobiales	MIN	9.5	0.978	30	6.72	0.827	0.6	-0.222
129. <i>Rhizobium meliloti</i>	Alphaproteobacteria: Rhizobiales	MIN	12	1.079	30	8.49	0.929	0.7	-0.155
130. <i>Rhodobacter sphaeroides</i>	Alphaproteobacteria: Rhodobacterales	MIN	17	1.230	28	13.81	1.140	0.8	-0.097
131. <i>Rhodococcus rhodochrous</i>	Actinobacteria: Actinomycetales	MIN	17	1.230	30	12.02	1.080		
132. <i>Rhodococcus</i> sp.	Actinobacteria: Actinomycetales	MIN	2.2	0.342	25	2.20	0.342		
133. <i>Rhodococcus</i> sp.	Actinobacteria: Actinomycetales	MIN	7	0.845	30	4.95	0.695		
134. <i>Rhodococcus</i> sp.	Actinobacteria: Actinomycetales	MIN	7.4	0.869	25	7.40	0.869		
135. <i>Rhodospirillum rubrum</i>	Alphaproteobacteria: Rhodospirillales	MIN	3.8	0.580	28	3.09	0.490	9	0.954
136. <i>Sallinivibrio costicola</i>	Gammaproteobacteria: "Vibrionales"	MIN	7	0.845	25	7.00	0.845	0.4	-0.398
137. <i>Salmonella typhimurium</i>	Gammaproteobacteria: "Enterobacteriales"	MIN	15	1.176	37	6.53	0.815	0.66	-0.180
138. <i>Serratia marcescens</i>	Gammaproteobacteria: "Enterobacteriales"	MIN	4	0.602	37	1.74	0.241	0.4	-0.398
139. <i>Shigella flexneri</i>	Gammaproteobacteria: "Enterobacteriales"	MIN	10	1.000	37	4.35	0.638		
140. <i>Sphaerotilus natans</i>	Betaproteobacteria: Burkholderiales	MIN	45	1.653	28	36.55	1.563	6.5	0.813
141. <i>Sporosarcina ureae</i>	"Bacilli": Bacillales	MIN	12	1.079	30	8.49	0.929	3.8	0.580
142. <i>Staphylococcus aureus</i>	"Bacilli": Bacillales	MIN	16	1.204	37	7	0.845	0.27	-0.569
143. <i>Staphylococcus epidermidis</i>	"Bacilli": Bacillales	MIN	27	1.431	30	19.09	1.281	0.5	-0.301
144. <i>Staphylococcus simulans</i>	"Bacilli": Bacillales	MIN	10	1.000	37	4.35	0.638		
145. <i>Staphylococcus</i> sp.	"Bacilli": Bacillales	MIN	17	1.230	30	12.02	1.080		
146. <i>Streptococcus agalactiae</i>	"Bacilli": "Lactobacillales"	MIN	1.7	0.230	37	0.74	-0.131	0.3	-0.523
147. <i>Streptococcus pneumoniae</i>	"Bacilli": "Lactobacillales"	MIN	1.7	0.230	30	1.20	0.079	0.4	-0.398
148. <i>Streptococcus pyogenes</i>	"Bacilli": "Lactobacillales"	MIN	3.3	0.519	30	2.33	0.367	0.2	-0.699
149. <i>Streptomyces coelicolor</i>	Actinobacteria: Actinomycetales	MIN	40	1.602	30	28.28	1.451		
150. <i>Streptomyces fradiae</i>	Actinobacteria: Actinomycetales	MIN	22	1.342	30	15.56	1.192		
151. <i>Streptomyces griseus</i>	Actinobacteria: Actinomycetales	MIN	18	1.255	30	12.73	1.105		
152. <i>Streptomyces longispororuber</i>	Actinobacteria: Actinomycetales	MIN	3.3	0.519	30	2.33	0.367		
153. <i>Streptomyces olivaceus</i>	Actinobacteria: Actinomycetales	MIN	9	0.954	37	3.92	0.593		
154. <i>Sulfolobus acidocaldarius</i>	Archaea: Thermoprotei (Crenarchaeota): Sulfolobales	MIN	15	1.176	60	1.33	0.124	0.4	-0.398
155. <i>Taylorella equigenitalis</i>	Betaproteobacteria: Burkholderiales	MIN	0.1	-1.000	30	0.07	-1.155	0.4	-0.398
156. <i>Thiobacillus ferrooxidans</i>	Betaproteobacteria: Hydrogenophilales	MIN	0.5	-0.301	25	0.50	-0.301	0.25	-0.602
157. <i>Thiobacillus intermedius</i>	Betaproteobacteria: Hydrogenophilales	MIN	11	1.041	30	7.78	0.891	0.4	-0.398
158. <i>Thiobacillus thiooxidans</i>	Betaproteobacteria: Hydrogenophilales	MIN	1.5	0.176	28	1.22	0.086	0.25	-0.602
159. <i>Thiocapsa roseopersicina</i>	Gammaproteobacteria: Chromatiales	MIN	5.6	0.748	30	3.96	0.598	1	0.000
160. <i>Thiocystis violacea</i>	Gammaproteobacteria: Chromatiales	MIN	2.5	0.398	30	1.77	0.248	11	1.041
161. <i>Thiorhodovibrio winogradskyi</i>	Gammaproteobacteria: Chromatiales	MIN	6.6	0.820	30	4.67	0.669		
162. Unidentified bacterium	Gammaproteobacteria: Pseudomonadales	MIN	11	1.041	30	7.78	0.891	0.6	-0.222
163. Unknown	Unknown	MIN	6.7	0.826	30	4.74	0.676		
164. Unknown	Unknown	MIN	30	1.477	30	21.21	1.327		
165. Unnamed rhizobiaceae	Alphaproteobacteria: Rhizobiales	MIN	42	1.623	25	42.00	1.623		
166. <i>Vibrio alginolyticus</i>	Gammaproteobacteria: "Vibrionales"	MIN	3.3	0.519	30	2.33	0.367	0.6	-0.222
167. <i>Vibrio fischeri</i>	Gammaproteobacteria: "Vibrionales"	MIN	22	1.342	20	31.11	1.493	0.6	-0.222
168. <i>Vibrio metschnikovii</i>	Gammaproteobacteria: "Vibrionales"	MIN	3.3	0.519	30	2.33	0.367	0.6	-0.222

169.Vibrio parahaemolyticus	Gammaproteobacteria: "Vibrionales"	MIN	1.7	0.230	30	1.20	0.079	0.6	-0.222
170.Vibrio sp.	Gammaproteobacteria: "Vibrionales"	MIN	0.11	-0.959	5	0.44	-0.357	0.14	-0.854
171.Vitreoscilla stercoraria	Betaproteobacteria: Neisseriales	MIN	35	1.544	30	24.75	1.394		
172.Xanthomonas axonopodis	Gammaproteobacteria: Xanthomonadales	MIN	37	1.568	30	26.16	1.418	0.25	-0.602
173.Yersinia pestis	Gammaproteobacteria: Enterobacteriales	MIN	4.8	0.681	28	3.90	0.591	0.55	-0.260

Dataset S2. Endogenous respiration rates in heterotrophic protozoa

Notes to Table S2a:

Data on endogenous respiration in heterotrophic protozoa mostly come from the compilation of Vladimirova & Zotin (1985). The data base for that work (Vladimirova & Zotin 1983, hereafter VZ83) is deposited at the All-Russian Institute of Scientific and Technological Information and was ordered from there. The data base contains more than 550 data entries for over 100 species (growing and starved cultures), of which 193 values of respiration in the absence of substrates (endogenous respiration) are presented below. Additionally, eight data entries were obtained from other sources (Ryley 1955a; Fenchel & Finlay 1983; Crawford et al. 1994). These data are presented in the end of the table with Source indicated as "other" and reference provided in the "**Reference**" column. Otherwise "**Source**" gives the original number of reference in the data base of Vladimirova & Zotin (1985); "**Reference**" is that reference itself; "**Culture age, stage or state**" gives literal translation of Vladimirova & Zotin's (1985) comments on data entries and/or relevant comments of other authors. Note that "Taxonomic group" is determined from various sources; this table should **not** be considered as an authoritative representation of protozoan complicated taxonomy.

"**Original units**" are the units of endogenous respiration rate measurements as given in the original publication (VZ83 or other); **qou** is the numeric value of endogenous respiration rate in the original units. In VZ83 data base all data are reported in ml oxygen consumed by 10^9 cells per hour, cell mass is simultaneously provided (column "**Mpg**").

qou is the numeric value of endogenous respiration rate in the original units.

qWkg is the original endogenous respiration rate **qou** converted to $W (kg WM)^{-1}$ (Watts per kg wet mass). For the data of VZ83, **qWkg** = $(qou / Mpg) \times (20 J/ml O_2) \times 10^6 / (3600 s) W (kg WM)^{-1}$.

Mpg: cell mass, pg ($1 pg = 10^{-12} g$). Where **Mpg** value is in brackets, it was characterized in VZ83 as "mean for the species" and apparently determined from different sources than the source of respiration rate. The same with the data of Fenchel & Finlay (1983). However, this should not bias the mass-specific metabolic rate value, because, as pointed out by VZ83, normally metabolic rates in unicells are reported on a mass-specific basis, so dividing **qou** by **Mpg** (irrespective of how the latter is determined) is equivalent to retrieving the original mass-specific value. Notably, independent analyses of protozoan metabolic rates by Fenchel & Finlay (1983) and Vladimirova & Zotin (1985) yielded similar results with respect to the mean mass-specific metabolic rate, as analysed by Makarieva et al. (2005).

TC: temperature in degrees Celsius. All data in VZ83 correspond to 20 °C, so TC = 20 is shown everywhere in the "**TC**" column.

For each species, the minimum qWkg value was chosen and converted to 25 °C. Data used in the analyses presented in Table 1 and Figures 1-3 in the paper are, for convenience, compiled below in a separate Table S2b.

Table S2a. Endogenous respiration rates in heterotrophic Protozoa.

1. Taxonomic group	Species	Original units	qou	qWkg	Mpg	TC	Culture age, stage or state	Source	Reference
2. Acanthamoebidae	Acanthamoeba (Hartmanella) castellani	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	2.69	11.2	1340	20	Log	114	Griffiths & Hughes 1968
3. Acanthamoebidae	Acanthamoeba (Hartmanella) castellani	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	33.7	28.2	[6700]	20	Log	297	Waidyasekera & Kitching 1975
4. Acanthamoebidae	Acanthamoeba (Hartmanella) castellani	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	5.8	4.8	[6700]	20	Log	86	Edwards & Lloyd 1977a
5. Acanthamoebidae	Acanthamoeba (Hartmanella) castellani	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	9.93	8.3	[6700]	20	Stationary	86	Edwards & Lloyd 1977a
6. Acanthamoebidae	Acanthamoeba (Hartmanella) castellani	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	4.53	3.8	[6700]	20	Log	87	Edwards & Lloyd 1977b
7. Acanthamoebidae	Acanthamoeba sp.	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	4.08	5.3	4320	20		210	Neff et al. 1958
8. Actinophryidae	Actinosphaerium eichhornii	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	1037	0.4	14561000	20		135	Howland & Bernstein 1931
9. Amoebae	Amoeba proteus	nl O ₂ (cell) ⁻¹ hr ⁻¹	0.248	1.5	[936000]	20	starved 2 d	OTHER	Fenchel & Finlay 1983
10. Amoebae	Amoeba proteus	nl O ₂ (cell) ⁻¹ hr ⁻¹	0.450	1	900000	20	starved 2-4 d	OTHER	Fenchel & Finlay 1983
11. Euglenida	Astasia klebsii	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	2.4	3.5	[3800]	20	Log	291	von Dach 1942
12. Euglenida	Astasia klebsii	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	1.2	1.8	[3800]	20	Stationary	291	von Dach 1942
13. Euglenida	Astasia klebsii	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	22	1.8	[3800]	20	Log	292	von Dach 1950
14. Euglenida	Astasia longa	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	2206	9.4	13500	20	Synchronized	223	Padilla 1960
15. Euglenida	Astasia longa	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	22.7	20	6400	20		224	Padilla & James 1960
16. Euglenida	Astasia longa	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	3.8	4.4	4800	20	Log	139	Hunter & Lee 1962
17. Euglenida	Astasia longa	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	6.0	4.9	6900	20	Log	305	Wilson 1963
18. Euglenida	Astasia longa	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	4.1	3.8	6000	20	Log	306	Wilson & James 1963
19. Euglenida	Astasia longa	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	7.1	2.9	13500	20	Synchronized	306	Wilson & James 1963
20. Ciliophora	Bresslaua insidiatrix	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	111	23.9	26000	20	Active	252	Scholander et al. 1952
21. Ciliophora	Bresslaua insidiatrix	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	66	21.7	17000	20	Cysts	252	Scholander et al. 1952
22. Amoebidae	Chaos chaos	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	17250	1.3	73870000	20	1 day starvation	133	Holter & Zeuthen 1948
23. Amoebidae	Chaos chaos	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	15596	1.7	50000000	20	4-5 days starvation	132	Holter 1950
24. Amoebidae	Chaos chaos	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	2297	1.3	9644000	20		252	Scholander et al. 1952
25. Amoebidae	Chaos chaos	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	6772	0.8	50000000	20		22	Albritton 1955
26. Cryptophyta	Chilomonas paramecium	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	6.4	16	2260	20	48 hr	191	Mast et al. 1936
27. Cryptophyta	Chilomonas paramecium	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	27.5	84	1840	20	Log	142	Hutchens et al. 1948
28. Cryptophyta	Chilomonas paramecium	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	24.4	61	[2260]	20		134	Holz 1954
29. Ciliophora	Coleps hirtus	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	68	4.2	91000	20	24 hr starvation	250	Sarojini & Nagabhushanam 1966
30. Ciliophora	Colpidium campylum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	104.5	10.8	[54400]	20	48 hr	230	Pitts 1932
31. Trypanosomatidae	Crithida (Strigomonas) fasciculata	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	2.90	7.8	[2078]	20		138	Hunter & Cosgrove 1956

32. Trypanosomatidae	Crithida (Strigomonas) fasciculata	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	2.96	8.0	[2078]	20	Log	70	Cosgrove 1959
33. Trypanosomatidae	Crithida (Strigomonas) fasciculata	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	2.72	7.3	[2078]	20	44-48 hr	137	Hunter 1960
34. Trypanosomatidae	Crithidia (Strigomonas) oncopelti	μl O ₂ (mg dry mass) ⁻¹ hr ⁻¹	12.4	10	[30]	30		OTHER	Ryley 1955a; [cell mass data: Holwill 1965, cylinder or pear shaped, 8.2×2.6 μm]
35. Dictyosteliida	Dictyostelium discodeum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.74	4.9	840	20	Amoebaform	112	Gregg 1950
36. Dictyosteliida	Dictyostelium discodeum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	1.33	7.6	980	20	Migration	112	Gregg 1950
37. Dictyosteliida	Dictyostelium discodeum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.80	7.7	580	20	Beginning of cumulation	112	Gregg 1950
38. Dictyosteliida	Dictyostelium discodeum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.83	9.5	490	20	Amoeba form	169,310	Liddel & Wright 1961; Wright 1964
39. Dictyosteliida	Dictyostelium discodeum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.48	6.9	390	20	Amoeba migration	169,310	Liddel & Wright 1961; Wright 1964
40. Dictyosteliida	Dictyostelium discodeum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.42	7.6	310	20	Precumulation	169,310	Liddel & Wright 1961; Wright 1964
41. Apicomplexa	Eimeria acervulina	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.40-1.52	2.0	2640	20	Sporulation	307	Wilson P. 1961
42. Apicomplexa	Eimeria stiedae	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	3.53	2.5	[7980]	20	Sporulation	296	Wagenbach & Burns 1969
43. Apicomplexa	Eimeria tenella	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	6.43	7.2	[5010]	20	Non-sporulating oocysts	262	Smith & Herrick 1944
44. Apicomplexa	Eimeria tenella	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	2.22	2.3	5440	20	Sporulation	296	Wagenbach & Burns 1969
45. Trypanosomatidae	Endotrypanum schaudinni	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.019	0.93	[114]	20	Log	312	Zeledon 1960a
46. Trypanosomatidae	Endotrypanum schaudinni	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.017	0.84	[114]	20	Log	313	Zeledon 1960b
47. Trypanosomatidae	Endotrypanum schaudinni	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.020	1.0	[114]	20	Log	314	Zeledon 1960c
48. Trypanosomatidae	Endotrypanum schaudinni	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.018	0.88	[114]	20	Log	315	Zeledon 1960d
49. Entamoebidae	Entamoeba histolytica	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	1.01	0.7	[8600]	20		202	Montalvo et al. 1971
50. Entamoebidae	Entamoeba histolytica	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	1.20	0.4	[8600]	20		235	Reeves 1971
51. Entamoebidae	Entamoeba histolytica	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	3.32	1.3	13900	20		299	Weinbach & Diamond 1974
52. Ciliophora	Frontonia leucas	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	49	0.4	745000	20		158	Laybourn & Finlay 1976
53. Trypanosomatidae	Leishmania brasiliensis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.076	33	13	20	Log	79	de Monge & Zeledon 1963
54. Trypanosomatidae	Leishmania brasiliensis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.076	33	13	20	Log	316	Zeledon & de Monge 1966
55. Trypanosomatidae	Leishmania brasiliensis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.032	22	8	20	Log	316	Zeledon & de Monge 1966
56. Trypanosomatidae	Leishmania brasiliensis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.074	59	13	20	Log	317	Zeledon & de Monge 1967
57. Trypanosomatidae	Leishmania brasiliensis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.028	22	7	20	Log	317	Zeledon & de Monge 1967
58. Trypanosomatidae	Leishmania brasiliensis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.032	20	9	20	Log	317	Zeledon & de Monge 1967
59. Trypanosomatidae	Leishmania brasiliensis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.040	28	8	20	Log	317	Zeledon & de Monge 1967
60. Trypanosomatidae	Leishmania donovani	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.039	12	[18]	20		102	Fulton & Joyner 1949
61. Trypanosomatidae	Leishmania donovani	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.055	17	[18]	20	3 days	56	Chatterjee & Ghosh 1959
62. Trypanosomatidae	Leishmania donovani	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.030	9.3	[18]	20	6 days	56	Chatterjee & Ghosh 1959
63. Trypanosomatidae	Leishmania donovani	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.249	77	[18]	20	3 days	108	Ghosh & Chatterjee 1961
64. Trypanosomatidae	Leishmania enrietti	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.015	7.0	[12]	20	Log	312	Zeledon 1960a
65. Trypanosomatidae	Leishmania enrietti	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.017	7.9	[12]	20	Log	313	Zeledon 1960b
66. Trypanosomatidae	Leishmania enrietti	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.022	10	[12]	20		111	Greenblatt & Glaser 1965
67. Paramoebidae	Mayorella palestinensis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	5.52	3.7	8300	20	1 day starvation	236	Reich 1948
68. Dinophyceae	Noctiluca miliaris	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	57897	1.4	223986000	20		232	Rajagopal 1962
69. Ciliophora	Paramecium aurelia	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	354	34.9	105000	20	10 days	219	Pace & Kimura 1944

70. Ciliophora	Paramecium aurelia	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	775	34.2	127000	20	5-7 days	215	Pace 1945
71. Ciliophora	Paramecium aurelia	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	81.5	2.9	160000	20	2 days starvation	258	Simonsen & Van Wagtendonk 1952
72. Ciliophora	Paramecium aurelia	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	266-280	9.6	160000	20	2 days starvation	165	Levine & Howard 1955
73. Ciliophora	Paramecium calkinsi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	158-177	10.0	[157000]	20	Capable of copulation	32,33	Boell & Woodruff 1940
74. Ciliophora	Paramecium calkinsi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	272-304	6.0	[157000]	20	Incapable of copulation	32,33	Boell & Woodruff 1940
75. Ciliophora	Paramecium calkinsi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	89-101	10.5	[157000]	20	Basal level	33	Woodruff 1940
76. Ciliophora	Paramecium calkinsi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	285	3.4	[157000]	20		31	Boell 1945
77. Ciliophora	Paramecium caudatum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	79	0.8	[526000]	20	Saturated	179	Lund 1918
78. Ciliophora	Paramecium caudatum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	22	0.2	[526000]	20	2 days starvation	179	Lund 1918
79. Ciliophora	Paramecium caudatum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	2713	30.3	501000	20		73	Cunningham & Kirk 1942
80. Ciliophora	Paramecium caudatum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	2110	17.7	668000	20	10 days	219	Pace & Kimura 1944
81. Ciliophora	Paramecium caudatum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	418	4.5	526000	20		61	Clark 1945
82. Ciliophora	Paramecium caudatum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	2804	25.4	618000	20	5 days	215	Pace 1945
83. Ciliophora	Paramecium caudatum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	2464	22.1	618000	20	15 days	215	Pace 1945
84. Ciliophora	Paramecium caudatum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	1709	15.5	618000	20	19 days	215	Pace 1945
85. Ciliophora	Paramecium caudatum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	394	4.2	526000	20		211	Nicol 1960
86. Ciliophora	Paramecium caudatum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	300	3.2	530000	20		17,18	Khlebovitch 1972; 1974
87. Ciliophora	Paramecium multimicronucleatum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	646	5.3	[685000]	20		191	Mast et al. 1936
88. Amoebidae	Pelomyxa carolinensis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	7050	1.1	36730000	20	3 days starvation	216	Pace & Belda 1944a
89. Amoebidae	Pelomyxa carolinensis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	6381	1.0	35400000	20		217	Pace & Belda 1944b
90. Amoebidae	Pelomyxa carolinensis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	5728	1.0	32830000	20		217	Pace & Belda 1944b
91. Amoebidae	Pelomyxa carolinensis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	6291	1.1	32010000	20		217	Pace & Belda 1944b
92. Amoebidae	Pelomyxa carolinensis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	5594	0.9	36060000	20		220	Pace & Kimura 1946
93. Amoebidae	Pelomyxa carolinensis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	4763	0.5	49708000	20		218	Pace & Frost 1948
94. Amoebidae	Pelomyxa carolinensis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	8041	0.9	49708000	20	Young	222	Pace & McCashland 1951
95. Amoebidae	Pelomyxa palustris	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	11955	0.7	90000000	20		163	Leiner et al. 1968
96. Apicomplexa	Plasmodium cathemerium	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.017	1.8	54	20	Intact blood, ¼ growth	184	Maier & Coggeshall 1941
97. Apicomplexa	Plasmodium cathemerium	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.051	3.2	89	20	Intact blood, ¾ growth	184	Maier & Coggeshall 1941
98. Apicomplexa	Plasmodium gallinaceum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.016	1.3	71	20	Erythrocytes extracted from blood	96	Evans 1946
99. Apicomplexa	Plasmodium gallinaceum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.014	1.1	71	20	Erythrocytes extracted from blood, 3 days after infection	264	Speck et al. 1946
100. Apicomplexa	Plasmodium gallinaceum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.010	0.8	71	20	Erythrocytes extracted from blood, 58% infected erythrocytes	186	Marshall 1948a

101.	Apicomplexa	Plasmodium gallinaceum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.031	2.4	71	20	Erythrocytes extracted from blood, 96% infected	187	Marshall 1948b
102.	Apicomplexa	Plasmodium knowlesi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.303	28.8	59	20	erythrocytes Erythrocytes extracted from blood, 7-12% infected	101	Fulton 1939
103.	Apicomplexa	Plasmodium knowlesi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.009	0.9	59	20	erythrocytes	184	Maier & Coggeshall 1941
104.	Apicomplexa	Plasmodium knowlesi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.010	1.5	37	20	Erythrocytes extracted from blood, 35% infected	199	McKee et al. 1946
105.	Kinetoplastida	Pleuromonas jaculans	nl O ₂ (cell) ⁻¹ hr ⁻¹	0.00005	12	25	20	erythrocytes starved	OTHER	Fenchel & Finlay 1983
106.	Ciliophora?	Podophrya fixa	nl O ₂ (cell) ⁻¹ hr ⁻¹	0.0077	0.52	14900	20	starved 96 hr	OTHER	Fenchel & Finlay 1983
107.	Trypanosomatidae	Schizotrypanum verpertilionis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.040	2.1	108	20	Log	312	Zeledon 1960a
108.	Ciliophora?	Spirostoma minus	nl O ₂ (cell) ⁻¹ hr ⁻¹	0.31	3.4	[500000]	20	starved	OTHER	Fenchel & Finlay 1983
109.	Ciliophora	Spirostomum ambiguum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	12700	5.9	12000000	20		17	Khlebiovich 1972
110.	Ciliophora	Spirostomum intermedium	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	500	2.1	1342000	20	6 days	226	Pigon 1955
111.	Ciliophora	Spirostomum intermedium	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	50	1.2	229000	20	113 days	226	Pigon 1955
112.	Ciliophora	Spirostomum teres	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	40	0.5	423000	20		158	Laybourn & Finlay 1976
113.	Ciliophora	Stentor coeruleus	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	1588	13.1	[680000]	20	3-5 days starvation	303	Whiteley 1960
114.	Ciliophora	Stentor coeruleus	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	350	1.8	1100000	20	Before division	156	Laybourn 1975
115.	Ciliophora	Stentor coeruleus	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	225	1.9	680000	20	3 days after division	156	Laybourn 1975
116.	Ciliophora	Tetrahymena geleii (pyriformis)	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	77	12.3	35000	20	Log	213	Ormsbee 1942
117.	Ciliophora	Tetrahymena geleii (pyriformis)	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	74	7.8	53000	20	Stationary	213	Ormsbee 1942
118.	Ciliophora	Tetrahymena geleii (pyriformis)	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	193	45.0	24000	20	3 days	221	Pace & Lyman 1947
119.	Ciliophora	Tetrahymena geleii (pyriformis)	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	128	29.9	24000	20	6-7 days	221	Pace & Lyman 1947
120.	Ciliophora	Tetrahymena geleii (pyriformis)	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	26.6	6.8	[22000]	20	Log	253	Seaman 1949
121.	Ciliophora	Tetrahymena geleii (pyriformis)	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	23	5.9	[22000]	20	3.5 days	254,255	Seaman 1950; Seaman & Noulhan 1950
122.	Ciliophora	Tetrahymena geleii (pyriformis)	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	59	15.0	[22000]	20	48 hr	89	Eichel 1953
123.	Ciliophora	Tetrahymena geleii (pyriformis)	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	62	15.8	[22000]	20	6-10 days	214	Ottova 1955
124.	Ciliophora	Tetrahymena geleii (pyriformis)	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	24	6.1	[22000]	20	20 days	214	Ottova 1955
125.	Ciliophora	Tetrahymena pyriformis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	95	24.2	[22000]	20	6 days	182	Lwoff 1934
126.	Ciliophora	Tetrahymena pyriformis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	82	20.9	[22000]	20	6 days	243	Ryley 1952
127.	Ciliophora	Tetrahymena pyriformis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	39	9.9	[22000]	20	4 days	241	Roth et al. 1954

128.	Ciliophora	Tetrahymena pyriformis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	31	7.9	[22000]	20	17 hr starvation	241	Roth et al. 1954
129.	Ciliophora	Tetrahymena pyriformis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	52.8	13.4	[22000]	20	3 days	240	Roth & Eichel 1955
130.	Ciliophora	Tetrahymena pyriformis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	42	13.8	17000	20	Log	120	Hamburger & Zeuthen 1957
131.	Ciliophora	Tetrahymena pyriformis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	53	6.1	49000	20	Synchronized	120	Hamburger & Zeuthen 1957
132.	Ciliophora	Tetrahymena pyriformis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	67	11.4	33000	20	3-7 days	276	Van de Vijver 1966
133.	Ciliophora	Tetrahymena pyriformis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	29	7.4	22000	20		62	Conner & Cline 1967
134.	Ciliophora	Tetrahymena pyriformis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	63.9	16.3	22000	20	5-10 days, saturated	29	Biczók 1969
135.	Ciliophora	Tetrahymena pyriformis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	53.2	13.5	22000	20	24 hr starvation	29	Biczók 1969
136.	Ciliophora	Tetrahymena pyriformis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	25.2	6.4	22000	20	1-2 hr starvation	121	Hamburger & Zeuthen 1971
137.	Ciliophora	Tetrahymena pyriformis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	48	12.2	22000	20	5-6 days	47	Burmeister 1972
138.	Ciliophora	Tetrahymena pyriformis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	185	94.2	11000	20		17,18	Khlebovitch 1972; 1974
139.	Ciliophora	Tetrahymena pyriformis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	40	10.2	22000	20	5-6 days	48	Burmeister 1976
140.	Ciliophora	Tracheloraphis sp.	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	969	16.0	340000	20	24 hr after collection	278	Vernberg & Coull 1974
141.	Trichomonada	Trichomonas (Tritrichomonas) foetus	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.67	3.6	670	20	48 hr	246	Ryley 1955b
142.	Trichomonada	Trichomonas (Tritrichomonas) foetus	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.282	2.7	[580]	20	48 hr	83	Doran 1957
143.	Trichomonada	Trichomonas (Tritrichomonas) foetus	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.47	4.5	[580]	20	24 hr	51	Čerkasovová 1970
144.	Trichomonada	Trichomonas batrachorum	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.192	1.9	[560]	20	48 hr	84	Doran 1958
145.	Trichomonada	Trichomonas nasai	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.305	8.1	[210]	20	48 hr	83	Doran 1957
146.	Trichomonada	Trichomonas vaginalis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.204	1.7	[690]	20	24-48 hr	233	Read & Rothman 1955
147.	Trichomonada	Trichomonas vaginalis	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.16	1.0	870	20	48 hr	309	Wirtschafter et al. 1956
148.	Amoebae	Trichosphaerium sieboldi	% cell carbon hr ⁻¹	0.27	4	100	20	starved 24 hr	OTHER	Crawford et al. 1994
149.	Trypanosomatidae	Trypanosoma (Schizotrypanum) cruzi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.028	1.5	[107]	20	14 days, cultured	209	Nakamura & Anderson 1951
150.	Trypanosomatidae	Trypanosoma (Schizotrypanum) cruzi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.058	4.0	82	20	cultured	282	von Brand & Agosin 1952
151.	Trypanosomatidae	Trypanosoma (Schizotrypanum) cruzi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.063	3.3	[107]	20	14 days,cultured	247	Ryley 1956
152.	Trypanosomatidae	Trypanosoma (Schizotrypanum) cruzi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.036	1.2	169	20	Blood form	247	Ryley 1956
153.	Trypanosomatidae	Trypanosoma (Schizotrypanum) cruzi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.040	2.1	[107]	20	cultured	312	Zeledon 1960a
154.	Trypanosomatidae	Trypanosoma (Schizotrypanum) cruzi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.041	2.1	[107]	20	cultured	313	Zeledon 1960b
155.	Trypanosomatidae	Trypanosoma (Schizotrypanum) cruzi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.044	2.3	[107]	20	Log, cultured	315	Zeledon 1960d
156.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.009	0.9	58	20	10-12 days	242,244	Ryley 1951; 1953
157.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.011	1.1	58	20	8-10 days	247	Ryley 1956
158.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.074	7.1	58	20	7-9 days	272	Thurston 1958
159.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.127	15.5	46	20	4 days	249	Sanchez & Dusanic 1968
160.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.216	20.9	58	20	8 days	249	Sanchez & Dusanic 1968
161.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.141	13.6	58	20	12 days	249	Sanchez & Dusanic 1968
162.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.289	28	58	20	mean	174	Lincicome & Warsi 1965
163.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.082	7.9	58	20	8 days	175	Lincicome & Warsi 1966

164.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.211	20.4	58	20	14 days	175	Lincicome & Warsi 1966
165.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.207	20.0	58	20	16 days	175	Lincicome & Warsi 1966
166.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.141	13.6	58	20	mean	175	Lincicome & Warsi 1966
167.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.063	6.1	58	20	8 days	171	Lincicome & Lee 1971
168.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.174	16.8	58	20	12 days	171	Lincicome & Lee 1971
169.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.063	6.1	58	20	18 days	171	Lincicome & Lee 1971
170.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.115	11.1	58	20	8 days	176	Lincicome & Warsi 1968
171.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.223	21.5	58	20	14 days	176	Lincicome & Warsi 1968
172.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.181	17.5	58	20	18 days	176	Lincicome & Warsi 1968
173.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.061	5.9	58	20	8 days	172	Lincicome & Smith 1964
174.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.110	10.6	58	20	10 days	172	Lincicome & Smith 1964
175.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.117	11.3	58	20	12 days	172	Lincicome & Smith 1964
176.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.014	1.7	46	20	6 days	170	Lincicome & Hill 1965
177.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.054	5.2	58	20	14 days	170	Lincicome & Hill 1965
178.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.059	5.7	58	20	17 days	170	Lincicome & Hill 1965
179.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.211	20	58	20	mean	174	Lincicome & Warsi 1965
180.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.055	5.3	58	20	6-8 days	173	Lincicome & Smith 1966
181.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.143	13.8	58	20	14 days	173	Lincicome & Smith 1966
182.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.054	5.2	58	20	17 days	173	Lincicome & Smith 1966
183.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.022	2.1	58	20	8 days	175	Lincicome & Warsi 1966
184.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.101	9.8	58	20	14 days	175	Lincicome & Warsi 1966
185.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.099	9.6	58	20	16 days	175	Lincicome & Warsi 1966
186.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.303	29	58	20	mean	175	Lincicome & Warsi 1966
187.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.150	14.5	58	20	8 days	176	Lincicome & Warsi 1968
188.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.319	30.8	58	20	14 days	176	Lincicome & Warsi 1968
189.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.232	22.4	58	20	18 days	176	Lincicome & Warsi 1968
190.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.134	12.9	58	20	6 days	171	Lincicome & Lee 1971
191.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.364	35.1	58	20	13 days	171	Lincicome & Lee 1971
192.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.282	27.2	58	20	18 days	171	Lincicome & Lee 1971
193.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.621	60.0	58	20	7 days	159	Lee & Barlow 1972
194.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.890	85.9	58	20	9 days	159	Lee & Barlow 1972
195.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.263	25.4	58	20	14 days	159	Lee & Barlow 1972
196.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.054	6.6	46	20	6 days	173	Lincicome & Smith 1966
197.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.136	13.1	58	20	12-14 days	173	Lincicome & Smith 1966
198.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.068	6.6	58	20	17 days	173	Lincicome & Smith 1966
199.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.082	7.9	58	20	8 days	176	Lincicome & Warsi 1968
200.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.183	17.7	58	20	13 days	176	Lincicome & Warsi 1968
201.	Trypanosomatidae	Trypanosoma lewisi	ml O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.122	11.8	58	20	18 days	176	Lincicome & Warsi 1968
202.	Ciliophora	Urostyla grandis	nl O ₂ (cell) ⁻¹ hr ⁻¹	1.7	57	[166000]	20	starved	OTHER	Fenchel & Finlay 1983

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Table **S2b**. Species' minimum endogenous respiration rates used in the analyses presented in Table 1 and Figures 1-3 in the paper

Note that when converting dry mass to wet mass both Vladimirova & Zotin (1983, 1985) and Fenchel & Finlay (1983) used a DM/WM ratio of 0.14. To convert the data to the reference DM/WM = 0.3 (crude mean for all taxa applied in the analysis (see SI Methods, Table S12a)), qWkg values from these sources were multiplied by a factor of 2. (The mean protozoan metabolic rate of 7.5 W kg^{-1} reported in Table 1 represent therefore a conservative estimate in that sense that per unit wet mass these small-sized species have an even lower metabolic rate than the mean of 7.5 W kg^{-1} .) After this, temperature conversion was performed from 20 to 25 °C using $Q_{10} = 2$, $q_{25\text{Wkg}} = q\text{Wkg} \times 2^{(25 - TC)/10}$, dimension W (kg WM)^{-1} . Note on temperature conversion: For unicells the interspecific comparisons by Robinson et al. (1983) [Robinson W.R., Peters R.H., Zimmermann J. (1983) The effects of body size and temperature on metabolic rate of organisms. Canadian Journal of Zoology 61, 281-288] yielded a Q_{10} of 1.6, although the temperature dependence was statistically insignificant. Vladimirova & Zotin (1985), based on the intraspecifically established formula $q/q_{20} = 0.166\exp(0.087 T)$, where q_{20} is metabolic rate at 20 °C and T is temperature in °C. It corresponds to $Q_{10} = 2.4$. Fenchel & Finlay (1983) used $Q_{10} = 2$ in the analysis of their extensive compilation of protozoan metabolic rates. Thus, we chose $Q_{10} = 2$ as a representative value for unicells.

The values of $q_{25\text{Wkg}}$ (a total of 52 values for 52 species) were used in our analysis. Log stands for the decimal logarithms of the corresponding variables. See Table **S2a** for other notations.

Species	qWkg	LogqWkg	TC	q25Wkg	Logq25Wkg	Mpg	LogMpg
1. Acanthamoeba (Hartmanella) castellani	3.8	0.580	20	10.740	1.031	6700	3.826
2. Acanthamoeba sp.	5.3	0.724	20	14.990	1.176	4320	3.635
3. Actinosphaerium eichhornii	0.4	-0.398	20	1.132	0.054	14561000	7.163
4. Amoeba proteus	1	0.000	20	2.828	0.451	900000	5.954
5. Astasia klebsii	1.8	0.255	20	5.092	0.707	3800	3.580
6. Astasia longa	2.9	0.462	20	8.202	0.914	13500	4.130
7. Bresslaia insidiatrix	21.7	1.336	20	61.376	1.788	17000	4.230
8. Chilomonas paramecium	16	1.204	20	45.254	1.656	2260	3.354
9. Coleps hirtus	4.2	0.623	20	11.880	1.075	91000	4.959
10. Colpidium campylum	10.8	1.033	20	30.548	1.485	54400	4.736
11. Crithidia (Strigomonas) fasciculata	7.3	0.863	20	20.648	1.315	2078	3.318
12. Crithidia (Strigomonas) oncopelti	10	1.000	30	7.071	0.849	30	1.477

13. Dictyostelium discodeum	4.9	0.690	20	13.860	1.142	840	2.924
14. Eimeria acervulina	2.0	0.301	20	5.656	0.753	2640	3.422
15. Eimeria stiedae	2.5	0.398	20	7.072	0.850	7980	3.902
16. Eimeria tenella	2.3	0.362	20	6.506	0.813	5440	3.736
17. Endotrypanum schaudinni	0.84	-0.076	20	2.376	0.376	114	2.057
18. Entamoeba histolytica	0.4	-0.398	20	1.132	0.054	8600	3.934
19. Frontonia leucas	0.4	-0.398	20	1.132	0.054	745000	5.872
20. Leishmania brasiliensis	20	1.301	20	56.568	1.753	9	0.954
21. Leishmania donovani	9.3	0.968	20	26.304	1.420	18	1.255
22. Leishmania enrietti	7.0	0.845	20	19.798	1.297	12	1.079
23. Mayorella palestinensis	3.7	0.568	20	10.466	1.020	8300	3.919
24. Noctiluca miliaris	1.4	0.146	20	3.960	0.598	223986000	8.350
25. Paramecium aurelia	2.9	0.462	20	8.202	0.914	160000	5.204
26. Paramecium calkinsi	3.4	0.531	20	9.616	0.983	157000	5.196
27. Paramecium caudatum	0.2	-0.699	20	0.566	-0.247	526000	5.721
28. Paramecium multimicronucleatum	5.3	0.724	20	14.990	1.176	685000	5.836
29. Pelomyxa carolinensis	0.5	-0.301	20	1.414	0.150	49708000	7.696
30. Pelomyxa palustris	0.7	-0.155	20	1.980	0.297	90000000	7.954
31. Plasmodium cathemerium	1.8	0.255	20	5.092	0.707	54	1.732
32. Plasmodium gallinaceum	0.8	-0.097	20	2.262	0.354	71	1.851
33. Plasmodium knowlesi	0.9	-0.046	20	2.546	0.406	59	1.771
34. Pleuromonas jaculans	12	1.079	20	33.942	1.531	25	1.398
35. Podophrya fixa	2.9	0.458	20	8.202	0.914	14900	4.173
36. Schizotrypanum verpertilionis	2.1	0.322	20	5.940	0.774	108	2.033
37. Spirostoma minus	3.4	0.531	20	9.616	0.983	500000	5.699
38. Spirostomum ambiguum	5.9	0.771	20	16.688	1.222	12000000	7.079
39. Spirostomum intermedium	1.2	0.079	20	3.394	0.531	229000	5.360
40. Spirostomum teres	0.5	-0.301	20	1.414	0.150	423000	5.626
41. Stentor coeruleus	1.8	0.255	20	5.092	0.707	1100000	6.041
42. Tetrahymena geleii (pyriformis)	5.9	0.771	20	16.688	1.222	22000	4.342
43. Tetrahymena pyriformis	6.1	0.785	20	17.254	1.237	49000	4.690
44. Tracheloraphis sp.	16.0	1.204	20	45.254	1.656	340000	5.531
45. Trichomonas (Tritrichomonas) foetus	2.7	0.431	20	7.636	0.883	580	2.763
46. Trichomonas batrachorum	1.9	0.279	20	5.374	0.730	560	2.748
47. Trichomonas nasai	8.1	0.908	20	22.910	1.360	210	2.322
48. Trichomonas vaginalis	1.0	0.000	20	2.828	0.451	870	2.940
49. Trypanosoma (Schizotrypanum) cruzi	1.2	0.079	20	3.394	0.531	169	2.228
50. Trypanosoma lewisi	0.9	-0.046	20	2.546	0.406	58	1.763
51. Urostyla grandis	57	1.756	20	161.220	2.207	166000	5.220
52. Trichosphaerium sieboldi	4	0.602	20	5.7	0.756	100	2.000

Dataset S3. Standard and routine respiration rates in aquatic invertebrates

Standard metabolic rates of insect species are presented. For details of data assembling procedure see Chown, S. L. et al. (2007) Scaling of insect metabolic rate is inconsistent with the nutrient supply network model. *Funct Ecol* 21: 282–290.

Notations:

M is body mass in mg, Q is whole-body standard metabolic rate in μW at 25 °C.

Analyses presented in Table 1 and Figs. 1-3 in the paper are based on mass-specific standard metabolic rate $q = (Q / M)$, dimension W kg^{-1} .

Species	Family	Order	M (mg)	Q (μW)
Unknown species*	Meinertellidae	Archaeognatha	12.75	38.3
Species 1 (Sutherland)*	Lepismatidae	Thysanura	23.04	36.9
Species 2 (Stellenbosch)*	Lepismatidae	Thysanura	26.64	23.0
Species 3 (Cederberg)*	Lepismatidae	Thysanura	17.80	42.3
<i>Anax junius</i> ¹	Aeschnidae	Odonata	1019.00	4148.4
<i>Brachymesia gravida</i> ¹	Libellulidae	Odonata	344.00	1555.6
<i>Erythemis simplicicollis</i> ¹	Libellulidae	Odonata	263.00	877.5
<i>Erythrodiplax berenice</i> ¹	Libellulidae	Odonata	125.00	438.8
<i>Erythrodiplax connata</i> ¹	Libellulidae	Odonata	52.00	191.5

<i>Libellula auripennis</i> ¹	Libellulidae	Odonata	464.00	1396.1
<i>Libellula needhami</i> ¹	Libellulidae	Odonata	518.00	1675.3
<i>Miathyria marcella</i> ¹	Libellulidae	Odonata	17.10	765.9
<i>Pachydiplax longipennis</i> ¹	Libellulidae	Odonata	200.00	1765.2
<i>Pantala flavescens</i> ¹	Libellulidae	Odonata	339.00	1675.3
<i>Perithemis tenera</i> ¹	Libellulidae	Odonata	61.00	331.1
<i>Tramea carolina</i> ¹	Libellulidae	Odonata	383.00	1795.0
<i>Coptotermes formosanus</i> ²	Rhinotermitidae	Isoptera	2.95	8.1
<i>Heterotermes tenuior</i> ³	Rhinotermitidae	Isoptera	1.40	1.9
<i>Reticulitermes flavipes</i> ²	Rhinotermitidae	Isoptera	2.97	12.0
<i>Schedorhinotermes javanicus</i> ³	Rhinotermitidae	Isoptera	2.70	2.9
<i>Schedorhinotermes sarawakensis</i> ³	Rhinotermitidae	Isoptera	8.30	17.9
<i>Labritermes kistneri</i> ³	Termitidae	Isoptera	0.70	2.0
<i>Dicuspitermes nemorosus</i> ³	Termitidae	Isoptera	3.50	1.7
<i>Dicuspitermes santschii</i> ³	Termitidae	Isoptera	2.60	0.6
<i>Homalotermes eleanorae</i> ³	Termitidae	Isoptera	2.20	1.5

<i>Pericapritermes nitobei</i> ³	Termitidae	Isoptera	1.20	1.2
<i>Pericapritermes semarangi</i> ³	Termitidae	Isoptera	1.30	0.8
<i>Procapritermes nr. sandakanensis</i> ³	Termitidae	Isoptera	5.90	2.4
<i>Syncapritermes</i> sp. A ³	Termitidae	Isoptera	6.70	3.1
<i>Termes borneensis</i> ³	Termitidae	Isoptera	3.00	2.1
<i>Globitermes globosus</i> ³	Termitidae	Isoptera	1.90	1.4
<i>Prohamitermes mirabilis</i> ³	Termitidae	Isoptera	3.50	1.9
<i>Aciculoiditermes</i> sp. A ³	Termitidae	Isoptera	1.80	1.8
<i>Bulbitermes</i> sp. C ³	Termitidae	Isoptera	2.80	2.3
<i>Havilanditermes atripennis</i> ³	Termitidae	Isoptera	8.00	8.0
<i>Hospitalitermes hospitalis</i> ³	Termitidae	Isoptera	7.70	5.6
<i>Nasutitermes longinasus</i> ³	Termitidae	Isoptera	2.60	1.5
<i>Proaciculitermes</i> sp. A ³	Termitidae	Isoptera	1.10	0.8
<i>Proaciculitermes</i> sp. E ³	Termitidae	Isoptera	2.70	1.4
<i>Hypotermes xenotermes</i> ³	Termitidae	Isoptera	2.60	2.9
<i>Macrotermes carbonarius</i> ⁴	Termitidae	Isoptera	11.93	51.8

<i>Macrotermes gilvus</i> ³	Termitidae	Isoptera	7.00	5.5
<i>Macrotermes malaccensis</i> ³	Termitidae	Isoptera	11.60	14.1
<i>Microcerotermes dubius</i> ³	Termitidae	Isoptera	2.20	2.9
<i>Microcerotermes serrula</i> ³	Termitidae	Isoptera	1.90	2.2
<i>Blatta orientalis</i> ⁵	Blattidae	Blattodea	325.00	555.9
<i>Eublaberus posticus</i> ⁶	Blattidae	Blattodea	2200.00	4219.5
<i>Periplaneta americana</i> ⁷	Blattidae	Blattodea	900.00	979.4
<i>Periplaneta orientalis</i> ⁸	Blattidae	Blattodea	165.00	307.9
<i>Blatella germanica</i> ⁹	Blattellidae	Blattodea	54.76	203.3
<i>Byrsotria fumagata</i> ⁶	Blattellidae	Blattodea	4950.00	5584.6
<i>Gromphadorhina chopardi</i> ⁶	Blattellidae	Blattodea	3400.00	2685.1
<i>Gromphadorhina portentosa</i> ¹⁰	Blattellidae	Blattodea	4950.00	4747.0
<i>Parcoblatta</i> sp. ¹¹	Blattellidae	Blattodea	73.00	92.0
<i>Blaberus discoidalis</i> ⁶	Blaberidae	Blattodea	4080.00	3912.6
<i>Blaberus giganteus</i> ¹²	Blaberidae	Blattodea	4330.00	3646.4
<i>Leucophaea maderae</i> ¹³	Blaberidae	Blattodea	2800.00	2903.9

<i>Nauphoeta cinerea</i> ¹³	Blaberidae	Blattodea	510.00	732.2
<i>Perisphaeria</i> sp. ¹⁴	Blaberidae	Blattodea	324.10	196.8
<i>Sphodromantis gastrica</i> *	Mantidae	Mantodea	335.70	611.6
<i>Karoophasma biedouwensis</i> *	Austrophasmatidae	Mantophasmatodea	103.95	317.9
<i>Oecanthus quadripunctatus</i> ¹⁵	Gryllidae	Orthoptera	50.00	134.3
<i>Enyaliopsis petersi</i> ¹⁶	Gryllidae	Orthoptera	750.00	1525.7
<i>Eugaster loricatus</i> ¹⁶	Gryllidae	Orthoptera	1580.00	1762.1
<i>Hophlosphyrum griseus</i> ¹⁷	Gryllidae	Orthoptera	36.20	79.9
<i>Pteronemobius fasciatus</i> ¹⁸	Gryllidae	Orthoptera	26.17	469.5
<i>Acheta domesticus</i> ¹⁹	Gryllidae	Orthoptera	369.00	1138.6
<i>Teleogryllus commodus</i> ⁷	Gryllidae	Orthoptera	950.00	5686.9
<i>Anurogryllis arboreus</i> ¹⁵	Gryllidae	Orthoptera	310.00	550.9
<i>Ceuthophilis fossor</i> ²⁰	Gryllacrididae	Orthoptera	70.15	817.6
<i>Ceuthophilis gracilipes</i> ¹¹	Gryllacrididae	Orthoptera	259.10	543.6
<i>Gryllotalpa australis</i> ²¹	Gryllotalpidae	Orthoptera	874.00	2235.6
<i>Conocephalus fasciatus</i> ¹⁸	Tettigoniidae	Orthoptera	51.37	717.1

<i>Euconocephalus nasutus</i> ²²	Tettigoniidae	Orthoptera	650.00	1026.7
<i>Insara covilleae</i> ²⁰	Tettigoniidae	Orthoptera	92.60	1633.1
<i>Neoconocephalus robustus</i> ²²	Tettigoniidae	Orthoptera	870.00	3729.9
<i>Requena verticalis</i> ¹⁵	Tettigoniidae	Orthoptera	370.00	683.0
<i>Anconia integra</i> ²⁰	Acrididae	Orthoptera	816.70	2644.1
<i>Bootettix punctatus</i> ²⁰	Acrididae	Orthoptera	47.01	364.7
<i>Encoptolophus s. costalis</i> ²³	Acrididae	Orthoptera	174.60	462.6
<i>Locusta migratoria migratorioides</i> ²⁴	Acrididae	Orthoptera	1500.00	4934.0
<i>Melanoplus bivittatus</i> ²⁵	Acrididae	Orthoptera	1650.00	5057.0
<i>Melanoplus complanipes</i> ²⁰	Acrididae	Orthoptera	141.55	624.3
<i>Oedalis instillatus</i> ¹⁶	Acrididae	Orthoptera	627.00	966.4
<i>Romalea guttata</i> ²⁶	Acrididae	Orthoptera	2874.00	2696.1
<i>Taeniopoda eques</i> ²⁶	Acrididae	Orthoptera	2043.00	2462.0
<i>Trimerotropis pallidipennis</i> ²⁷	Acrididae	Orthoptera	221.35	743.3
<i>Trimerotropis saxatilis</i> ²⁸	Acrididae	Orthoptera	155.00	321.8
<i>Trimerotropis</i> sp. ²⁰	Acrididae	Orthoptera	145.90	1007.5

<i>Trimerotropis suffusa</i> ²⁷	Acrididae	Orthoptera	297.50	906.2
<i>Xanthippus corallipes</i> ²⁹	Acrididae	Orthoptera	2302.20	7508.5
Unknown genus ¹⁶	Acrididae	Orthoptera	2948.0	1354.6
<i>Tanaocerus koebeli</i> ²⁰	Tanaoceridae	Orthoptera	102.50	625.9
<i>Xiphoceriana</i> sp. ¹⁶	Acridiidae	Orthoptera	1568.00	743.0
<i>Euborellia annulipes</i> *	Carcinophoridae	Dermaptera	31.76	82.9
Unknown genus ²⁰	Psyllidae	Hemiptera	0.16	5.6
<i>Pseudococcus citri</i> ³⁰	Pseudococcidae	Hemiptera	1.30	11.3
<i>Neophilaenus lineatus</i> ³¹	Cercopidae	Hemiptera	1.91	30.6
<i>Philaenus spumarius</i> ³²	Cercopidae	Hemiptera	3.93	59.5
<i>Cystosoma saundersii</i> ³³	Cicadidae	Hemiptera	1158.00	2345.1
<i>Diceroprocta apache</i> ³⁴	Cicadidae	Hemiptera	622.00	1605.2
<i>Fidicina mannifera</i> ³⁵	Cicadidae	Hemiptera	2840.00	9689.906
<i>Multareoides bifurcatus</i> ²⁰	Membracidae	Hemiptera	1.34	15.2
<i>Hysteropterum</i> sp. ²⁰	Issidae	Hemiptera	1.71	40.3
<i>Phytocoris nigripubescens</i> ²⁰	Miridae	Hemiptera	1.27	30.8

<i>Slaterocoris</i> sp. ²⁰	Miridae	Hemiptera	0.86	17.2
Unknown genus ²⁰	Miridae	Hemiptera	0.51	12.0
Unknown genus ²⁰	Miridae	Hemiptera	3.56	72.9
Unknown genus ²⁰	Miridae	Hemiptera	3.85	35.0
<i>Rhodnius prolixus</i> ³⁶	Reduviidae	Hemiptera	66.75	92.7
Unknown genus ²⁰	Nabidae	Hemiptera	1.89	25.9
<i>Dendrocoris contaminatus</i> ²⁰	Pentatomidae	Hemiptera	11.00	26.2
Unknown genus ²⁰	Coreidae	Hemiptera	66.70	410.5
<i>Lygus kalmii</i> ²⁰	Lygaeidae	Hemiptera	10.17	146.3
Unknown genus ²⁰	Neididae	Hemiptera	0.95	28.1
<i>Corydalis cornutus</i> ³⁷	Corydalidae	Megaloptera	326.00	4724.5
<i>Saprinus</i> sp. ²⁰	Histeridae	Coleoptera	8.21	65.4
<i>Sphaeridium lunatum</i> ³⁸	Hydrophilidae	Coleoptera	34.00	175.5
Species 3 ³⁹	Staphilinidae	Coleoptera	0.089	0.34
Species 2 ³⁹	Carabidae	Coleoptera	0.89	1.34
<i>Reichebnachia</i> sp. ³⁹	Pselaphidae	Coleoptera	0.077	0.82

<i>Popilius disjunctus</i> ¹¹	Passalidae	Coleoptera	1630.50	1611.6
<i>Popilius</i> sp. ⁴⁰	Passalidae	Coleoptera	563.00	799.4
<i>Omorgus radula</i> ⁴¹	Trogidae	Coleoptera	207.00	114.1
<i>Geotrupes</i> sp. ³⁹	Geotrupiidae	Coleoptera	160.30	679.8
<i>Geotrupes spiniger</i> ³⁸	Geotrupidae	Coleoptera	825.00	4946.1
<i>Aphodius contaminatus</i> ³⁸	Aphodiidae	Coleoptera	14.00	103.7
<i>Aphodius fossor</i> ⁴²	Aphodiidae	Coleoptera	121.30	268.8
<i>Aphodius rufipes</i> ³⁸	Aphodiidae	Coleoptera	84.00	382.9
<i>Aphodius distinctus</i> ⁴³	Scarabaeidae	Coleoptera	7.80	104.9
<i>Aphodius fimetarius</i> ⁴³	Scarabaeidae	Coleoptera	30.90	187.3
<i>Aphodius prodromus</i> ⁴³	Scarabaeidae	Coleoptera	13.80	167.0
<i>Aphodius rufus</i> ⁴³	Scarabaeidae	Coleoptera	209.00	170.4
<i>Anomala</i> sp. ⁴⁰	Scarabaeidae	Coleoptera	100.00	137.0
<i>Pelidnota</i> sp. ⁴⁰	Scarabaeidae	Coleoptera	1091.50	1511.5
<i>Coeloesis biloba</i> ⁴⁰	Scarabaeidae	Coleoptera	3735.00	3724.2
<i>Strategus aloeus</i> ⁴⁰	Scarabaeidae	Coleoptera	5050.00	5149.6

<i>Cyclocephala</i> sp. ⁴⁰	Scarabaeidae	Coleoptera	246.00	592.5
<i>Dyscinetus</i> sp. ⁴⁰	Scarabaeidae	Coleoptera	390.00	469.6
<i>Phileurus</i> sp. ⁴⁰	Scarabaeidae	Coleoptera	3501.00	1562.6
<i>Sisiyphys fasciculatus</i> ⁴⁴	Scarabaeidae	Coleoptera	136.00	375.1
<i>Anachalcos convexus</i> ⁴⁴	Scarabaeidae	Coleoptera	1421.00	1539.8
<i>Circellium bacchus</i> ⁴⁴	Scarabaeidae	Coleoptera	7285.00	4017.3
<i>Pachylomerus femoralis</i> ⁴⁵	Scarabaeidae	Coleoptera	4915.00	4282.2
<i>Scarabaeus flavicornis</i> ⁴⁴	Scarabaeidae	Coleoptera	322.00	483.7
<i>Scarabaeus galenus</i> ⁴⁶	Scarabaeidae	Coleoptera	1681.00	881.7
<i>Scarabaeus westwoodi</i> ⁴⁷	Scarabaeidae	Coleoptera	1780.00	1324.7
<i>Scarabaeus rusticus</i> ⁴⁷	Scarabaeidae	Coleoptera	1070.00	696.2
<i>Scarabaeus gariepinus</i> ⁴⁷	Scarabaeidae	Coleoptera	1140.00	288.0
<i>Scarabaeus striatum</i> ⁴⁷	Scarabaeidae	Coleoptera	790.00	282.6
<i>Scarabaeus hippocrates</i> ⁴⁷	Scarabaeidae	Coleoptera	2010.00	622.1
<i>Pleocoma australis</i> ⁴⁸	Scarabaeidae	Coleoptera	940.00	3238.8
<i>Leucocelis elegans</i> ¹⁶	Scarabaeidae	Coleoptera	80.00	337.0

<i>Simplocaria metallica</i> ⁴⁹	Byrrhidae	Coleoptera	2.85	0.5
<i>Agrypnus bocande</i> ¹⁶	Elateridae	Coleoptera	166.20	71.6
<i>Rhizopertha dominica</i> ⁵⁰	Bostrichidae	Coleoptera	1.40	14.8
<i>Melyrid</i> sp. ²⁰	Melyridae	Coleoptera	1.16	20.9
<i>Hippodamia convergens</i> ⁵¹	Coccinellidae	Coleoptera	17.00	193.2
<i>Hydromedion sparsutum</i> ⁵²	Perimylopidae	Coleoptera	23.50	83.5
<i>Perimylops antarcticus</i> ⁵²	Perimylopidae	Coleoptera	14.50	48.6
<i>Adesmia baccata</i> ⁵³	Tenebrionidae	Coleoptera	407.00	179.2
<i>Anepsius brunneus</i> ²⁰	Tenebrionidae	Coleoptera	6.15	24.8
<i>Centrioptera muricata</i> ²⁰	Tenebrionidae	Coleoptera	132.60	468.6
<i>Erodium nanus</i> ⁵⁴	Tenebrionidae	Coleoptera	21.90	101.3
<i>Euschides luctata</i> ²⁰	Tenebrionidae	Coleoptera	83.80	464.8
<i>Peristepus</i> sp. ⁵³	Tenebrionidae	Coleoptera	55.00	54.5
<i>Phrynocolus auriculatus</i> ⁵³	Tenebrionidae	Coleoptera	193.00	169.1
<i>Phrynocolus petrosus</i> ⁵³	Tenebrionidae	Coleoptera	1109.00	355.7
<i>Phrynocolus</i> sp. ⁵³	Tenebrionidae	Coleoptera	204.00	191.4

<i>Physosterna cribripes</i> ⁵⁵	Tenebrionidae	Coleoptera	1226.00	2443.7
<i>Physadesmia globosa</i> ⁵⁶	Tenebrionidae	Coleoptera	516.00	599.6
<i>Sphaeriontis dilatata</i> ²⁰	Tenebrionidae	Coleoptera	46.00	150.7
<i>Blaps gigas</i> ⁵⁴	Tenebrionidae	Coleoptera	1776.00	4508.3
<i>Nyctobates procerus</i> ⁴⁰	Tenebrionidae	Coleoptera	2198.00	1753.3
<i>Pterohelaeus</i> sp. ⁵⁷	Tenebrionidae	Coleoptera	245.00	481.6
<i>Tenebrio molitor</i> ³⁰	Tenebrionidae	Coleoptera	100.00	372.2
<i>Zophobas</i> sp. ⁴⁰	Tenebrionidae	Coleoptera	535.00	1269.6
<i>Tribolium castaneum</i> *	Tenebrionidae	Coleoptera	2.40	13.0
<i>Tribolium confusum</i> ⁵⁸	Tenebrionidae	Coleoptera	2.01	16.3
<i>Trogoderus costatus</i> ²⁰	Tenebrionidae	Coleoptera	29.87	93.1
<i>Vieta bulbifera</i> ⁵³	Tenebrionidae	Coleoptera	479.00	538.8
<i>Vieta muscosa</i> ⁵³	Tenebrionidae	Coleoptera	140.00	135.4
<i>Cryptoglossa verrucosa</i> ⁵⁹	Tenebrionidae	Coleoptera	700.00	418.8
<i>Edrotes ventricosus</i> ²⁰	Tenebrionidae	Coleoptera	24.35	91.9
<i>Pimelia cenchronota</i> ⁵³	Tenebrionidae	Coleoptera	1474.00	620.9

<i>Pimelia grandis</i> ⁶⁰	Tenebrionidae	Coleoptera	2098.00	3734.1
<i>Pimelia obsoleta</i> ⁵⁴	Tenebrionidae	Coleoptera	861.00	3011.3
<i>Rhytinota praelonga</i> ⁵³	Tenebrionidae	Coleoptera	105.00	100.0
<i>Triorophus laevis</i> ²⁰	Tenebrionidae	Coleoptera	8.46	107.7
<i>Helius waiti</i> ⁵⁷	Tenebrionidae	Coleoptera	672.00	1675.1
<i>Cardiosis fairmarei</i> ⁵⁶	Tenebrionidae	Coleoptera	32.00	29.6
<i>Zophosis orbicularis</i> ⁵⁶	Tenebrionidae	Coleoptera	103.00	121.4
<i>Psammodes striatus</i> ⁵⁶	Tenebrionidae	Coleoptera	3010.00	4271.5
<i>Epiphysa arenicola</i> ⁵⁶	Tenebrionidae	Coleoptera	1237.00	469.4
<i>Onymacris plana</i> ⁵⁶	Tenebrionidae	Coleoptera	767.00	925.9
<i>Onymacris laeviceps</i> ⁵⁶	Tenebrionidae	Coleoptera	525.00	427.9
<i>Onymacris rugatipennis rugatipennis</i> ⁵⁶	Tenebrionidae	Coleoptera	496.00	446.2
<i>Onymacris rugatipennis albotesselata</i> ⁵⁶	Tenebrionidae	Coleoptera	573.00	578.91
<i>Onymacris unguicularis</i> ⁶¹	Tenebrionidae	Coleoptera	737.00	176.0
<i>Stenocara gracilipes</i> ⁵⁶	Tenebrionidae	Coleoptera	268.00	476.3
<i>Eusattus dubius</i> ²⁰	Tenebrionidae	Coleoptera	27.22	282.6

<i>Eleodes armata</i> ⁵⁹	Tenebrionidae	Coleoptera	917.00	823.6
<i>Eleodes grandicollis</i> ²⁰	Tenebrionidae	Coleoptera	521.93	2537.0
<i>Eleodes</i> sp. ²⁰	Tenebrionidae	Coleoptera	88.85	356.2
<i>Eleodes tenebrosa</i> ²⁰	Tenebrionidae	Coleoptera	51.53	247.4
<i>Meloid</i> sp. ²⁰	Meloidae	Coleoptera	17.34	406.7
<i>Acanthoderes circumflexa</i> ⁴⁰	Cerambycidae	Coleoptera	139.00	241.8
<i>Acrocinus longimanus</i> ⁴⁰	Cerambycidae	Coleoptera	5383.00	16565.8
<i>Dryoctenes scrupulosa</i> ⁴⁰	Cerambycidae	Coleoptera	1823.00	2825.3
<i>Oncideres putator</i> ⁴⁰	Cerambycidae	Coleoptera	681.00	1095.9
<i>Taeinotes scalaris</i> ⁴⁰	Cerambycidae	Coleoptera	397.00	952.3
<i>Acanthophorus confinis</i> ¹⁶	Cerambycidae	Coleoptera	3490.00	6180.9
<i>Macrodontia dejeani</i> ⁴⁰	Cerambycidae	Coleoptera	4860.00	7799.2
<i>Stenodontes molaria</i> ⁴⁰	Cerambycidae	Coleoptera	2944.00	6393.9
<i>Stenodontes</i> sp. ⁴⁰	Cerambycidae	Coleoptera	1043.00	2040.6
<i>Trachysomus peregrinus</i> ⁴⁰	Cerambycidae	Coleoptera	547.00	603.8
<i>Brasilanus batus</i> ⁴⁰	Cerambycidae	Coleoptera	1965.00	6232.9

<i>Eburia</i> sp. ⁴⁰	Cerambycidae	Coleoptera	582.00	1493.2
<i>Nyssicus setosus</i> ⁴⁰	Cerambycidae	Coleoptera	996.00	715.5
<i>Phorocantha recurva</i> ⁶²	Cerambycidae	Coleoptera	260.00	800.2
<i>Phorocantha semipunctata</i> ⁶²	Cerambycidae	Coleoptera	300.00	701.7
<i>Xenambyx laticauda</i> ⁴⁰	Cerambycidae	Coleoptera	1217.00	1571.9
<i>Diplocapsis</i> sp. ²⁰	Chrysomelidae	Coleoptera	5.34	89.1
<i>Griburius</i> sp. 1 ²⁰	Chrysomelidae	Coleoptera	3.31	60.1
<i>Griburius</i> sp. 2 ²⁰	Chrysomelidae	Coleoptera	9.42	43.7
<i>Leptinotarsa decemlineata</i> ⁶³	Chrysomelidae	Coleoptera	150.00	693.9
<i>Monoxia</i> sp. ²⁰	Chrysomelidae	Coleoptera	4.63	79.2
<i>Smicronyx imbricata</i> ²⁰	Curculionidae	Coleoptera	0.32	3.6
<i>Ectemnorhinus marioni</i> ⁶⁴	Curculionidae	Coleoptera	9.60	27.3
<i>Ectemnorhinus similis</i> ⁶⁴	Curculionidae	Coleoptera	13.40	32.5
<i>Ips acuminatus</i> ⁶⁵	Scolytidae	Coleoptera	2.84	36.3
<i>Calandra oryzae</i> ⁵⁰	Curculionidae	Coleoptera	1.70	17.7
<i>Sitophilus granarius</i> ⁶⁶	Curculionidae	Coleoptera	3.68	139.6

<i>Microcerus</i> sp. ¹⁶	Curculionidae	Coleoptera	132.00	189.6
<i>Ophryastes varius</i> ²⁰	Curculionidae	Coleoptera	27.26	119.3
<i>Miloderes</i> sp. ²⁰	Curculionidae	Coleoptera	3.40	65.7
<i>Hylobius abietis</i> ⁶⁷	Curculionidae	Coleoptera	183.00	418.1
<i>Hipporhinus tenuegranosus</i> ¹⁶	Curculionidae	Coleoptera	1080.00	548.0
<i>Eucyllus vagans</i> ²⁰	Curculionidae	Coleoptera	7.43	31.8
<i>Eucyllus unicolor</i> ²⁰	Curculionidae	Coleoptera	2.59	13.2
<i>Bothrometopus randi</i> ⁶⁴	Curculionidae	Coleoptera	24.60	51.1
<i>Bothrometopus parvulus</i> ⁶⁴	Curculionidae	Coleoptera	3.60	19.4
<i>Bothrometopus elongatus</i> ⁶⁴	Curculionidae	Coleoptera	1.70	15.2
<i>Palirhoeus eatoni</i> ⁶⁸	Curculionidae	Coleoptera	6.70	24.3
<i>Lixus bisulcatus</i> ¹⁶	Curculionidae	Coleoptera	405.00	1357.0
<i>Rhynchaenus flagellum</i> ⁴⁹	Curculionidae	Coleoptera	0.51	0.2
<i>Rhytonomus isobellina</i> ⁵⁴	Curculionidae	Coleoptera	8.10	93.7
<i>Hypera postica</i> ⁶⁹	Curculionidae	Coleoptera	5.80	159.5
<i>Canonopsis sericeus</i> ⁶⁴	Curculionidae	Coleoptera	58.90	175.1

Unknown genus ²⁰	Curculionidae	Coleoptera	0.3	1.63
Unknown genus 1 ¹⁶	Curculionidae	Coleoptera	193	120.1
Unknown genus 3 ¹⁶	Curculionidae	Coleoptera	10.5	7.0
<i>Cerotalis</i> sp. ⁵⁷	Carabidae	Coleoptera	562.00	474.6
<i>Chilanthia cavernosa</i> ⁵³	Carabidae	Coleoptera	1026.00	1571.5
<i>Thermophilum babaulti</i> ⁵³	Carabidae	Coleoptera	1295.00	3223.9
<i>Thermophilum hexastictum</i> ⁵³	Carabidae	Coleoptera	2025.00	2423.2
<i>Triaenogeius scupturatus</i> ⁵³	Carabidae	Coleoptera	170.00	341.8
<i>Anthia fabricii</i> ⁷⁰	Carabidae	Coleoptera	2250.00	1875.1
<i>Cypholoba bihamata</i> ⁵³	Carabidae	Coleoptera	245.00	961.6
<i>Cypholoba chanleri</i> ⁵³	Carabidae	Coleoptera	326.00	780.2
<i>Cypholoba</i> sp. ⁵³	Carabidae	Coleoptera	192.00	937.4
<i>Cypholoba tenuicollis</i> ⁵³	Carabidae	Coleoptera	59.00	245.7
<i>Cypholoba tetrastigma</i> ⁵³	Carabidae	Coleoptera	132.00	802.4
<i>Cypholoba trilunata</i> ⁵³	Carabidae	Coleoptera	213.00	530.2
<i>Amara quenseli</i> ⁴⁹	Carabidae	Coleoptera	15.00	1.4

<i>Evarthrus sodalis</i> ¹¹	Carabidae	Coleoptera	164.10	186.7
<i>Carinum</i> sp. ⁵⁷	Carabidae	Coleoptera	4240.00	2428.9
<i>Crepidogaster bioculata</i> ⁵³	Carabidae	Coleoptera	171.00	286.5
<i>Sphaeroderus stenostomus</i> ¹¹	Carabidae	Coleoptera	169.60	262.1
<i>Calosoma affine</i> ⁷	Carabidae	Coleoptera	620.00	787.2
<i>Calosoma</i> sp. ²⁰	Carabidae	Coleoptera	151.23	838.6
<i>Campalita chlorostictum</i> ⁵³	Carabidae	Coleoptera	521.00	798.0
Unknown genus ¹⁶	Carabidae	Coleoptera	79.0	533.2
<i>Cicindela longilabris</i> ⁷¹	Cicindelidae	Coleoptera	124.30	269.6
<i>Cicindela repanda</i> ⁷²	Cicindelidae	Coleoptera	63.40	145.5
<i>Paractora dreuxi</i> ⁷³	Helcomyzidae	Diptera	19.17	243.9
<i>Paractora trichosterna</i> ⁷³	Helcomyzidae	Diptera	13.26	261.3
<i>Antrops truncipennis</i> ⁷³	Sphaeroceridae	Diptera	2.21	51.6
<i>Drosophila americana</i> ⁷⁴	Drosophilidae	Diptera	1.10	15.3
<i>Drosophila melanogaster</i> ⁷⁵	Drosophilidae	Diptera	0.95	32.0
<i>Drosophila mimica</i> ⁷⁵	Drosophilidae	Diptera	2.44	49.5

<i>Drosophila nikananu</i> ⁷⁵	Drosophilidae	Diptera	0.49	19.4
<i>Drosophila repleta</i> ⁷⁶	Drosophilidae	Diptera	3.43	32.3
<i>Drosophila virilis</i> ⁷⁵	Drosophilidae	Diptera	1.67	45.7
<i>Glossina morsitans</i> ⁷⁷	Glossinidae	Diptera	22.34	123.5
<i>Glossina pallidipes</i> ⁷⁸	Glossinidae	Diptera	38.45	374.3
<i>Musca autumnalis</i> ⁷⁹	Muscidae	Diptera	22.00	360.8
<i>Musca domestica</i> ⁸⁰	Muscidae	Diptera	18.00	587.6
<i>Phormia regina</i> ⁸¹	Calliphoridae	Diptera	50.00	541.5
<i>Protophormia terraenovae</i> ⁸²	Calliphoridae	Diptera	25.00	225.2
<i>Pilica formidolosa</i> ⁸³	Asilidae	Diptera	200.00	789.8
<i>Promachus</i> sp. 2 ⁸³	Asilidae	Diptera	180.00	1447.4
<i>Tabanus affinis</i> ⁸⁴	Tabanidae	Diptera	161.70	114.5
<i>Pantophthalmus tabaninus</i> ⁸⁵	Pantophthalmidae	Diptera	1746.00	3036.9
<i>Simulium venustum</i> ⁸⁴	Simuliidae	Diptera	2.53	86.0
<i>Chironomus riparius</i> ⁸⁶	Chironomidae	Diptera	1.00	16.1
<i>Aedes campestris</i> ⁸⁴	Culicidae	Diptera	6.72	295.0

<i>Culex tarsalis</i> ⁸⁷	Culicidae	Diptera	2.23	23.6
<i>Xenopsilla ramesis</i> ⁸⁸	Pulicidae	Siphonaptera	0.16	0.6
<i>Vanessa io</i> ³⁰	Nymphalidae	Lepidoptera	230.00	922.2
<i>Thais cassandra</i> ³⁰	Papilionidae	Lepidoptera	130.00	3488.4
<i>Apetaloides firmiana</i> ⁸⁹	Notodontidae	Lepidoptera	169.00	828.3
<i>Hapigia simplex</i> ⁸⁹	Notodontidae	Lepidoptera	478.00	1584.0
<i>Lirimiris</i> sp. ⁸⁹	Notodontidae	Lepidoptera	564.00	1765.4
<i>Mamestra configurata</i> ⁹⁰	Noctuidae	Lepidoptera	139.05	465.3
<i>Bombyx mori</i> ³⁰	Bombicidae	Lepidoptera	490.00	3492.6
<i>Deilephila elpenor</i> ³⁰	Sphingidae	Lepidoptera	600.00	893.2
<i>Deilephila euphorbiae</i> ³⁰	Sphingidae	Lepidoptera	700.00	2347.9
<i>Xylophanes chiron</i> ⁸⁹	Sphingidae	Lepidoptera	708.00	1348.2
<i>Xylophanes libya</i> ⁸⁹	Sphingidae	Lepidoptera	559.00	2551.4
<i>Xylophanes pluto</i> ⁸⁹	Sphingidae	Lepidoptera	829.00	2258.2
<i>Enyo ocypete</i> ⁸⁹	Sphingidae	Lepidoptera	453.33	1148.7
<i>Erinnyis ello</i> ⁸⁹	Sphingidae	Lepidoptera	1210.00	4135.4

<i>Erinnyis oenotrus</i> ⁸⁹	Sphingidae	Lepidoptera	964.00	3615.5
<i>Madoryx oecus</i> ⁸⁹	Sphingidae	Lepidoptera	1699.00	5368.8
<i>Oryba achemenides</i> ⁸⁹	Sphingidae	Lepidoptera	2808.50	6058.0
<i>Pachygonia drucei</i> ⁸⁹	Sphingidae	Lepidoptera	702.00	1481.3
<i>Pachylia ficus</i> ⁸⁹	Sphingidae	Lepidoptera	3225.00	5114.9
<i>Perigonia lusca</i> ⁸⁹	Sphingidae	Lepidoptera	558.33	2265.2
<i>Metopsilus porcellus</i> ³⁰	Sphingidae	Lepidoptera	285.00	1361.8
<i>Mimas tiliae</i> ³⁰	Sphingidae	Lepidoptera	320.00	1666.8
<i>Protambulyx strigilis</i> ⁸⁹	Sphingidae	Lepidoptera	1095.33	2251.1
<i>Manduca corallina</i> ⁸⁹	Sphingidae	Lepidoptera	1618.25	3239.1
<i>Manduca lefeburei</i> ⁸⁹	Sphingidae	Lepidoptera	571.00	925.0
<i>Manduca rustica</i> ⁸⁹	Sphingidae	Lepidoptera	2810.00	6704.9
<i>Sphinx ligustri</i> ³⁰	Sphingidae	Lepidoptera	1400.00	7513.4
<i>Adeloneivaia boisduvalii</i> ⁸⁹	Saturniidae	Lepidoptera	1034.00	1795.6
<i>Adeloneivaia subungulata</i> ⁸⁹	Saturniidae	Lepidoptera	487.00	1850.1
<i>Eacles imperialis</i> ⁸⁹	Saturniidae	Lepidoptera	1105.00	3282.9

<i>Sphingicampa quadrilineata</i> ⁸⁹	Saturniidae	Lepidoptera	818.00	2134.2
<i>Syssphinx molina</i> ⁸⁹	Saturniidae	Lepidoptera	1757.00	4431.7
<i>Antheraea pernyi</i> ³⁰	Saturniidae	Lepidoptera	1210.00	6134.5
<i>Automerina auletes</i> ⁸⁹	Saturniidae	Lepidoptera	720.00	1813.8
<i>Automeris fieldi</i> ⁸⁹	Saturniidae	Lepidoptera	394.00	1088.3
<i>Automeris hamata</i> ⁸⁹	Saturniidae	Lepidoptera	564.00	1995.2
<i>Automeris jacunda</i> ⁸⁹	Saturniidae	Lepidoptera	653.25	1549.3
<i>Automeris zugana</i> ⁸⁹	Saturniidae	Lepidoptera	539.75	1103.4
<i>Dirphea agis</i> ⁸⁹	Saturniidae	Lepidoptera	197.00	483.7
<i>Hylesia praeda</i> ⁸⁹	Saturniidae	Lepidoptera	146.00	816.2
<i>Hylesia</i> sp. ⁸⁹	Saturniidae	Lepidoptera	239.14	682.3
<i>Hyperchirica nausica</i> ⁸⁹	Saturniidae	Lepidoptera	199.50	786.0
<i>Artace</i> sp. ⁸⁹	Lasiocampidae	Lepidoptera	132.00	332.5
<i>Euglyphis</i> sp. ⁸⁹	Lasiocampidae	Lepidoptera	87.00	411.1
<i>Odonestis pruni</i> ³⁰	Lasiocampidae	Lepidoptera	250.00	1130.0
<i>Galleria mellonella</i> ⁹¹	Pyrilidae	Lepidoptera	53.60	635.0

<i>Ostrinia nubilalis</i> ⁹²	Pyrilidae	Lepidoptera	48.40	484.2
<i>Neocossus</i> sp. ⁸⁹	Cossidae	Lepidoptera	1771.63	4008.5
<i>Megalpyge</i> sp. ⁸⁹	Megalpygidae	Lepidoptera	627.00	1559.9
<i>Apis mellifera ligustica</i> ⁹³	Apidae	Hymenoptera	94.40	606.9
<i>Bombus terrestris</i> ⁹⁴	Apidae	Hymenoptera	740.00	821.8
<i>Xylocopa capitata</i> ⁹⁵	Xylocopidae	Hymenoptera	1300.00	6835.4
<i>Dasylabris</i> sp. ¹⁶	Mutillidae	Hymenoptera	86.50	37.3
<i>Dasymutilla gloriosa</i> ⁹⁶	Mutillidae	Hymenoptera	76.45	1869.5
Unknown genus ²⁰	Pompilidae	Hymenoptera	14.14	202.0
<i>Leptothorax acerovorum</i> ⁹⁷	Formicidae	Hymenoptera	0.37	5.1
<i>Leptothorax unifasciatus</i> ⁹⁸	Formicidae	Hymenoptera	0.49	2.4
<i>Atta laevigata</i> ⁹⁹	Formicidae	Hymenoptera	15.00	35.2
<i>Atta sexdens</i> ¹⁰⁰	Formicidae	Hymenoptera	15.00	43.1
<i>Myrmica alaskensis</i> ⁹⁷	Formicidae	Hymenoptera	0.91	10.7
<i>Myrmica rubra</i> ¹⁰¹	Formicidae	Hymenoptera	2.76	6.8
<i>Pogonomyrmex californicus</i> ¹⁰²	Formicidae	Hymenoptera	5.92	9.3

<i>Pogonomyrmex maricopa</i> ¹⁰³	Formicidae	Hymenoptera	11.07	40.6
<i>Pogonomyrmex occidentalis</i> ¹⁰²	Formicidae	Hymenoptera	7.96	12.8
<i>Pogonomyrmex rugosus</i> ¹⁰²	Formicidae	Hymenoptera	14.30	27.6
<i>Pogonomyrmex</i> sp. ²⁰	Formicidae	Hymenoptera	3.74	12.6
<i>Aphaenogaster cockerelli</i> ⁹⁷	Formicidae	Hymenoptera	4.72	82.5
<i>Messor capensis</i> *	Formicidae	Hymenoptera	13.70	38.3
<i>Messor capitatus</i> ¹⁰⁴	Formicidae	Hymenoptera	3.14	16.2
<i>Messor julianus</i> ¹⁰⁵	Formicidae	Hymenoptera	5.09	7.1
<i>Messor pergande</i> ¹⁰⁵	Formicidae	Hymenoptera	7.19	14.0
<i>Chelaner rothsteini</i> ¹⁰⁶	Formicidae	Hymenoptera	0.25	1.1
<i>Solenopsis invicta</i> ¹⁰⁷	Formicidae	Hymenoptera	2.96	5.6
<i>Tetramorium caespitum</i> ¹⁰¹	Formicidae	Hymenoptera	0.69	2.4
<i>Camponotus detritus</i> ¹⁰⁸	Formicidae	Hymenoptera	42.90	78.0
<i>Camponotus fulvopilosus</i> ¹⁰⁹	Formicidae	Hymenoptera	43.00	63.4
<i>Camponotus laevigatus</i> ⁹⁷	Formicidae	Hymenoptera	19.15	117.8
<i>Camponotus maculatus</i> *	Formicidae	Hymenoptera	42.49	51.1

<i>Camponotus sericeiventris</i> ¹¹⁰	Formicidae	Hymenoptera	40.20	73.2
<i>Camponotus</i> sp. ⁹⁷	Formicidae	Hymenoptera	15.41	78.2
<i>Camponotus vafer</i> ⁹⁷	Formicidae	Hymenoptera	4.51	58.8
<i>Camponotus vicinus</i> ¹¹¹	Formicidae	Hymenoptera	107.00	143.5
<i>Cataglyphis bicolor</i> ¹¹²	Formicidae	Hymenoptera	34.00	37.8
<i>Formica exsecta</i> ¹⁰¹	Formicidae	Hymenoptera	4.05	21.9
<i>Formica fusca</i> var. <i>subaenescens</i> ⁹⁷	Formicidae	Hymenoptera	1.35	31.3
<i>Formica fusca</i> ¹⁰¹	Formicidae	Hymenoptera	4.41	9.65
<i>Formica occulta</i> ⁹⁷	Formicidae	Hymenoptera	1.30	19.5
<i>Formica pratensis</i> ¹⁰¹	Formicidae	Hymenoptera	7.17	31.4
<i>Lasius alienus</i> ¹⁰¹	Formicidae	Hymenoptera	1.38	6.0
<i>Lasius flavus</i> ¹⁰¹	Formicidae	Hymenoptera	2.58	11.5
<i>Lasius niger</i> ¹⁰¹	Formicidae	Hymenoptera	1.74	6.1
<i>Lasius sitiens</i> ⁹⁷	Formicidae	Hymenoptera	0.29	3.2
<i>Forelius foetidus</i> ⁹⁷	Formicidae	Hymenoptera	0.10	1.3
<i>Anoplolepis steinergroeveri</i> *	Formicidae	Hymenoptera	4.94	14.1

<i>Leptogenys attenuata</i> ¹¹³	Formicidae	Hymenoptera	4.00	12.3
<i>Leptogenys nitida</i> ¹¹³	Formicidae	Hymenoptera	1.72	5.2
<i>Leptogenys schwabi</i> ¹¹³	Formicidae	Hymenoptera	8.96	21.0
<i>Paraponera clavata</i> ¹¹⁴	Formicidae	Hymenoptera	190.00	430.8
<i>Eciton hamatum</i> ¹¹⁵	Formicidae	Hymenoptera	6.00	25.3
Ponaria sp. ³⁹	Formicidae	Hymenoptera	0.19	1.35
Unknown genus ²⁰	Braconidae	Hymenoptera	1.70	27.8
Unknown genus ²⁰	Chalcidoidea	Hymenoptera	0.16	3.5
Unknown genus ¹⁶	Formicidae	Hymenoptera	46.4	120.2

* Chown lab, unpublished data.

‡ Wing status: 1 = winged; 0 = no wings

Ten data points added to the data set of Chown et al. (2007) are shown in green color.

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Dataset S4. Standard and routine respiration rates in aquatic invertebrates

Notes to Table S4:

Group (as shown in Table 1 in the paper): I – copepods & krill (also includes seven branchiopods and two *Balanus* spp.), Crustacea; II – peracarids (amphipods, isopods, mysids, two cumaceans, as well as five (non-peracarid) ostracod species), Crustacea; III – decapods, Crustacea; IV – cephalopods, Mollusca; V – gelatinous invertebrates: chaetognaths and medusae.

Higher taxon, Family: taxonomic status, including family, was determined from various sources, including PubMed Taxonomy; World of Copepods (Smithsonian National Museum of Natural History), Integrated Taxonomic Information System (www.itis.gov) and metabolic data sources. Table S4 should NOT, however, be used as an authoritative reference for detailed invertebrate taxonomy.

MIN: for each species, indicates the minimum value of mass-specific metabolic rate corresponding to 25 °C. Data from rows marked “MIN” were used in the analyses shown in Table 1 and Figures 1-3 in the paper (a total of 376 values for 376 species).

qWkg is standard or routine respiration rate converted to $W \text{ (kg WM)}^{-1}$ (Watts per kg wet mass) using the energy conversion factor of $1 \text{ ml O}_2 = 20 \text{ J}$. For crustaceans, the basis for this data set was formed by the data base of Alekseeva (2001) deposited at Koltzov Institute of Developmental Biology, Moscow, Russia and kindly provided to the authors by T.A. Alekseeva. This database contained 200 species, to which about one hundred species was added by the present authors by literature search. Usually, in metabolic studies animals were kept in filtered water for 12-24 hours or until their guts were empty. However, for some species, especially small marine ones (Ikeda & Skjoldal 1982) it was not possible to ensure that stomachs were empty. Where possible, respiration rates were measured on calm inactive animals. For each species, the minimum reported value was always used. To get an idea of how the reported metabolic rates are elevated above the "true" standard metabolic rate (sensu, e.g., Steffensen 2002), see SI Methods. Taking into account that in copepods the dry mass to wet mass ratio is somewhat less than the crude mean of 0.3 applied throughout the analysis (see SI Methods, Table S12a) (copepod DM/WM is about 0.17-0.20, see column WC below), even if "true" standard metabolic rate is approximately twice less than the measured group mean, while DM/WM ratio is about 1.5-2 times lower than 0.3, this means that if correction is made for the lower DM/WM content in copepods, our value of 3 W kg^{-1} is comparable to the other group means in the database. In gelatinous species (group V) (medusae and chaetognaths) with DM/WM ratio significantly smaller (by 7 and 3 times, respectively (Ikeda & Skjoldal 1989; Hirst & Lucas 1998)) than the crude mean DM/WM = 0.3 for the non-gelatinous groups, wet-mass based metabolic rates were multiplied by a factor of 7 and 3, respectively, to be comparable with the rest of the data base analysed assuming DM/WM = 0.3. For these species qWkg stands for the actual value multiplied by the corresponding factor (7 or 3). In cephalopods, minimum mass-specific metabolic rates for each of the 38 species studied by Seibel (2007) were included into the analysis. The data base of Seibel (2007) contains 218 values for 38 species, of which only minimum values for each species are reported here, those used for the analysis.

TC is ambient temperature during measurements, degrees Celsius.

q25Wkg is metabolic rate converted to 25 °C using $Q_{10} = 2$, $q25Wkg = qWkg \times 2^{(25 - TC)/10}$, dimension W (kg WM)⁻¹. For each species rows are arranged in the order of increasing **q25Wkg**.

Mg is wet body mass in grams; **DMg** is dry body mass in grams; **N/DM** is the nitrogen to dry mass ratio, where available; **WC** is water content in wet mass, %: $WC = (1 - DM/WM) \times 100$, values with questions (e.g., ?80) indicate that **qWkg** was obtained from dry-mass based measurements using an assumed WC determined from a different source than the source of metabolic data for that species.

Group	Higher Taxon	Family	Species	MIN	q25Wkg	qWkg	Mg	TC	DMg	N/DM	WC	Source
I	1.Copepoda	Acartiidae	Acartia clausi	MIN	5.2520	2.626	0.0000275	15	0.0000055		80	Nival et al. 1972
I	2.Copepoda	Acartiidae	Acartia clausi		7.5	2.0	0.000025	5.9	0.0000049		?80	Conover 1960
I	3.Copepoda	Acartiidae	Acartia clausi		8.3145	4.1	0.0000425	14.8	0.0000085		?80	Ikeda et al. 2001
I	4.Copepoda	Acartiidae	Acartia clausi		13.6	3.4	0.000021	5	0.0000041		?80	Conover 1960
I	5.Copepoda	Acartiidae	Acartia clausi		14.4	3.6	0.000024	5	0.0000047		?80	Conover 1960
I	6.Copepoda	Acartiidae	Acartia clausi		15.8	4.2	0.000022	5.9	0.0000044		?80	Conover 1960
I	7.Copepoda	Acartiidae	Acartia clausi		16.3	4.25	0.000037	5.6	0.0000073		?80	Conover 1960
I	8.Copepoda	Acartiidae	Acartia clausi		17.6	4.4	0.000036	5	0.0000072		?80	Conover 1960
I	9.Copepoda	Acartiidae	Acartia latisetosa	MIN	11.7	8.3	0.0000297	20				Pavlova 1961
I	10.Copepoda	Acartiidae	Acartia longiremis	MIN	11.1665	3.84	0.0000405	9.6	0.0000081		?80	Ikeda et al. 2001
I	11.Copepoda	Acartiidae	Acartia pacifica	MIN	6.1580	6.6	0.000039	26	0.0000078		?80	Ikeda et al. 2001
I	12.Copepoda	Acartiidae	Acartia tonsa	MIN	10.9572	8.9	0.000035	22	0.000007		?80	Ikeda et al. 2001
I	13.Copepoda	Arietellidae	Arietellus cf. plumifer	MIN	0.880	0.220	0.0095	5				Thuesen et al. 1998
I	14.Copepoda	Megacalanidae	Bathycalanus bradyi	MIN	0.5788	0.135	0.0299	4	0.0084	0.069	72	Childress 1975
I	15.Copepoda	Megacalanidae	Bathycalanus bradyi		0.816	0.204	0.0555	5				Thuesen et al. 1998
I	16.Copepoda	Megacalanidae	Bathycalanus bradyi		1.5862	0.370	0.0299	4	0.0084	0.069	72	Childress 1975
I	17.Copepoda	Megacalanidae	Bathycalanus bradyi		0.736	0.184	0.0756	5				Thuesen et al. 1998
I	18.Copepoda	Megacalanidae	Bathycalanus bradyi		1.071	0.210	0.0530	1.5				Thuesen et al. 1998
I	19.Copepoda	Megacalanidae	Bathycalanus princeps	MIN	0.2280	0.059	0.0355	5.5	0.0070	0.096	80.2	Childress 1975
I	20.Copepoda	Megacalanidae	Bathycalanus princeps		1.2634	0.327	0.0355	5.5	0.0070	0.096	80.2	Childress 1975
I	21.Copepoda	Megacalanidae	Bathycalanus richard	MIN	0.732	0.183	0.0380	5				Thuesen et al. 1998
I	22.Copepoda	Megacalanidae	Bathycalanus richardi		0.532	0.133	0.0560	5				Thuesen et al. 1998
I	23.Copepoda	Megacalanidae	Bathycalanus richardi		1.193	0.234	0.0377	1.5				Thuesen et al. 1998
I	24.Copepoda	Megacalanidae	Bathycalanus sp.	MIN	1.67	0.42	0.0895	5.1	0.0179		?80	Conover 1960
I	25.Copepoda	Megacalanidae	Bathycalanus sp. A	MIN	0.816	0.204	0.0572	5				Thuesen et al. 1998
I	26.Copepoda	Calanidae	Calanoides acutus	MIN	0.9108	0.161	0.003329	0	0.00079		76.3	Kawall et al. 2001
I	27.Copepoda	Calanidae	Calanoides acutus		3.3841	0.59	0.002	-0.2	0.000394		?80	Ikeda et al. 2001
I	28.Copepoda	Calanidae	Calanoides acutus		3.8652	0.66	0.0011	-0.5	0.000218		?80	Ikeda et al. 2001
I	29.Copepoda	Pontellidae	Calanopia elliptica	MIN	4.7079	5.56	0.00028	27.4	0.000056		?80	Ikeda et al. 2001
I	30.Copepoda	Calanidae	Calanus cristatus	MIN	2.8473	1.094	0.00862	11.2	0.00169			Ikeda 1971
I	31.Copepoda	Calanidae	Calanus cristatus		2.9639	1.100	0.00908	10.7	0.00178			Ikeda 1971
I	32.Copepoda	Calanidae	Calanus cristatus		3.1012	1.167	0.00704	10.9	0.00138			Ikeda 1971
I	33.Copepoda	Calanidae	Calanus cristatus		6.0946	1.850	0.00918	7.8	0.0018			Ikeda 1971
I	34.Copepoda	Calanidae	Calanus cristatus		7.0211	2.161	0.00755	8	0.00148			Ikeda 1971
I	35.Copepoda	Calanidae	Calanus cristatus		7.4097	2.394	0.0076	8.7	0.00149			Ikeda 1971
I	36.Copepoda	Calanidae	Calanus finmarchicus	MIN	1.6	0.50	0.0017	8	0.000344		?80	Conover 1960
I	37.Copepoda	Calanidae	Calanus finmarchicus		3.6	1.06	0.00094	7.5	0.000188		?80	Conover 1960

I	38.Copepoda	Calanidae	Calanus finmarchicus		7	1.25	0.0015	0.1	0.000387	0.0763	73.5	Ikeda & Skjoldal 1989
I	39.Copepoda	Calanidae	Calanus glacialis	MIN	7.8	1.73	0.002	3.5	0.000410	0.1121	79.7	Ikeda & Skjoldal 1989
I	40.Copepoda	Calanidae	Calanus glacialis		7.83	1.34	0.0024	-0.5	0.000662	0.0969	76.6	Ikeda & Skjoldal 1989
I	41.Copepoda	Calanidae	Calanus helgolandicus	MIN	7.8180	3.909	0.000415	15	0.000086		83	Nival et al. 1972
I	42.Copepoda	Calanidae	Calanus helgolandicus		12.6912	4.487	0.000595	10	0.000119		80	Nival et al. 1972
I	43.Copepoda	Calanidae	Calanus hyperboreus	MIN	1.05	0.37	0.013	10	0.00368		??1	Conover 1960
I	44.Copepoda	Calanidae	Calanus hyperboreus		1.36	0.48	0.012	10	0.00362		??1	Conover 1960
I	45.Copepoda	Calanidae	Calanus hyperboreus		2.23	0.79		10	0.00155		??1	Conover 1960
I	46.Copepoda	Calanidae	Calanus hyperboreus		3.7	0.67	0.012	1.3	0.00395	0.0715	68	Ikeda & Skjoldal 1989
I	47.Copepoda	Calanidae	Calanus hyperboreus		3.72	0.70	0.0081	0.9	0.00194	0.0737	76.1	Ikeda & Skjoldal 1989
I	48.Copepoda	Calanidae	Calanus hyperboreus		3.9	0.68	0.0085	-0.3	0.00268	0.0670	68.6	Ikeda & Skjoldal 1989
I	49.Copepoda	Calanidae	Calanus pacificus	MIN	8.3190	4.778	0.000513	17	0.00009		82.5	Ikeda 1970
I	50.Copepoda	Calanidae	Calanus propinquus	MIN	1.6631	0.294	0.004718	0	0.00101		78.5	Kawall et al. 2001
I	51.Copepoda	Calanidae	Calanus propinquus		5.4	0.89	0.0042	-1	0.000847	0.123	??80	Ikeda & Mitchell 1982
I	52.Copepoda	Calanidae	Calanus propinquus		7.0936	1.17	0.0056	-1	0.00104	0.125	81.4	Ikeda 1988
I	53.Copepoda	Calanidae	Calanus propinquus		7.6	1.26	0.0052	-1	0.00104	0.125	??80	Ikeda & Mitchell 1982
I	54.Copepoda	Calanidae	Calanus tonsus	MIN	13.2049	2.178	0.001	-1	0.0002		90	Opalinski 1991
I	55.Copepoda	Calanidae	Calanus vulgaris	MIN	16.6955	23.611	0.00106	30	0.00022		79.2	Ikeda 1970
I	56.Copepoda	Candaciidae	Candacia aethiopica	MIN	16.4076	21.278	0.00046	28.75	0.000065		85.9	Ikeda 1970
I	57.Copepoda	Candaciidae	Candacia columbiae	MIN	0.7091	0.194	0.00367	6.3	0.00042		88.6	Ikeda 1970
I	58.Copepoda	Candaciidae	Candacia columbiae		1.9341	0.544	0.00312	6.7	0.0006		80.8	Ikeda 1970
I	59.Copepoda	Candaciidae	Candacia sp.	MIN	20.4300	10.215	0.000155	15	0.000031		80	Nival et al. 1972
I	60.Copepoda	Centropagidae	Centropages abdominalis	MIN	10.5227	5.6	0.000085	15.9	0.000017		??80	Ikeda et al. 2001
I	61.Copepoda	Centropagidae	Centropages brachiatus	MIN	13.4364	6.4	0.0001	14.3	0.00002		??80	Ikeda et al. 2001
I	62.Copepoda	Centropagidae	Centropages kroyeri	MIN	9.4	6.7	0.000039	20				Pavlova 1961
I	63.Copepoda	Centropagidae	Centropages trispinosus	MIN	34.6199	24.48	0.000066	20				Rajagopal 1962
I	64.Copepoda	Centropagidae	Centropages trispinosus		35.5964	46.97	0.000066	29	0.000013		80.4	Rajagopal 1962
I	65.Copepoda	Clausocalanidae	Clausocalanus furcatus	MIN	20.844	14.739	0.000019	20				Pavlova 1975
I	66.Copepoda	Halocyprididae	Conchoecia sp.	MIN	17.709	12.522	0.000045	20				Pavlova 1975
I	67.Copepoda	Corycaidae	Corycaeus typicus	MIN	7.7660	3.883	0.0000515	15	0.0000103		80	Nival et al. 1972
I	68.Copepoda	Heterorhabdidae	Disseta grandis	MIN	0.692	0.173	0.0114	5				Thuesen et al. 1998
I	69.Copepoda	Heterorhabdidae	Disseta scopularis	MIN	0.332	0.083	0.0195	5				Thuesen et al. 1998
I	70.Copepoda	Heterorhabdidae	Disseta scopularis		0.693	0.136	0.0257	1.5				Thuesen et al. 1998
I	71.Copepoda	Aetideidae	Euatideus giesbrechti	MIN	5.940	4.200	0.000215	20				Pavlova 1975
I	72.Copepoda	Augaptilidae	Euaugaptilus antarcticus	MIN	0.512	0.128	0.0225	5				Thuesen et al. 1998
I	73.Copepoda	Augaptilidae	Euaugaptilus magnus	MIN	0.288	0.072	0.0309	5				Thuesen et al. 1998
I	74.Copepoda	Augaptilidae	Euaugaptilus nodifrons	MIN	0.280	0.070	0.0597	5				Thuesen et al. 1998
I	75.Copepoda	Eucalanidae	Eucalanus attenuatus	MIN	2.3980	1.199	0.001389	15	0.00025		82	Nival et al. 1972
I	76.Copepoda	Eucalanidae	Eucalanus attenuatus		6.2151	7.34	0.000685	27.4	0.000137		??80	Ikeda et al. 2001
I	77.Copepoda	Eucalanidae	Eucalanus bungii	MIN	1.8496	0.506	0.00822	6.3	0.00093		88.7	Ikeda 1970
I	78.Copepoda	Eucalanidae	Eucalanus bungii		2.2367	0.589	0.00807	5.75	0.00081		90	Ikeda 1970
I	79.Copepoda	Eucalanidae	Eucalanus bungii		3.2470	0.87	0.005	6	0.00101		??80	Ikeda et al. 2001
I	80.Copepoda	Eucalanidae	Eucalanus crassus	MIN	10.8093	15.500	0.00207	30.2	0.00036		82.6	Ikeda 1970
I	81.Copepoda	Eucalanidae	Eucalanus elongatus	MIN	1.870	1.322	0.001786	20				Pavlova 1975
I	82.Copepoda	Eucalanidae	Eucalanus elongatus		5.2060	2.603	0.00054	15	0.000108		80	Nival et al. 1972
I	83.Copepoda	Eucalanidae	Eucalanus marina	MIN	8.4832	10.444	0.00143	28	0.000234		83.7	Ikeda 1970
I	84.Copepoda	Eucalanidae	Eucalanus marina		11.6954	15.167	0.00082	28.75	0.000119		85.5	Ikeda 1970
I	85.Copepoda	Eucalanidae	Eucalanus monachus	MIN	16.8143	24.111	0.00145	30.2	0.00022		79.2	Ikeda 1970
I	86.Copepoda	Eucalanidae	Eucalanus mucronatus	MIN	3.8106	5.389	0.00072	30	0.00018		80.6	Ikeda 1970
I	87.Copepoda	Eucalanidae	Eucalanus subcrassus	MIN	7.5024	7.0	0.00052	24	0.000104		??80	Ikeda et al. 2001
I	88.Copepoda	Eucalanidae	Eucalanus subcrassus		18.2278	12.89	0.000191	20				Rajagopal 1962
I	89.Copepoda	Eucalanidae	Eucalanus subcrassus		19.0457	25.13	0.000191	29	0.0000375		80.4	Rajagopal 1962

I	90.Copepoda	Euchaetidae	Euchaeta antarctica	MIN	1.0069	0.178	0.003	0	0.0006		80	Opalinski 1991
I	91.Copepoda	Euchaetidae	Euchaeta marina	MIN	5.4660	5.1	0.00146	24	0.000291		?80	Ikeda et al. 2001
I	92.Copepoda	Euchaetidae	Euchaeta plana	MIN	1.1160	0.7891	0.003	20				Opalinski 1991
I	93.Copepoda	Euchaetidae	Euchaeta plana		12.3641	15.222	0.00095	28	0.000158		83.4	Ikeda 1970
I	94.Copepoda	Lucitiidae	Euchirella amoena	MIN	9.1606	11.278	0.00266	28	0.00056		79	Ikeda 1970
I	95.Copepoda	Lucitiidae	Euchirella bitumida	MIN	1.196	0.299	0.0094	5				Thuesen et al. 1998
I	96.Copepoda	Lucitiidae	Euchirella bitumida		1.417	0.278	0.0128	1.5				Thuesen et al. 1998
I	97.Copepoda	Lucitiidae	Euchirella maxima	MIN	1.012	0.253	0.0188	5				Thuesen et al. 1998
I	98.Copepoda	Lucitiidae	Euchirella rostrata	MIN	3.2340	1.617	0.00158	15	0.000316		80	Nival et al. 1972
I	99.Copepoda	Lucitiidae	Euchirella rostrata		4.3	1.2	0.0045	6.5	0.000890		?80	Conover 1960
I	100.Copepoda	Lucitiidae	Euchirella rostrata		4.5	1.25	0.0041	6.5	0.000820		?80	Conover 1960
I	101.Copepoda	Lucitiidae	Euchirella rostrata		5.7	1.6	0.0047	6.5	0.000947		?80	Conover 1960
I	102.Copepoda	Tachidiidae	Euterpina acutifrons	MIN	6.0220	3.011	0.0000107	15	0.00000214		80	Nival et al. 1972
I	103.Copepoda	Aetideidae	Gaetanus antarcticus	MIN	0.459	0.090	0.0257	1.5				Thuesen et al. 1998
I	104.Copepoda	Aetideidae	Gaetanus antarcticus		0.628	0.157	0.0257	5				Thuesen et al. 1998
I	105.Copepoda	Aetideidae	Gaetanus kruppi	MIN	0.688	0.172	0.0072	5				Thuesen et al. 1998
I	106.Copepoda	Aetideidae	Gaetanus pileatus	MIN	2.132	0.533	0.0085	5				Thuesen et al. 1998
I	107.Copepoda	Aetideidae	Gaetanus tenuispinus	MIN	0.3168	0.056	0.00364	0	0.00545		85.0	Kawall et al. 2001
I	108.Copepoda	Metridiidae	Gaussia princeps	MIN	0.2925	0.084	0.0328	7	0.0054	0.098	83.7	Childress 1975
I	109.Copepoda	Metridinidae	Gaussia princeps		0.956	0.239	0.0356	5				Thuesen et al. 1998
I	110.Copepoda	Metridinidae	Gaussia princeps		1.5008	0.431	0.0328	7	0.0054	0.098	83.7	Childress 1975
I	111.Copepoda	Metridinidae	Gaussia princeps (males)		1.032	0.258	0.0346	5				Thuesen et al. 1998
I	112.Copepoda	Augaptilidae	Haloptilus longicornis	MIN	1.706	1.206	0.00022	20				Pavlova 1975
I	113.Copepoda	Augaptilidae	Haloptilus longicornis		3.244	2.294	0.00016	20				Pavlova 1975
I	114.Copepoda	Heterorhabdidae	Hemirhabdus grimaldii	MIN	0.252	0.063	0.0382	5				Thuesen et al. 1998
I	115.Copepoda	Heterorhabdidae	Heterorhabdus farrani	MIN	2.2062	0.39	0.003215	0	0.00034		89.5	Kawall et al. 2001
I	116.Copepoda	Pontellidae	Labidocera acuta	MIN	7.8458	10	0.00114	28.5	0.000228		?80	Ikeda et al. 2001
I	117.Copepoda	Pontellidae	Labidocera acuta		9.6460	9.0	0.00117	24	0.000233		?80	Ikeda et al. 2001
I	118.Copepoda	Pontellidae	Labidocera acuta		11.3533	16.056	0.00099	30	0.00017		82.8	Ikeda 1970
I	119.Copepoda	Pontellidae	Labidocera detruncata	MIN	34.8814	42.944	0.00093	28	0.00015		83.9	Ikeda 1970
I	120.Copepoda	Pontellidae	Labidocera jollae	MIN	3.7538	2.156	0.042	17	0.00645		84.5	Childress 1975
I	121.Copepoda	Pontellidae	Labidocera jollae		3.8414	2.7163	0.0416	20				Childress 1975
I	122.Copepoda	Pontellidae	Labidocera sp.	MIN	12.8849	18.222	0.00082	30	0.00018		78	Ikeda 1970
I	123.Copepoda	Scolecithricidae	Landrums gigas		1.136	0.284	0.0309	5				Thuesen et al. 1998
I	124.Copepoda	Lucitiidae	Lucicutia bicornuta	MIN	0.548	0.137	0.0105	5				Thuesen et al. 1998
I	125.Copepoda	Lucitiidae	Lucicutia flavicornis	MIN	8.085	5.717	0.000129	20				Pavlova 1975
I	126.Copepoda	Lucitiidae	Lucicutia maxima	MIN	0.696	0.174	0.0118	5				Thuesen et al. 1998
I	127.Copepoda	Lucitiidae	Lucicutia maxima		0.749	0.147	0.0147	1.5				Thuesen et al. 1998
I	128.Copepoda	Lucitiidae	Lucicutia maxima		0.973	0.344	0.0104	10				Thuesen et al. 1998
I	129.Copepoda	Megacalanidae	Megacalanus princeps	MIN	0.820	0.205	0.0556	5				Thuesen et al. 1998
I	130.Copepoda	Megacalanidae	Megacalanus sp. A	MIN	0.880	0.220	0.0349	5				Thuesen et al. 1998
I	131.Copepoda	Megacalanidae	Megacalanus sp. A		0.744	0.186	0.0345	5				Thuesen et al. 1998
I	132.Copepoda	Megacalanidae	Megacalanus sp. A		0.918	0.180	0.0356	1.5				Thuesen et al. 1998
I	133.Copepoda	Megacalanidae	Megacalanus sp. A		1.071	0.210	0.0345	1.5				Thuesen et al. 1998
I	134.Copepoda	Megacalanidae	Megacalanus sp. A		1.321	0.467	0.0326	10				Thuesen et al. 1998
I	135.Copepoda	Megacalanidae	Megacalanus sp. B	MIN	0.640	0.160	0.0372	5				Thuesen et al. 1998
I	136.Copepoda	Calanidae	Mesocalanus tenuicornis	MIN	14.9269	7.889	0.000155	15.8	0.000031		?80	Ikeda et al. 2001
I	137.Copepoda	Metridinidae	Metridia gerlachei	MIN	2.3250	0.411	0.001409	0	0.00025		82.2	Kawall et al. 2001
I	138.Copepoda	Metridinidae	Metridia gerlachei		6.7	1.1	0.0011	-1	0.000214	0.111	?80	Ikeda & Mitchell 1982
I	139.Copepoda	Metridinidae	Metridia gerlachei		8.3526	1.34	0.0013	-1.4	0.000265	0.114	?80	Ikeda & Mitchell 1982
I	140.Copepoda	Metridinidae	Metridia longa	MIN	8.2	1.46	0.00155	0.1	0.000353	0.1044	77.2	Ikeda & Skjoldal 1989
I	141.Copepoda	Metridinidae	Metridia pacifica	MIN	4.2937	1.34	0.00074	8.2	0.000148		?80	Ikeda et al. 2001

I	142. Copepoda	Metridinidae	Metridia pacifica		7.4436	3.24	0.00047	13	0.000094		?80	Ikeda et al. 2001
I	143. Copepoda	Metridinidae	Metridia princeps	MIN	0.808	0.202	0.0111	5				Thuesen et al. 1998
I	144. Copepoda	Metridinidae	Metridia princeps		1.300	0.255	0.0125	1.5				Thuesen et al. 1998
I	145. Copepoda	Calanidae	Nannocalanus minor	MIN	5.0843	5.8	0.0002	26.9	0.00004		?80	Ikeda et al. 2001
I	146. Copepoda	Calanidae	Neocalanus cristatus	MIN	0.5466	0.111	0.023	2	0.00832	0.0675	63.9	Ikeda et al. 2004
I	147. Copepoda	Calanidae	Neocalanus cristatus		0.5762	0.117	0.0172	2	0.00611	0.0833	63.7	Ikeda et al. 2004
I	148. Copepoda	Calanidae	Neocalanus cristatus		0.6008	0.122	0.0264	2	0.00933	0.0741	64.5	Ikeda et al. 2004
I	149. Copepoda	Calanidae	Neocalanus cristatus		0.6550	0.133	0.0242	2	0.00797	0.0586	67.2	Ikeda et al. 2004
I	150. Copepoda	Calanidae	Neocalanus cristatus		0.6550	0.133	0.0269	2	0.00908	0.0609	65.6	Ikeda et al. 2004
I	151. Copepoda	Calanidae	Neocalanus cristatus		0.6845	0.139	0.021	2	0.0068	0.0754	67.6	Ikeda et al. 2004
I	152. Copepoda	Calanidae	Neocalanus cristatus		0.7387	0.150	0.0236	2	0.00865	0.0689	63.5	Ikeda et al. 2004
I	153. Copepoda	Calanidae	Neocalanus cristatus		0.7929	0.161	0.0260	2	0.010	0.0687	61.3	Ikeda et al. 2004
I	154. Copepoda	Calanidae	Neocalanus cristatus		0.8470	0.172	0.0195	2	0.00579	0.0810	70.3	Ikeda et al. 2004
I	155. Copepoda	Calanidae	Neocalanus cristatus		0.8470	0.172	0.0221	2	0.00727	0.0697	67.8	Ikeda et al. 2004
I	156. Copepoda	Calanidae	Neocalanus cristatus		0.9012	0.183	0.0206	2	0.00355	0.0901	82.9	Ikeda et al. 2004
I	157. Copepoda	Calanidae	Neocalanus cristatus		0.9195	0.283	0.0192	8	0.00209	0.104	89.4	Ikeda et al. 2004
I	158. Copepoda	Calanidae	Neocalanus cristatus		0.9554	0.194	0.0299	2	0.00663	0.085	77.7	Ikeda et al. 2004
I	159. Copepoda	Calanidae	Neocalanus cristatus		1.2016	0.244	0.0153	2	0.00265	0.0863	83.1	Ikeda et al. 2004
I	160. Copepoda	Calanidae	Neocalanus cristatus		1.2241	0.328	0.0166	6	0.00172	0.0960	89.8	Ikeda et al. 2004
I	161. Copepoda	Calanidae	Neocalanus cristatus		1.2607	0.256	0.0226	2	0.00618	0.0685	72.2	Ikeda et al. 2004
I	162. Copepoda	Calanidae	Neocalanus cristatus		2.2161	0.450	0.0202	2	0.00524	0.0775	74.1	Ikeda et al. 2004
I	163. Copepoda	Calanidae	Neocalanus cristatus		2.2456	0.456	0.0090	2	0.0017	0.108	81.5	Ikeda et al. 2004
I	164. Copepoda	Calanidae	Neocalanus cristatus		4.2767	1.17	0.008	6.3	0.001587		?80	Ikeda et al. 2001
I	165. Copepoda	Calanidae	Neocalanus gracilis	MIN	5.2559	3.64	0.0025	19.7	0.0005		?80	Ikeda et al. 2001
I	166. Copepoda	Calanidae	Neocalanus gracilis		6.2275	7.667	0.00212	28	0.00039		81.6	Ikeda 1970
I	167. Copepoda	Calanidae	Neocalanus plumchrus	MIN	3.2741	0.96	0.004	7.3	0.000785		?80	Ikeda et al. 2001
I	168. Copepoda	Calanidae	Neocalanus plumchrus		4.1710	2.1	0.0015	15.1	0.000294		?80	Ikeda et al. 2001
I	169. Copepoda	Calanidae	Neocalanus plumchrus		4.5277	1.18	0.0013	5.6	0.000263		?80	Ikeda et al. 2001
I	170. Copepoda	Oithonidae	Oithona davisae	MIN	9.8500	9.85	0.0000030	25	0.000000575		81	Castellani et al. 2005
I	171. Copepoda	Oithonidae	Oithona minuta	MIN	29.9	21.1	0.0000039	20				Pavlova 1961
I	172. Copepoda	Oithonidae	Oithona nana	MIN	17.2360	7.0	0.0000026	12	0.0000005		81	Castellani et al. 2005
I	173. Copepoda	Oithonidae	Oithona plumifera	MIN	1.9292	1.8	0.0000089	24	0.0000017		81	Castellani et al. 2005
I	174. Copepoda	Oithonidae	Oithona setigera	MIN	6.4306	6.0	0.0000086	24	0.0000016		81	Castellani et al. 2005
I	175. Copepoda	Oithonidae	Oithona similis	MIN	0.4800	0.48	0.000007	25	0.0000014		?80	Castellani et al. 2005
I	176. Copepoda	Oithonidae	Oithona similis		7.8000	7.8	0.000007	25	0.0000014		?80	Castellani et al. 2005
I	177. Copepoda	Oithonidae	Oithona tenuis	MIN	5.8948	5.5	0.0000067	24	0.0000013		81	Castellani et al. 2005
I	178. Copepoda	Oncaeidae	Oncaea conifera	MIN	5.665	4.006	0.00006	20				Pavlova 1975
I	179. Copepoda	Oncaeidae	Oncaea mediterranea	MIN	6.655	4.706	0.000062	20				Pavlova 1975
I	180. Copepoda	Phaennidae	Onchocalanus magnus	MIN	1.680	0.420	0.0124	5				Thuesen et al. 1998
I	181. Copepoda	Augaupiliidae	Pachyptilus pacificus	MIN	0.692	0.173	0.0158	5				Thuesen et al. 1998
I	182. Copepoda	Augaupiliidae	Pachyptilus pacificus		0.795	0.156	0.0158	1.5				Thuesen et al. 1998
I	183. Copepoda	Paracalanidae	Paracalanus parvus	MIN	19.5556	9.06	0.000019	13.9	0.0000038		?80	Ikeda et al. 2001
I	184. Copepoda	Euchaetidae	Paraeuchaeta antarctica	MIN	0.7241	0.128	0.019551	0	0.004		79.3	Kawall et al. 2001
I	185. Copepoda	Euchaetidae	Paraeuchaeta birostrata	MIN	1.021	0.361	0.0129	10				Thuesen et al. 1998
I	186. Copepoda	Euchaetidae	Paraeuchaeta birostrata		1.156	0.289	0.0126	5				Thuesen et al. 1998
I	187. Copepoda	Euchaetidae	Paraeuchaeta brevicauda	MIN	0.912	0.228	0.0264	5				Thuesen et al. 1998
I	188. Copepoda	Euchaetidae	Paraeuchaeta californica	MIN	1.204	0.301	0.0137	5				Thuesen et al. 1998
I	189. Copepoda	Euchaetidae	Paraeuchaeta japonica	MIN	2.3892	0.672	0.0093	6.7	0.00201		78.4	Ikeda 1970
I	190. Copepoda	Euchaetidae	Paraeuchaeta norvegica	MIN	1.39	0.49	0.0223	10	0.00459		?80	Conover 1960
I	191. Copepoda	Euchaetidae	Paraeuchaeta norvegica		1.58	0.56	0.0194	10	0.00388		?80	Conover 1960
I	192. Copepoda	Euchaetidae	Paraeuchaeta norvegica		1.81	0.64	0.0190	10	0.00380		?80	Conover 1960
I	193. Copepoda	Euchaetidae	Paraeuchaeta rubra	MIN	0.848	0.212	0.0127	5				Thuesen et al. 1998

I	194. Copepoda	Euchaetidae	Paraeuchaeta rubra		1.6251	0.33	0.0131	2	0.00383	0.0729	69.1	Ikeda et al. 2004
I	195. Copepoda	Euchaetidae	Paraeuchaeta sesquipedalis	MIN	1.020	0.255	0.0249	5				Thuesen et al. 1998
I	196. Copepoda	Euchaetidae	Paraeuchaeta sp.	MIN	6.0641	1.072	0.017	0	0.003		83	Opalinski 1991
I	197. Copepoda	Euchaetidae	Paraeuchaeta tonsa	MIN	2.164	0.541	0.0066	5				Thuesen et al. 1998
I	198. Copepoda	Metridinidae	Pleuromamma abdominalis	MIN	4.895	3.461	0.00123	20				Pavlova 1975
I	199. Copepoda	Metridinidae	Pleuromamma abdominalis		10.0140	5.007	0.000415	15	0.000083		80	Nival et al. 1972
I	200. Copepoda	Metridinidae	Pleuromamma abdominalis		10.120	7.156	0.0008	20				Pavlova 1975
I	201. Copepoda	Metridinidae	Pleuromamma gracilis	MIN	5.500	3.889	0.000270	20				Pavlova 1975
I	202. Copepoda	Metridinidae	Pleuromamma gracilis		11.2580	5.629	0.0001875	15	0.0000375		80	Nival et al. 1972
I	203. Copepoda	Metridinidae	Pleuromamma robusta	MIN	7.1	2.5	0.0014	8	0.000283		780	Conover 1960
I	204. Copepoda	Metridinidae	Pleuromamma robusta		10.1	3.1	0.0014	8	0.000283		780	Conover 1960
I	205. Copepoda	Pontellidae	Pontella danae	MIN	4.9369	5.44	0.0035	26.4	0.0007		780	Ikeda et al. 2001
I	206. Copepoda	Pontellidae	Pontella sp.	MIN	11.5880	5.794	0.000345	15	0.000069		80	Nival et al. 1972
I	207. Copepoda	Pseudocalanidae	Pseudocalanus elongatus	MIN	8.6	6.1	0.000059	20				Pavlova 1961
I	208. Copepoda	Pseudocalanidae	Pseudocalanus elongatus		11.5316	3.7	0.00006	8.6	0.000012		780	Ikeda et al. 2001
I	209. Copepoda	Pseudocalanidae	Pseudocalanus elongatus		17.2086	6.589	0.00004	11.15	0.0000012		70	Ikeda 1970
I	210. Copepoda	Lucitiidae	Pseudochirella polyspina	MIN	0.856	0.214	0.0117	5				Thuesen et al. 1998
I	211. Copepoda	Pseudodiaptomidae	Pseudodiaptomus marinus	MIN	9.2375	4.4	0.00007	14.3	0.000014		780	Ikeda et al. 2001
I	212. Copepoda	Eucalanidae	Rhincalanus gigas	MIN	0.5657	0.100	0.011692	0	0.00157		86.6	Kawall et al. 2001
I	213. Copepoda	Eucalanidae	Rhincalanus gigas		2.8639	0.45	0.009	-1.7	0.00180		780	Ikeda et al. 2001
I	214. Copepoda	Eucalanidae	Rhincalanus nasutus	MIN	0.5091	0.3600	0.024	20				Opalinski 1991
I	215. Copepoda	Eucalanidae	Rhincalanus nasutus		1.4281	1.0098	0.009	20				Opalinski 1991
I	216. Copepoda	Eucalanidae	Rhincalanus nasutus		1.4651	1.0360	0.015	20				Opalinski 1991
I	217. Copepoda	Eucalanidae	Rhincalanus nasutus		2.38	0.66	0.0054	6.5	0.001079		780	Conover 1960
I	218. Copepoda	Eucalanidae	Rhincalanus nasutus		2.46	0.68	0.0051	6.5	0.001019		780	Conover 1960
I	219. Copepoda	Eucalanidae	Rhincalanus nasutus		2.63	0.73	0.00413	6.5	0.000826		780	Conover 1960
I	220. Copepoda	Sapphirinidae	Sapphirina gemma	MIN	7.5786	10.000	0.00071	29	0.00007		90.1	Ikeda 1970
I	221. Copepoda	Sapphirinidae	Sapphirina gemma		29.9735	42.389	0.001	30	0.00008		92	Ikeda 1970
I	222. Copepoda	Scolecithrichidae	Scolecithrix danae	MIN	3.080	2.178	0.00300	20				Pavlova 1975
I	223. Copepoda	Temoridae	Temora longicornis	MIN	36.6506	48.36	0.000061	29	0.000012		80.4	Rajagopal 1962
I	224. Copepoda	Tortanidae	Tortanus discaudatus	MIN	8.1484	2.34	0.000285	7	0.000057		780	Ikeda et al. 2001
I	225. Copepoda	Tortanidae	Tortanus discaudatus		16.5565	6.3393	0.0002	11.15	0.00005		75	Ikeda 1970
I	226. Copepoda	Tortanidae	Tortanus gracilis	MIN	6.1767	6.62	0.00012	26	0.000024		780	Ikeda et al. 2001
I	227. Copepoda	Calanidae	Undinula vulgaris	MIN	6.4355	5.80	0.000925	23.5	0.000185		780	Ikeda et al. 2001
I	228. Copepoda	Calanidae	Undinula vulgaris		7.7663	9.3	0.000838	27.6	0.000167		780	Ikeda et al. 2001
I	229. Calanoida	Euchaetidae	Valdiviella oligartha	MIN	0.752	0.188	0.0194	5				Thuesen et al. 1998
I	230. Euphausiacea	Euphausiidae	Euphausia crystallorophias	MIN	1.8213	0.311	0.05	-0.5	0.01		80	Opalinski 1991
I	231. Euphausiacea	Euphausiidae	Euphausia crystallorophias		3.4145	0.544	0.09	-1.5	0.018		80	Opalinski 1991
I	232. Euphausiacea	Euphausiidae	Euphausia crystallorophias		3.6714	0.589	0.03	-1.4	0.006		80	Opalinski 1991
I	233. Euphausiacea	Euphausiidae	Euphausia crystallorophias		4.1750	0.656	0.211	-1.7	0.047		78	Opalinski 1991
I	234. Euphausiacea	Euphausiidae	Euphausia crystallorophias		4.3468	0.683	0.079	-1.7	0.017		78.5	Opalinski 1991
I	235. Euphausiacea	Euphausiidae	Euphausia crystallorophias		5.3396	0.839	0.249	-1.7	0.05		80	Opalinski 1991
I	236. Euphausiacea	Euphausiidae	Euphausia crystallorophias		6.8025	1.122	0.053	-1	0.011		79.3	Opalinski 1991
I	237. Euphausiacea	Euphausiidae	Euphausia mutica	MIN	14.5222	18.833	0.00326	28.75	0.00042		87.1	Ikeda 1970
I	238. Euphausiacea	Euphausiidae	Euphausia pacifica	MIN	1.3378	0.473	0.0338	10	0.0070	0.123	79.4	Childress 1975
I	239. Euphausiacea	Euphausiidae	Euphausia pacifica		4.5504	2.650	0.0177	17.2	0.0033		81.4	Ikeda 1970
I	240. Euphausiacea	Euphausiidae	Euphausia pacifica		9.1754	3.244	0.0338	10	0.0070	0.123	79.4	Childress 1975
I	241. Euphausiacea	Euphausiidae	Euphausia superba	MIN	2.9102	0.48	1.43	-1.0	0.35378	0.103	75.3	Ikeda 1988
I	242. Euphausiacea	Euphausiidae	Euphausia superba		3.0647	0.502	0.145	-1.1	0.02703	0.110	81.4	Ikeda 1988
I	243. Euphausiacea	Euphausiidae	Euphausia superba		3.0647	0.502	1.01	-1.1	0.23768	0.102	76.4	Ikeda 1988
I	244. Euphausiacea	Euphausiidae	Euphausia superba		3.1197	0.511	0.369	-1.1	0.07677	0.102	79.2	Ikeda 1988
I	245. Euphausiacea	Euphausiidae	Euphausia superba		3.2051	0.525	0.616	-1.1	0.14286	0.099	76.8	Ikeda 1988

I	246.Euphausiacea	Euphausiidae	Euphausia tricantha	MIN	2.4718	0.44	0.32	0.1	0.08283	0.107	74.1	Ikeda 1988
I	247.Euphausiacea	Euphausiidae	Thysanoessa inermis	MIN	1.2	1.004	0.017	0.1	0.00457	0.1004	74.1	Ikeda & Skjoldal 1989
I	248.Euphausiacea	Euphausiidae	Thysanoessa inermis		3.5	0.71	0.108	1.9	0.03478	0.0744	67.8	Ikeda & Skjoldal 1989
I	249.Euphausiacea	Euphausiidae	Thysanoessa inermis		5.8987	2.306	0.00801	11.45	0.00119		85.1	Ikeda 1970
I	250.Euphausiacea	Euphausiidae	Thysanoessa macrura	MIN	0.9780	0.167	0.05	-0.5	0.01		80	Opalinski 1991
I	251.Euphausiacea	Euphausiidae	Thysanoessa macrura		2.0189	0.333	0.006	-1	0.001		83	Opalinski 1991
I	252.Euphausiacea	Euphausiidae	Thysanoessa macrura		2.0856	0.344	0.009	-1	0.002		78	Opalinski 1991
I	253.Euphausiacea	Euphausiidae	Thysanoessa macrura		2.1556	0.400	0.05	0.7	0.01		80	Opalinski 1991
I	254.Euphausiacea	Euphausiidae	Thysanoessa macrura		2.7287	0.517	0.09	1	0.019		79	Opalinski 1991
I	255.Euphausiacea	Euphausiidae	Thysanoessa macrura		2.8344	0.600	0.034	2.6	0.007		79.4	Opalinski 1991
I	256.Euphausiacea	Euphausiidae	Thysanoessa macrura		2.9926	0.511	0.05	-0.5	0.01		80	Opalinski 1991
I	257.Euphausiacea	Euphausiidae	Thysanoessa raschii	MIN	3.0116	0.850	0.107	6.75	0.0185		82.7	Ikeda 1970
I	258.Euphausiacea	Euphausiidae	Thysanoessa raschii		3.8371	1.083	0.108	6.75	0.0175		83.4	Ikeda 1970
I	259.Euphausiacea	Euphausiidae	Thysanoessa raschii		4.5829	1.289	0.0466	6.7	0.008		82.8	Ikeda 1970
I	260.Euphausiacea	Euphausiidae	Thysanopoda cornuta	MIN	0.8400	0.21	5.88	5				Cowles et al. 1991
I	261.Branchiopoda	Daphnidae	Daphnia ambigua	MIN	9.6225	6.006	0.0000197	18.2	0.00000387		80.4	Armitage & Lei 1979
I	262.Branchiopoda	Daphnidae	Daphnia ambigua		13.045	11.278	0.000019	22.9			80.4	Armitage & Lei 1979
I	263.Branchiopoda	Daphnidae	Daphnia ambigua		13.050	11.283	0.000019	22.9	0.00000372		80.4	Armitage & Lei 1979
I	264.Branchiopoda	Daphnidae	Daphnia ambigua		13.643	15.244	0.0000184	26.6	0.00000362		80.4	Armitage & Lei 1979
I	265.Branchiopoda	Daphnidae	Daphnia ambigua		13.918	12.033	0.0000308	22.9	0.00000573		80.4	Armitage & Lei 1979
I	266.Branchiopoda	Daphnidae	Daphnia ambigua		14.041	12.139	0.0000178	22.9	0.00000349		80.4	Armitage & Lei 1979
I	267.Branchiopoda	Daphnidae	Daphnia ambigua		14.368	12.422	0.0000287	22.9	0.00000563		80.4	Armitage & Lei 1979
I	268.Branchiopoda	Daphnidae	Daphnia ambigua		16.573	14.328	0.0000221	22.9	0.00000434		80.4	Armitage & Lei 1979
I	269.Branchiopoda	Daphnidae	Daphnia ambigua		16.759	14.489	0.0000265	22.9	0.00000520		80.4	Armitage & Lei 1979
I	270.Branchiopoda	Daphnidae	Daphnia ambigua		17.196	14.867	0.0000136	22.9	0.00000266		80.4	Armitage & Lei 1979
I	271.Branchiopoda	Daphnidae	Daphnia ambigua		17.775	7.900	0.0000251	13.3	0.0000049		80.4	Armitage & Lei 1979
I	272.Branchiopoda	Daphnidae	Daphnia ambigua		25.254	21.833	0.00000893	22.9	0.00000175		80.4	Armitage & Lei 1979
I	273.Branchiopoda	Daphnidae	Daphnia ambigua		26.435	7.333	0.0000047	6.5				Armitage & Lei 1979
I	274.Branchiopoda	Daphnidae	Daphnia ambigua		26.597	7.378	0.0000258	6.5	0.00000507		80.4	Armitage & Lei 1979
I	275.Branchiopoda	Daphnidae	Daphnia ambigua		29.620	10.400	0.0000267	9.9	0.00000524		80.4	Armitage & Lei 1979
I	276.Branchiopoda	Daphnidae	Daphnia ambigua		31.320	27.078	0.00000597	22.9	0.00000117		80.4	Armitage & Lei 1979
I	277.Branchiopoda	Daphnidae	Daphnia ambigua		32.898	12.728	0.0000254	11.3	0.00000498		80.4	Armitage & Lei 1979
I	278.Branchiopoda	Daphnidae	Daphnia ambigua		33.608	29.056	0.00000587	22.9	0.00000150		80.4	Armitage & Lei 1979
I	279.Branchiopoda	Daphnidae	Daphnia magma	MIN	26.713	18.889	0.00015	20				Schindler 1968
I	280.Branchiopoda	Daphnidae	Daphnia magma		28.288	7.072	0.000146	5				Goss & Bunting 1980
I	281.Branchiopoda	Daphnidae	Daphnia magma		29.855	21.111	0.0001	20				Schindler 1968
I	282.Branchiopoda	Daphnidae	Daphnia magma		32.997	23.333	0.00005	20				Schindler 1968
I	283.Branchiopoda	Daphnidae	Daphnia magma		35.386	12.511	0.000118	10				Goss & Bunting 1980
I	284.Branchiopoda	Daphnidae	Daphnia magma		36.195	25.594	0.000061	20				Goss & Bunting 1980
I	285.Branchiopoda	Daphnidae	Daphnia magma		38.034	19.017	0.000087	15				Goss & Bunting 1980
I	286.Branchiopoda	Daphnidae	Daphnia magma		47.600	47.600	0.000066	25				Goss & Bunting 1980
I	287.Branchiopoda	Daphnidae	Daphnia pulex	MIN	37.668	9.417	0.000581	5				Goss & Bunting 1980
I	288.Branchiopoda	Daphnidae	Daphnia pulex		42.285	14.950	0.000481	10				Goss & Bunting 1980
I	289.Branchiopoda	Daphnidae	Daphnia pulex		51.147	36.167	0.000055	20				Ikeda 1970
I	290.Branchiopoda	Daphnidae	Daphnia pulex		58.139	82.222	0.000147	30				Goss & Bunting 1980
I	291.Branchiopoda	Daphnidae	Daphnia pulex		64.067	64.067	0.0002	25				Goss & Bunting 1980
I	292.Branchiopoda	Daphnidae	Daphnia pulex		65.548	46.350	0.000208	20				Goss & Bunting 1980
I	293.Branchiopoda	Daphnidae	Daphnia pulex		65.556	32.778	0.000374	15				Goss & Bunting 1980
I	294.Branchiopoda	Daphnidae	Daphnia pulex		132.489	103.9	0.0006	21.5				Ikeda 1970
I	295.Branchiopoda	Sididae	Penilia avirostris	MIN	3.1	2.2	0.00004	20				Pavlova 1961
I	296.Branchiopoda	Podonidae	Podon polyphemoides	MIN	4.2	2.9	0.00012	20				Pavlova 1961
I	297.Branchiopoda	Polyartemiidae	Polyartemia forcipata	MIN	2.8284	2.000	0.00769	20				Ivanova & Korobtsova 1974

I	298.Branchiopoda	Triopsidae	Triops cancriformis	MIN	1.6	1.1	0.230	20	0.019		91.7	Tscherbakov & Muragina 1953
I	299.Branchiopoda	Triopsidae	Triops cancriformis		1.7	1.2	0.340	20	0.028		91.8	Tscherbakov & Muragina 1953
I	300.Branchiopoda	Triopsidae	Triops cancriformis		1.75	1.24	0.410	20	0.035		91.5	Tscherbakov & Muragina 1953
I	301.Branchiopoda	Triopsidae	Triops cancriformis		3.1	2.2	0.190	20	0.0185		90.3	Tscherbakov & Muragina 1953
I	302.Branchiopoda	Triopsidae	Triops cancriformis		3.5	2.5	0.160	20	0.0135		91.6	Tscherbakov & Muragina 1953
I	303.Thoracica	Balanidae	Balanus balanoides	MIN	2.3480	0.5870	0.0046	5	0.00122		73.7	Barnes & Barnes 1969
I	304.Thoracica	Balanidae	Balanus balanoides		3.1113	2.2000	0.005	20	0.00131		73.7	Barnes & Barnes 1969
I	305.Thoracica	Balanidae	Balanus balanoides		3.5790	1.7895	0.0076	15	0.002		73.7	Barnes & Barnes 1969
I	306.Thoracica	Balanidae	Balanus balanoides		3.5915	1.2698	0.0063	10	0.00167		73.7	Barnes & Barnes 1969
I	307.Thoracica	Balanidae	Balanus balanus	MIN	0.7238	0.5118	0.1129	20				Barnes & Barnes 1969
I	308.Thoracica	Balanidae	Balanus balanus		0.7328	0.5182	0.11	20	0.0297		73.7	Barnes & Barnes 1969
I	309.Thoracica	Balanidae	Balanus balanus		0.7520	0.1880	0.117	5	0.0309		73.7	Barnes & Barnes 1969
I	310.Thoracica	Balanidae	Balanus balanus		0.9348	0.4674	0.092	15	0.0242		73.7	Barnes & Barnes 1969
I	311.Thoracica	Balanidae	Balanus balanus		0.9704	0.3431	0.102	10	0.0269		73.7	Barnes & Barnes 1969
I	312.Thoracica	Balanidae	Balanus balanus		0.9745	0.6891	0.092	20				Barnes & Barnes 1969
I	313.Thoracica	Balanidae	Balanus balanus		3.2022	2.2643	0.00498	20				Barnes & Barnes 1969
I	314.Thoracica	Balanidae	Balanus balanus		3.7059	2.6205	0.0076	20				Barnes & Barnes 1969
II	315.Isopoda	Aegidae	Aega sp.	MIN	4.5373	2.606	0.026	17	0.00664		74.4	Ikeda 1970
II	316.Mysidacea	Mysidae	Antarctomysis maxima	MIN	1.6065	1.1360	0.721	20				Opalinski 1991
II	317.Isopoda	Anuropidae	Anuropus bathypelagicus	MIN	0.0889	0.023	3.45	5.5	0.95	0.029	72.4	Childress 1975
II	318.Isopoda	Anuropidae	Anuropus bathypelagicus		0.4984	0.129	3.45	5.5	0.95	0.029	72.4	Childress 1975
II	319.Mysidacea	Mysidae	Archaeomysis grebnitzkii	MIN	1.6541	1.1696	0.0106	20				Jawed 1973
II	320.Mysidacea	Mysidae	Archaeomysis grebnitzkii		1.7225	1.2180	0.00935	20				Jawed 1973
II	321.Mysidacea	Mysidae	Archaeomysis grebnitzkii		1.8013	1.2737	0.00806	20				Jawed 1973
II	322.Mysidacea	Mysidae	Archaeomysis grebnitzkii		1.8451	1.3047	0.00645	20				Jawed 1973
II	323.Mysidacea	Mysidae	Archaeomysis grebnitzkii		1.8661	1.3195	0.0071	20				Jawed 1973
II	324.Mysidacea	Mysidae	Archaeomysis grebnitzkii		2.3404	1.6549	0.00339	20				Jawed 1973
II	325.Mysidacea	Mysidae	Archaeomysis grebnitzkii		2.5070	1.7727	0.00269	20				Jawed 1973
II	326.Mysidacea	Mysidae	Archaeomysis grebnitzkii		3.1352	2.2169	0.00124	20				Jawed 1973
II	327.Mysidacea	Mysidae	Archaeomysis grebnitzkii		3.3057	2.3375	0.00108	20				Jawed 1973
II	328.Isopoda	Armadillidae	Armadillidium nasatum	MIN	1.0235	0.7237	0.03	20				Reichle 1968
II	329.Isopoda	Armadillidae	Armadillidium pallasii	MIN	0.707	0.500	0.27	20				Byzova 1973
II	330.Isopoda	Armadillidae	Armadillidium pallasii		0.731	0.517	0.1	20				Byzova 1973 (data of Müller 1943)
II	331.Isopoda	Armadillidae	Armadillidium pallasii		0.747	0.528	0.16	20				Byzova 1973 (data of Müller 1943)
II	332.Isopoda	Armadillidae	Armadillidium pallasii		0.817	0.578	0.13	20				Byzova 1973 (data of Müller 1943)
II	333.Isopoda	Armadillidae	Armadillidium pallasii		0.935	0.661	0.05	20				Byzova 1973 (data of Müller 1943)
II	334.Isopoda	Armadillidae	Armadillidium pallasii		1.100	0.778	0.09	20				Byzova 1973
II	335.Isopoda	Armadillidae	Armadillidium pallasii		1.242	0.878	0.03	20				Byzova 1973 (data of Müller 1943)
II	336.Isopoda	Armadillidae	Armadillidium pallasii		1.414	1.000	0.015	20				Byzova 1973 (data of Müller 1943)
II	337.Isopoda	Armadillidae	Armadillidium pallasii		1.894	1.339	0.053	20				Byzova 1973
II	338.Isopoda	Armadillidae	Armadillidium pallasii		2.106	1.489	0.053	20				Byzova 1973
II	339.Isopoda	Armadillidae	Armadillidium pallasii		2.908	2.056	0.024	20				Byzova 1973
II	340.Isopoda	Armadillidae	Armadillidium pallasii		3.615	2.556	0.0025	20				Byzova 1973
II	341.Isopoda	Armadillidae	Armadillidium pallasii		4.400	3.111	0.0009	20				Byzova 1973
II	342.Isopoda	Armadillidae	Armadillidium vulgare	MIN	0.7863	0.556	0.144	20				Byzova 1973 (data of Reichle 1968)
II	343.Isopoda	Armadillidae	Armadillidium vulgare		1.1469	0.811	0.065	20				Byzova 1973 (data of Edney 1964)
II	344.Isopoda	Armadillidae	Armadillidium vulgare		1.4142	1.000	0.12	20				Byzova 1973 (data of Reichle 1968)
II	345.Isopoda	Armadillidae	Armadillidium vulgare		4.0857	2.889	0.004	20				Byzova 1973 (data of Engelmann 1961)
II	346.Isopoda	Armadillidae	Armadillidium vulgare		11.2345	7.944	0.00064	20				Byzova 1973 (data of Edney & Spencer 1956)
II	347.Isopoda	Asellidae	Asellus aquaticus	MIN	3.1	1.11	0.0168	10				Simčič & Brancelj 2006
II	348.Amphipoda	Pontoporeiidae	Bathyporeia pelgica	MIN	0.0062	0.0044	0.0022	20				Fish & Preece 1970
II	349.Amphipoda	Pontoporeiidae	Bathyporeia pelgica		6.7908	4.8018	0.00222	20				Fish & Preece 1970

II	350. Amphipoda	Pontoporeiidae	Bathyporeia pelgica		7.1562	5.0602	0.00133	20									Fish & Preece 1970
II	351. Amphipoda	Pontoporeiidae	Bathyporeia pelgica		8.9144	6.3034	0.00089	20									Fish & Preece 1970
II	352. Amphipoda	Pontoporeiidae	Bathyporeia pilosa	MIN	0.0057	0.004	0.00219	20									Fish & Preece 1970
II	353. Amphipoda	Pontoporeiidae	Bathyporeia pilosa		3.2202	2.2770	0.00444	20									Fish & Preece 1970
II	354. Amphipoda	Pontoporeiidae	Bathyporeia pilosa		6.7908	4.8018	0.00222	20									Fish & Preece 1970
II	355. Amphipoda	Pontoporeiidae	Bathyporeia pilosa		8.9531	6.3308	0.00133	20									Fish & Preece 1970
II	356. Amphipoda	Pontoporeiidae	Bathyporeia pilosa		10.6940	7.5618	0.00089	20									Fish & Preece 1970
II	357. Mysidacea	Mysidae	Boreomysis californica	MIN	0.3323	0.086	0.035	5.5	0.0061	0.079	82.6	Childress 1975					
II	358. Mysidacea	Mysidae	Boreomysis californica		1.2519	0.324	0.035	5.5	0.0061	0.079	82.6	Childress 1975					
II	359. Amphipoda	Eusiridae	Bovallia gigantea	MIN	2.9346	0.556	0.528	1	0.104		80	Opalinski 1991					
II	360. Amphipoda	Lycaeidae	Brachyscelus latipes	MIN	15.2837	20.167	0.00128	29	0.00019		85.2	Ikeda 1970					
II	361. Amphipoda	Ampeliscidae	Byblis securiger	MIN	4.5792	0.956	0.239	2.4	0.067		72	Opalinski 1991					
II	362. Isopoda	Cylisticidae	Cylisticus convexus	MIN	1.8574	1.3134	0.0384	20				Reiche 1968					
II	363. Amphipoda	Cyphocarididae	Cyphocaris richardi	MIN	1.3526	0.217	1.062	-1.4	0.287		73	Opalinski 1991					
II	364. Amphipoda	Cyphocarididae	Cyphocaris sp	MIN	1.7351	0.311	0.32	0.2	0.08142	0.070	74.9	Ikeda 1988					
II	365. Cumacea	Diastylidae	Diastylis sp.	MIN	5.0004	2.872	0.0006	17	0.00018		70	Ikeda 1970					
II	366. Amphipoda	Gammaridae	Dikerogammarus haemobaphes	MIN	4.5123	3.1907	0.008	20				Kititsina 1980					
II	367. Amphipoda	Eusiridae	Djerboa furcipes	MIN	2.3872	0.422	0.1	0			80	Opalinski 1991					
II	368. Amphipoda	Eurycopidae	Eurymera monticulosa	MIN	3.3991	0.644	0.235	1	0.042		82.2	Opalinski 1991					
II	369. Amphipoda	Eurycopidae	Eurythenes gryllus	MIN	3.0678	0.506	0.95	-1	0.173		81.8	Opalinski 1991					
II	370. Amphipoda	Eusiridae	Eusirus perdentatus	MIN	0.8064	0.133	1.628	-1	0.344		79	Opalinski 1991					
II	371. Amphipoda	Hyperiidae	Euthemisto compressa	MIN	3.3	0.77	0.022	4	0.00435		780	Conover 1960					
II	372. Amphipoda	Hyperiidae	Euthemisto compressa		3.6	0.83	0.024	4	0.00487		780	Conover 1960					
II	373. Amphipoda	Hyperiidae	Euthemisto compressa		3.7	0.86	0.025	4	0.00497		780	Conover 1960					
II	374. Amphipoda	Hyperiidae	Euthemisto libellula	MIN	4.7	0.83	0.017	-0.1	0.00298	0.0805	82.4	Ikeda & Skjoldal 1989					
II	375. Amphipoda	Hyperiidae	Euthemisto libellula		4.7239	1.244	0.00791	5.75	0.00153		80.7	Ikeda 1970					
II	376. Amphipoda	Hyperiidae	Euthemisto libellula		11.0608	3.111	0.00455	6.7	0.0008		82.5	Ikeda 1970					
II	377. Amphipoda	Gammaridae	Gammarus duebeni	MIN	3.1113	1.1	0.05	10				Bulnheim 1979					
II	378. Amphipoda	Gammaridae	Gammarus fossarum	MIN	2.0	0.722	0.0189	10				Simčič & Brancelj 2006					
II	379. Amphipoda	Gammaridae	Gammarus locusta	MIN	2.3220	1.6419	0.015	20	0.00349		77.4	Ivlev & Sutschenya 1961					
II	380. Amphipoda	Gammaridae	Gammarus locusta		2.6286	1.8587	0.015	20	0.00349		77.4	Ivlev & Sutschenya 1961					
II	381. Amphipoda	Gammaridae	Gammarus locusta		4.8000	2.4	0.05	15				Bulnheim 1979					
II	382. Amphipoda	Gammaridae	Gammarus oceanicus	MIN	2.8000	1.4	0.05	15				Bulnheim 1979					
II	383. Amphipoda	Gammaridae	Gammarus salinus	MIN	3.3941	1.2	0.05	10				Bulnheim 1979					
II	384. Amphipoda	Gammaridae	Gammarus zaddachi	MIN	3.1113	1.1	0.05	10				Bulnheim 1979					
II	385. Isopoda	Chaetiliidae	Glyptonotus antarcticus	MIN	0.3224	0.057	33	0				Robertson et al. 2003					
II	386. Mysidacea	Lophogastridae	Gnathophausia gigas	MIN	0.3440	0.086	1.01	5				Cowles et al. 1991					
II	387. Mysidacea	Lophogastridae	Gnathophausia gigas		0.4459	0.104	1.052	4	0.193	0.071	79.5	Childress 1975					
II	388. Mysidacea	Lophogastridae	Gnathophausia gigas		0.7811	0.14	0.77	0.2	0.12729	0.073	83.4	Ikeda 1988					
II	389. Mysidacea	Lophogastridae	Gnathophausia gigas		4.8444	1.13	1.052	4	0.193	0.071	79.5	Childress 1975					
II	390. Mysidacea	Lophogastridae	Gnathophausia gracilis	MIN	0.2658	0.062	2.05	4	0.38	0.070	81.6	Childress 1975					
II	391. Mysidacea	Lophogastridae	Gnathophausia gracilis		0.4520	0.113	2.45	5				Cowles et al. 1991					
II	392. Mysidacea	Lophogastridae	Gnathophausia gracilis		1.8092	0.422	2.05	4	0.38	0.070	81.6	Childress 1975					
II	393. Mysidacea	Lophogastridae	Gnathophausia ingens	MIN	0.0810	0.0189	52.42	4	4.875	0.054	0.91	Childress 1975					
II	394. Mysidacea	Lophogastridae	Gnathophausia ingens		0.3593	0.093	5.542	5.5	1.514	0.063	72.6	Childress 1975					
II	395. Mysidacea	Lophogastridae	Gnathophausia ingens		0.5374	0.19	5.11	10				Cowles et al. 1991					
II	396. Mysidacea	Lophogastridae	Gnathophausia ingens		1.0800	0.27	5.11	5				Cowles et al. 1991					
II	397. Mysidacea	Lophogastridae	Gnathophausia ingens		3.2417	0.839	5.542	5.5	1.514	0.063	72.6	Childress 1975					
II	398. Mysidacea	Lophogastridae	Gnathophausia zoea	MIN	0.3111	0.11	3.99	10				Cowles et al. 1991					
II	399. Mysidacea	Lophogastridae	Gnathophausia zoea		0.3593	0.093	5.2	5.5				Childress 1975					
II	400. Mysidacea	Lophogastridae	Gnathophausia zoea		0.4675	0.121	5.2	5.5				Childress 1975					
II	401. Mysidacea	Lophogastridae	Gnathophausia zoea		0.8000	0.20	3.99	5				Cowles et al. 1991					

II	402. Amphipoda	Eusiridae	Gondogeneia antarctica	MIN	10.5561	2.000	0.029	1	0.005	83	Opalinski 1991
II	403. Isopoda	Hemioniscidae	Hemilepistus elegans	MIN	0.7382	0.522	0.275	20			Byzova 1973
II	404. Isopoda	Hemioniscidae	Hemilepistus elegans		1.5316	1.083	0.115	20			Byzova 1973
II	405. Mysidacea	Mysidae	Hypererythrosp. sp.	MIN	7.5929	4.361	0.00111	17	0.000542	52.2	Ikeda 1970
II	406. Amphipoda	Lanceolidae	Hyperia galba	MIN	0.88	0.23	0.0824	5.6	0.00824	?90	Conover 1960
II	407. Amphipoda	Lanceolidae	Hyperia galba		0.99	0.26	0.01015	5.6	0.001015	?90	Conover 1960
II	408. Amphipoda	Lanceolidae	Hyperia galba		1.2021	0.425	0.0357	10	0.0041	0.067	88.5 Childress 1975
II	409. Amphipoda	Lanceolidae	Hyperia galba		1.2444	0.383	0.2527	8	0.0286		88.7 Ikeda 1970
II	410. Amphipoda	Lanceolidae	Hyperia galba		1.30	0.34	0.0352	5.6	0.00352	?90	Conover 1960
II	411. Amphipoda	Lanceolidae	Hyperia galba		1.34	0.35	0.0462	5.6	0.00462	?90	Conover 1960
II	412. Amphipoda	Lanceolidae	Hyperia galba		2.0641	0.850	0.0497	12.2	0.00616		87.6 Ikeda 1970
II	413. Amphipoda	Lanceolidae	Hyperia galba		2.2458	0.794	0.0357	10	0.0041	0.067	88.5 Childress 1975
II	414. Amphipoda	Lanceolidae	Hyperia gaudichaudii	MIN	1.4351	0.24	0.525	-0.8	0.10530	0.071	?80 Ikeda 1988
II	415. Amphipoda	Lanceolidae	Hyperia sp.	MIN	2.9047	0.817	0.00927	6.7	1.2		87.1 Ikeda 1970
II	416. Amphipoda	Lanceolidae	Hyperia sp.		5.3104	3.050	0.00131	17	0.00019		85.5 Ikeda 1970
II	417. Amphipoda	Lanceolidae	Hyperia sp.		6.7903	3.900	0.000428	17	0.000072		83.2 Ikeda 1970
II	418. Amphipoda	Hyperlopsidae	Hyperiella antarctica	MIN	1.2496	0.8836	0.02	20			Opalinski 1991
II	419. Isopoda	Idoteidae	Idotea baltica basteri	MIN	2.1	1.45	0.1385	20			Khmeleva 1973
II	420. Cumacea	Bodotriidae	Iphinoe sp.	MIN	0.7579	1.000	0.00664	29	0.00109	83.6	Ikeda 1970
II	421. Cumacea	Bodotriidae	Iphinoe sp.		2.1998	3.111	0.00062	30	0.0001	83.9	Ikeda 1970
II	422. Isopoda	Ligiidae	Ligia oceanica	MIN	1.2572	0.889	0.8	20			Byzova 1973 (data of Edney & Spencer 1955)
II	423. Isopoda	Ligiidae	Ligia oceanica		1.3350	0.944	0.8	20			Byzova 1973 (data of Ellenby 1951)
II	424. Isopoda	Ligiidae	Ligidium japonica	MIN	1.8074	1.278	0.025	20			Byzova 1973 (data of Saito 1965)
II	425. Isopoda	Ligiidae	Ligidium japonica		3.1424	2.222	0.0045	20			Byzova 1973 (data of Saito 1965)
II	426. Isopoda	Oniscidae	Mesidothea entomon	MIN	1.2885	0.9111	10	20			Romanova & Khmeleva 1978
II	427. Isopoda	Oniscidae	Mesidothea entomon		1.4503	1.0255	5	20			Romanova & Khmeleva 1978
II	428. Isopoda	Oniscidae	Mesidothea entomon		1.6190	1.1448	3.2	20			Romanova & Khmeleva 1978
II	429. Isopoda	Oniscidae	Mesidothea entomon		1.6859	1.1921	2	20			Romanova & Khmeleva 1978
II	430. Isopoda	Oniscidae	Mesidothea entomon		2.1579	1.5259	0.5	20			Romanova & Khmeleva 1978
II	431. Mysidacea	Mysidae	Mesopodopsis slabberi	MIN	1.4500	1.0253	0.0325	20			Braginski 1957
II	432. Mysidacea	Mysidae	Mysis relicta	MIN	3.2052	2.2664	0.01	20			Foulds & Roff 1976
II	433. Mysidacea	Lophogastridae	Neomysis awatschensis	MIN	2.0290	1.4347	0.0235	20			Jawed 1973
II	434. Mysidacea	Lophogastridae	Neomysis awatschensis		2.1807	1.5420	0.0195	20			Jawed 1973
II	435. Mysidacea	Lophogastridae	Neomysis awatschensis		2.3507	1.6622	0.0162	20			Jawed 1973
II	436. Mysidacea	Lophogastridae	Neomysis awatschensis		2.4176	1.7095	0.0148	20			Jawed 1973
II	437. Mysidacea	Lophogastridae	Neomysis awatschensis		2.4910	1.7614	0.0136	20			Jawed 1973
II	438. Mysidacea	Lophogastridae	Neomysis awatschensis		2.7249	1.9268	0.01086	20			Jawed 1973
II	439. Mysidacea	Lophogastridae	Neomysis awatschensis		2.9786	2.1062	0.00855	20			Jawed 1973
II	440. Mysidacea	Lophogastridae	Neomysis awatschensis		3.5358	2.5002	0.00543	20			Jawed 1973
II	441. Mysidacea	Lophogastridae	Neomysis awatschensis		4.2360	2.9953	0.00339	20			Jawed 1973
II	442. Mysidacea	Lophogastridae	Neomysis awatschensis		4.6741	3.3051	0.00258	20			Jawed 1973
II	443. Mysidacea	Lophogastridae	Neomysis mirabilis	MIN	5.4931	3.8842	0.0118	20			Kuzmicheva & Kukina 1974
II	444. Mysidacea	Lophogastridae	Neomysis mirabilis		5.6948	4.0268	0.0118	20			Kuzmicheva & Kukina 1974
II	445. Mysidacea	Lophogastridae	Neomysis mirabilis		6.5317	4.6186	0.0118	20			Klyashtorin & Kuzmicheva 1975
II	446. Amphipoda	Niphargidae	Niphargus krameri	MIN	1.10	0.39	0.0149	10			Simčič et al. 2005
II	447. Amphipoda	Niphargidae	Niphargus rhenorhodanensis	MIN	1.2	0.44	0.013	11			Hervant et al. 1997
II	448. Amphipoda	Niphargidae	Niphargus sphagnicolus	MIN	4.2	1.5	0.0059	10			Simčič & Brancelj 2006
II	449. Amphipoda	Niphargidae	Niphargus stygius	MIN	0.93	0.33	0.0304	10			Simčič et al. 2005
II	450. Amphipoda	Niphargidae	Niphargus virei	MIN	0.45	0.17	0.093	11			Hervant et al. 1997
II	451. Isopoda	Oniscidae	Oniscus asellus	MIN	0.9433	0.667	0.15	20			Byzova 1973 (data of Phillipson & Watson 1965)
II	452. Isopoda	Oniscidae	Oniscus asellus		1.5712	1.111	0.1	20			Byzova 1973 (data of Phillipson & Watson 1965)
II	453. Isopoda	Oniscidae	Oniscus asellus		2.3575	1.667	0.02	20			Byzova 1973 (data of Phillipson & Watson 1965)

II	454. Isopoda	Oniscidae	Oniscus asellus		2.8284	2.000	0.06	20					Byzova 1973 (data of Phillipson & Watson 1965)
II	455. Isopoda	Oniscidae	Oniscus asellus		3.1424	2.222	0.018	20					Byzova 1973 (data of Edney & Spencer 1955)
II	456. Isopoda	Oniscidae	Oniscus asellus		6.6001	4.667	0.01	20					Byzova 1973 (data of Will 1952)
II	457. Isopoda	Oniscidae	Oniscus asellus		7.0711	5.000	0.0061	20					Byzova 1973 (data of Edwards 1946)
II	458. Isopoda	Oniscidae	Oniscus asellus		12.2570	8.667	0.0018	20					Byzova 1973 (data of Morrison 1946)
II	459. Amphipoda	Talitridae	Orchestia bottae	MIN	0.9977	0.7055	0.11	20	0.0245		77.1		Ivlev & Sutschenya 1961
II	460. Amphipoda	Talitridae	Orchestia bottae		2.8593	2.0218	0.11	20	0.0245		77.1		Ivlev & Sutschenya 1961
II	461. Amphipoda	Lysianassidae	Orchomene sp.	MIN	15.7189	22.000	0.00362	29.85	0.00091		75		Ikeda 1970
II	462. Amphipoda	Lysianassidae	Orchomenella chilensis	MIN	3.6430	0.644	0.05	0					Opalinski 1991
II	463. Amphipoda	Lysianassidae	Orchomenella plebs	MIN	1.7593	0.311	0.161	0	0.032		80.2		Opalinski 1991
II	464. Amphipoda	Lysianassidae	Orchomenella plebs		1.9634	0.372	0.105	1	0.021		80		Opalinski 1991
II	465. Amphipoda	Lysianassidae	Orchomenella plebs		2.3840	0.372	0.194	-1.8	0.039		80		Opalinski 1991
II	466. Amphipoda	Lysianassidae	Orchomenella plebs		2.3888	0.394	0.148	-1	0.03		80		Opalinski 1991
II	467. Amphipoda	Lysianassidae	Orchomenella plebs		2.4929	0.389	0.091	-1.8	0.023		75		Opalinski 1991
II	468. Amphipoda	Lysianassidae	Orchomenella plebs		2.6543	0.539	0.072	2	0.014		80.6		Opalinski 1991
II	469. Amphipoda	Lysianassidae	Orchomenella plebs		3.1658	0.494	0.097	-1.8					Opalinski 1991
II	470. Amphipoda	Lysianassidae	Orchomenella plebs		3.2748	0.511	0.088	-1.8	0.029		67		Opalinski 1991
II	471. Amphipoda	Lysianassidae	Orchomenella plebs		3.3837	0.528	0.073	-1.8	0.019		74		Opalinski 1991
II	472. Amphipoda	Lysianassidae	Orchomenella sp.	MIN	3.0981	0.511	0.206	-1	0.041		80		Opalinski 1991
II	473. Amphipoda	Calliopiidae	Paracallisoma coecus	MIN	0.7727	0.2	0.2	5.5	0.063	0.045	68.2		Childress 1975
II	474. Amphipoda	Calliopiidae	Paracallisoma coecus		1.1205	0.29	0.2	5.5	0.063	0.045	68.2		Childress 1975
II	475. Amphipoda	Potogeneiidae	Paramoera walkeri	MIN	1.3180	0.233	0.044	0	0.009		79.6		Opalinski 1991
II	476. Amphipoda	Potogeneiidae	Paramoera walkeri		1.7940	0.278	0.02	-1.9	0.004		80		Opalinski 1991
II	477. Amphipoda	Potogeneiidae	Paramoera walkeri		1.8554	0.328	0.044	0	0.009		79.6		Opalinski 1991
II	478. Amphipoda	Potogeneiidae	Paramoera walkeri		2.1383	0.378	0.044	0	0.009		79.6		Opalinski 1991
II	479. Amphipoda	Potogeneiidae	Paramoera walkeri		2.2627	0.400	0.02	0	0.005		75		Opalinski 1991
II	480. Amphipoda	Potogeneiidae	Paramoera walkeri		2.3215	0.478	0.028	2.2	0.006		79		Opalinski 1991
II	481. Amphipoda	Potogeneiidae	Paramoera walkeri		2.7381	0.556	0.02	2	0.005		75		Opalinski 1991
II	482. Amphipoda	Potogeneiidae	Paramoera walkeri		3.0369	0.494	0.021	-1.2	0.004		80		Opalinski 1991
II	483. Mysidacea	Mysidae	Paramysis kessleri	MIN	2.3272	1.6456	0.03	20					Braginski 1957
II	484. Mysidacea	Mysidae	Paramysis kessleri		3.9899	2.8213	0.0515	20					Braginski 1957
II	485. Amphipoda	Paraphronimidae	Parandania boeckii	MIN	1.9424	0.389	0.395	1.8	0.076		80.8		Opalinski 1991
II	486. Amphipoda	Paraphronimidae	Parathemisto gaudichaudii	MIN	2.6677	0.44	0.07	-1	0.01312	0.074	780		Ikeda 1988
II	487. Amphipoda	Paraphronimidae	Parathemisto gaudichaudii		4.4248	0.74	0.02	-0.8	0.00360	0.086	780		Ikeda 1988
II	488. Amphipoda	Phronimidae	Phronima sedentaria	MIN	0.2036	0.072	0.086	10	0.0637	0.057	92.6		Childress 1975
II	489. Amphipoda	Phronimidae	Phronima sedentaria		0.9588	0.339	0.086	10	0.0637	0.057	92.6		Childress 1975
II	490. Amphipoda	Phronimidae	Phronima sp.	MIN	1.1351	1.472	0.145	28.75	8.17		94.4		Ikeda 1970
II	491. Amphipoda	Pontogeneiidae	Pontogeneia antarctica	MIN	10.9132	1.800	0.005	-1	0.001		80		Opalinski 1991
II	492. Isopoda	Porcellionidae	Porcellio laevis	MIN	0.7071	0.500	0.18	20					Byzova 1973 (data of Edney 1964)
II	493. Isopoda	Porcellionidae	Porcellio laevis		1.5712	1.111	0.012	20					Byzova 1973 (data of Edney 1964)
II	494. Isopoda	Porcellionidae	Porcellio scaber	MIN	0.9433	0.667	0.06	20					Byzova 1973 (data of Edney & Spencer 1955)
II	495. Isopoda	Porcellionidae	Porcellio scaber		1.5712	1.111	0.1	20					Byzova 1973 (data of Wieser 1963, 1965)
II	496. Isopoda	Porcellionidae	Porcellio scaber		1.8851	1.333	0.25	20					Byzova 1973 (data of Edney 1964)
II	497. Isopoda	Porcellionidae	Porcellio scaber		3.1424	2.222	0.007	20					Byzova 1973 (data of Will 1952)
II	498. Isopoda	Serolidae	Serolis cornuta	MIN	0.3076	0.049	2.775	-1.5					Luxmoore 1984
II	499. Isopoda	Serolidae	Serolis cornuta		0.3224	0.057	2.59	0					Luxmoore 1984
II	500. Isopoda	Serolidae	Serolis cornuta		0.4181	0.082	1.64	1.5					Luxmoore 1984
II	501. Isopoda	Serolidae	Serolis cornuta		0.4870	0.106	1.4	3					Luxmoore 1984
II	502. Isopoda	Serolidae	Serolis cornuta		1.7261	0.275	2.775	-1.5	0.555		80		Opalinski 1991
II	503. Isopoda	Serolidae	Serolis cornuta		1.8215	0.322	2.59	0	0.518		80		Opalinski 1991
II	504. Isopoda	Serolidae	Serolis cornuta		2.3554	0.462	1.64	1.5	0.328		80.5		Opalinski 1991
II	505. Isopoda	Serolidae	Serolis cornuta		2.7293	0.594	1.4	3	0.28		80		Opalinski 1991

II	506.Isopoda	Serolidae	Serolis polita	MIN	0.4205	0.067	0.345	-1.5	0.069		80	Opalinski 1991
II	507.Isopoda	Serolidae	Serolis polita		0.4503	0.098	0.35	3	0.07		80	Opalinski 1991
II	508.Isopoda	Serolidae	Serolis polita		0.4639	0.082	0.305	0	0.061		80	Opalinski 1991
II	509.Isopoda	Serolidae	Serolis polita		0.4792	0.094	0.3	1.5	0.06		80	Opalinski 1991
II	510.Isopoda	Asellidae	Stenasellus virei	MIN	0.82	0.31	0.012	11				Hervant et al. 1997
II	511.Amphipoda	Synopiidae	Synopia ultramarina	MIN	80.5855	115.6	0.0007	30.2	0.00009		87.1	Ikeda 1970
II	512.Amphipoda	Talitridae	Talitrus sylvaticus	MIN	0.7780	0.778	0.05	25				Byzova 1973 (data of Clark 1955)
II	513.Amphipoda	Talitridae	Talitrus sylvaticus		1.0720	1.072	0.021	25				Byzova 1973 (data of Clark 1955)
II	514.Amphipoda	Talitridae	Talitrus sylvaticus		1.3890	1.389	0.0085	25				Byzova 1973 (data of Clark 1955)
II	515.Amphipoda	Talitridae	Talitrus sylvaticus		2.5000	2.500	0.0015	25				Byzova 1973 (data of Clark 1955)
II	516.Amphipoda	Talitridae	Talorchestia megalophtalma	MIN	0.9977	0.839	0.388	22.5				Byzova 1973 (data of Edwards & Irving 1943)
II	517.Amphipoda	Talitridae	Talorchestia megalophtalma		1.1428	0.961	0.291	22.5				Byzova 1973 (data of Edwards & Irving 1943)
II	518.Amphipoda	Talitridae	Talorchestia megalophtalma		1.3081	1.100	0.23	22.5				Byzova 1973 (data of Edwards & Irving 1943)
II	519.Amphipoda	Talitridae	Talorchestia megalophtalma		1.3545	1.139	0.278	22.5				Byzova 1973 (data of Edwards & Irving 1943)
II	520.Amphipoda	Talitridae	Talorchestia megalophtalma		1.4663	1.233	0.225	22.5				Byzova 1973 (data of Edwards & Irving 1943)
II	521.Amphipoda	Talitridae	Talorchestia megalophtalma		1.5460	1.300	0.2	22.5				Byzova 1973 (data of Edwards & Irving 1943)
II	522.Amphipoda	Talitridae	Talorchestia megalophtalma		1.7041	1.433	0.155	22.5				Byzova 1973 (data of Edwards & Irving 1943)
II	523.Amphipoda	Talitridae	Talorchestia megalophtalma		1.7446	1.467	0.19	22.5				Byzova 1973 (data of Edwards & Irving 1943)
II	524.Amphipoda	Talitridae	Talorchestia megalophtalma		1.8635	1.567	0.12	22.5				Byzova 1973 (data of Edwards & Irving 1943)
II	525.Amphipoda	Talitridae	Talorchestia megalophtalma		2.2202	1.867	0.11	22.5				Byzova 1973 (data of Edwards & Irving 1943)
II	526.Amphipoda	Vibiliidae	Viblia antarctica	MIN	0.6716	0.11	0.061	-1.1	0.01214	0.082	780	Ikeda 1988
II	527.Amphipoda	Lysianassidae	Waldeckia obesa	MIN	0.8485	0.150	0.218	0	0.044		80	Opalinski 1991
II	528.Amphipoda	Lysianassidae	Waldeckia obesa		0.8498	0.161	0.238	1	0.048		80	Opalinski 1991
II	529.Amphipoda	Lysianassidae	Waldeckia obesa		1.0391	0.211	0.252	2	0.05		80	Opalinski 1991
II	530.Amphipoda	Lysianassidae	Waldeckia obesa		1.3907	0.217	0.295	-1.8	0.059		80	Opalinski 1991
II	531.Amphipoda	Lysianassidae	Waldeckia obesa		1.6188	0.267	0.229	-1	0.046		80	Opalinski 1991
II	532.Ostracoda	Halocyprididae	Conchoecia sp.	MIN	8.799	6.222	0.000079	20				Pavlova 1975
II	533.Ostracoda	Cypridinidae	Cypridina hilgendorffii	MIN	9.6244	13.611	0.00198	30	0.00035		82.3	Ikeda 1970
II	534.Ostracoda	Cypridinidae	Gigantocypris agassizii	MIN	0.0141	0.0033	5.7	4	0.26	0.030	95.4	Childress 1975
II	535.Ostracoda	Cypridinidae	Gigantocypris agassizii		0.0514	0.012	5.7	4	0.26	0.030	95.4	Childress 1975
II	536.Ostracoda	Cypridinidae	Gigantocypris mulleri	MIN	0.0909	0.015	1.84	-1.0	0.13966	0.084	92.4	Ikeda 1988
II	537.Ostracoda	Cypridinidae	Gigantocypris mulleri		0.1264	0.021	0.74	-0.9	0.04797	0.082	93.5	Ikeda 1988
II	538.Ostracoda	Cypridinidae	Gigantocypris mulleri		0.1339	0.024	1.88	0.2	0.16008	0.093	91.5	Ikeda 1988
II	539.Ostracoda	Cypridinidae	Gigantocypris mulleri		0.1953	0.035	0.81	0.2	0.05486	0.082	93.2	Ikeda 1988
II	540.Ostracoda	Cypridinidae	Pyrocypis sp.	MIN	8.8388	12.500	0.00045	30	0.00008		82.2	Ikeda 1970
III	541.Decapoda	Oplophoridae	Acanthephyra acutifrons	MIN	0.5968	0.211	8.64	10				Cowles et al. 1991
III	542.Decapoda	Oplophoridae	Acanthephyra acutifrons		0.8240	0.206	8.64	5				Cowles et al. 1991
III	543.Decapoda	Oplophoridae	Acanthephyra curtirostris	MIN	0.0662	0.0234	2.29	10				Cowles et al. 1991
III	544.Decapoda	Oplophoridae	Acanthephyra curtirostris		0.0904	0.0226	2.29	5				Cowles et al. 1991
III	545.Decapoda	Oplophoridae	Acanthephyra curtirostris		0.1468	0.038	3.53	5.5	0.85	0.085	75.9	Childress 1975
III	546.Decapoda	Oplophoridae	Acanthephyra curtirostris		3.1683	0.82	3.53	5.5	0.85	0.085	75.9	Childress 1975
III	547.Decapoda	Oplophoridae	Acanthephyra smithi	MIN	1.6405	0.58	3.84	10				Cowles et al. 1991
III	548.Decapoda	Oplophoridae	Acanthephyra smithi		1.9799	1.4	3.84	20				Cowles et al. 1991
III	549.Decapoda	Oplophoridae	Acanthephyra smithi		3.6400	0.91	3.84	5				Cowles et al. 1991
III	550.Decapoda	Axiidae	Calastacus quinqueseriatus	MIN	0.1665	0.035	3.21	2.5				Childress et al. 1990
III	551.Decapoda	Axiidae	Calastacus quinqueseriatus		0.4000	0.10	3.13	5				Childress et al. 1990
III	552.Decapoda	Axiidae	Calastacus quinqueseriatus		0.4808	0.17	1.73	10				Childress et al. 1990
III	553.Decapoda	Axiidae	Calastacus quinqueseriatus		0.5400	0.27	1.38	15				Childress et al. 1990
III	554.Decapoda	Callinideidae	Callinectes sapidus	MIN	0.5739	0.4058	142.4	20				Lewis & Haefner 1976
III	555.Decapoda	Callinideidae	Callinectes sapidus		0.6863	0.4853	134.1	20				Lewis & Haefner 1976
III	556.Decapoda	Callinideidae	Callinectes sapidus		0.7689	0.5437	158.9	20				Lewis & Haefner 1976
III	557.Decapoda	Callinideidae	Callinectes sapidus		0.7771	0.5495	107.2	20				Lewis & Haefner 1976

III	558.Decapoda	Cancridae	Cancer magister	MIN	0.5565	0.3935	948	20										Cameron 1975
III	559.Decapoda	Cancridae	Cancer pagurus	MIN	0.093	0.033	460	10										Aldrich 1975
III	560.Decapoda	Portunidae	Carcinus maenas	MIN	0.3718	0.2629	52.8	20										Taylor et al. 1977
III	561.Decapoda	Portunidae	Carcinus maenas		0.3847	0.2720	52.8	20										Taylor et al. 1977
III	562.Decapoda	Portunidae	Carcinus maenas		0.5261	0.372	13.3	20										Ivlev & Sutschenya 1961
III	563.Decapoda	Portunidae	Carcinus maenas		0.6180	0.437	12.7	20										Ivlev & Sutschenya 1961
III	564.Decapoda	Parastacidae	Cherax destructor	MIN	0.4243	0.30	53	20										Ellis & Morris 1995
III	565.Decapoda	Hyppolitidae	Chorismus antarcticus	MIN	0.8465	0.133	2.273	-1.7	0.593							74		Opalinski 1991
III	566.Decapoda	Hyppolitidae	Chorismus antarcticus		1.5759	0.320	5	2	1							80		Opalinski 1991
III	567.Decapoda		Cleistosoma edwardsii	MIN	1.6973	1.2002	0.818	20										Dye & van der Veen 1980
III	568.Decapoda		Cleistosoma edwardsii		1.8429	1.3031	0.818	20										Dye & van der Veen 1980
III	569.Decapoda	Coenobitidae	Coenobita compressus	MIN	0.7979	0.5642	3.5	20										Herreid & Full 1986
III	570.Decapoda	Coenobitidae	Coenobita compressus		1.0936	0.7733	3.7	20										Herreid & Full 1986
III	571.Decapoda	Crangonidae	Crangon abyssorum	MIN	0.2624	0.078	1.20	7.5										Childress et al. 1990
III	572.Decapoda	Crangonidae	Crangon abyssorum		0.4383	0.089	1.28	2.0										Childress et al. 1990
III	573.Decapoda	Crangonidae	Crangon affinis	MIN	1.4159	1.0012	0.0506	20										Ikeda 1970
III	574.Decapoda	Crangonidae	Crangon communis	MIN	0.5346	0.189	2.91	10										Childress et al. 1990
III	575.Decapoda	Crangonidae	Crangon communis		0.7120	0.178	4.69	5										Childress et al. 1990
III	576.Decapoda	Crangonidae	Crangon communis		0.9780	0.489	2.14	15										Childress et al. 1990
III	577.Decapoda	Xanthidae	Eriphia spinifrons	MIN	0.27	0.193	350	20										Sutschenya & Abolmasova 1973
III	578.Decapoda	Gecarcinidae	Gecarcinus lateralis	MIN	0.0987	0.0698	40	20										Taylor & Davies 1981
III	579.Decapoda	Gecarcinidae	Gecarcinus lateralis		0.1754	0.1240	20	20										Taylor & Davies 1981
III	580.Decapoda	Gecarcinidae	Gecarcinus lateralis		0.3179	0.2248	140	20										Cameron 1975
III	581.Decapoda	Gecarcinidae	Gecarcinus lateralis		0.4387	0.3102	10	20										Taylor & Davies 1981
III	582.Decapoda	Gecarcinidae	Gecarcinus lateralis		0.5920	0.4186	53.4	20										Herreid et al. 1983
III	583.Decapoda	Aristeidae/Sergestidae	Gecarcoidea natalis	MIN	0.7000	0.70	160	25										Adamczewska & Morris 1994
III	584.Decapoda	Aristeidae/Sergestidae	Gennadas kempi /Petalidium foliaceum	MIN	1.1635	0.21	1.64	0.3	0.49434		0.074	69.8						Ikeda 1988
III	585.Decapoda	Aristeidae/Sergestidae	Gennadas kempi /Petalidium foliaceum		1.4505	0.26	0.96	0.2	0.28837		0.078	70.1						Ikeda 1988
III	586.Decapoda	Aristeidae/Sergestidae	Gennadas kempi /Petalidium foliaceum		1.5063	0.27	0.55	0.2	0.16089		0.080	71.0						Ikeda 1988
III	587.Decapoda	Aristeidae	Gennadas propinquus	MIN	0.4134	0.107	1.62	5.5	0.378		0.089	76.7						Childress 1975
III	588.Decapoda	Aristeidae	Gennadas propinquus		1.0432	0.27	1.62	5.5	0.378		0.089	76.7						Childress 1975
III	589.Decapoda	Glyphocrangonidae	Glyphocrangon vicaria	MIN	0.5278	0.20	9.85	11										Childress et al. 1990
III	590.Decapoda	Glyphocrangonidae	Glyphocrangon vicaria		0.5909	0.12	9.71	2										Childress et al. 1990
III	591.Decapoda	Grapsidae	Hemigrapsus nudus	MIN	0.0106	0.0075	20	20										Dehnel 1960
III	592.Decapoda	Grapsidae	Hemigrapsus nudus		0.0151	0.0107	20	20										Dehnel 1960
III	593.Decapoda	Grapsidae	Hemigrapsus oregonensis	MIN	0.0150	0.0106	20	20										Dehnel 1960
III	594.Decapoda	Grapsidae	Hemigrapsus oregonensis		0.0199	0.0141	20	20										Dehnel 1960
III	595.Decapoda	Sundathelphusidae	Holthuisana transversa	MIN	0.1481	0.1047	30	20										MacMillen 1978
III	596.Decapoda	Sundathelphusidae	Holthuisana transversa		0.1921	0.1358	20	20										MacMillen 1978
III	597.Decapoda	Sundathelphusidae	Holthuisana transversa		0.2249	0.1590	15	20										MacMillen 1978
III	598.Decapoda	Sundathelphusidae	Holthuisana transversa		0.2572	0.1819	12	20										MacMillen 1978
III	599.Decapoda	Sundathelphusidae	Holthuisana transversa		0.3015	0.2132	10	20										MacMillen 1978
III	600.Decapoda	Sundathelphusidae	Holthuisana transversa		0.4332	0.3063	5	20										MacMillen 1978
III	601.Decapoda	Sundathelphusidae	Holthuisana transversa		0.7577	0.5358	2	20										MacMillen 1978
III	602.Decapoda	Nephropidae	Homarus americanus	MIN	0.3454	0.2442	180	20										Penkoff & Thurberg 1982
III	603.Decapoda	Oplophoridae	Hymenodora frontalis	MIN	0.3516	0.091	1.4	5.5	0.507		0.059	63.8						Childress 1975
III	604.Decapoda	Oplophoridae	Hymenodora frontalis		0.6993	0.181	1.4	5.5	0.507		0.059	63.8						Childress 1975
III	605.Decapoda	Oplophoridae	Janicella spinicauda	MIN	1.7819	0.63	0.35	10										Cowles et al. 1991
III	606.Decapoda	Oplophoridae	Janicella spinicauda		2.7153	1.92	0.35	20										Cowles et al. 1991
III	607.Decapoda	Sergestidae	Lucifer typus	MIN	10.9115	6.267	0.000466	17	0.000068			85.4						Ikeda 1970
III	608.Decapoda	Hippolytidae	Lysmata sp.	MIN	5.0678	7.167	0.0109	30	0.00147			86.5						Ikeda 1970
III	609.Decapoda	Palaemonidae	Macrobrachium acanthurus	MIN	1.3011	0.92	1.7	20	0.5			771						Moreira et al. 1983

III	610.Decapoda	Palaemonidae	Macrobrachium heterochirus	MIN	1.2587	0.89	2.6	20	0.75		771	Moreira et al. 1983
III	611.Decapoda	Palaemonidae	Macrobrachium olfersi	MIN	0.8485	0.60	1.2	20	0.35		771	Moreira et al. 1983
III	612.Decapoda	Palaemonidae	Macrobrachium potiuna	MIN	0.6930	0.49	0.68	20	0.2		771	Moreira et al. 1983
III	613.Decapoda	Majidae	Maja squinado	MIN	0.063	0.022	591	10				Aldrich 1975
III	614.Decapoda	Xanthidae	Menippe mercenaria	MIN	0.1732	0.1225	31.6	20				Leffler 1973
III	615.Decapoda	Xanthidae	Menippe mercenaria		3.2227	2.2788	0.0242	20				Moots & Epifanio 1974
III	616.Decapoda	Xanthidae	Menippe mercenaria		6.5721	4.6472	0.000303	20				Moots & Epifanio 1974
III	617.Decapoda	Panaeidae	Metapenaeus pruinus	MIN	2.6248	1.856	0.184	20				Reiche 1968
III	618.Decapoda	Galatheididae	Munidopsis verrilli	MIN	0.1427	0.03	5.62	2.5				Childress et al. 1990
III	619.Decapoda	Crangonidae	Notocrangon antarcticus	MIN	2.6543	0.539	1	2	0.4		60	Opalinski 1991
III	620.Decapoda	Oplophoridae	Notostomus elegans	MIN	0.6680	0.167	22.54	5				Cowles et al. 1991
III	621.Decapoda	Oplophoridae	Notostomus elegans		1.0126	0.358	22.54	10				Cowles et al. 1991
III	622.Decapoda	Oplophoridae	Notostomus gibbosus	MIN	0.2400	0.06	10.40	5				Cowles et al. 1991
III	623.Decapoda	Oplophoridae	Notostomus gibbosus		0.4525	0.16	10.40	10				Cowles et al. 1991
III	624.Decapoda	Oplophoridae	Notostomus sp.	MIN	0.1158	0.027	40	4	2.8	0.086	93.0	Childress 1975
III	625.Decapoda	Oplophoridae	Notostomus sp.		0.1801	0.042	40	4	2.8	0.086	93.0	Childress 1975
III	626.Decapoda	Ocypodidae	Ocypode platytarsis	MIN	0.7846	0.5548	10	20				Veerannan 1974
III	627.Decapoda	Ocypodidae	Ocypode platytarsis		1.0964	0.7753	5	20				Veerannan 1974
III	628.Decapoda	Ocypodidae	Ocypode platytarsis		1.3620	0.9631	3	20				Veerannan 1974
III	629.Decapoda	Ocypodidae	Ocypode platytarsis		1.6542	1.1697	2	20				Veerannan 1974
III	630.Decapoda	Ocypodidae	Ocypode platytarsis		1.7603	2.022	3.58	27				Veerannan 1974
III	631.Decapoda	Ocypodidae	Ocypode platytarsis		2.2691	1.6045	1	20				Veerannan 1974
III	632.Decapoda	Ocypodidae	Ocypode platytarsis		3.1735	2.2440	0.5	20				Veerannan 1974
III	633.Decapoda	Ocypodidae	Ocypode quadrata	MIN	0.4131	0.2921	70.9	20				Full 1987
III	634.Decapoda	Ocypodidae	Ocypode quadrata		0.7084	0.5009	26.9	20				Full 1987
III	635.Decapoda	Ocypodidae	Ocypode quadrata		0.8690	0.8108	33.3	24				Full 1987
III	636.Decapoda	Ocypodidae	Ocypode quadrata		1.4771	1.0445	2.1	20				Full 1987
III	637.Decapoda	Oplophoridae	Oplophorus gracilorostris	MIN	3.7335	1.32	2.58	10				Cowles et al. 1991
III	638.Decapoda	Oplophoridae	Oplophorus gracilorostris		4.0871	2.89	2.58	20				Cowles et al. 1991
III	639.Decapoda	Oplophoridae	Oplophorus gracilorostris		6.5200	1.63	2.58	5				Cowles et al. 1991
III	640.Decapoda	Oplophoridae	Oplophorus spinosus	MIN	1.7876	0.632	2.92	10				Cowles et al. 1991
III	641.Decapoda	Oplophoridae	Oplophorus spinosus		4.1295	2.92	2.92	20				Cowles et al. 1991
III	642.Decapoda	Oplophoridae	Oplophorus spinosus		4.2000	1.05	2.92	5				Cowles et al. 1991
III	643.Decapoda	Grapsidae	Pachygrapsus marmoratus	MIN	0.5791	0.4095	14.7	20				Abolmasova 1970
III	644.Decapoda	Grapsidae	Pachygrapsus marmoratus		0.6817	0.482	5.3	20				Ivlev & Sutschenya 1961
III	645.Decapoda	Grapsidae	Pachygrapsus marmoratus		0.6913	0.4888	7.77	20				Abolmasova 1970
III	646.Decapoda	Grapsidae	Pachygrapsus marmoratus		0.6995	0.4946	0.152	20				Ivlev & Sutschenya 1961
III	647.Decapoda	Grapsidae	Pachygrapsus marmoratus		0.7623	0.5390	5.09	20				Abolmasova 1970
III	648.Decapoda	Grapsidae	Pachygrapsus marmoratus		0.8573	0.6062	3.23	20				Abolmasova 1970
III	649.Decapoda	Grapsidae	Pachygrapsus marmoratus		1.1936	0.844	4.4	20				Ivlev & Sutschenya 1961
III	650.Decapoda	Palaemonidae	Palaemon adspersus	MIN	1.5322	1.0834	3.2	20				Romanova & Khmeleva 1978
III	651.Decapoda	Palaemonidae	Palaemon adspersus		1.6787	1.1870	3.28	20				Romanova & Khmeleva 1978
III	652.Decapoda	Palaemonidae	Palaemon adspersus		3.4019	2.4055	0.132	20				Romanova & Khmeleva 1978
III	653.Decapoda	Palaemonidae	Palaemon adspersus		3.5702	2.5245	0.1	20				Romanova & Khmeleva 1978
III	654.Decapoda	Pandalidae	Pandalopsis ampla	MIN	0.2994	0.089	9.55	7.5				Childress et al. 1990
III	655.Decapoda	Pandalidae	Pandalopsis ampla		0.5054	0.11	5.96	3				Childress et al. 1990
III	656.Decapoda	Pandalidae	Pandalus jordani	MIN	0.6800	0.34	12.80	15				Childress et al. 1990
III	657.Decapoda	Pandalidae	Pandalus jordani		0.9617	0.34	12.67	10				Childress et al. 1990
III	658.Decapoda	Pandalidae	Pandalus jordani		1.0000	0.25	12.08	5				Childress et al. 1990
III	659.Decapoda	Pandalidae	Pandalus platyceros	MIN	0.8200	0.41	32.26	15				Childress et al. 1990
III	660.Decapoda	Pandalidae	Pandalus platyceros		0.9200	0.23	25.40	5				Childress et al. 1990
III	661.Decapoda	Pandalidae	Pandalus platyceros		0.9334	0.33	40.89	10				Childress et al. 1990

III	662.Decapoda	Xanthidae	Panopeus herbstii	MIN	0.4110	0.2906	10	20									Leffler 1973
III	663.Decapoda	Xanthidae	Panopeus herbstii		0.5483	0.3877	1	20									Leffler 1973
III	664.Decapoda	Lithodidae	Paralomis multispina	MIN	0.3079	0.067	7.64	3									Childress et al. 1990
III	665.Decapoda	Parathelphusidae	Parathelphusa hydrodromus	MIN	0.3079	0.2177	10	20									Kotaiah & Rajabai 1972
III	666.Decapoda	Parathelphusidae	Parathelphusa hydrodromus		0.3610	0.2553	10	20									Kotaiah & Rajabai 1972
III	667.Decapoda	Pasiphaeidae	Pasiphaea chacei	MIN	1.4194	0.422	1.73	7.5	0.353		0.101	79.5	Childress 1975				
III	668.Decapoda	Pasiphaeidae	Pasiphaea chacei		2.4117	0.717	1.73	7.5	0.353		0.101	79.5	Childress 1975				
III	669.Decapoda	Pasiphaeidae	Pasiphaea emarginata	MIN	0.2705	0.07	11.07	5.5	2.58		0.077	76.7	Childress 1975				
III	670.Decapoda	Pasiphaeidae	Pasiphaea emarginata		2.7819	0.72	11.07	5.5	2.58		0.077	76.7	Childress 1975				
III	671.Decapoda	Pasiphaeidae	Pasiphaea pacifica	MIN	1.3791	0.41	1.91	7.5	0.37			80.4	Childress 1975				
III	672.Decapoda	Pasiphaeidae	Pasiphaea pacifica		1.8163	0.54	1.91	7.5	0.37			80.4	Childress 1975				
III	673.Decapoda	Pasiphaeidae	Pasiphaea scotiae	MIN	1.3297	0.24	0.61	0.3	0.19591		0.062	68.0	Ikeda 1988				
III	674.Decapoda	Pasiphaeidae	Pasiphaea scotiae		1.3390	0.24	1.26	0.2	0.40320		0.052	68.0	Ikeda 1988				
III	675.Decapoda	Penaidae	Penaeus sp.	MIN	3.0088	1.239	0.103	12.2	0.01558			84.9	Ikeda 1970				
III	676.Decapoda	Pandalidae	Plesionika sp.	MIN	2.0478	0.53	1.92	5.5	0.43		0.099	77.7	Childress 1975				
III	677.Decapoda	Pandalidae	Plesionika sp.		4.1342	1.07	1.92	5.5	0.43		0.099	77.7	Childress 1975				
III	678.Decapoda	Galatheidae	Pleuroncodes planipes	MIN	1.1200	0.56	1.9	15	0.47		0.065	75.4	Childress 1975				
III	679.Decapoda	Crangonidae	Sabinea septemcarinata	MIN	1.0748	0.19	4.1	0					Schmid 1996				
III	680.Decapoda	Crangonidae	Sclerocrangon ferox	MIN	0.3790	0.067	13	0					Schmid 1996				
III	681.Decapoda	Scyllaridae	Scylla serrata	MIN	1.0711	0.7574	10	20					Veerannan 1972				
III	682.Decapoda	Scyllaridae	Scylla serrata		1.3964	0.9874	5	20					Veerannan 1972				
III	683.Decapoda	Scyllaridae	Scylla serrata		1.6636	1.911	3.58	27					Veerannan 1974				
III	684.Decapoda	Scyllaridae	Scylla serrata		1.6661	1.1781	3	20					Veerannan 1972				
III	685.Decapoda	Scyllaridae	Scylla serrata		1.9041	1.3464	2	20					Veerannan 1972				
III	686.Decapoda	Scyllaridae	Scylla serrata		1.9332	1.367	3.58	20					Veerannan 1972				
III	687.Decapoda	Scyllaridae	Scylla serrata		2.5230	1.7840	1	20					Veerannan 1972				
III	688.Decapoda	Scyllaridae	Scylla serrata		3.1577	2.2328	0.5	20					Veerannan 1972				
III	689.Decapoda	Sergestidae	Sergestes bisulcatus	MIN	1.2800	0.32	2.54	5					Cowles et al. 1991				
III	690.Decapoda	Sergestidae	Sergestes bisulcatus		1.3011	0.92	2.54	20					Cowles et al. 1991				
III	691.Decapoda	Sergestidae	Sergestes bisulcatus		1.6405	0.58	2.54	10					Cowles et al. 1991				
III	692.Decapoda	Sergestidae	Sergestes fulgens	MIN	1.8800	0.47	1.06	5					Cowles et al. 1991				
III	693.Decapoda	Sergestidae	Sergestes fulgens		2.4042	0.85	1.06	10					Cowles et al. 1991				
III	694.Decapoda	Sergestidae	Sergestes fulgens		3.5355	2.5	1.06	20					Cowles et al. 1991				
III	695.Decapoda	Sergestidae	Sergestes phorcus	MIN	0.4868	0.126	3.951	5.5	0.89		0.103	77.5	Childress 1975				
III	696.Decapoda	Sergestidae	Sergestes phorcus		1.5880	0.411	3.951	5.5	0.89		0.103	77.5	Childress 1975				
III	697.Decapoda	Sergestidae	Sergestes similis	MIN	1.3633	0.482	0.57	10	0.133		0.110	76.6	Childress 1975				
III	698.Decapoda	Sergestidae	Sergestes similis		4.5255	1.60	0.57	10	0.133		0.110	76.6	Childress 1975				
III	699.Decapoda	Sergestidae	Sergestes tenuiremis	MIN	0.7600	0.19	2.61	5					Cowles et al. 1991				
III	700.Decapoda	Sergestidae	Sergestes tenuiremis		0.8344	0.59	2.61	20					Cowles et al. 1991				
III	701.Decapoda	Sergestidae	Sergestes tenuiremis		1.4142	0.50	2.61	10					Cowles et al. 1991				
III	702.Decapoda	Grapsidae	Sesarma quadratus	MIN	1.1821	0.8359	5	20					Veerannan 1974				
III	703.Decapoda	Grapsidae	Sesarma quadratus		1.4810	1.0472	3	20					Veerannan 1974				
III	704.Decapoda	Grapsidae	Sesarma quadratus		1.7731	1.2538	2	20					Veerannan 1974				
III	705.Decapoda	Grapsidae	Sesarma quadratus		2.3880	1.6886	1	20					Veerannan 1974				
III	706.Decapoda	Grapsidae	Sesarma quadratus		3.3163	2.3450	0.5	20					Veerannan 1974				
III	707.Decapoda	Sicyoniidae	Sicyonia igentis	MIN	0.6788	0.24	19.26	10					Childress et al. 1990				
III	708.Decapoda	Sicyoniidae	Sicyonia igentis		1.3000	0.65	19.03	15					Childress et al. 1990				
III	709.Decapoda	Polychelidae	Stereomastis sculpta	MIN	0.1189	0.025	16.76	2.5					Childress et al. 1990				
III	710.Decapoda	Grapsidae	Systellaspis cristata	MIN	0.3130	0.081	1.39	5.5	0.378		0.065	72.8	Childress 1975				
III	711.Decapoda	Grapsidae	Systellaspis cristata		3.0485	0.789	1.39	5.5	0.378		0.065	72.8	Childress 1975				
III	712.Decapoda	Oplophoridae	Systellaspis debilis	MIN	1.2000	0.30	1.22	5					Cowles et al. 1991				
III	713.Decapoda	Oplophoridae	Systellaspis debilis		1.3718	0.97	1.22	20					Cowles et al. 1991				

III	714. Decapoda	Oplophoridae	Systellaspis debilis		1.5274	0.54	1.22	10	Cowles et al. 1991
III	715. Decapoda	Ocypodidae	Uca leptodactyla	MIN	0.5319	0.3761	0.27	20	Vernberg 1959
III	716. Decapoda	Ocypodidae	Uca minax	MIN	0.2703	0.1911	6.37	20	Vernberg 1959
III	717. Decapoda	Ocypodidae	Uca mordax	MIN	0.4766	0.3370	2.58	20	Vernberg 1959
III	718. Decapoda	Ocypodidae	Uca pugnax	MIN	0.2516	0.1779	2.21	20	Silverthorn 1975
III	719. Decapoda	Ocypodidae	Uca pugnax		0.3240	0.2291	2.26	20	Silverthorn 1975
III	720. Decapoda	Ocypodidae	Uca pugnax		0.4134	0.2923	2.61	20	Silverthorn 1975
III	721. Decapoda	Ocypodidae	Uca pugnax		0.4284	0.3029	3	20	Vernberg 1959
III	722. Decapoda	Ocypodidae	Uca pugnax		0.4718	0.3336	2.27	20	Silverthorn
III	723. Decapoda	Ocypodidae	Uca rapax	MIN	0.5234	0.3701	3.85	20	Vernberg 1959
III	724. Decapoda	Ocypodidae	Uca rapax		0.5951	0.4208	2.48	20	Vernberg 1959
III	725. Decapoda	Ocypodidae	Uca thayeri	MIN	0.3651	0.2582	4.65	20	Vernberg 1959
III	726. Decapoda	Xanthidae	Xantho hydrophilus	MIN	0.3294	0.2329	8.31	20	Abolmasova 1969
III	727. Decapoda	Xanthidae	Xantho hydrophilus		0.3462	0.2448	16.52	20	Abolmasova 1969
III	728. Decapoda	Xanthidae	Xantho hydrophilus		0.3953	0.2795	5.8	20	Abolmasova 1969
III	729. Decapoda	Xanthidae	Xantho hydrophilus		0.3956	0.2797	13.7	20	Abolmasova 1969
III	730. Decapoda	Xanthidae	Xantho hydrophilus		0.4121	0.2914	11.3	20	Abolmasova 1969
III	731. Decapoda	Xanthidae	Xantho hydrophilus		0.4837	0.3420	6.2	20	Abolmasova 1969
III	732. Decapoda	Xanthidae	Xantho hydrophilus		0.4871	0.3444	4.61	20	Abolmasova 1969
III	733. Decapoda	Xanthidae	Xantho hydrophilus		0.5030	0.3557	3.06	20	Abolmasova 1969
III	734. Decapoda	Xanthidae	Xantho hydrophilus		0.5268	0.3725	2.5	20	Abolmasova 1969
III	735. Decapoda	Xanthidae	Xantho hydrophilus		0.5432	0.3841	3.9	20	Abolmasova 1969
III	736. Decapoda	Xanthidae	Xantho hydrophilus		0.6107	0.4318	1.91	20	Abolmasova 1969
III	737. Decapoda	Xanthidae	Xantho hydrophilus		0.7662	0.5418	0.906	20	Abolmasova 1969
IV	738. Cephalopoda	Octopodidae	Bathypolypus articus	MIN	1.4625	0.2340	3.0	5	Seibel 2007
IV	739. Cephalopoda	Cranchidae	Cranchia scabra	MIN	0.2256	0.0361	35.39	5	Seibel 2007
IV	740. Cephalopoda	Ommastrephidae	Dosidicus gigas	MIN	2.7225	0.4356	12200	5	Seibel 2007
IV	741. Cephalopoda	Octopodidae	Eledone cirrhosa	MIN	0.9100	0.1456	150	5	Seibel 2007
IV	742. Cephalopoda	Bolitaenidae	Eledonella pygmaea	MIN	0.0388	0.0062	20.2	5	Seibel 2007
IV	743. Cephalopoda	Cranchidae	Galiteuthis phyllura	MIN	0.4744	0.0759	5.19	5	Seibel 2007
IV	744. Cephalopoda	Gonatidae	Gonatus onyx	MIN	1.9525	0.3124	0.117	5	Seibel 2007
IV	745. Cephalopoda	Gonatidae	Gonatus pyros	MIN	2.4513	0.3922	3.840	5	Seibel 2007
IV	746. Cephalopoda	Cranchidae	Helicocranchia pfeferi	MIN	0.3188	0.0510	3.60	5	Seibel 2007
IV	747. Cephalopoda	Histioteuthidae	Histioteuthis heteropsis	MIN	0.3500	0.0560	36.98	5	Seibel 2007
IV	748. Cephalopoda	Histioteuthidae	Histioteuthis hoylei	MIN	0.8813	0.1410	8.51	5	Seibel 2007
IV	749. Cephalopoda	Ommastrephidae	Illex illecebrosus	MIN	3.1113	0.4978	443	5	Seibel 2007
IV	750. Cephalopoda	Bolitaenidae	Japetella diaphana	MIN	0.0313	0.0050	242.17	5	Seibel 2007
IV	751. Cephalopoda	Bolitaenidae	Japetella heathi	MIN	0.0231	0.0037	162.50	5	Seibel 2007
IV	752. Cephalopoda	Cranchidae	Liocranchia pacificus	MIN	0.1869	0.0299	2.12	5	Seibel 2007
IV	753. Cephalopoda	Cranchidae	Liocranchia valdivia	MIN	0.2100	0.0336	21.28	5	Seibel 2007
IV	754. Cephalopoda	Loliginidae	Loligo forbesi	MIN	3.2513	0.5202	937	5	Seibel 2007
IV	755. Cephalopoda	Loliginidae	Loligo opalescens	MIN	4.3013	0.6882	30.0	5	Seibel 2007
IV	756. Cephalopoda	Loliginidae	Loligo pealei	MIN	5.9888	0.9582	100	5	Seibel 2007
IV	757. Cephalopoda	Loliginidae	Loliguncula brevis	MIN	3.0025	0.4804	41.10	5	Seibel 2007
IV	758. Cephalopoda	Cranchidae	Megalocranchia sp.	MIN	0.3031	0.0485	47.9	5	Seibel 2007
IV	759. Cephalopoda	Octopodidae	Octopus bimaculoides	MIN	0.4431	0.0709	360	5	Seibel 2007
IV	760. Cephalopoda	Octopodidae	Octopus briareus	MIN	0.5081	0.0813	345.5	5	Seibel 2007
IV	761. Cephalopoda	Octopodidae	Octopus californicus	MIN	0.3031	0.0485	330	5	Seibel 2007
IV	762. Cephalopoda	Octopodidae	Octopus cyanea	MIN	0.7700	0.1232	1150	5	Seibel 2007
IV	763. Cephalopoda	Octopodidae	Octopus doffeini	MIN	0.2331	0.0373	8200	5	Seibel 2007
IV	764. Cephalopoda	Octopodidae	Octopus maya	MIN	1.9056	0.3049	45.1	5	Seibel 2007
IV	765. Cephalopoda	Octopodidae	Octopus micropyrsus	MIN	1.2831	0.2053	4.65	5	Seibel 2007

IV	766.Cephalopoda	Octopodidae	Octopus rubescens	MIN	5.7556	0.9209	0.096	5					Seibel 2007
IV	767.Cephalopoda	Octopodidae	Octopus sp.	MIN	15.8275	2.5324	0.0068	5					Seibel 2007
IV	768.Cephalopoda	Octopodidae	Octopus sp.	MIN	10.4225	1.6676	0.0012	5					Seibel 2007
IV	769.Cephalopoda	Octopodidae	Octopus tuberculata	MIN	3.2431	0.5189	1.21	5					Seibel 2007
IV	770.Cephalopoda	Octopodidae	Octopus vulgaris	MIN	0.2956	0.0473	2160	5					Seibel 2007
IV	771.Cephalopoda	Octopodidae	Paralelone characoti	MIN	0.3500	0.0560	136.7	5					Seibel 2007
IV	772.Cephalopoda	Loliginidae	Sepioteuthis lessoniana	MIN	2.2325	0.3572	68.5	5					Seibel 2007
IV	773.Cephalopoda	Ommastrephidae	Sthenoteuthis oualaniensis	MIN	1.8744	0.2999	750	5					Seibel 2007
IV	774.Cephalopoda	Ommastrephidae	Sthenoteuthis pteropus	MIN	2.2556	0.3609	1300	5					Seibel 2007
IV	775.Cephalopoda	Vampyroteuthidae	Vampyroteuthis infernalis	MIN	0.0194	0.0031	1050.0	5					Seibel 2007
V	776.Medusae	Aeginidae	Aegina citrea	MIN	0.644	0.023	1.9	5					Thuesen & Childress 1994
V	777.Medusae	Rhopalonematidae	Aglantha digitale	MIN	1.89	0.051	0.25	1.1	0.014	0.043	94.7		Ikeda & Skjoldal 1982
V	778.Medusae	Atollidae	Atolla vanhoeffeni	MIN	0.7	0.025	0.60	5					Thuesen & Childress 1994
V	779.Medusae	Atollidae	Atolla wyvillei	MIN	0.476	0.017	1.94	5					Thuesen & Childress 1994
V	780.Medusae	Halicreatidae	Botrynum brucei	MIN	0.476	0.017	1.33	5					Thuesen & Childress 1994
V	781.Medusae	Rhopalonematidae	Colobema sericeum	MIN	0.308	0.011	4.2	5					Thuesen & Childress 1994
V	782.Medusae	Rhopalonematidae	Crossota alba	MIN	1.148	0.041	0.53	5					Thuesen & Childress 1994
V	783.Medusae	Rhopalonematidae	Crossota rufobrunnea	MIN	0.532	0.019	0.26	5					Thuesen & Childress 1994
V	784.Medusae	Rhopalonematidae	Crossota sp. A	MIN	0.588	0.021	0.14	5					Thuesen & Childress 1994
V	785.Medusae	Eirenidae	Eirene mollis	MIN	1.358	0.097	0.22	15					Thuesen & Childress 1994
V	786.Medusae	Halicreatidae	Haliscera bigelowi	MIN	0.448	0.016	0.47	5					Thuesen & Childress 1994
V	787.Medusae	Halicreatidae	Halitrehees maasi	MIN	0.1596	0.0057	19	5					Thuesen & Childress 1994
V	788.Medusae	Pandaeidae	Leukartiara octona	MIN	1.428	0.051	0.15	5					Thuesen & Childress 1994
V	789.Medusae	Nausithoidae	Nausithoe rubra	MIN	0.756	0.027	2.6	5					Thuesen & Childress 1994
V	790.Medusae	Rhopalonematidae	Pantachogon sp. A	MIN	0.896	0.032	0.51	5					Thuesen & Childress 1994
V	791.Medusae	Peryphyllinidae	Paraphyllina ransonii	MIN	1.148	0.041	0.25	5					Thuesen & Childress 1994
V	792.Medusae	Periphyllidae	Periphylla peryphylla	MIN	0.336	0.012	11	5					Thuesen & Childress 1994
V	793.Medusae	Rhopalonematidae	Tetrorchis erythrogaster	MIN	0.42	0.015	0.44	5					Thuesen & Childress 1994
V	794.Medusae	Olindiidae	Vallentinia adherens	MIN	3.36	0.24	0.026	15					Thuesen & Childress 1994
V	795.Medusae	Rhopalonematidae	Vampyrocrossota childressi	MIN	0.476	0.017	0.34	5					Thuesen & Childress 1994
V	796.Chaetognatha	Sagittidae	Caecosagitta macrocephala	MIN	1.164	0.097	0.021	5					Thuesen & Childress 1993
V	797.Chaetognatha	Sagittidae	Decipisagitta decipiens	MIN	2.652	0.221	0.0054	5					Thuesen & Childress 1993
V	798.Chaetognatha	Eukrohnidae	Eukrohnia bathypelagica	MIN	0.6954	0.060	0.0266	5.5					Thuesen & Childress 1993
V	799.Chaetognatha	Eukrohnidae	Eukrohnia fowleri	MIN	0.516	0.043	0.094	5					Thuesen & Childress 1993
V	800.Chaetognatha	Eukrohnidae	Eukrohnia hamata	MIN	0.6375	0.055	0.040	5.5					Thuesen & Childress 1993
V	801.Chaetognatha	Sagittidae	Flaccisagitta hexaptera	MIN	0.876	0.073	0.193	5					Thuesen & Childress 1993
V	802.Chaetognatha	Heterokrohnidae	Heterokrohnia murina	MIN	0.708	0.059	0.204	5					Thuesen & Childress 1993
V	803.Chaetognatha	Sagittidae	Parasagitta elegans	MIN	1.9422	0.151	0.020	4					Thuesen & Childress 1993
V	804.Chaetognatha	Sagittidae	Parasagitta euneritica	MIN	3.108	0.518	0.0023	15					Thuesen & Childress 1993
V	805.Chaetognatha	Sagittidae	Pseudosagitta gazellae	MIN	0.3252	0.018	0.387	-0.9					Thuesen & Childress 1993
V	806.Chaetognatha	Sagittidae	Pseudosagitta lyra	MIN	0.384	0.032	0.237	5					Thuesen & Childress 1993
V	807.Chaetognatha	Sagittidae	Pseudosagitta maxima	MIN	0.42	0.035	0.235	5					Thuesen & Childress 1993
V	808.Chaetognatha	Sagittidae	Sagitta elegans	MIN	3.3	0.185	0.042	-0.3	0.0045	0.1037	89.4		Ikeda & Skjoldal 1982
V	809.Chaetognatha	Sagittidae	Solidosagitta zetesios	MIN	1.044	0.087	0.071	5					Thuesen & Childress 1993

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Dataset S5. Standard metabolic rates in ectothermic vertebrates

Standard metabolic rates of amphibia and reptilia were taken from White, C. R., Phillips, N. R. & Seymour, R. S. The scaling and temperature dependence of vertebrate metabolism. *Biol. Lett.* **2**, 125–127 (2006). Minimum mass-specific value for each species was taken, resulting in 158 values for 158 amphibian species (out of 682 values available in White et al. (2006)) and 156 values for 156 reptilian species (out of 483 values available in White et al. (2006)). These minimum values are presented in Table **S5a** (amphibians) and Table **S5b** (reptiles) below.

Database www.fishbase.org was searched for fish metabolism data, of which only routine and standard values obtained with non-stressed animals were considered. This yielded a total of 6,333 values of mass-specific metabolic rate for 266 species. For each species the minimum value was taken, listed below in Table **S5c**.

Notations to Tables **S5a**, **S5b**, **S5c**:

qWkg is **standard** (or, in some cases in fish, routine) metabolic rate in Watts per kilogram (converted from oxygen uptake rates at 1 ml O₂ = 20 J); **TC** is ambient temperature at which measurements were taken, degrees Celsius; **q25Wkg** is metabolic rate converted to 25 °C using $Q_{10} = 1.65, 2.21$ and 2.44 for fish, amphibians and reptiles, respectively (White et al. 2006), $q25Wkg = qWkg \times Q_{10}^{(25 - TC)/10}$, dimension W (kg WM)⁻¹; **Mg** is wet body mass in grams; **Log** stands for decimal logarithms of the corresponding variables.

For a discussion of the differences between standard and routine metabolic rates in aquatic versus terrestrial ectotherms see SI Methods.

Table **S5a**. Standard metabolic rates in amphibians (after White et al. 2006)

Species	q25Wkg	Logq25Wkg	Mg	LogMg	TC	qWkg
1. <i>Acris crepitans</i>	0.369	-0.433	1.59	0.201	15	0.167
2. <i>Agalychnis callidryas</i>	0.492	-0.308	5.65	0.752	20	0.331
3. <i>Ambystoma gracile</i>	0.256	-0.592	30.39	1.483	15	0.116
4. <i>Ambystoma jeffersonianum</i>	0.435	-0.362	8.92	0.950	15	0.197
5. <i>Ambystoma macrodactylum</i>	0.396	-0.402	3.414	0.533	15	0.179
6. <i>Ambystoma maculatum</i>	0.281	-0.551	15.83	1.199	15	0.127
7. <i>Ambystoma mexicanum</i>	0.105	-0.979	21.3	1.328	22	0.083
8. <i>Ambystoma opacum</i>	0.246	-0.609	4.65	0.667	14	0.103
9. <i>Ambystoma talpoideum</i>	0.981	-0.008	7	0.845	15	0.444
10. <i>Ambystoma tigrinum</i>	0.157	-0.804	10.78	1.033	15	0.071
11. <i>Amphiuma means</i>	0.068	-1.167	352	2.547	5	0.014
12. <i>Amphiuma tridactylum</i>	0.046	-1.337	652	2.814	15	0.021
13. <i>Aneides ferreus</i>	0.334	-0.476	2.73	0.436	15	0.151
14. <i>Aneides flavipunctatus</i>	0.323	-0.491	4.57	0.660	15	0.146
15. <i>Aneides hardii</i>	0.437	-0.360	0.97	-0.013	20	0.294
16. <i>Aneides lugubris</i>	0.329	-0.483	5.58	0.747	15	0.149
17. <i>Batrachoseps attenuatus</i>	0.479	-0.320	0.93	-0.032	25	0.479
18. <i>Bolitoglossa franklini</i>	0.212	-0.674	3.18	0.502	15	0.096
19. <i>Bolitoglossa morio</i>	0.261	-0.583	2.09	0.320	15	0.118
20. <i>Bolitoglossa occidentalis</i>	0.256	-0.592	0.97	-0.013	15	0.116
21. <i>Bolitoglossa subpalmata</i>	0.264	-0.578	1.63	0.212	5	0.054
22. <i>Bombina orientalis</i>	0.462	-0.335	2.6	0.415	20	0.311
23. <i>Boulengerula taitanus</i>	0.433	-0.364	5	0.699	35	0.956
24. <i>Bufo alvaris</i>	0.966	-0.015	150.8	2.178	15	0.437
25. <i>Bufo americanus</i>	0.229	-0.640	50	1.699	20	0.154
26. <i>Bufo boreas</i>	0.150	-0.824	57.8	1.762	20	0.101
27. <i>Bufo bufo</i>	0.460	-0.337	76.13	1.882	15	0.208
28. <i>Bufo calamita</i>	0.479	-0.320	8.7	0.940	20	0.322
29. <i>Bufo cognatus</i>	0.409	-0.388	40.2	1.604	25	0.409
30. <i>Bufo debilis</i>	1.571	0.196	0.66	-0.180	15	0.711
31. <i>Bufo marinus</i>	0.130	-0.886	145	2.161	15	0.059
32. <i>Bufo terrestris</i>	0.629	-0.201	19.8	1.297	25	0.629

33.	<i>Bufo viridis</i>	0.207	-0.684	35	1.544	20	0.139
34.	<i>Bufo woodhousii</i>	0.221	-0.656	56.3	1.751	25	0.221
35.	<i>Ceratophrys calcarata</i>	0.297	-0.527	55.5	1.744	25	0.297
36.	<i>Chiromantis petersi</i>	0.394	-0.405	11.2	1.049	25	0.394
37.	<i>Chiropterotriton bromeliacia</i>	0.301	-0.521	0.59	-0.229	15	0.136
38.	<i>Colostethus inguinalis</i>	0.718	-0.144	1.57	0.196	20	0.483
39.	<i>Colostethus nubicola</i>	0.895	-0.048	0.27	-0.569	25	0.895
40.	<i>Colostethus trinitatus</i>	1.117	0.048	1	0.000	25	1.117
41.	<i>Conraua goliath</i>	0.417	-0.380	251	2.400	25	0.417
42.	<i>Crinia parinsignifera</i>	1.413	0.150	0.452	-0.345	30	2.101
43.	<i>Crinia signifera</i>	1.333	0.125	0.621	-0.207	25	1.333
44.	<i>Cryptobranchus alleganiensis</i>	0.172	-0.764	423	2.626	25	0.172
45.	<i>Cyclorana platycephala</i>	0.659	-0.181	11	1.041	15	0.298
46.	<i>Dendrobates auratus</i>	0.500	-0.301	1.77	0.248	25	0.500
47.	<i>Desmognathus fuscus</i>	0.354	-0.451	2.09	0.320	20	0.238
48.	<i>Desmognathus monticola</i>	0.351	-0.455	3.43	0.535	20	0.236
49.	<i>Desmognathus ochrophaes</i>	0.268	-0.572	1.6	0.204	17.5	0.148
50.	<i>Desmognathus quadramaculatus</i>	0.190	-0.721	23.41	1.369	5	0.039
51.	<i>Dicamptodon ensatus</i>	0.159	-0.799	101.25	2.005	15	0.072
52.	<i>Discoglossus pictus</i>	0.306	-0.514	30.7	1.487	20	0.206
53.	<i>Eleutherodactylus coqui</i>	0.340	-0.469	4.06	0.609	20	0.229
54.	<i>Eleutherodactylus portoricensis</i>	0.851	-0.070	3.7	0.568	15	0.385
55.	<i>Ensatina eschscholtzi</i>	0.374	-0.427	5.3	0.724	25	0.374
56.	<i>Eurycea bislineata</i>	0.376	-0.425	1.14	0.057	5	0.077
57.	<i>Eurycea longicauda</i>	0.393	-0.406	1.57	0.196	15	0.178
58.	<i>Eurycea multiplicata</i>	0.267	-0.573	0.71	-0.149	15	0.121
59.	<i>Eurycea nana</i>	0.476	-0.322	0.154	-0.812	25	0.476
60.	<i>Eurycea neotenes</i>	0.590	-0.229	0.243	-0.614	25	0.590
61.	<i>Eurycea pterophila?</i>	0.499	-0.302	0.196	-0.708	25	0.499
62.	<i>Gastrophryne carolinensis</i>	0.430	-0.367	1.9	0.279	20	0.289
63.	<i>Geotrypetes seraphini</i>	0.306	-0.514	1.93	0.286	20	0.206
64.	<i>Gyrinophilus danielsi</i>	0.274	-0.562	14.44	1.160	15	0.124
65.	<i>Gyrinophilus porphyricus</i>	0.178	-0.750	12.4	1.093	15.5	0.084
66.	<i>Hydromantes sp.</i>	0.471	-0.327	3.23	0.509	15	0.213
67.	<i>Hyla arborea</i>	2.922	0.466	7.78	0.891	18.5	1.745
68.	<i>Hyla arenicolor</i>	0.736	-0.133	3.37	0.528	20	0.495
69.	<i>Hyla chrysoscelis</i>	0.693	-0.159	3.9	0.591	10	0.211
70.	<i>Hyla cinerea</i>	0.567	-0.246	4.5	0.653	25	0.567
71.	<i>Hyla crepitans</i>	0.644	-0.191	9.9	0.996	25	0.644
72.	<i>Hyla crucifer</i>	0.908	-0.042	1.15	0.061	20	0.611
73.	<i>Hyla gratiosa</i>	0.384	-0.416	13.9	1.143	29	0.528
74.	<i>Hyla maxima</i>	0.528	-0.277	41.4	1.617	25	0.528
75.	<i>Hyla regilla</i>	0.299	-0.524	2.27	0.356	21	0.218
76.	<i>Hyla squirella</i>	0.783	-0.106	2.2	0.342	27	0.917
77.	<i>Hyla versicolor</i>	0.697	-0.157	8.62	0.936	19	0.433
78.	<i>Hyperolius marmoratus</i>	0.555	-0.256	1	0.000	20	0.373
79.	<i>Hyperolius parallelus</i>	0.632	-0.199	1	0.000	20	0.425
80.	<i>Hyperolius tuberilinguis</i>	0.560	-0.252	1	0.000	20	0.377
81.	<i>Hyperolius viridiflavus</i>	0.776	-0.110	0.9	-0.046	20	0.522
82.	<i>Kaloula pulchra</i>	0.214	-0.670	30.7	1.487	20	0.144
83.	<i>Kassina senegalensis</i>	0.623	-0.206	3.02	0.480	20	0.419
84.	<i>Kassina weali</i>	0.421	-0.376	6.25	0.796	20	0.283
85.	<i>Lepidobatrachus llanensis</i>	0.539	-0.268	88.5	1.947	25	0.539
86.	<i>Leptodactylus typhoniis</i>	0.639	-0.194	5.1	0.708	25	0.639
87.	<i>Microhyla carolinensis</i>	0.811	-0.091	3.5	0.544	15	0.367
88.	<i>Molga torosa?</i>	1.452	0.162	17.5	1.243	18.5	0.867
89.	<i>Necturus maculosus</i>	0.110	-0.959	125	2.097	25	0.110
90.	<i>Notophthalmus viridescens</i>	0.372	-0.429	3	0.477	20	0.250
91.	<i>Occidozyga martensii</i>	0.291	-0.536	9.3	0.968	35	0.644
92.	<i>Odontophrynus americanus</i>	0.297	-0.527	15.24	1.183	20	0.200
93.	<i>Osteopilus septentrionalis</i>	0.546	-0.263	5	0.699	20	0.367
94.	<i>Phyllomedusa sauvagei</i>	0.529	-0.277	17.5	1.243	10	0.161
95.	<i>Physalaemus pustulosus</i>	0.650	-0.187	2.9	0.462	25	0.650
96.	<i>Plethodon cinereus</i>	0.283	-0.548	0.6	-0.222	17.5	0.156
97.	<i>Plethodon dorsalis</i>	0.192	-0.717	0.69	-0.161	25	0.192
98.	<i>Plethodon glutinosus</i>	0.275	-0.561	5.01	0.700	20	0.185
99.	<i>Plethodon jordani</i>	0.234	-0.631	3.1	0.491	5	0.048
100.	<i>Plethodon neomexicanus</i>	0.339	-0.470	2.47	0.393	25	0.339
101.	<i>Plethodon spp.</i>	0.403	-0.395	3.25	0.512	22	0.318
102.	<i>Pseudacris nigrita</i>	0.900	-0.046	1	0.000	25	0.900
103.	<i>Pseudacris triseriata</i>	0.405	-0.393	0.94	-0.027	5	0.083
104.	<i>Pseudobranchius striatus</i>	0.283	-0.548	2.19	0.340	25	0.283
105.	<i>Pseudoeurycea belli</i>	0.163	-0.788	24.45	1.388	25	0.163
106.	<i>Pseudoeurycea brunnata</i>	2.420	0.384	3.33	0.522	15	1.095
107.	<i>Pseudoeurycea cephalica</i>	0.301	-0.521	1.45	0.161	15	0.136
108.	<i>Pseudoeurycea cochranae</i>	0.252	-0.599	2.23	0.348	15	0.114
109.	<i>Pseudoeurycea gadovii</i>	0.362	-0.441	2.88	0.459	15	0.164
110.	<i>Pseudoeurycea goebeli</i>	0.259	-0.587	3.78	0.577	5	0.053
111.	<i>Pseudoeurycea leprosa</i>	0.278	-0.556	2.46	0.391	15	0.126

112.	<i>Pseudoeurycea rex</i>	0.259	-0.587	1.86	0.270	15	0.117
113.	<i>Pseudoeurycea smithii</i>	0.219	-0.660	6.66	0.823	15	0.099
114.	<i>Pseudotriton ruber</i>	0.171	-0.767	10.81	1.034	5	0.035
115.	<i>Pternophyla fodiens</i>	0.265	-0.577	15.1	1.179	20	0.178
116.	<i>Pyxicephalus adspersus</i>	0.210	-0.678	562.3	2.750	20	0.141
117.	<i>Rana arvalis</i>	0.674	-0.171	17	1.230	5	0.138
118.	<i>Rana aspersa</i>	0.524	-0.281	563	2.751	18.5	0.313
119.	<i>Rana berlandieri</i>	0.406	-0.391	70	1.845	29	0.558
120.	<i>Rana blythi</i>	0.274	-0.562	88.7	1.948	25	0.274
121.	<i>Rana cancrivora</i>	0.413	-0.384	20.45	1.311	30	0.614
122.	<i>Rana catesbeiana</i>	0.137	-0.863	262	2.418	5	0.028
123.	<i>Rana chalconota</i>	0.530	-0.276	4.1	0.613	25	0.530
124.	<i>Rana clamitans</i>	0.462	-0.335	32.5	1.512	25	0.462
125.	<i>Rana cyanophlyctis</i>	2.968	0.472	0.294	-0.532	29	4.076
126.	<i>Rana erythraea</i>	0.115	-0.939	19	1.279	15	0.052
127.	<i>Rana esculenta</i>	0.108	-0.967	15.2	1.182	23	0.092
128.	<i>Rana hexadactyla</i>	0.934	-0.030	51.9	1.715	29	1.283
129.	<i>Rana magna</i>	0.311	-0.507	34.2	1.534	30	0.463
130.	<i>Rana muscosa</i>	0.217	-0.664	13.5	1.130	4	0.041
131.	<i>Rana nicobariensis</i>	0.547	-0.262	2.6	0.415	25	0.547
132.	<i>Rana palustris</i>	0.748	-0.126	36	1.556	21.5	0.567
133.	<i>Rana pipiens</i>	0.256	-0.592	34.8	1.542	10	0.078
134.	<i>Rana ridibunda</i>	0.265	-0.577	35	1.544	20	0.178
135.	<i>Rana sylvatica</i>	0.600	-0.222	6	0.778	25	0.600
136.	<i>Rana temporaria</i>	0.088	-1.056	39	1.591	15	0.040
137.	<i>Rana virgatipes</i>	0.542	-0.266	7	0.845	5	0.111
138.	<i>Rhyacotriton olympicus</i>	1.178	0.071	2.6	0.415	15	0.533
139.	<i>Salamandra maculosa</i>	1.197	0.078	28.32	1.452	18.5	0.715
140.	<i>Salamandra salamandra</i>	0.171	-0.767	75	1.875	16	0.084
141.	<i>Scaphiopus bombifrons</i>	1.708	0.232	13.3	1.124	15	0.773
142.	<i>Scaphiopus couchii</i>	0.564	-0.249	24.6	1.391	21	0.411
143.	<i>Scaphiopus hammondi</i>	0.687	-0.163	12.74	1.105	21	0.500
144.	<i>Scaphiopus holbrooki</i>	1.028	0.012	14	1.146	15	0.465
145.	<i>Siren intermedia</i>	0.183	-0.738	13.7	1.137	25	0.183
146.	<i>Siren lacertina</i>	0.063	-1.201	269	2.430	25	0.063
147.	<i>Smilisca baudinii</i>	0.820	-0.086	23.7	1.375	15	0.371
148.	<i>Taricha granulosa</i>	0.301	-0.521	6.56	0.817	15	0.136
149.	<i>Taricha rivularis</i>	0.340	-0.469	10.83	1.035	15	0.154
150.	<i>Taricha torosa</i>	0.164	-0.785	9.87	0.994	10	0.050
151.	<i>Telmatobius culeus</i>	0.256	-0.592	122.3	2.087	10	0.078
152.	<i>Telmatobius marmoratus</i>	0.913	-0.040	18.4	1.265	10	0.278
153.	<i>Thorius sp.</i>	0.316	-0.500	0.31	-0.509	15	0.143
154.	<i>Triturus cristatus</i>	1.005	0.002	7	0.845	10	0.306
155.	<i>Triturus vulgaris</i>	1.033	0.014	8.75	0.942	25	1.033
156.	<i>Typhlonectes compressicauda</i>	0.262	-0.582	30.62	1.486	25	0.262
157.	<i>Xenopus laevis</i>	0.227	-0.644	139	2.143	20	0.153
158.	<i>Xenopus mulleri</i>	1.096	0.040	24	1.380	17	0.581

Table S5b. Standard metabolic rates in reptiles (after White et al. 2006)

Species		q25Wkg	Logq25Wkg	Mg	LogMg	TC	qWkg
1.	<i>Acabthodactylus boskianus</i>	0.437	-0.360	7.8	0.892	40	1.667
2.	<i>Acanthodactylus erythrurus</i>	0.797	-0.099	9	0.954	35	1.945
3.	<i>Acanthodactylus opheodurus</i>	0.740	-0.131	3.8	0.580	30	1.156
4.	<i>Acanthodactylus pardalis</i>	1.020	0.009	9.7	0.987	40	3.889
5.	<i>Acanthodactylus schmidtii</i>	0.505	-0.297	14.5	1.161	35	1.233
6.	<i>Acanthodactylus schreiberi</i>	0.407	-0.390	10.9	1.037	40	1.550
7.	<i>Acanthodactylus scutellatus</i>	0.671	-0.173	6.6	0.820	40	2.556
8.	<i>Acanthophis praelongus</i>	0.167	-0.777	105.5	2.023	30	0.261
9.	<i>Acontias meleagris</i>	0.253	-0.597	7.3	0.863	33	0.517
10.	<i>Acrantophis dumerili</i>	0.066	-1.180	2548.5	3.406	20	0.042
11.	<i>Acrochordus araduræ</i>	0.084	-1.076	1047.7	3.020	30	0.131
12.	<i>Aligator mississippiensis</i>	0.118	-0.928	1287	3.110	10	0.031
13.	<i>Amblyrhynchus cristatus</i>	0.203	-0.693	747.5	2.874	25	0.203
14.	<i>Amphibolurus barbatus</i>	0.267	-0.573	373	2.572	37	0.778
15.	<i>Amphibolurus nuchalis</i>	0.351	-0.455	24.65	1.392	40	1.339
16.	<i>Anarbylus switaki</i>	0.410	-0.387	9.48	0.977	25	0.410
17.	<i>Anguis fragilis</i>	0.209	-0.680	12.067	1.082	30	0.327
18.	<i>Anniella pulchra</i>	0.363	-0.440	4.8466	0.685	25	0.363
19.	<i>Anolis acutus</i>	1.031	0.013	4.3	0.633	30	1.611
20.	<i>Anolis bonariensis</i>	0.302	-0.520	12	1.079	27	0.361
21.	<i>Anolis carolinensis</i>	0.669	-0.175	4.5	0.653	30	1.045
22.	<i>Anolis limifrons</i>	0.694	-0.159	1.5	0.176	20	0.444
23.	<i>Antaresia childreni</i>	0.212	-0.674	331.7	2.521	24	0.194
24.	<i>Antaresia stimsoni</i>	0.203	-0.693	349.9	2.544	30	0.317
25.	<i>Aspidites melanocephalus</i>	0.223	-0.652	1027.5	3.012	24	0.204

26.	<i>Blanus cinereus</i>	0.626	-0.203	2.4	0.380	30	0.978
27.	<i>Boa constrictor</i>	0.058	-1.237	7815.5	3.893	30	0.090
28.	<i>Bunopus tuberculatus</i>	2.330	0.367	2.5	0.398	35	5.684
29.	<i>Candoia carinatus</i>	0.064	-1.194	521.9	2.718	20	0.041
30.	<i>Chalcides ocellatus</i>	0.256	-0.592	22.06	1.344	33	0.522
31.	<i>Chelydra serpentina</i>	0.072	-1.143	3473	3.541	10	0.019
32.	<i>Chironius quadricarinatus</i>	0.295	-0.530	61	1.785	20	0.189
33.	<i>Cnemidophorus murinus</i>	0.267	-0.573	85	1.929	40	1.017
34.	<i>Cnemidophorus tigris</i>	0.498	-0.303	18	1.255	30	0.778
35.	<i>Coleonyx variegatus</i>	0.814	-0.089	3.43	0.535	25	0.814
36.	<i>Coluber constrictor</i>	0.146	-0.836	262	2.418	35	0.356
37.	<i>Corallus caninus</i>	0.077	-1.114	556	2.745	20	0.049
38.	<i>Corallus enhydris</i>	0.083	-1.081	802	2.904	30	0.130
39.	<i>Cosymbotus platyrus</i>	0.465	-0.333	3.5	0.544	27	0.556
40.	<i>Crotalus viridis</i>	0.146	-0.836	301	2.479	35	0.356
41.	<i>Crotaphytus collaris</i>	0.534	-0.272	30	1.477	37	1.556
42.	<i>Ctenotus labillardieri</i>	0.620	-0.208	2.8	0.447	20	0.397
43.	<i>Cyclagras gigas</i>	0.278	-0.556	2680	3.428	20	0.178
44.	<i>Diadophis punctatus</i>	0.469	-0.329	4.4	0.643	30	0.733
45.	<i>Diplometopon zarudnyi</i>	0.348	-0.458	6.34	0.802	35	0.850
46.	<i>Dipsas albifrons</i>	0.312	-0.506	22	1.342	20	0.200
47.	<i>Dipsosaurus dorsalis</i>	0.233	-0.633	40.36	1.606	45	1.389
48.	<i>Egernia cunninghami</i>	0.295	-0.530	261	2.417	20	0.189
49.	<i>Elaphe guttata</i>	0.628	-0.202	800	2.903	25	0.628
50.	<i>Epicrates cenchria</i>	0.104	-0.983	416	2.619	30	0.162
51.	<i>Eryx colubrinus</i>	0.128	-0.893	85.85	1.934	20	0.082
52.	<i>Eumeces fasciatus</i>	0.853	-0.069	7	0.845	30	1.333
53.	<i>Eumeces inexpectatus</i>	0.544	-0.264	9.6	0.982	30	0.850
54.	<i>Eumeces obsoletus</i>	0.434	-0.363	30	1.477	20	0.278
55.	<i>Eunectes murinus</i>	0.183	-0.738	1130	3.053	20	0.117
56.	<i>Eunectes notaeus</i>	0.130	-0.886	14400	4.158	20	0.083
57.	<i>Garthia gaudichaudi</i>	0.531	-0.275	0.805	-0.094	25	0.531
58.	<i>Gekko gecko</i>	0.343	-0.465	61.5	1.789	30	0.536
59.	<i>Gerrhonotus multicarinatus</i>	0.494	-0.306	29	1.462	20	0.316
60.	<i>Gonotodes antillensis</i>	0.413	-0.384	1.8	0.255	34	0.922
61.	<i>Helicops modestus</i>	0.196	-0.708	196	2.292	30	0.306
62.	<i>Hemidactylus frenatus</i>	0.488	-0.312	2	0.301	27	0.583
63.	<i>Iguana iguana</i>	0.286	-0.544	795	2.900	37	0.833
64.	<i>Klauberina riversiana</i>	0.045	-1.347	19	1.279	35	0.111
65.	<i>Lacerta agilis</i>	1.109	0.045	8.4	0.924	35	2.706
66.	<i>Lacerta sicula</i>	0.472	-0.326	9	0.954	20	0.302
67.	<i>Lacerta trilineata</i>	0.198	-0.703	71	1.851	20	0.127
68.	<i>Lacerta viridis</i>	0.234	-0.631	31	1.491	20	0.150
69.	<i>Lacerta vivipara</i>	0.843	-0.074	3.95	0.597	30	1.317
70.	<i>Lampropeltis getulus</i>	0.143	-0.845	1217	3.085	26	0.156
71.	<i>Lampropeltis miliaris</i>	0.267	-0.573	401	2.603	25	0.267
72.	<i>Leimadophis poecilogyrus</i>	0.347	-0.460	42	1.623	20	0.222
73.	<i>Lepidophyma gaigeae</i>	0.366	-0.437	5	0.699	15	0.150
74.	<i>Lepidophyma smithi</i>	0.249	-0.604	25	1.398	30	0.389
75.	<i>Liasis fuscus</i>	0.078	-1.108	1306.9	3.116	30	0.122
76.	<i>Liasis olivaceus</i>	0.091	-1.041	3000.7	3.477	24	0.083
77.	<i>Lichanura roseofusca</i>	0.208	-0.682	314	2.497	32	0.389
78.	<i>Lichanura trivirgata</i>	0.114	-0.943	182	2.260	20	0.073
79.	<i>Masticodryas bifossatus</i>	0.373	-0.428	735	2.866	20	0.239
80.	<i>Masticophis flagellum</i>	1.936	0.287	262	2.418	35	4.723
81.	<i>Morelia spilota variegata</i>	0.092	-1.036	2173.5	3.337	30	0.144
82.	<i>Natrix maura</i>	0.214	-0.670	22.5	1.352	5	0.036
83.	<i>Natrix natrix helretica</i>	0.131	-0.883	82.5	1.916	5	0.022
84.	<i>Nerodia rhombifera</i>	0.338	-0.471	238	2.377	30	0.528
85.	<i>Ophisaurus ventralis</i>	0.214	-0.670	32.185	1.508	25	0.214
86.	<i>Oxyrhopus trigeminus</i>	0.303	-0.519	98	1.991	20	0.194
87.	<i>Pelamis platurus</i>	0.213	-0.672	116	2.064	30	0.333
88.	<i>Philodryas olfersii</i>	0.339	-0.470	176	2.246	20	0.217
89.	<i>Philodryas patagoniensis</i>	0.442	-0.355	388	2.589	20	0.283
90.	<i>Philodryas serra</i>	0.303	-0.519	135	2.130	20	0.194
91.	<i>Phrynosoma cornutum</i>	0.378	-0.423	35	1.544	35	0.922
92.	<i>Phrynosoma douglassi</i>	0.389	-0.410	28	1.447	35	0.950
93.	<i>Phrynosoma m'calli</i>	0.524	-0.281	16	1.204	37	1.528
94.	<i>Physignathus lesueurii</i>	0.267	-0.573	504	2.702	37	0.778
95.	<i>Pituophis melanoleucus</i>	0.166	-0.780	548	2.739	20	0.106
96.	<i>Pituphophis catenifer affinis</i>	0.142	-0.848	548	2.739	30	0.222
97.	<i>Podarcis hispanica</i>	0.423	-0.374	3.43	0.535	10	0.111
98.	<i>Podarcis lilfordi brauni</i>	0.634	-0.198	7.4	0.869	20	0.406
99.	<i>Podarcis muralis</i>	0.972	-0.012	5.5	0.740	35	2.372
100.	<i>Psammotromus algerus</i>	1.067	0.028	5.2	0.716	20	0.683
101.	<i>Pseudemys scripta</i>	0.080	-1.097	305	2.484	10	0.021
102.	<i>Pseudonaja nuchalis</i>	0.201	-0.697	214.1	2.331	33	0.411
103.	<i>Ptyodactylus hasselquistii</i>	0.519	-0.285	8.5	0.929	30	0.811
104.	<i>Python curtis</i>	0.064	-1.194	2373.5	3.375	20	0.041

105.	<i>Python molurus</i>	0.044	-1.357	29777.25	4.474	20	0.028
106.	<i>Python regius</i>	0.077	-1.114	787	2.896	20	0.049
107.	<i>Python reticulatus</i>	0.078	-1.108	14326	4.156	20	0.050
108.	<i>Python sebae</i>	0.061	-1.215	16140	4.208	20	0.039
109.	<i>Salvadora hexalepis</i>	0.380	-0.420	65	1.813	30	0.594
110.	<i>Sauromalus hispidus</i>	0.155	-0.810	574	2.759	20	0.099
111.	<i>Sauromalus obesus</i>	0.251	-0.600	150	2.176	20	0.161
112.	<i>Sceloporus graciosus</i>	0.489	-0.311	5	0.699	25	0.489
113.	<i>Sceloporus occidentalis</i>	0.600	-0.222	10.105	1.005	25	0.600
114.	<i>Sceloporus olivaceus</i>	0.572	-0.243	24	1.380	20	0.366
115.	<i>Sceloporus undulatus</i>	0.339	-0.470	3.8	0.580	20	0.217
116.	<i>Sceloporus variabilis</i>	3.495	0.543	12.215	1.087	10	0.917
117.	<i>Scelotes gronovii</i>	0.618	-0.209	1.1	0.041	33	1.261
118.	<i>Scincella lateralis</i>	1.042	0.018	1	0.000	20	0.667
119.	<i>Scinus mitranus</i>	0.341	-0.467	14.6	1.164	35	0.833
120.	<i>Sibynomorphis mikanii</i>	1.067	0.028	11	1.041	20	0.683
121.	<i>Spalerosophis cliffordi</i>	0.433	-0.364	300	2.477	35	1.056
122.	<i>Sphaerodactylus beattyi</i>	0.800	-0.097	0.4	-0.398	30	1.250
123.	<i>Sphaerodactylus cinereus</i>	0.465	-0.333	0.4	-0.398	27	0.556
124.	<i>Sphaerodactylus macrolepis</i>	0.925	-0.034	0.5	-0.301	30	1.445
125.	<i>Sphenodon punctatum</i>	0.172	-0.764	430	2.633	25	0.172
126.	<i>Sphenops sepsoides</i>	0.281	-0.551	7.4	0.869	30	0.439
127.	<i>Storeria dekayi</i>	0.576	-0.240	7.2	0.857	30	0.900
128.	<i>Tarentola mauritanica</i>	0.414	-0.383	6.6	0.820	35	1.011
129.	<i>Terrapene ornata ornata</i>	0.015	-1.824	354	2.549	10	0.004
130.	<i>Thamnodyastes strigatus</i>	0.356	-0.449	55	1.740	20	0.228
131.	<i>Thamnophis butleri</i>	0.342	-0.466	19.02	1.279	25	0.342
132.	<i>Thamnophis proximus</i>	0.533	-0.273	31	1.491	30	0.833
133.	<i>Thamnophis sirtalis</i>	0.128	-0.893	200	2.301	29	0.183
134.	<i>Tiliqua rugosa</i>	0.328	-0.484	508.6	2.706	35	0.800
135.	<i>Tiliqua scincoides</i>	0.234	-0.631	493	2.693	20	0.150
136.	<i>Trachydosaurus rugosus</i>	0.261	-0.583	461	2.664	20	0.167
137.	<i>Trogonophis weigmanni</i>	0.214	-0.670	4.985	0.698	25	0.214
138.	<i>Uromastyx microlepis</i>	0.105	-0.979	289.84	2.462	20	0.067
139.	<i>Uta mearnsi</i>	0.339	-0.470	14	1.146	20	0.217
140.	<i>Uta stansburiana</i>	0.572	-0.243	3.625	0.559	37	1.667
141.	<i>Varanus albigularis</i>	0.353	-0.452	963	2.984	35	0.861
142.	<i>Varanus bengalensis</i>	0.181	-0.742	3440	3.537	30	0.283
143.	<i>Varanus exanthematicus</i>	2.743	0.438	3836	3.584	25	2.743
144.	<i>Varanus giganteus</i>	0.220	-0.658	2496	3.397	25.9	0.238
145.	<i>Varanus gilleni</i>	0.371	-0.431	27.5	1.439	37	1.083
146.	<i>Varanus gouldi</i>	0.208	-0.682	674	2.829	20	0.133
147.	<i>Varanus mertensi</i>	0.177	-0.752	904	2.956	35	0.433
148.	<i>Varanus panoptes</i>	0.227	-0.644	2005	3.302	20.5	0.152
149.	<i>Varanus varius</i>	0.178	-0.750	4410	3.644	30	0.278
150.	<i>Varnus rosenbergi</i>	0.300	-0.523	1269.3	3.104	35	0.733
151.	<i>Vipera berus</i>	0.500	-0.301	63	1.799	25	0.500
152.	<i>Xantusia henshawi</i>	0.503	-0.298	3.5	0.544	15	0.206
153.	<i>Xantusia vigilis</i>	0.614	-0.212	1.5	0.176	30	0.959
154.	<i>Xenodon guentheri</i>	0.303	-0.519	50	1.699	20	0.194
155.	<i>Xenodon merremii</i>	0.373	-0.428	502	2.701	20	0.239
156.	<i>Xenodon neuwiedii</i>	0.322	-0.492	53	1.724	20	0.206

Table S5c. Standard and routine metabolic rates in fish (after www.fishbase.org)

Note: N is the number of metabolic rate values available for each species at www.fishbase.org; qWkg is the minimum of the N values.

Species		q25Wkg	Logq25Wkg	Mg	LogMg	TC	qWkg	N
1.	Abramis brama	0.746	-0.127	102.90	2.012	18.0	0.5252	40
2.	Acanthopagrus schlegelii schlegelii	0.117	-0.932	254.00	2.405	25.0	0.1167	21
3.	Acipenser gueldenstaedtii	0.387	-0.412	208.00	2.318	18.0	0.2723	8
4.	Acipenser nudiiventris	0.714	-0.146	89.00	1.949	19.0	0.5290	1
5.	Acipenser ruthenus	0.175	-0.757	258.00	2.412	20.0	0.1362	4
6.	Acipenser stellatus	0.345	-0.462	6500.00	3.813	20.0	0.2684	22
7.	Acipenser transmontanus	0.019	-1.721	950.00	2.978	15.0	0.0117	10
8.	Aequidens pulcher	0.475	-0.323	5.00	0.699	25.0	0.4746	1
9.	Alburnus alburnus	0.729	-0.137	9.00	0.954	8.0	0.3112	4
10.	Ambassis interrupta	0.315	-0.502	5.00	0.699	25.0	0.3151	10
11.	Ameiurus melas	0.424	-0.373	50.00	1.699	17.0	0.2840	3
12.	Ameiurus natalis	0.334	-0.476	20.00	1.301	15.0	0.2023	6
13.	Ameiurus nebulosus	0.165	-0.783	116.00	2.064	10.0	0.0778	90
14.	Anabas testudineus	0.017	-1.770	27.62	1.441	28.0	0.0195	86
15.	Anarhichas minor	0.458	-0.339	27.50	1.439	1.35	0.1400	1
16.	Anguilla anguilla	0.128	-0.893	71.10	1.852	14.0	0.0739	151
17.	Anguilla australis australis	0.110	-0.959	700.00	2.845	20.0	0.0856	2
18.	Anguilla japonica	0.132	-0.879	325.00	2.512	14.5	0.0778	34

19.	Anguilla rostrata	0.135	-0.870	315.00	2.498	15.0	0.0817	19
20.	Anoplogaster cornuta	0.032	-1.495	36.70	1.565	5.0	0.0117	14
21.	Aphanius dispar dispar	1.940	0.288	0.56	-0.252	17.0	1.2993	5
22.	Arapaima gigas	0.084	-1.076	2300.00	3.362	28.0	0.0973	35
23.	Aristostomias lunifer	0.159	-0.799	21.10	1.324	5.0	0.0584	1
24.	Bajacalifornia burragei	0.085	-1.071	24.90	1.396	5.0	0.0311	1
25.	Balistes capriscus	0.476	-0.322	320.00	2.505	17.5	0.3268	3
26.	Bathylagus antarcticus	0.349	-0.457	27.15	1.434	0.25	0.1011	1
27.	Bathylagus stibius	0.371	-0.431	8.55	0.932	5.0	0.1362	1
28.	Bathylagus wesethi	1.121	0.050	1.50	0.176	10.0	0.5290	1
29.	Benthalbella elongata	0.712	-0.148	35.30	1.548	0.25	0.2062	1
30.	Boreogadus saida	0.233	-0.633	110.00	2.041	1.0	0.0700	7
31.	Borostomias panamensis	0.265	-0.577	110.25	2.042	5.0	0.0973	2
32.	Brevoortia tyrannus	0.841	-0.075	78.40	1.894	10.0	0.3968	21
33.	Callionymus lyra	0.161	-0.793	105.00	2.021	11.5	0.0817	28
34.	Campostoma anomalum	1.038	0.016	19.25	1.284	17.9	0.7274	1
35.	Caranx hippos	1.466	0.166	38.30	1.583	14.7	0.8753	1
36.	Carassius auratus auratus	0.127	-0.896	3.80	0.580	5.0	0.0467	310
37.	Carassius carassius	0.106	-0.975	12.50	1.097	5.0	0.0389	32
38.	Catostomus commersonii	0.289	-0.539	100.00	2.000	10.0	0.1362	48
39.	Catostomus tahoensis	0.574	-0.241	43.73	1.641	8.0	0.2451	6
40.	Centropristis striata	0.389	-0.410	84.00	1.924	25.0	0.3890	5
41.	Chaenocephalus aceratus	0.221	-0.656	1130.00	3.053	2.0	0.0700	24
42.	Channa marulius	0.121	-0.917	93.00	1.968	30.0	0.1556	16
43.	Channa orientalis	0.164	-0.785	30.00	1.477	30.0	0.2101	52
44.	Channa punctata	0.067	-1.174	12.50	1.097	26.0	0.0700	84
45.	Channa striata	0.130	-0.886	82.00	1.914	30.0	0.1673	11
46.	Channichthys rhinocerus	0.713	-0.147	200.00	2.301	5.5	0.2684	2
47.	Chanos chanos	2.692	0.430	0.70	-0.155	25.0	2.6919	56
48.	Chiasmodon niger	0.468	-0.330	76.60	1.884	2.5	0.1517	1
49.	Chiloscyllium plagiosum	0.202	-0.695	880.00	2.944	23.0	0.1828	3
50.	Chromis chromis	0.812	-0.090	10.40	1.017	16.0	0.5174	4
51.	Cichlasoma bimaculatum	0.267	-0.573	11.00	1.041	22.0	0.2295	38
52.	Cirrhinus cirrhosus	0.213	-0.672	1821.00	3.260	21.5	0.1789	87
53.	Citharichthys stigmaeus	0.513	-0.290	15.00	1.176	15.0	0.3112	1
54.	Clarias batrachus	0.204	-0.690	77.00	1.886	26.0	0.2140	9
55.	Clinocottus analis	0.610	-0.215	37.00	1.568	20.0	0.4746	1
56.	Colisa fasciata	0.348	-0.458	0.92	-0.036	28.0	0.4046	42
57.	Colossoma macropomum	0.195	-0.710	1760.00	3.246	25.0	0.1945	115
58.	Conger conger	0.610	-0.215	545.00	2.736	13.0	0.3345	5
59.	Coregonus autumnalis	1.309	0.117	234.00	2.369	7.2	0.5368	4
60.	Coregonus fera	4.049	0.607	0.11	-0.959	14.0	2.3340	26
61.	Coregonus sardinella	0.935	-0.029	365.00	2.562	9.2	0.4240	22
62.	Coryphaena equiselis	6.114	0.786	0.03	-1.523	23.0	5.5316	71
63.	Coryphaena hippurus	3.114	0.493	0.90	-0.046	13.5	1.7505	1
64.	Coryphaenoides armatus	0.047	-1.328	1200.00	3.079	3.0	0.0156	3
65.	Cottus gobio	1.961	0.292	2.90	0.462	18.0	1.3810	1
66.	Ctenopharyngodon idella	0.497	-0.304	16.05	1.205	22.0	0.4279	8
67.	Cyclothone acclinidens	0.281	-0.551	0.87	-0.060	3.0	0.0934	9
68.	Cyclothone microdon	0.309	-0.510	0.78	-0.108	0.25	0.0895	1
69.	Cyprinodon variegatus variegatus	0.661	-0.180	2.40	0.380	25.0	0.6613	18
70.	Cyprinus carpio carpio	0.140	-0.854	174.00	2.241	10.0	0.0661	208
71.	Dactylopterus volitans	14.340	1.157	0.02	-1.699	23.0	12.9732	14
72.	Dasyatis sabina	0.282	-0.550	503.00	2.702	22.2	0.2451	10
73.	Diaphus theta	1.620	0.210	2.65	0.423	5.0	0.5952	2
74.	Diplodus sargus sargus	1.275	0.106	25.75	1.411	14.0	0.7352	2
75.	Dorosoma cepedianum	0.417	-0.380	87.70	1.943	13.4	0.2334	52
76.	Echiichthys vipera	1.239	0.093	10.50	1.021	18.5	0.8947	2
77.	Electrona antarctica	0.806	-0.094	7.90	0.898	0.25	0.2334	1
78.	Embiotoca lateralis	0.687	-0.163	599.00	2.777	15.0	0.4162	2
79.	Encheliophis homei	0.260	-0.585	7.50	0.875	30.0	0.3345	6
80.	Engraulis japonicus	1.263	0.101	628.79	2.799	16.2	0.8130	15
81.	Epinephelus akaara	0.175	-0.757	281.00	2.449	25.0	0.1751	30
82.	Erimyzon oblongus	0.824	-0.084	25.80	1.412	18.5	0.5952	1
83.	Erpetichthys calabaricus	0.405	-0.393	27.40	1.438	27.0	0.4474	1
84.	Esomus danricus	1.661	0.220	0.62	-0.208	27.5	1.8828	20
85.	Esox lucius	0.212	-0.674	600.00	2.778	5.0	0.0778	26
86.	Esox masquinongy	1.070	0.029	12.20	1.086	5.0	0.3929	9
87.	Etheostoma blennioides	1.189	0.075	12.35	1.092	17.5	0.8169	1
88.	Euthynnus affinis	2.057	0.313	2260.00	3.354	24.0	1.9567	1
89.	Exodon paradoxus	0.190	-0.721	3.85	0.585	20.0	0.1478	4
90.	Fundulus grandis	0.035	-1.456	31.50	1.498	25.0	0.0350	22
91.	Fundulus heteroclitus heteroclitus	0.138	-0.860	9.00	0.954	5.0	0.0506	18
92.	Fundulus parvipinnis	0.030	-1.523	6.00	0.778	20.0	0.0233	11
93.	Fundulus similis	0.031	-1.509	21.00	1.322	25.0	0.0311	22
94.	Gadus morhua	0.328	-0.484	6650.00	3.823	3.0	0.1089	140
95.	Gadus ogac	0.871	-0.060	180.00	2.255	0.0	0.2490	1
96.	Gambusia affinis	1.377	0.139	0.24	-0.620	10.0	0.6496	27
97.	Gambusia holbrooki	0.303	-0.519	0.30	-0.523	27.0	0.3345	16

98.	Gasterosteus aculeatus aculeatus	0.907	-0.042	1.95	0.290	10.0	0.4279	22
99.	Genyagnus monopterygius	0.058	-1.237	164.00	2.215	17.0	0.0389	21
100.	Gilchristella aestuaria	1.046	0.020	1.44	0.158	15.0	0.6341	20
101.	Gillichthys mirabilis	0.030	-1.523	16.70	1.223	20.0	0.0233	75
102.	Girella nigricans	0.475	-0.323	210.00	2.322	18.0	0.3345	5
103.	Glossogobius giuris	0.126	-0.900	15.00	1.176	26.0	0.1323	42
104.	Gnathonemus petersii	0.377	-0.424	5.85	0.767	26.0	0.3968	1
105.	Gobio gobio gobio	1.096	0.040	17.00	1.230	12.0	0.5718	6
106.	Gobionotothen gibberifrons	0.309	-0.510	470.00	2.672	0.25	0.0895	1
107.	Gobius paganellus	0.937	-0.028	10.50	1.021	23.1	0.8519	1
108.	Gymnocephalus cernuus	0.825	-0.084	64.80	1.812	17.0	0.5524	2
109.	Gymnodraco acuticeps	0.484	-0.315	87.20	1.941	-0.3	0.1362	2
110.	Gymnoscopelus braueri	0.510	-0.292	11.40	1.057	0.25	0.1478	1
111.	Gymnoscopelus opisthopterus	0.430	-0.367	23.55	1.372	0.25	0.1245	1
112.	Harpagifer georgianus	0.025	-1.602	4.13	0.616	1.5	0.0078	1
113.	Hemichromis bimaculatus	0.751	-0.124	3.00	0.477	25.0	0.7508	1
114.	Heteropneustes fossilis	0.204	-0.690	45.00	1.653	18.0	0.1439	36
115.	Hippocampus hippocampus	0.712	-0.148	10.00	1.000	18.0	0.5018	5
116.	Hippoglossoides platessoides	0.194	-0.712	390.00	2.591	3.5	0.0661	5
117.	Hoplerythrinus unitaeniatus	0.127	-0.896	243.00	2.386	28.5	0.1517	55
118.	Ichthyomyzon fossor	0.845	-0.073	3.78	0.577	18.0	0.5952	1
119.	Ictalurus punctatus	0.232	-0.635	825.00	2.916	18.0	0.1634	29
120.	Katsuwonus pelamis	0.881	-0.055	632.00	2.801	23.5	0.8169	47
121.	Kuhlia sandvicensis	0.254	-0.595	55.97	1.748	23.0	0.2295	62
122.	Labeo calbasu	0.196	-0.708	0.30	-0.523	27.5	0.2217	8
123.	Labeo capensis	0.355	-0.450	357.96	2.554	8.0	0.1517	49
124.	Labeo rohita	0.764	-0.117	5.00	0.699	29.0	0.9336	10
125.	Labeobarbus aeneus	0.140	-0.854	325.50	2.513	10.0	0.0661	28
126.	Labrus bergylta	0.470	-0.328	125.00	2.097	18.0	0.3307	4
127.	Lagodon rhomboides	0.226	-0.646	13.50	1.130	25.0	0.2256	14
128.	Lampetra fluviatilis	0.185	-0.733	1.43	0.155	4.4	0.0661	62
129.	Lampetra planeri	0.146	-0.836	2.79	0.446	5.3	0.0545	17
130.	Leiostomus xanthurus	0.218	-0.662	11.10	1.045	25.0	0.2178	85
131.	Lepidocephalichthys guntea	0.573	-0.242	1.27	0.104	24.0	0.5446	8
132.	Lepidogalaxias salamandroides	0.270	-0.569	0.62	-0.208	20.0	0.2101	31
133.	Lepomis cyanellus	0.834	-0.079	10.00	1.000	15.0	0.5057	3
134.	Lepomis gibbosus	0.244	-0.613	30.00	1.477	5.0	0.0895	9
135.	Lepomis macrochirus	0.151	-0.821	133.00	2.124	30.0	0.1945	250
136.	Leporinus fasciatus	0.541	-0.267	3.95	0.597	25.0	0.5407	3
137.	Leucaspis delineatus	2.199	0.342	1.52	0.182	20.0	1.7116	1
138.	Leuciscus cephalus	0.719	-0.143	14.00	1.146	15.0	0.4357	18
139.	Leuciscus idus	0.212	-0.674	600.00	2.778	5.0	0.0778	15
140.	Leuciscus leuciscus	2.143	0.331	9.95	0.998	16.5	1.4004	1
141.	Limanda limanda	0.159	-0.799	400.00	2.602	5.0	0.0584	7
142.	Lipolagus ochotensis	0.989	-0.005	3.40	0.531	10.0	0.4668	1
143.	Lipophrys pholis	0.549	-0.260	20.20	1.305	16.0	0.3501	109
144.	Liza dumerili	0.387	-0.412	42.50	1.628	18.0	0.2723	45
145.	Liza macrolepis	0.478	-0.321	8.00	0.903	29.0	0.5835	59
146.	Liza richardsonii	0.915	-0.039	39.20	1.593	13.0	0.5018	22
147.	Lota lota	0.572	-0.243	213.00	2.328	11.3	0.2879	1
148.	Lutjanus campechanus	0.469	-0.329	365.50	2.563	15.0	0.2840	18
149.	Lycodichthys dearborni	0.117	-0.932	44.61	1.649	-1.5	0.0311	16
150.	Macrogathus aculeatus	0.233	-0.633	53.00	1.724	21.0	0.1906	43
151.	Melamphaes acanthomus	0.191	-0.719	17.40	1.241	5.0	0.0700	2
152.	Melanocetus johnsonii	0.288	-0.541	50.55	1.704	2.5	0.0934	1
153.	Melanogrammus aeglefinus	0.280	-0.553	155.90	2.193	10.0	0.1323	12
154.	Melanonus zugmayeri	0.265	-0.577	31.50	1.498	5.0	0.0973	1
155.	Melanostigma gelatinosum	0.403	-0.395	47.20	1.674	0.25	0.1167	1
156.	Melanostigma pammelas	0.117	-0.932	10.00	1.000	3.0	0.0389	8
157.	Micropterus salmoides	0.122	-0.914	178.00	2.250	15.0	0.0739	84
158.	Microstomus kitt	0.307	-0.513	229.00	2.360	5.0	0.1128	6
159.	Misgurnus fossilis	0.679	-0.168	28.50	1.455	18.0	0.4785	1
160.	Monopterus cuchia	0.113	-0.947	172.88	2.238	25.0	0.1128	10
161.	Mugil cephalus	0.342	-0.466	221.00	2.344	14.5	0.2023	72
162.	Mugil curema	0.500	-0.301	140.00	2.146	20.0	0.3890	12
163.	Myoxocephalus octodecemspinosus	0.330	-0.481	200.00	2.301	10.0	0.1556	16
164.	Myoxocephalus scorpius	0.519	-0.285	87.50	1.942	2.4	0.1673	5
165.	Mystus armatus	0.469	-0.329	9.20	0.964	30.0	0.6030	10
166.	Mystus cavasius	0.576	-0.240	45.00	1.653	29.0	0.7041	17
167.	Mystus gulio	0.549	-0.260	12.00	1.079	27.0	0.6068	2
168.	Mystus vittatus	0.545	-0.264	7.40	0.869	20.0	0.4240	60
169.	Myxine glutinosa	0.460	-0.337	38.40	1.584	7.0	0.1867	2
170.	Nannobranchium regale	0.169	-0.772	2.90	0.462	5.0	0.0622	1
171.	Nannobranchium ritteri	0.625	-0.204	1.80	0.255	5.0	0.2295	2
172.	Naukrates ductor	8.888	0.949	0.07	-1.155	23.0	8.0406	50
173.	Notothenia coriiceps	0.390	-0.409	1000.00	3.000	0.25	0.1128	3
174.	Notothenia cyanobrancha	1.194	0.077	200.00	2.301	4.5	0.4279	1
175.	Notothenia rossii	0.304	-0.517	159.70	2.203	3.0	0.1011	12
176.	Oligolepis acutipennis	0.115	-0.939	6.30	0.799	26.0	0.1206	12

177.	Oncorhynchus mykiss	0.413	-0.384	450.00	2.653	5.0	0.1517	548
178.	Oncorhynchus nerka	0.434	-0.363	36.70	1.565	5.0	0.1595	87
179.	Oncorhynchus tshawytscha	1.050	0.021	26.00	1.415	16.0	0.6691	1
180.	Oneirodes acanthias	0.127	-0.896	4.20	0.623	5.0	0.0467	1
181.	Ophiodon elongatus	0.186	-0.730	1591.00	3.202	12.1	0.0973	16
182.	Opsanus tau	1.999	0.301	325.00	2.512	20.0	1.5560	1
183.	Oreochromis aureus	0.244	-0.613	697.00	2.843	26.0	0.2567	6
184.	Oreochromis mossambicus	0.226	-0.646	144.30	2.159	16.0	0.1439	206
185.	Oreochromis niloticus niloticus	0.211	-0.676	310.00	2.491	26.0	0.2217	50
186.	Orthodon microlepidotus	0.239	-0.622	650.00	2.813	12.0	0.1245	22
187.	Oryzias latipes	0.720	-0.143	0.27	-0.569	25.0	0.7197	59
188.	Osphronemus goramy	0.298	-0.526	12.50	1.097	28.0	0.3462	2
189.	Pagetopsis macropterus	0.327	-0.485	76.00	1.881	0.0	0.0934	1
190.	Pagothenia borchgrevinki	0.398	-0.400	90.15	1.955	0.5	0.1167	2
191.	Paranotothenia magellanica	0.983	-0.007	200.00	2.301	5.25	0.3657	2
192.	Parophrys vetulus	0.642	-0.192	70.00	1.845	15.0	0.3890	1
193.	Parvilux ingens	0.191	-0.719	9.40	0.973	5.0	0.0700	1
194.	Perca fluviatilis	0.140	-0.854	11.00	1.041	16.0	0.0895	17
195.	Petromyzon marinus	0.561	-0.251	22.80	1.358	5.0	0.2062	7
196.	Pimephales promelas	0.732	-0.135	2.00	0.301	15.0	0.4435	9
197.	Platichthys flesus	0.104	-0.983	350.00	2.544	9.0	0.0467	11
198.	Platichthys stellatus	0.139	-0.857	1290.00	3.111	9.0	0.0622	23
199.	Plecoglossus altivelis altivelis	4.208	0.624	10.70	1.029	19.0	3.1159	1
200.	Pleuronectes platessa	0.212	-0.674	288.60	2.460	5.0	0.0778	104
201.	Poecilia latipinna	0.132	-0.879	5.60	0.748	25.0	0.1323	14
202.	Pollachius pollachius	1.293	0.112	870.00	2.940	18.5	0.9336	2
203.	Pomadasys commersonnii	0.289	-0.539	2627.00	3.419	15.0	0.1751	30
204.	Pomoxis annularis	0.518	-0.286	11.50	1.061	17.4	0.3540	1
205.	Poromitra crassiceps	0.169	-0.772	17.10	1.233	5.0	0.0622	2
206.	Protopterus annectens annectens	0.242	-0.616	368.00	2.566	30.0	0.3112	19
207.	Psenes whiteleggii	4.497	0.653	1.30	0.114	13.5	2.5285	1
208.	Psetta maxima	0.732	-0.135	320.00	2.505	15.0	0.4435	31
209.	Pseudobathylagus milleri	0.169	-0.772	41.10	1.614	5.0	0.0622	1
210.	Pseudochaenichthys georgianus	0.531	-0.275	35.70	1.553	0.5	0.1556	2
211.	Pseudopleuronectes americanus	0.030	-1.523	26.15	1.417	12.0	0.0156	68
212.	Pterophyllum scalare	0.307	-0.513	19.03	1.279	25.0	0.3073	31
213.	Rhinogobiops nicholsii	0.199	-0.701	4.04	0.606	15.0	0.1206	21
214.	Rhodeus amarus	1.184	0.073	0.71	-0.149	16.0	0.7547	1
215.	Rhodeus sericeus	1.365	0.135	3.00	0.477	12.0	0.7119	3
216.	Rutilus rutilus	0.684	-0.165	27.00	1.431	8.0	0.2918	11
217.	Sagamichthys abei	0.244	-0.613	5.70	0.756	5.0	0.0895	1
218.	Salmo salar	0.556	-0.255	25.00	1.398	7.0	0.2256	48
219.	Salmo trutta fario	0.478	-0.321	350.00	2.544	10.0	0.2256	8
220.	Salmo trutta trutta	0.363	-0.440	575.00	2.760	10.0	0.1712	52
221.	Salvelinus alpinus alpinus	0.513	-0.290	210.00	2.322	15.0	0.3112	2
222.	Salvelinus fontinalis	0.181	-0.742	690.00	2.839	10.0	0.0856	147
223.	Salvelinus namaycush	0.197	-0.706	82.80	1.918	9.2	0.0895	117
224.	Sander lucioperca	0.318	-0.498	600.00	2.778	5.0	0.1167	8
225.	Sander vitreus	0.325	-0.488	291.00	2.464	20.0	0.2529	35
226.	Sarda chiliensis lineolata	7.794	0.892	2530.00	3.403	22.0	6.7064	1
227.	Sarotherodon galilaeus galilaeus	0.259	-0.587	283.00	2.452	26.0	0.2723	6
228.	Scardinius erythrophthalmus	1.084	0.035	53.00	1.724	20.0	0.8441	1
229.	Scopelogadus tristis	0.138	-0.860	49.80	1.697	5.0	0.0506	1
230.	Scopelogadus mizolepis mizolepis	0.212	-0.674	3.60	0.556	5.0	0.0778	1
231.	Scophthalmus rhombus	0.700	-0.155	145.00	2.161	18.5	0.5057	3
232.	Scorpaena porcus	0.330	-0.481	50.00	1.699	20.0	0.2567	3
233.	Scyliorhinus canicula	0.211	-0.676	857.00	2.933	7.0	0.0856	25
234.	Scyliorhinus stellaris	0.022	-1.658	2530.00	3.403	18.3	0.0156	14
235.	Sebastes diploproa	0.965	-0.015	2.00	0.301	10.0	0.4551	19
236.	Sebastolobus altivelis	0.041	-1.387	198.00	2.297	5.7	0.0156	6
237.	Seriola quinqueradiata	0.770	-0.114	989.00	2.995	19.2	0.5757	1
238.	Serranus scriba	1.013	0.006	4.10	0.613	16.0	0.6457	1
239.	Solea solea	0.709	-0.149	185.00	2.267	14.0	0.4085	6
240.	Sparus aurata	1.121	0.050	78.00	1.892	18.0	0.7897	2
241.	Spinachia spinachia	1.955	0.291	0.30	-0.523	18.0	1.3771	2
242.	Squalus acanthias	0.182	-0.740	1812.00	3.258	9.0	0.0817	23
243.	Stenobranchius leucopsarus	0.635	-0.197	4.30	0.633	5.0	0.2334	2
244.	Stomias atriventer	0.478	-0.321	9.30	0.968	10.0	0.2256	1
245.	Stomias danae	0.504	-0.298	13.80	1.140	2.5	0.1634	1
246.	Symbolophorus californiensis	0.911	-0.040	0.80	-0.097	5.0	0.3345	1
247.	Synbranchus marmoratus	0.140	-0.854	151.00	2.179	25.0	0.1400	18
248.	Syngnathus acus	1.045	0.019	7.60	0.881	18.5	0.7547	2
249.	Tarletonbeania crenularis	1.248	0.096	3.29	0.517	8.0	0.5329	2
250.	Tautoglabrus adspersus	0.710	-0.149	50.00	1.699	20.0	0.5524	2
251.	Theragra chalcogramma	0.427	-0.370	70.00	1.845	1.0	0.1284	23
252.	Thorichthys meeki	0.319	-0.496	15.00	1.176	25.0	0.3190	2
253.	Thymallus arcticus arcticus	0.757	-0.121	283.00	2.452	4.0	0.2645	16
254.	Tilapia rendalli	0.488	-0.312	50.00	1.699	17.0	0.3268	7
255.	Tilapia zillii	0.218	-0.662	315.00	2.498	26.0	0.2295	30

256.	Tinca tinca	0.379	-0.421	541.00	2.733	16.0	0.2412	21
257.	Torpedo marmorata	0.134	-0.873	448.00	2.651	16.0	0.0856	3
258.	Torpedo torpedo	0.417	-0.380	315.00	2.498	15.0	0.2529	2
259.	Trematomus bernacchii	0.176	-0.754	55.20	1.742	0.1	0.0506	2
260.	Trematomus hansonii	0.939	-0.027	145.00	2.161	-0.9	0.2567	1
261.	Trematomus pennellii	0.626	-0.203	183.00	2.262	-0.9	0.1712	1
262.	Trichogaster trichopterus	0.588	-0.231	7.97	0.901	27.0	0.6496	3
263.	Triphoturus mexicanus	0.381	-0.419	1.15	0.061	5.0	0.1400	2
264.	Typhlogobius californiensis	0.013	-1.886	3.54	0.549	15.0	0.0078	34
265.	Xiphophorus hellerii	0.833	-0.079	2.00	0.301	25.0	0.8325	2
266.	Zoarces viviparus	0.763	-0.117	0.29	-0.538	5.0	0.2801	8

Dataset S6. Basal metabolic rates in birds

McKechnie & Wolf (2004, Table A1) analyzed whole-body basal metabolic rates Q ($W \text{ ind}^{-1}$) as compiled by Reynolds & Lee (1996) for 254 bird species, to find that in 42 cases the conditions for measurements of basal metabolic rate had not been met. Those 42 values were not analyzed in our study. Additionally, McKechnie & Wolf (2004, Table A2) compiled literature data on basal metabolic rate in 60 bird species.

In some cases where no sufficient details were given in the published study, McKechnie & Wolf (2004) contacted the authors to ensure that basal metabolic rate conditions were strictly met. Most data of V. Gavrilov (Gavrilov 1974; Kendeigh, Dol'nik & Gavrilov 1977, ~60 species), who had not been contacted, were presented in Table A1 of McKechnie & Wolf (2004) as having an unknown or small ($n < 3$) number of individuals measured.

Below in Table **S6a** the following data are presented. 314 values from McKechnie & Wolf (2004) ($254-42+60$) were pooled with 113 values of basal metabolic rate of 113 bird species measured by V. Gavrilov. As no sufficient details about the data of V. Gavrilov are present in the international literature, these data are provided below in a separate Table **S6b** with a detailed description of the measurement procedure, the number of individuals studied, time and season of measurements and reference publications.

In the resulting compilation of 385 values, some species were represented twice, like where McKechnie & Wolf (2004) cited the work of Gavrilov (1974) or Kendeigh, Dol'nik & Gavrilov (1977), or where two or more studies investigated one and the same species. In the former case the original value provided by V. Gavrilov in Table **S6b** was taken. In the latter case the lowest value for the species was chosen. This yielded a total of 321 BMR values for 321 species that are presented in Table **S6a**. These values were used in the analyses presented in Table 1 and Figures 1-3 in the paper.

References:

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- McKechnie A.E., Wolf B.O. (2004) The allometry of avian basal metabolic rate: good predictions need good data. *Physiological and Biochemical Zoology* 77: 502-521.
- Reynolds P.S., Lee, R.M. (1996) Phylogenetic analysis of avian energetics: passerines and non-passerines do not differ. *American Naturalist* 147: 735-759.

Table S6a. Basal metabolic rate in birds

Notations: qWkg — mass-specific basal metabolic rate, $W \text{ kg}^{-1}$; LogqWkg — decimal logarithm of qWkg; Mg — body mass, g; LogMg — decimal logarithm of Mg; Src — source of data, G: experimental data of V. Gavrilov (Table S6b); MW: data compiled from the literature by McKechnie & Wolf (2004).

Species	qWkg	LogqWkg	Mg	LogMg	Src
1. <i>Acanthis cannabina</i>	20.1	1.302	16.9	1.228	G
2. <i>Acanthis flammea</i>	20.4	1.310	14.0	1.146	G
3. <i>Acanthorhynchus tenuirostris</i>	25.7	1.409	9.7	0.987	MW
4. <i>Accipiter nisus</i>	7.0	0.848	135	2.130	G
5. <i>Acrocephalus palustris</i>	18.9	1.276	10.8	1.033	G
6. <i>Acridotheres cristatellus</i>	11.0	1.042	109.4	2.039	MW
7. <i>Acrocephalus arundinaceus</i>	11.7	1.069	21.9	1.340	MW
8. <i>Acrocephalus bistrigiceps</i>	16.6	1.220	7.9	0.898	MW
9. <i>Acrocephalus palustris</i>	18.8	1.274	10.8	1.033	MW
10. <i>Acrocephalus schoenobaenus</i>	18.9	1.277	11.5	1.061	G
11. <i>Aegithalos caudatus</i>	22.4	1.350	8.9	0.949	G
12. <i>Aegolius acadicus</i>	5.3	0.722	124	2.093	MW
13. <i>Aethopyga christinae</i>	22.7	1.356	5.2	0.716	MW
14. <i>Agapornis fisheri</i>	9.3	0.969	56.7	1.754	G
15. <i>Agapornis roseicollis</i>	9.6	0.983	48.4	1.685	G
16. <i>Agelaius phoeniceus</i>	14.7	1.167	56.7	1.754	MW
17. <i>Agleactis cupripennis</i>	35.0	1.544	7.2	0.857	MW
18. <i>Aix sponsa</i>	5.0	0.701	448	2.651	G
19. <i>Alaemon alaudipes</i>	11.3	1.054	37.7	1.576	MW
20. <i>Alauda arvensis</i>	22.8	1.357	31.7	1.501	MW
21. <i>Alcedo atthis</i>	11.0	1.043	34.3	1.535	G
22. <i>Alectoris graeca</i>	4.0	0.602	633	2.801	G
23. <i>Alophoixus bres</i>	10.1	1.005	35	1.544	MW
24. <i>Amadina erythrocephala</i>	9.5	0.978	22.4	1.350	MW
25. <i>Amadina fasciata</i>	12.4	1.095	17.2	1.236	MW
26. <i>Ammodramus savannarum</i>	12.9	1.111	13.8	1.140	MW
27. <i>Amphispiza bilineata</i>	17.0	1.230	11.6	1.064	MW
28. <i>Anas acuta</i>	6.1	0.782	721	2.858	MW
29. <i>Anas penelope</i>	3.9	0.592	723	2.859	G
30. <i>Anas platyrhynchos</i>	4.0	0.601	1020	3.009	G
31. <i>Anas strepera</i>	7.8	0.894	791	2.898	MW
32. <i>Anhinga rufa</i> (anhinga)	3.1	0.487	1040	3.017	MW
33. <i>Anser anser</i>	3.3	0.524	3250	3.512	G
34. <i>Anthracothorax nigricollis</i>	39.6	1.598	7.7	0.886	MW
35. <i>Anthus campestris</i>	17.6	1.245	21.8	1.338	G
36. <i>Anthus pratensis</i>	15.9	1.201	18.9	1.276	MW
37. <i>Anthus trivialis</i>	17.2	1.236	19.7	1.294	G
38. <i>Aptenodytes patagonica</i>	2.0	0.298	11080	4.045	G
39. <i>Apteryx australis</i>	1.7	0.229	2380	3.377	MW
40. <i>Apteryx haasti</i>	1.7	0.232	2450	3.389	MW
41. <i>Apteryx owenii</i>	1.9	0.276	1096	3.040	MW
42. <i>Apus apus</i>	9.7	0.988	44.9	1.652	G
43. <i>Arachnothera longirostra</i>	14.5	1.163	13	1.114	MW
44. <i>Ardea herodias</i>	3.3	0.520	1870	3.272	MW
45. <i>Arenaria interpres</i>	10.2	1.010	90	1.954	MW
46. <i>Asio flammeus</i>	3.2	0.501	406	2.609	MW
47. <i>Asio otus</i>	3.8	0.578	252	2.401	MW
48. <i>Authus pratensis</i>	15.9	1.202	18.9	1.276	G
49. <i>Barnardius zonarius</i>	5.2	0.719	137	2.137	MW
50. <i>Bombycilla garrulus</i>	13.2	1.120	72.5	1.860	G
51. <i>Bonasa umbellus</i>	3.7	0.568	644	2.809	MW
52. <i>Botaurus lentiginosus</i>	4.5	0.655	600	2.778	MW
53. <i>Bubo virginianus</i>	3.6	0.557	1450	3.161	MW

54. Buteo buteo	3.7	0.570	1012	3.005	MW
55. Buteo lineatus	3.2	0.506	658	2.818	MW
56. Cacactua tenuirostris	5.8	0.760	549.9	2.740	MW
57. Cacatua galerita	4.4	0.644	776.1	2.890	MW
58. Cacomantis variolosus	5.1	0.706	23.8	1.377	MW
59. Calidris canutus	6.8	0.831	130	2.114	MW
60. Callipepla gambelii	6.0	0.777	126.1	2.101	MW
61. Caprimulgus europaeus	8.3	0.921	77.4	1.889	G
62. Cardinalis cardinalis	12.3	1.090	41	1.613	MW
63. Cardinalis sinuata	12.3	1.088	32	1.505	MW
64. Carduelis carduelis	21.1	1.325	16.5	1.217	G
65. Carduelis pinus	20.8	1.318	14	1.146	MW
66. Carduelis tristis	24.6	1.390	13.6	1.134	MW
67. Carpodacus cassinii	12.4	1.092	27.4	1.438	MW
68. Carpodacus erythrinus	16.6	1.220	21.6	1.334	G
69. Carpodacus mexicanus	15.2	1.182	20.4	1.310	MW
70. Casurarius bennetti	1.4	0.152	17600	4.246	MW
71. Catharactus skua	4.9	0.690	970	2.987	MW
72. Centropus senegalensis	8.6	0.935	175	2.243	MW
73. Certhilauda erythrochlamys	15.1	1.179	27.3	1.436	MW
74. Chalcophaps inidica	6.4	0.806	124	2.093	MW
75. Charadius dubius	10.9	1.038	44	1.643	G
76. Chauna chavaria	2.6	0.419	2620	3.418	MW
77. Chloris chloris	16.8	1.226	28.2	1.450	G
78. Chloropsis sonnerati	9.5	0.979	39.7	1.599	MW
79. Chlorostilbon mellisugus	50.0	1.699	2.9	0.462	MW
80. Chordeiles minor	6.1	0.787	72	1.857	MW
81. Cinclus mexicanus	9.2	0.962	50.2	1.701	MW
82. Coccythraustes coccythraustes	14.5	1.160	48.3	1.684	G
83. Coereba flaveola	21.5	1.332	10	1.000	MW
84. Coleus monedula	7.3	0.861	209.0	2.320	G
85. Colinus virginianus	5.7	0.759	194	2.288	MW
86. Colius castanotus	15.0	1.176	69	1.839	MW
87. Colius colius	5.0	0.703	35.1	1.545	MW
88. Colius striatus	4.6	0.665	51	1.708	MW
89. Columba leucomela	5.3	0.728	456	2.659	MW
90. Columba livia	4.5	0.654	368	2.566	G
91. Columba palumbus	4.0	0.604	493	2.693	G
92. Columba unicincta	5.4	0.732	318	2.502	MW
93. Contopus virens	18.5	1.267	13.9	1.143	MW
94. Copsychus saularis	6.9	0.840	33.5	1.525	MW
95. Cornus ruficollis	5.1	0.712	660.0	2.820	G
96. Corvus brachyrhynchos	8.5	0.931	384.8	2.585	MW
97. Corvus corax	4.6	0.661	1203.0	3.080	G
98. Corvus corone cornix	6.4	0.807	518.0	2.714	G
99. Corvus frugilegus	6.7	0.827	390.0	2.591	G
100. Coturnix chinensis	8.2	0.914	44.9	1.652	MW
101. Coturnix coturnix	7.6	0.881	109	2.037	G
102. Coturnix japonica	8.5	0.930	115	2.061	MW
103. Coturnix pectoralis	6.6	0.821	95.8	1.981	MW
104. Crax alberti	2.4	0.371	2800	3.447	MW
105. Crax daubentoni	2.6	0.409	2800	3.447	MW
106. Crex crex	8.2	0.915	96	1.982	G
107. Cuculus canorus	7.5	0.876	111.6	2.048	G
108. Cyanocitta cristata	10.3	1.013	80.8	1.907	MW
109. Cygnus buccinator	2.3	0.362	8800	3.944	MW
110. Dendragapus obscurus	4.4	0.642	1131	3.053	MW
111. Dendrocopus major	8.9	0.950	117.0	2.068	G
112. Dendroica coronata	16.4	1.216	11.5	1.061	MW
113. Dendroica dominica	16.3	1.213	9.8	0.991	MW
114. Dendroica palmarum	15.8	1.199	9.8	0.991	MW
115. Dendroica pinus	14.9	1.174	12	1.079	MW
116. Diomedea chrysostoma	2.3	0.355	3753	3.574	MW

117.	Diomedea exulans	2.5	0.398	8130	3.910	MW
118.	Diomedea immutabilis	3.0	0.471	2522	3.402	MW
119.	Emberiza citrinella	16.3	1.212	26.8	1.428	G
120.	Emberiza hortulana	15.1	1.179	27.0	1.431	G
121.	Emberiza schoeniclus	17.1	1.233	17.6	1.246	G
122.	Empidonax virescens	14.6	1.163	12.3	1.090	MW
123.	Eolophus roseicapillus	4.6	0.667	268.7	2.429	MW
124.	Eremalauda dunni	13.5	1.130	20.6	1.314	MW
125.	Eremophila alpestris	11.9	1.076	26	1.415	MW
126.	Erithacus rubecula	16.0	1.204	17.6	1.246	G
127.	Estrilda melpoda	17.5	1.242	7.5	0.875	MW
128.	Estrilda troglodytes	20.1	1.302	7.5	0.875	G
129.	Eudocimus albus (Guara alba)	4.4	0.641	940	2.973	MW
130.	Eudynamys scolopacea	8.8	0.942	188	2.274	MW
131.	Eudytes chrysolophus	2.2	0.349	3870	3.588	G
132.	Eudytes cristatus	2.5	0.399	2330	3.367	G
133.	Eudytula minor	3.9	0.590	960	2.982	MW
134.	Eurostopodus argus (Eurostopodus guttatus)	4.6	0.665	88	1.944	MW
135.	Excalfactoria chinensis	9.2	0.966	44	1.643	G
136.	Falco sparverius	7.2	0.858	117	2.068	MW
137.	Falco subbuteo	6.2	0.795	208	2.318	G
138.	Falco tinnunculus	5.9	0.772	131	2.117	G
139.	Ficedula hypoleuca	19.9	1.299	11.7	1.068	G
140.	Fregata magnificens	2.6	0.411	1078	3.033	MW
141.	Fringilla coelebs	17.7	1.249	21.0	1.322	G
142.	Fringilla montifringilla	18.2	1.261	21.0	1.322	G
143.	Fulica atra	5.0	0.695	412	2.615	G
144.	Gallus gallus	2.2	0.346	2710	3.433	MW
145.	Garrulus glandarius	9.1	0.957	153.0	2.185	G
146.	Geococcyx californianus	5.1	0.711	284.7	2.454	MW
147.	Geopelia cuneata	6.8	0.834	39	1.591	MW
148.	Geopelia placida	6.8	0.834	52	1.716	MW
149.	Geophaps plumifera	4.9	0.687	81	1.908	MW
150.	Geophaps smithii	4.4	0.644	198	2.297	MW
151.	Glaucidium cuculoides	5.3	0.726	163	2.212	MW
152.	Glaucidium gnoma	8.2	0.912	54	1.732	MW
153.	Grus canadensis	2.1	0.321	3890	3.590	MW
154.	Grus paradisea	2.6	0.422	4030	3.605	MW
155.	Gypaetus barbatus	2.2	0.338	5070	3.705	MW
156.	Haematopus ostralegus	5.3	0.720	554	2.744	MW
157.	Himatione sanguinea	22.2	1.347	13.5	1.130	MW
158.	Hippolais icterina	20.2	1.305	12.5	1.097	G
159.	Hirundo rustica	16.4	1.214	18.4	1.265	G
160.	Hirundo tahitica	12.7	1.104	14.1	1.149	MW
161.	Icterus galbula	13.4	1.128	37.5	1.574	MW
162.	Jabiru mycteria	2.4	0.382	5470	3.738	MW
163.	Junco hyemalis	16.4	1.215	18	1.255	MW
164.	Lagopus lagopus	5.1	0.705	567	2.754	G
165.	Lagopus leucurus	7.2	0.860	326	2.513	MW
166.	Lanius collurio	14.2	1.152	27.0	1.431	G
167.	Lanius excubitor	11.2	1.051	72.4	1.860	G
168.	Larus argentatus	4.8	0.682	1000	3.000	MW
169.	Larus atricilla	6.8	0.833	275.6	2.440	MW
170.	Larus canus	5.2	0.718	431	2.634	G
171.	Larus ridibundus	6.1	0.784	306	2.486	G
172.	Leptoptilos javanicus	2.6	0.416	5710	3.757	MW
173.	Leptotila verreauxi	6.8	0.830	131	2.117	MW
174.	Leucosarcia melanoleuca	3.8	0.581	445	2.648	MW
175.	Lichenostomus virescens	14.2	1.151	25	1.398	MW
176.	Lichmera indistincta	23.1	1.364	9	0.954	MW
177.	Lonchura fuscans	10.2	1.009	9.5	0.978	MW
178.	Lonchura maja	11.7	1.069	12.8	1.107	MW

179.	Lonchura malacca	11.9	1.074	11.8	1.072	MW
180.	Lonchura striata	19.7	1.295	10.1	1.004	G
181.	Loxia curvirostra	15.2	1.183	39.4	1.595	G
182.	Loxia pytiopsittacus	14.9	1.173	53.7	1.730	G
183.	Loxoides baileui	12.9	1.109	36	1.556	MW
184.	Lullula arborea	14.7	1.169	33.2	1.521	G
185.	Luscinia svecica	17.3	1.237	20.8	1.318	G
186.	Macronectes giganteus	2.8	0.446	4780	3.679	MW
187.	Malacopteron cinereum	13.5	1.130	15.8	1.199	MW
188.	Manacus vitellinus	15.0	1.175	15.5	1.190	MW
189.	Megadyptes antipodes	2.4	0.380	4800	3.681	MW
190.	Meitihreptus lunatus	17.4	1.241	14.3	1.155	MW
191.	Melanerpes formicivorus	10.1	1.004	73	1.863	MW
192.	Melopsittacus undulatus	9.8	0.992	33.6	1.526	G
193.	Melospiza georgiana	14.2	1.151	14.9	1.173	MW
194.	Melospiza melodia	13.1	1.117	19.1	1.281	MW
195.	Merops viridis	8.7	0.941	33.8	1.529	MW
196.	Motacilla alba	15.5	1.189	18.2	1.260	G
197.	Motacilla flava	17.5	1.243	14.7	1.167	G
198.	Muscicapa striata	17.1	1.234	14.4	1.158	G
199.	Myiarchus crinitus	11.3	1.053	33.9	1.530	MW
200.	Nectarinia venusta	19.7	1.295	7.1	0.851	MW
201.	Neophema petrophila	13.1	1.117	48.4	1.685	MW
202.	Nucifraga caryocatactes	9.2	0.962	147.0	2.167	G
203.	Nyctea scandiaca	2.1	0.318	2026	3.307	MW
204.	Nymphicus hollandicus	8.0	0.906	85.6	1.932	G
205.	Oceanodroma furcata	10.1	1.004	44.6	1.649	MW
206.	Ocyphaps lophotes	5.8	0.764	187	2.272	MW
207.	Oreotrochilus estella	22.9	1.359	8.4	0.924	MW
208.	Oriolus oriolus	10.0	1.000	64.9	1.812	G
209.	Otus asio	3.5	0.548	166	2.220	MW
210.	Otus trichopsis	3.7	0.570	120	2.079	MW
211.	Padda oryzivora	12.1	1.084	25.4	1.405	MW
212.	Parus ater	22.0	1.342	10.8	1.033	G
213.	Parus atricapillus	24.5	1.389	10.3	1.013	MW
214.	Parus major	20.1	1.303	16.4	1.215	G
215.	Parus varius	20.3	1.307	17.7	1.248	G
216.	Passer domesticus bactrianus	15.9	1.200	23.2	1.365	G
217.	Passer montanus	17.9	1.253	22.0	1.342	G
218.	Passerculus sandwichensis	13.9	1.143	15.9	1.201	MW
219.	Patagona gigas	15.0	1.175	19.1	1.281	MW
220.	Pelacanoides urinatrix	10.3	1.011	136	2.134	MW
221.	Pelecanus conspicillatus	3.6	0.551	5090	3.707	MW
222.	Pelecanus occidentalis	3.4	0.533	3038	3.483	MW
223.	Penelope purpurescens	2.7	0.425	2040	3.310	MW
224.	Perdix perdix	4.3	0.634	501	2.700	G
225.	Perisoreus canadensis	9.5	0.979	71.2	1.852	MW
226.	Pernis apivorus	3.6	0.555	652	2.814	G
227.	Phalacrocorax auritus	4.1	0.616	1330	3.124	MW
228.	Phalaenoptilus nuttalli	4.4	0.646	35	1.544	MW
229.	Phaps chalcoptera	5.0	0.702	304	2.483	MW
230.	Phaps elegans	6.5	0.814	190	2.279	MW
231.	Phaps histrionica	5.0	0.703	257	2.410	MW
232.	Philidonyris novaehollandiae	18.3	1.263	17.3	1.238	MW
233.	Phoenicopterus ruber	5.0	0.701	3040	3.483	MW
(Phoenicopterus antiquorum)						
234.	Phoeniculus purpureus	2.3	0.371	74.07	1.870	MW
235.	Phoenicurus ochruros	17.4	1.241	13.9	1.143	G
236.	Phoenicurus phoenicurus	17.9	1.253	13.0	1.114	G
237.	Phylidonyris melanops	15.6	1.193	18.8	1.274	MW
238.	Phylloscopus collybita	20.0	1.302	8.2	0.914	G
239.	Phylloscopus sibilatrix	19.0	1.279	9.2	0.964	G
240.	Phylloscopus trochilus	19.5	1.289	10.7	1.029	G

241.	<i>Pica nuttalli</i>	9.7	0.985	151.9	2.182	MW
242.	<i>Pica pica</i>	7.5	0.877	158.9	2.201	MW
243.	<i>Picoides major</i>	8.9	0.949	117	2.068	MW
244.	<i>Picoides pubescens</i>	17.7	1.247	21.7	1.336	MW
245.	<i>Pinicola enucleator</i>	13.8	1.141	78.4	1.894	G
246.	<i>Pipra mentalis</i>	15.8	1.198	12.3	1.090	MW
247.	<i>Pluvialis dominica</i>	5.5	0.737	118	2.072	MW
248.	<i>Pluvialis squatarola</i>	7.9	0.896	226	2.354	MW
249.	<i>Podargus ocellatus</i>	3.9	0.592	145	2.161	MW
250.	<i>Podargus strigoides</i>	2.7	0.434	380.3	2.580	MW
251.	<i>Pooectes gramineus</i>	12.6	1.101	21.5	1.332	MW
252.	<i>Protonotaria citrea</i>	15.5	1.192	12.8	1.107	MW
253.	<i>Prunella modularis</i>	19.4	1.287	16.8	1.225	G
254.	<i>Psaltiriparus minimus</i>	22.0	1.342	5.5	0.740	MW
255.	<i>Pterocles orientalis</i>	5.0	0.702	386.4	2.587	MW
256.	<i>Pterodroma phaeopygia</i>	12.4	1.092	425	2.628	MW
257.	<i>Ptilinopus melanospila</i>	5.0	0.697	98	1.991	MW
258.	<i>Ptilinopus superbus</i>	6.3	0.798	120.4	2.081	MW
259.	<i>Puffinus griseus</i>	3.9	0.591	740	2.869	MW
260.	<i>Pycnonotus finlaysoni</i>	8.4	0.924	26.3	1.420	MW
261.	<i>Pycnonotus goiavier</i>	8.6	0.936	28.6	1.456	MW
262.	<i>Pygoscelis adeliae</i>	3.1	0.489	3970	3.599	MW
263.	<i>Pygoscelis papua</i>	3.0	0.470	6290	3.799	MW
264.	<i>Pyrrhula pyrrhula</i>	18.2	1.259	30.4	1.483	G
265.	<i>Regulus regulus</i>	26.5	1.423	5.5	0.740	G
266.	<i>Riparia riparia</i>	17.1	1.233	13.6	1.134	G
267.	<i>Saxicola rubetra</i>	16.9	1.228	14.3	1.155	G
268.	<i>Sayornis phoebe</i>	15.9	1.202	21.6	1.334	MW
269.	<i>Scardefella inca</i>	6.2	0.794	40.5	1.607	MW
270.	<i>Scolopax minor</i>	6.8	0.833	156.7	2.195	MW
271.	<i>Scolopax rusticola</i>	5.0	0.701	430	2.633	G
272.	<i>Sephanoides sephaniodes</i>	17.8	1.250	5.74	0.759	MW
273.	<i>Serinus canaria</i>	17.1	1.234	13.3	1.124	G
274.	<i>Sialia mexicana</i>	15.4	1.187	27.5	1.439	MW
275.	<i>Spermestes cucullatus</i>	7.3	0.866	10.62	1.026	MW
276.	<i>Spheniscus humboldti</i>	2.5	0.390	3870	3.588	MW
277.	<i>Spinus spinus</i>	20.8	1.317	14.0	1.146	G
278.	<i>Spizella arborea</i>	19.8	1.297	16.6	1.220	MW
279.	<i>Spizella passerina</i>	16.3	1.212	11.9	1.076	MW
280.	<i>Sterna maxima</i>	6.7	0.828	373	2.572	MW
281.	<i>Streptopelia senegalensis</i>	7.9	0.895	108	2.033	G
282.	<i>Streptopelia turtur</i>	7.4	0.869	154	2.188	G
283.	<i>Strix aluco</i>	4.0	0.602	520	2.716	MW
284.	<i>Strix occidentalis</i>	4.7	0.671	571	2.757	MW
285.	<i>Struthio camelus</i>	0.6	-0.200	100000	5.000	MW
286.	<i>Sturnus vulgaris</i>	12.0	1.078	75.0	1.875	G
287.	<i>Sula dactylatra</i>	4.3	0.631	1289	3.110	MW
288.	<i>Sylvia atricapilla</i>	19.0	1.279	21.9	1.340	G
289.	<i>Sylvia borin</i>	16.8	1.225	24.8	1.394	G
290.	<i>Sylvia curruca</i>	18.8	1.274	10.6	1.025	G
291.	<i>Sylvia nisoria</i>	15.1	1.180	21.4	1.330	G
292.	<i>Taeniopygia castanotis</i>	19.5	1.290	11.7	1.068	G
293.	<i>Tarsiger cyanurus</i>	16.0	1.205	14.8	1.170	G
294.	<i>Tetrao urogallus</i>	2.9	0.470	4010	3.603	G
295.	<i>Thamnophilus punctatus</i>	16.4	1.214	21	1.322	MW
296.	<i>Thinocorus rumicivorus</i>	5.6	0.747	55.5	1.744	MW
297.	<i>Tiaris canora</i>	19.9	1.299	7.8	0.892	G
298.	<i>Tringa ochropus</i>	10.2	1.010	90	1.954	MW
299.	<i>Troglodytes troglodytes</i>	23.7	1.374	9.0	0.954	G
300.	<i>Trogon rufus</i>	8.1	0.910	53	1.724	MW
301.	<i>Tudus viscivorus</i>	10.2	1.009	108.2	2.034	MW
302.	<i>Turdus iliacus</i>	12.5	1.095	58.0	1.763	G
303.	<i>Turdus merula</i>	11.3	1.052	82.6	1.917	G

304.	<i>Turdus philomelos</i>	11.6	1.063	62.8	1.798	G
305.	<i>Turdus viscivorus</i>	10.2	1.009	108.2	2.034	G
306.	<i>Turnix suscitator</i>	6.7	0.824	58.1	1.764	MW
307.	<i>Tyrannus tyrannus</i>	12.2	1.087	35.7	1.553	MW
308.	<i>Upupa epops</i>	8.2	0.916	67	1.826	MW
309.	<i>Uraeginthus bengalis</i>	17.0	1.232	9.1	0.959	G
310.	<i>Uria aalge</i>	7.1	0.852	956	2.980	MW
311.	<i>Uria lomvia</i>	6.9	0.838	989	2.995	MW
312.	<i>Vermivora pinus</i>	19.2	1.284	7.8	0.892	MW
313.	<i>Vidua paradisaea</i>	18.5	1.267	10.5	1.021	MW
314.	<i>Vultur gryphus</i>	1.6	0.217	10320	4.014	MW
315.	<i>Xiphorhynchus guttatus</i>	9.9	0.994	45.2	1.655	MW
316.	<i>Yynx torquilla</i>	11.3	1.052	31.8	1.502	G
317.	<i>Zenaida macroura</i>	6.0	0.777	123	2.090	MW
318.	<i>Zonotrichia querula</i>	13.4	1.127	33.3	1.522	MW
319.	<i>Zonotrichia albicollis</i>	13.8	1.139	20.2	1.305	MW
320.	<i>Zonotrichia leucophrys</i>	12.9	1.110	26.1	1.417	MW
321.	<i>Zosterops lateralis</i>	13.5	1.132	11	1.041	MW

Table **S6b**. Basal metabolic rate in birds (experimental data obtained by V. Gavrilov)

Notes on measurements procedure. Basal metabolism is the rate of energy utilization by a bird at complete rest, without the energetic costs of digestion and assimilation of food or cold stress, i.e. the rate of energy utilization of fasting, inactive birds in the zone of thermoneutrality. Basal metabolism was determined by measuring rates of oxygen consumption in closed boxes of different volumes, where CO₂ was absorbed using NaOH or KOH (closed-circuit respirometry). Basal oxygen consumption was measured in birds under postabsorptive conditions at night (N), as indicated in Table **S6a**, at different ambient temperatures that allowed strict determination of the thermoneutrality zone. The larger birds were deprived of food from mid-day, and the smaller birds from 3 to 4 hours before the nightfall. The birds were placed singly in small cages, which were then placed in plexiglass chambers in the dark. The chamber sizes varied from 3 to 25 liters, depending on the size of the bird. The flow of air through the chamber was controlled, and after stabilizing chamber temperature, the outflow was connected to an oxygen measuring instrument. Actual measurements of oxygen consumption were never started earlier than 1-3 hours after dusk and always ended 1 to 2 hours before dawn. Each experiment lasted from 1 to 4 hours. Tests were done at the beginning and end of the experiments to make certain that the boxes were properly sealed. Measurements were also done in the daytime on birds with different amounts of food in the alimentary tract. **The number of individuals measured varied from two (*Tetrao urogallus*) or three-four (in some large nonpasserine birds) to eight-twelve (in most other species) and up to 20-35 (in well-studied species, like for example *Parus major*, *Fringilla coelebs*).** The birds were weighed early at the beginning and at the end of the experiment. If the change of body mass was small (not more than 1.5-2% of body mass), corrections were made using a caloric equivalent equal to 25.1 kJ g⁻¹. If the change of body mass was greater, the data were not used.

Notations:

Mg — mean body mass, g; *N* — number of measured birds;

D — measurements were made during the active (day-time) phase of the avian circadian cycle;

N — measurements were made during the resting (night-time) phase of the avian circadian cycle;

W — measurements were made during the nonproductive "winter" phase of the avian annual cycle;
S — measurements were made during the reproductive "summer" phase of the avian annual cycle;
A — measurements were made during the nonproductive "autumnal" (postmoult) phase of the avian annual cycle;
V — measurements were made during the early part of the productive "vernal" phase of the avian annual cycle;
Y — measurements were made during the whole year;
QkJday — whole-body basal metabolic rate, kJ ind⁻¹ day⁻¹.

Note: data in Table S6a marked "G" are taken from Table S6b and correspond to the minimum night-time (N) mass-specific basal metabolic rate for the corresponding species, qWkg = (QkJday/Mg)×10⁶/24/3600 W kg⁻¹.

Species	N	Mg	Season	Time	QkJday	References
<u>Sphenisciformes</u>						
Eudyptes cristatus	4	2330	W	N	504.5	Gavrilov, 1977
Eudyptes chrysolophus	4	3870	W	N	747.8	Gavrilov, 1977
Aptenodytes patagonica	3	11080	W	N	1899.2	Gavrilov, 1977
<u>Anseriformes</u>						
Aix sponsa	8	448	S	N	194.3	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Aix sponsa	8	448	S	D	221.5	Gavrilov, 1985ab, 1997
Aix sponsa	8	468	W	N	205.6	Gavrilov, 1981, 1997, 1999ab
Aix sponsa	8	468	W	D	273.4	Gavrilov, 1981, 1997
Anas penelope	4	723	S	N	244.1	Gavrilov, 1980abc, 1982ab, 1997
Anas penelope	4	718	W	N	260.4	Gavrilov, 1980abc, 1982ab, 1997
Anas platyrhynchos	12	1020	S	N	351.7	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Anas platyrhynchos	12	1020	S	D	415.6	Gavrilov, 1985ab, 1997
Anas platyrhynchos	12	1132	W	N	435.4	Gavrilov, 1981, 1997, 1999ab
Anas platyrhynchos	12	1132	W	D	566.9	Gavrilov, 1981, 1997
Anser anser	4	3250	S	N	937.9	Gavrilov, Dolnik, 1985
<u>Falconiformes</u>						
Falco tinnunculus	4	131	A	N	67.0	Gavrilov, 1985ab
Accipiter nisus	6	135	S	N	82.1	Gavrilov, Dolnik, 1985
Falco subbuteo	4	208	A	N	112.2	Gavrilov, Dolnik, 1985
Pernis apivorus	2	652	S	N	202.2	Gavrilov, Dolnik, 1985
<u>Galliformes</u>						
Excalfactoria chinensis	6	44	S	N	35.15	Gavrilov, 1985ab
Excalfactoria chinensis	6	44	S	D	35.56	Gavrilov, 1985ab
Excalfactoria chinensis	6	41	W	N	50.7	Gavrilov, 1981
Excalfactoria chinensis	6	41	W	D	60.3	Gavrilov, 1981
Coturnix coturnix	4	97	S	N	77.0	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Coturnix coturnix	4	97	S	D	83.2	Gavrilov, 1985ab, 1997
Coturnix coturnix	4	109	W	N	71.6	Gavrilov, 1981, 1997, 1999ab
Coturnix coturnix	4	109	W	D	85.4	Gavrilov, 1981, 1997
Perdix perdix	5	483	S	N	207.3	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Perdix perdix	5	483	S	D	225.9	Gavrilov, 1985ab, 1997
Perdix perdix	5	501	W	N	186.3	Gavrilov, 1981, 1997, 1999ab
Perdix perdix	5	501	W	D	235.3	Gavrilov, 1981, 1997
Lagopus lagopus	6	524	S	N	268.8	Gavrilov, 1980abc, 1982ab, 1997, 1999ab

Lagopus lagopus	6	524	S	D	306.4	Gavrilov, 1985ab, 1997
Lagopus lagopus	6	567	W	N	248.3	Gavrilov, 1981, 1997, 1999ab
Lagopus lagopus	6	567	W	D	328.7	Gavrilov, 1981, 1997
Alectoris graeca	4	620	S	N	246.6	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Alectoris graeca	4	633	W	N	219.0	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Tetrao urogallus ♀	2	3900	S	N	1030.0	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Tetrao urogallus ♀	2	4010	W	N	1021.6	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
<u>Gruiformes</u>						
Crex crex	4	96	S	N	68.2	Gavrilov, Dolnik, 1985
Fulica atra	3	412	S	N	176.3	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Fulica atra	3	436	W	N	204.3	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
<u>Charadriiformes</u>						
Charadrius dubius	4	36	S	N	36.0	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Charadrius dubius	4	44	W	N	41.5	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Larus ridibundus	5	285	S	N	173.3	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Larus ridibundus	5	285	S	D	194.1	Gavrilov, 1985ab, 1997
Larus ridibundus	5	306	W	N	160.8	Gavrilov, 1981, 1997, 1999ab
Larus ridibundus	5	306	W	D	193.0	Gavrilov, 1981, 1997
Larus canus	3	428	S	N	201.0	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Larus canus	3	428	S	D	215.0	Gavrilov, 1985ab, 1997
Larus canus	3	431	W	N	194.3	Gavrilov, 1981, 1997, 1999ab
Larus canus	3	431	W	D	251.2	Gavrilov, 1981, 1997
Scolopax rusticola	4	430	S	N	186.7	Gavrilov, 1981
<u>Columbiformes</u>						
Streptopelia senegalensis	3	108	S	N	73.3	Gavrilov, Dolnik, 1985
Streptopelia turtur	4	154	A	N	98.4	Gavrilov, Dolnik, 1985
Columba livia	6	353	W	N	160.4	Gavrilov, 1981
Columba livia	6	353	W	D	178.8	Gavrilov, 1981
Columba livia	6	368	S	N	143.2	Gavrilov, 1985ab
Columba livia	6	368	S	D	154.4	Gavrilov, 1985ab
Columba palumbus	4	493	A	N	171.3	Gavrilov, Dolnik, 1985
<u>Psittaciformes</u>						
Melopsittacus undulatus	18	25,2	S	N	26.0	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Melopsittacus undulatus	18	25,2	S	D	28.0	Gavrilov, 1985ab, 1997
Melopsittacus undulatus	18	33,6	W	N	28.5	Gavrilov, 1981, 1997, 1999ab
Melopsittacus undulatus	18	33,6	W	D	31.4	Gavrilov, 1981, 1997
Agapornis roseicollis	6	48.1	S	N	40.2	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Agapornis roseicollis	6	48.1	S	D	44.0	Gavrilov, 1985ab, 1997
Agapornis roseicollis	6	48.4	W	N	40.2	Gavrilov, 1981, 1997, 1999ab
Agapornis roseicollis	6	48.4	W	D	53.2	Gavrilov, 1981, 1997
Agapornis fisheri	3	56.7	W	N	45.6	Gavrilov, Dolnik, 1985
Nymphicus hollandicus	5	85.6	S	N	59.5	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Nymphicus hollandicus	5	85.6	S	D	65.4	Gavrilov, 1985ab, 1997
Nymphicus hollandicus	5	94.3	W	N	74.5	Gavrilov, 1981, 1997, 1999ab
Nymphicus hollandicus	5	94.3	W	D	88.3	Gavrilov, 1981, 1997
<u>Cuculiformes</u>						
Cuculus canorus	4	111.6	S	N	72.4	Gavrilov, Dolnik, 1985
<u>Strigiformes</u>						
Asio otus	6	236	S	N	113.0	Gavrilov, Dolnik, 1985
<u>Caprimulgiformes</u>						
Caprimulgus europeus	3	77.4	S	N	55.7	Gavrilov, Dolnik, 1985
<u>Apodiformes</u>						
Apus apus	6	44.9	S	N	37.7	Gavrilov, 1985ab
<u>Coraciiformes</u>						
Alcedo atthis	4	34.3	S	N	32.7	Gavrilov, Dolnik, 1985

<u>Piciformes</u>						
Yynx torquilla	6	31.8	S	N	31.0	Gavrilov, Dolnik, 1985
Dendrocopus major	7	98.0	S	N	77.5	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Dendrocopus major	7	117.0	W	N	90.0	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
<u>Passeriformes</u>						
Regulus regulus	22	5.5	S	N	12.6	Gavrilov, 1997, 1999ab
Regulus regulus	22	5.5	W	N	15.9	Gavrilov, 1997, 1999ab
Estrilda troglodytes	6	7.5	S	N	13.0	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Estrilda troglodytes	6	7.5	S	D	14.0	Gavrilov, 1985ab, 1997
Estrilda troglodytes	6	7.7	W	N	13.4	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Estrilda troglodytes	6	7.7	W	D	14.6	Gavrilov, 1985ab, 1997
Tiaris canora	4	7.6	S	N	13.4	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Tiaris canora	4	7.8	W	N	13.4	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Phylloscopus collybita	6	8.2	A	N	14.2	Gavrilov, Dolnik, 1985
Aegithalos caudatus	17	8.9	S	N	17.2	Gavrilov, 1974, 1980abc, 1982ab, 1997, 1999ab
Aegithalos caudatus	17	8.8	W	N	21.8	Gavrilov, 1974, 1980abc, 1982ab, 1997, 1999ab
Troglodytes troglodytes	16	9.0	S	N	18.4	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Troglodytes troglodytes	16	9.2	W	N	20.9	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Uraeginthus bengalis	5	9.1	S	N	13.4	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Uraeginthus bengalis	5	9.2	W	N	14.2	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Phylloscopus sibilatrix	4	9.2	S	N	15.1	Gavrilov, Dolnik, 1985
Lonchura striata	6	10.1	S	N	17.2	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Lonchura striata	6	10.3	W	N	18.4	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Sylvia curruca	8	10.6	S	N	17.2	Gavrilov, Dolnik, 1985
Phylloscopus trochilus	7	10.7	W	N	18.0	Gavrilov, Dolnik, 1985
Acrcocephalus palustris	4	10.8	S	N	17.6	Gavrilov, Dolnik, 1985
Parus ater	18	10.8	S	N	20.5	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Parus ater	18	10.8	S	D	22.6	Gavrilov, 1985ab, 1997
Parus ater	18	11.0	W	N	23.4	Gavrilov, 1981, 1997, 1999ab
Parus ater	18	11.0	W	D	27.7	Gavrilov, 1981, 1997
Taeniopygia castanotis	14	11.7	S	N	19.7	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Taeniopygia castanotis	14	11.8	S	D	20.3	Gavrilov, 1985ab, 1997
Taeniopygia castanotis	14	11.8	W	N	20.1	Gavrilov, 1981, 1997, 1999ab
Taeniopygia castanotis	14	11.8	W	D	22.6	Gavrilov, 1981, 1997
Acrocephalus schoenobaenus	3	11.5	S	N	18.8	Gavrilov, Dolnik, 1985
Ficedula hypoleuca	9	11.7	A	N	20.1	Gavrilov, Dolnik, 1985, Gavrilov et al., 1995b, 1998
Hippolais icterina	6	12.5	S	N	21.8	Gavrilov, Dolnik, 1985
Acanthis flammea	16	14.0	S	N	24.7	Gavrilov, 1974, 1980abc, 1982ab, 1997, 1999ab
Acanthis flammea	16	14.3	W	N	29.3	Gavrilov, 1974, 1980abc, 1982ab, 1997, 1999ab
Phoenicurus phoenicurus	4	13.0	S, A	N	20.1	Gavrilov, Dolnik, 1985
Serinus canaria	5	13.3	A	N	19.7	Gavrilov, Dolnik, 1985
Riparia riparia	3	13.6	A	N	20.1	Gavrilov, 1986
Phoenicurus ochruros	3	13.9	S	N	20.9	Gavrilov, Dolnik, 1985
Spinus spinus	18	14.0	S	N	25.1	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Spinus spinus	18	14.0	S	D	27.6	Gavrilov, 1985ab, 1997
Spinus spinus	18	14.2	W	N	28.5	Gavrilov, 1981, 1997, 1999ab
Spinus spinus	18	14.2	W	D	31.4	Gavrilov, 1981, 1997
Saxicola rubetra	4	14.3	S	N	20.9	Gavrilov, Dolnik, 1985
Muscicapa striata	3	14.4	S	N	21.3	Gavrilov, Dolnik, 1985
Motacilla flava	2	14.7	S	N	22.2	Gavrilov, Dolnik, 1985
Tarsiger cyanurus	5	14.8	W	N	20.5	Gavrilov, 1985ab

Parus major	20	16.4	S	N	28.5	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Parus major	20	16.4	S	D	31.6	Gavrilov, 1985ab, 1997, Gavrilov et al. 1995a
Parus major	20	17.1	W	N	32.2	Gavrilov, 1981, 1997, 1999ab
Parus major	20	17.1	W	D	35.6	Gavrilov, 1981, 1997, Gavrilov et al. 1995a
Carduelis carduelis	6	16.5	W	N	30.1	Gavrilov, 1982b
Prunella modularis	4	16.8	A	N	28.1	Gavrilov, Dolnik, 1985
Acanthis cannabina	4	16.9	A	N	29.3	Gavrilov, 1982b
Emberiza schoeniclus	3	17.6	A	N	26.0	Gavrilov, 1982b
Erithacus rubecula	18	17.6	S	N	26.0	Gavrilov, 1980abc, 1982ab
Erithacus rubecula	18	17.6	S	D	29.1	Gavrilov, 1985ab
Erithacus rubecula	18	17.6	W	N	24.3	Gavrilov, 1981
Erithacus rubecula	18	17.6	W	D	26.4	Gavrilov, 1981
Parus varius	5	17.7	W	N	31.0	Gavrilov, 1985ab
Parus varius	5	17.7	W	D	37.2	Gavrilov, 1985ab
Hirundo rustica	4	18.4	S	N	26.0	Gavrilov, 1986
Motacilla alba	8	18.0	S	N	26.0	Gavrilov, 1980abc, 1982ab
Motacilla alba	8	18.2	W	N	24.3	Gavrilov, 1980abc, 1982ab
Authus pratensis	3	18.9	S	N	26.0	Gavrilov, 1982b
Anthus trivialis	5	19.7	A	N	29.3	Gavrilov, 1982b
Luscinia svecica	3	20.8	S	N	31.0	Gavrilov, 1982b
Fringilla coelebs	35	21.0	S	N	32.2	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Fringilla coelebs	35	21.0	S	D	39.0	Gavrilov, 1985ab, 1997
Fringilla coelebs	35	20.8	W	N	38.1	Gavrilov, 1981, 1997, 1999ab
Fringilla coelebs	35	20.8	W	D	41.5	Gavrilov, 1981, 1997, 1999ab
Fringilla montifringilla	12	21.0	A	N	33.1	Gavrilov, 1982b
Sylvia nisoria	3	21.3	S	N	33.1	Gavrilov, 1982b
Sylvia nisoria	3	21.4	W	N	28.0	Gavrilov, 1982b
Carpodacus erythrinus	14	21.2	S	N	31.8	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Carpodacus erythrinus	14	21.2	S	D	36.6	Gavrilov, 1985ab, 1997
Carpodacus erythrinus	14	21.6	W	N	31.0	Gavrilov, 1981, 1997, 1999ab
Carpodacus erythrinus	14	21.6	W	D	33.1	Gavrilov, 1981, 1997
Anthus campestris	2	21.8	S	N	33.1	Gavrilov, 1981
Sylvia atricapilla	8	21.9	A	N	36.0	Gavrilov, 1981
Emberiza hortulana	8	24.3	S	N	36.0	Gavrilov, 1980abc, 1982ab
Emberiza hortulana	8	27.0	W	N	35.2	Gavrilov, 1980abc, 1982ab
Passer montanus	7	22.0	S	N	34.0	Gavrilov, 1981
Passer montanus	7	22.3	A	N	35.2	Gavrilov, 1981
Passer domesticus bactrianus	32	23.0	S	N	31.8	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Passer domesticus bactrianus	32	23.2	W	N	31.8	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Sylvia borin	12	24.8	A	N	36.0	Gavrilov, 1982b
Passer domesticus	33	26.5	S	N	41.0	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Passer domesticus	33	26.5	S	D	47.2	Gavrilov, 1985ab, 1997
Passer domesticus	33	26.4	W	N	42.3	Gavrilov, 1981, 1997, 1999ab
Passer domesticus	33	26.4	W	D	44.8	Gavrilov, 1981, 1997
Emberiza citrinella	27	26.8	S	N	37.7	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Emberiza citrinella	27	26.8	S	D	43.3	Gavrilov, 1985ab, 1997
Emberiza citrinella	27	27.4	W	N	43.1	Gavrilov, 1981, 1997, 1999ab
Emberiza citrinella	27	27.4	W	D	49.4	Gavrilov, 1981, 1997
Lanius collurio	4	27.0	S	N	33.1	Gavrilov, 1982b
Chloris chloris	17	28.2	S	N	41.0	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Chloris chloris	17	28.2	S	D	46.4	Gavrilov, 1985ab, 1997
Chloris chloris	17	29.0	W	N	48.1	Gavrilov, 1981, 1997, 1999ab
Chloris chloris	17	29.0	W	D	51.9	Gavrilov, 1981, 1997
Loxia curvirostra	9	39.4	S	N	51.9	Gavrilov, 1980abc, 1982ab, 1997, 1999ab
Loxia curvirostra	9	42.7	W	N	58.2	Gavrilov, 1980abc, 1982ab, 1997, 1999ab

<i>Pyrrhula pyrrhula</i>	11	30.4	W	N	47.7	Gavrilov, 1982b
<i>Lullula arborea</i>	7	33.2	A	N	42.3	Gavrilov, 1982b
<i>Coccothraustes coccothraustes</i>	4	48.3	A	N	60.3	Gavrilov, 1982b
<i>Loxia pytiopsittacus</i>	6	53.7	W	N	69.1	Gavrilov, 1982b
<i>Turdus iliacus</i>	9	58.0	W	N	62.4	Gavrilov, 1979ab, 1981
<i>Turdus iliacus</i>	9	58.0	W	D	72.8	Gavrilov, 1981
<i>Turdus philomelos</i>	12	62.8	S	N	62.8	Gavrilov, 1979ab, 1980abc, 1982ab, 1997, 1999ab
<i>Turdus philomelos</i>	12	62.8	S	D	71.0	Gavrilov, 1985ab, 1997
<i>Turdus philomelos</i>	12	64.0	W	N	65.3	Gavrilov, 1979ab, 1980abc, 1982ab, 1997, 1999ab
<i>Turdus philomelos</i>	12	64.0	W	D	74.9	Gavrilov, 1982ab, 1997
<i>Oriolus oriolus</i>	3	64.9	S	N	56.1.	Gavrilov, 1982b
<i>Lanius excubitor</i>	4	72.4	A	N	70.3	Gavrilov, 1982b
<i>Bombycilla garrulus</i>	6	72.5	A	N	82.5	Gavrilov, Dolnik, 1985
<i>Sturnus vulgaris</i>	13	75.0	A	N	77.5	Gavrilov, 1982b
<i>Pinicola enucleator</i>	5	78.4	W	N	93.8	Gavrilov, Dolnik, 1985
<i>Turdus merula</i>	12	82.6	S	N	80.4	Gavrilov, 1979ab, 1980abc, 1982ab, 1997, 1999ab
<i>Turdus merula</i>	12	82.6	S	D	93.3	Gavrilov, 1997
<i>Turdus merula</i>	12	83.0	W	N	89.6	Gavrilov, 1979ab, 1980abc, 1982ab, 1997, 1999ab
<i>Turdus merula</i>	12	83.0	W	D	105.5	Gavrilov, 1981, 1997, 1999ab
<i>Turdus viscivorus</i>	9	108.2	W	N	95.5	Gavrilov, 1979ab
<i>Nucifraga caryocatactes</i>	11	147.0	W	N	116.4	Gavrilov, 1979ab
<i>Garrulus glandarius</i>	13	153.0	W	N	119.7	Gavrilov, 1979ab
<i>Pica pica</i>	6	202.0	W	N	148.6	Gavrilov, 1979ab
<i>Coleus monedula</i>	9	209.0	S	N	131.2	Gavrilov, 1985ab
<i>Coleus monedula</i>	9	209.0	S	D	151.2	Gavrilov, 1985ab
<i>Coleus monedula</i>	9	215.0	W	N	160.8	Gavrilov, 1979ab, 1981
<i>Coleus monedula</i>	9	215.0	W	D	167.5	Gavrilov, 1981
<i>Corvus frugilegus</i>	5	390.0	W	N	226.1	Gavrilov, 1979ab
<i>Corvus corone cornix</i>	11	518.0	S	N	286.8	Gavrilov, 1979ab, 1980abc, 1982ab, 1997, 1999ab
<i>Corvus corone cornix</i>	11	518.0	S	D	329.8	Gavrilov, 1985ab, 1997
<i>Corvus corone cornix</i>	11	540.0	W	N	330.8	Gavrilov, 1979ab, 1980abc, 1981, 1982ab, 1997, 1999ab
<i>Corvus corone cornix</i>	11	540.0	W	D	386.9	Gavrilov, 1981, 1997
<i>Cornus ruficollis</i>	4	660.0	W	N	293.5	Gavrilov, 1979ab
<i>Corvus corax</i>	7	1203.0	S	N	476.1	Gavrilov, 1979ab, 1980abc, 1981, 1982ab, 1997, 1999ab
<i>Corvus corax</i>	7	1203.0	S	D	518.9	Gavrilov, 1985ab, 1997
<i>Corvus corax</i>	7	1208	W	N	518.3	Gavrilov, 1979ab, 1980abc, 1982ab, 1997, 1999ab
<i>Corvus corax</i>	7	1208	W	D	618.0	Gavrilov, 1981, 1997

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Dataset S7. Dark respiration rates in cyanobacteria

Notes to Table S7:

Data on dark respiration rates in cyanobacteria (unicellular, filamentous and mat-forming species) are presented. Taxonomic status (the “**Order**” column) was determined for each genus following www.algaebase.org.

Abbreviations and universal conversions: DM – dry mass; WM – wet mass; N – nitrogen mass; Chl a – Chl a mass; C – carbon mass; Pr – protein mass; X/Y – X by Y mass ratio in the cell, e.g. DM/WM is the ratio of dry to wet cell mass; $1 \text{ W} = 1 \text{ J s}^{-1}$; $1 \text{ mol O}_2 = 32 \text{ g O}_2$.

Column “**U**” (mass units of respiration rate measurements): D – dry mass or Chl a mass with known Chl a/DM ratio; W – wet mass without information on DM/WM ratio; Chl – chlorophyll mass without information on Chl a/DM ratio; Pr – protein mass.

“**Original units**” are the units of dark respiration rate measurements as given in the original publication (“**Source**”); **qou** is the numeric value of dark respiration rate in the original units. E.g., if it is “ $\text{mg O}_2 (\text{g DM})^{-1} \text{ hr}^{-1}$ ” in the column “**Original units**” and “1.1” in the column “**qou**”, this means that dark respiration rate of the corresponding species, as given in the original publication indicated in the column “**Source**”, is $1.1 \text{ mg O}_2 (\text{g DM})^{-1} \text{ hr}^{-1}$.

qWkg is the original dark respiration rate **qou** converted to W (kg WM)^{-1} (Watts per kg wet mass) using the following conversion factors: $\text{C/DM} = 0.5$ (Kratz & Myers 1955; Bratbak & Dundas 1984; Gordillo et al. 1999; Stal & Moezelaar 1997), $\text{Chl a/DM} = 0.015$ (APHA 1992), $\text{Pr/DM} = 0.5$ (Otte et al. 1999; Zubkov et al. 1999; Stal & Moezelaar 1997) and $\text{DM/WM} = 0.3$ as a crude mean for all taxa applied in the analysis (SI Methods, Table S12a). If the Chl a/DM ratio is known (shown in the “**Comments**” column), while **qou** is per unit Chl a mass, the dark respiration rate is first calculated per unit dry mass and then converted to **qWkg** using the reference $\text{DM/WM} = 0.3$. Energy conversion: $1 \text{ ml O}_2 = 20 \text{ J}$.

TC is ambient temperature during measurements, degrees Celsius.

q25Wkg is dark respiration rate converted to 25°C using $Q_{10} = 2$, $\text{q25Wkg} = \text{qWkg} \times 2^{(25 - \text{TC})/10}$, dimension W (kg WM)^{-1} . For each species rows are arranged in the order of increasing **q25Wkg**.

Mpg: estimated cell mass, pg ($1 \text{ pg} = 10^{-12} \text{ g}$). In most cases it is estimated from linear dimensions (using geometric mean of the available linear size range) assuming spherical cell shape. For filamentous cyanobacteria **Mpg** is estimated from linear width as if it were a spherical cell of the same diameter, to be comparable to unicellular species. As argued in the paper, for plant it is the minimal linear

size (e.g., leaf thickness) rather than total mass that is energetically relevant. Square brackets around **Mpg** value indicate that this value was obtained from a different source than the source of dark respiration rate data. When converting cell volume to cell mass, cell density of 1 g ml⁻¹ was assumed.

Source: the first, unbracketed reference in this column is where the value of **qou** is taken from; references and data in square brackets refer to cell size determination.

Comments: this column provides relevant information on culture conditions and cellular composition of the studied species. C/N — carbon to cell nitrogen mass ratio; C/DM — carbon mass to dry mass ratio; DM/WM — dry mass to wet mass ratio; DM/V — dry mass to volume ratio (pg/μm³); C/Chl — carbon to chlorophyll mass ratio; C/V — carbon mass to cell volume ratio (pg/μm³); C/cell — C per cell (pg/cell); Pr/cell — pg protein per cell (pg/cell); AFDM/WM — ash-free dry mass to wet mass ratio; ODM — organic dry matter.

Log₁₀-transformed values of **q25Wkg** (W (kg WM)⁻¹), minimum for each species, were used in the analyses shown in Figures 1 and 2 and Table 1 in the paper (a total of 25 values for n = 25 species). The corresponding rows are highlighted in **blue**.

References within Table **S7** to Tables, Figures etc. refer to the corresponding items in the original literature indicated in the **Source** column.

Table **S7**. Dark respiration rates in cyanobacteria.

Species	U	Original units	qou	qWkg	q25Wkg	TC	Mpg	Order	Source	Comments
1. <i>Anabaena flos-aquae</i>	Chl	μmol O ₂ (mg Chl) ⁻¹ hr ⁻¹	2.4	1.3	1.30	25	[22]	Nostocales	Rubin et al. 1977 [estimated from images at UTEX Culture Collection (http://www.zo.utexas.edu/research/utex/), diam 3.5 μm, filamentous]	
2. <i>Anabaena variabilis</i>	D	μl O ₂ (mg DM) ⁻¹ hr ⁻¹	1.7	2.8	1.06	39	[14]	Nostocales	Kratz & Myers 1955 [estimated from image of ATCC 29413, diam 3 μm, filamentous, displayed at http://www.ibvf.cartuja.csic.es/Cultivos/Seccion_IV.htm (Instituto de Bioquímica Vegetal y Fotosíntesis, Cevilla, Spain)]	Cells stored in darkness for 24 hr before measurements

3.	Anabaena variabilis	D	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	8.4	14	5.31	39	[14]	Nostocales	Kratz & Myers 1955 [estimated from image of ATCC 29413, diam 3 μm , filamentous, displayed at http://www.ibvf.cartuja.csic.es/Cultivos/Seccion_IV.htm (Instituto de Bioquímica Vegetal y Fotosíntesis, Cevilla, Spain)]	growing cells ($\log_{10}k/\text{day}=0.55$) harvested and prepared for dark respiration measurements in less than 35 min
4.	Anabaena variabilis	Pr	$\mu\text{mol O}_2 (95 \text{ mg protein})^{-1} \text{ hr}^{-1}$	93	18	19.29	24	[14]	Nostocales	Haury & Spiller 1981 [estimated from image of ATCC 29413, diam 3 μm , filamentous, displayed at http://www.ibvf.cartuja.csic.es/Cultivos/Seccion_IV.htm (Instituto de Bioquímica Vegetal y Fotosíntesis, Cevilla, Spain)]	protein/packed cell volume=95 mg/ml Chl/packed cell volume=3.4 mg/ml; carbon-starved; respiration is characterized as "high" compared to the strain studied by Kratz & Myers (1955)
5.	Anacystis nidulans PCC 6301 (Synechococcus leopoliensis)	D	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	0.3	0.5	0.50	25	[4]	Chroococcales	Kratz & Myers 1955 [estimated from linear dimensions 1.6×2.2 μm assuming cylindrical form]	C/DM=0.483 H/DM=0.067 N/DM=0.094 ash/DM=0.044 [when grown at $\log_{10}k/\text{day}=1.0$]; cells stored in darkness for 24 hr before measurements
6.	Anacystis nidulans PCC 6301 (Synechococcus leopoliensis)	D	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	1.9	3.2	1.21	39	[4]	Chroococcales	Kratz & Myers 1955 [estimated from linear dimensions 1.6×2.2 μm assuming cylindrical form]	C/DM=0.483 H/DM=0.067 N/DM=0.094 ash/DM=0.044 [when grown at $\log_{10}k/\text{day}=1.0$]; cells stored in darkness for 24 hr before measurements
7.	Anacystis nidulans PCC 6301 (Synechococcus leopoliensis)	D	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	1.6	2.7	2.70	25	[4]	Chroococcales	Kratz & Myers 1955 [estimated from linear dimensions 1.6×2.2 μm assuming cylindrical form]	C/DM=0.483 H/DM=0.067 N/DM=0.094 ash/DM=0.044 [when grown at $\log_{10}k/\text{day}=1.0$]; growing cells ($\log_{10}k/\text{day}=0.3$) harvested and prepared for dark respiration measurements in less than 35 min
8.	Anacystis nidulans PCC 6301 (Synechococcus leopoliensis)	D	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	4.7	7.8	2.96	39	[4]	Chroococcales	Kratz & Myers 1955 [estimated from linear dimensions 1.6×2.2 μm assuming cylindrical form]	C/DM=0.483 H/DM=0.067 N/DM=0.094 ash/DM=0.044 [when grown at $\log_{10}k/\text{day}=1.0$]; growing cells ($\log_{10}k/\text{day}=0.55$) harvested and prepared for dark respiration measurements in less than 35 min

9. <i>Anacystis nidulans</i> PCC 6301 (<i>Synechococcus leopoliensis</i>)	D	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	7.5	12.5	4.74	39	[4]	Chroococcales	Kratz & Myers 1955 [estimated from linear dimensions $1.6 \times 2.2 \mu\text{m}$ assuming cylindrical form]	C/DM=0.483 H/DM=0.067 N/DM=0.094 ash/DM=0.044 [when grown at $\log_{10}k/\text{day}=1.0$]; growing cells ($\log_{10}k/\text{day}=2.50$) harvested and prepared for dark respiration measurements in less than 35 min
10. <i>Anacystis nidulans</i> PCC 6301 (<i>Synechococcus leopoliensis</i>)	D	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	2.9	4.8	4.80	25	[4]	Chroococcales	Kratz & Myers 1955 [estimated from linear dimensions $1.6 \times 2.2 \mu\text{m}$ assuming cylindrical form]	C/DM=0.483 H/DM=0.067 N/DM=0.094 ash/DM=0.044 [when grown at $\log_{10}k/\text{day}=1.0$]; cells stored in darkness for 24 hr before measurements nitrogen-deprived cells
11. <i>Anacystis nidulans</i> PCC 6301 (<i>Synechococcus leopoliensis</i>)	Chl	$\mu\text{mol O}_2 (\text{mg Chl})^{-1} \text{ hr}^{-1}$	32	18	6.36	40	[4]	Chroococcales	Romero et al. 1989 [estimated from linear dimensions $1.6 \times 2.2 \mu\text{m}$ (Kratz & Myers 1955) assuming cylindrical form]	
12. <i>Anacystis nidulans</i> PCC 6301 (<i>Synechococcus leopoliensis</i>)	Chl	$\mu\text{mol O}_2 (\text{mg Chl})^{-1} \text{ hr}^{-1}$	41	23	8.13	40	[4]	Chroococcales	Romero et al. 1989 [estimated from linear dimensions $1.6 \times 2.2 \mu\text{m}$ (Kratz & Myers 1955) assuming cylindrical form]	KNO ₃ added
13. <i>Anacystis nidulans</i> PCC 6301 (<i>Synechococcus leopoliensis</i>)	D	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	4.3-5.9	8.5	8.50	25	[4]	Chroococcales	Biggins 1969 [estimated from linear dimensions $1.6 \times 2.2 \mu\text{m}$ (Kratz & Myers 1955) assuming cylindrical form]	stable respiration during 8 hr in darkness
14. <i>Anacystis nidulans</i> PCC 6301 (<i>Synechococcus leopoliensis</i>)	D	$\mu\text{l O}_2 (\text{mg DM})^{-1} (40 \text{ min})^{-1}$	10	25	9.47	39	[4]	Chroococcales	Doolittle & Singer 1974, Fig. 6 [estimated from linear dimensions $1.6 \times 2.2 \mu\text{m}$ (Kratz & Myers 1955) assuming cylindrical form]	DM/cell=1.1 pg; cells harvested in log-phase
15. <i>Anacystis nidulans</i> PCC 6301 (<i>Synechococcus leopoliensis</i>)	Chl	$\mu\text{mol O}_2 (\text{mg Chl})^{-1} \text{ hr}^{-1}$	51	29	10.25	40	[4]	Chroococcales	Romero et al. 1989 [estimated from linear dimensions $1.6 \times 2.2 \mu\text{m}$ (Kratz & Myers 1955) assuming cylindrical form]	NH ₄ Cl added
16. <i>Aphanocapsa</i> PCC 6714	W	$\text{nmol O}_2 (\text{mg WM})^{-1} \text{ min}^{-1}$	<0.02	0.15	0.15	25	[10]	<i>Synechococcales</i>	Pelroy & Bassham 1973 [Stanier et al. 1971, Fig. 6]	stabilized respiration of cells harvested during late-log phase
17. <i>Coccochloris penicystis</i>	Chl	$\mu\text{mol O}_2 (\text{mg Chl})^{-1} \text{ hr}^{-1}$	32.3	18	6.36	40	[1]	Chroococcales	Coleman & Colman 1980 [Stanier et al. 1971, Fig. 2 PCC 6307, ellipsoid $1 \times 2 \mu\text{m}$]	
18. <i>Coccochloris penicystis</i>	Chl	$\mu\text{mol O}_2 (\text{mg Chl})^{-1} \text{ hr}^{-1}$	24.4	14	9.90	30	[1]	Chroococcales	Coleman & Colman 1980 [Stanier et al. 1971, Fig. 2 PCC 6307, ellipsoid $1 \times 2 \mu\text{m}$]	

19. <i>Coccochloris penicystis</i>	Chl	$\mu\text{mol O}_2 (\text{mg Chl})^{-1} \text{ hr}^{-1}$	15.2	8.5	12.02	20	[1]	Chroococcales	Coleman & Colman 1980 [Stanier et al. 1971, Fig. 2 PCC 6307, ellipsoid 1×2 μm]	
20. <i>Gloeobacter violaceus</i>	Chl	$\mu\text{mol O}_2 (\text{mg Chl})^{-1} \text{ hr}^{-1}$	50	28	28.00	25	[0.7]	Chroococcales	Koenig & Schmidt 1995 [Pasteur Culture Collection, www.pasteur.fr , width/diameter 1-1.2 μm]	
21. <i>Gloeotheca</i> sp. PCC 6909	Pr	$\mu\text{mol O}_2 (\text{mg protein})^{-1} \text{ hr}^{-1}$	0.208	4	5.66	20	[180]	Chroococcales	Ortega-Calvo & Stal 1994 [Stanier et al. 1971, Fig. 7, ellipsoid 6×10 μm]	
22. <i>Gloeotheca</i> sp. PCC 6909	Pr	$\mu\text{mol O}_2 (\text{mg protein})^{-1} \text{ hr}^{-1}$	0.696	13	18.39	20	[180]	Chroococcales	Ortega-Calvo & Stal 1994 [Stanier et al. 1971, Fig. 7, ellipsoid 6×10 μm]	
23. <i>Gloeotheca</i> sp. PCC 6909	Pr	$\mu\text{mol O}_2 (\text{mg protein})^{-1} \text{ hr}^{-1}$	0.856	16	22.63	20	[180]	Chroococcales	Ortega-Calvo & Stal 1994 [Stanier et al. 1971, Fig. 7, ellipsoid 6×10 μm]	
24. <i>Gloeotheca</i> sp. PCC 6909	Pr	$\mu\text{mol O}_2 (\text{mg protein})^{-1} \text{ hr}^{-1}$	1.061	20	28.28	20	[180]	Chroococcales	Ortega-Calvo & Stal 1994 [Stanier et al. 1971, Fig. 7, ellipsoid 6×10 μm]	
25. <i>Gloeotheca</i> sp. PCC 6909	Pr	$\mu\text{mol O}_2 (\text{mg protein})^{-1} \text{ hr}^{-1}$	1.714	32	45.26	20	[180]	Chroococcales	Ortega-Calvo & Stal 1994 [Stanier et al. 1971, Fig. 7, ellipsoid 6×10 μm]	
26. <i>Gloeotheca</i> sp. PCC 6909	Pr	$\mu\text{mol O}_2 (\text{mg protein})^{-1} \text{ hr}^{-1}$	1.927	36	50.91	20	[180]	Chroococcales	Ortega-Calvo & Stal 1994 [Stanier et al. 1971, Fig. 7, ellipsoid 6×10 μm]	
27. <i>Gloeotheca</i> sp. PCC 6909	Pr	$\mu\text{mol O}_2 (\text{mg protein})^{-1} \text{ hr}^{-1}$	2.149	40	56.57	20	[180]	Chroococcales	Ortega-Calvo & Stal 1994 [Stanier et al. 1971, Fig. 7, ellipsoid 6×10 μm]	
28. <i>Nostoc commune</i>	D	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.067	2	0.6	0.52	27	[8]	Nostocales	Scherer et al. 1984 [Pereira et al. 2005, Fig. 8, cell width 2.5 μm]	Chl a/WM=0.0060-0.0098%; DM/WM=0.067; 6 hrs incubation
29. <i>Nostoc commune</i> var. <i>flagelliforme</i>	D	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.172	5.5	1.4	1.22	27	[8]	Nostocales	Scherer et al. 1984 [Pereira et al. 2005, Fig. 8, cell width 2.5 μm]	DM/WM=0.172; 6 hrs incubation
30. <i>Nostoc muscorum</i> G.	D	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	1.1	1.8	1.80	25	[33]	Nostocales	Kratz & Myers 1955 [estimated from images at UTEX Culture Collection (http://www.zo.utexas.edu/research/utex/), diam 4 μm , filamentous]	Cells stored in darkness for 24 hr before measurements
31. <i>Nostoc muscorum</i> G.	D	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	4.4	7.3	7.30	25	[33]	Nostocales	Kratz & Myers 1955 [estimated from images at UTEX Culture Collection (http://www.zo.utexas.edu/research/utex/), diam 4 μm , filamentous]	growing cells harvested and prepared for dark respiration measurements in less than 35 min
32. <i>Nostoc</i> sp. strain Mac	D	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	23	26	9.85	39		Nostocales	Ingram et al. 1973	
33. <i>Nostoc</i> sp. strain Mac	D	$\mu\text{mol O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	0.90	34	12.88	39		Nostocales	Bottomley & van Baalen 1978	

34. <i>Nostoc sphaeroides</i>	D	$\mu\text{mol O}_2 (\text{mg Chl a})^{-1} \text{ hr}^{-1}$	21.3	4.5	4.50	25		Nostocales	Li & Gao 2004	Chl a/DM=0.0056 DM/WM=0.037; colony diam 0.08 cm
35. <i>Nostoc sphaeroides</i>	D	$\mu\text{mol O}_2 (\text{mg Chl a})^{-1} \text{ hr}^{-1}$	20.3	5.2	5.20	25		Nostocales	Li & Gao 2004	Chl a/DM=0.0068 DM/WM=0.013; colony diam 0.30 cm
36. <i>Nostoc sphaeroides</i>	D	$\mu\text{mol O}_2 (\text{mg Chl a})^{-1} \text{ hr}^{-1}$	23.8	7.2	7.20	25		Nostocales	Li & Gao 2004	Chl a/DM=0.0081 DM/WM=0.015; colony diam 0.67 cm
37. <i>Nostoc sphaeroides</i>	D	$\mu\text{mol O}_2 (\text{mg Chl a})^{-1} \text{ hr}^{-1}$	35.7	9.5	9.50	25		Nostocales	Li & Gao 2004	Chl a/DM=0.0071 DM/WM=0.014; colony diam 0.20 cm
38. <i>Oscillatoria (Limnothrix) redekei</i> van Goor	C	$\text{pg C} (\text{pg cell C})^{-1} \text{ day}^{-1}$	0.13	8	16.00	15		Oscillatoriales	Geider & Osborne 1989 (data of Foy & Gibson 1982)	
39. <i>Oscillatoria (Limnothrix) redekei</i> van Goor	C	$\text{pg C} (\text{pg cell C})^{-1} \text{ day}^{-1}$	0.18	12	24.00	15		Oscillatoriales	Geider & Osborne 1989 (data of Foy & Gibson 1982)	
40. <i>Oscillatoria terebriformis</i>	D	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	17	29	7.25	45	65	Oscillatoriales	Richardson & Castenholz 1987 [cell diam 5 μm , Fig. 2A, filamentous]	In nature this thermophilic bacterium moves to anaerobic environments during the night, where it can live without external fructose for 3-4 days; survives aerobically in darkness no more than 1-2 days
41. <i>Phormidium autumnale</i> (Ag.) Gom.	D	$\text{mg C} (\text{g ash-free DM})^{-1} \text{ hr}^{-1}$	0.04	0.12	0.59	2	[30]	Oscillatoriales	Davey 1989 [Broady 1991, trichome width 3-5 μm]	Ash/DM=0.60; AFDM/WM=0.04-0.067
42. <i>Phormidium autumnale</i> (Ag.) Gom.	D	$\text{mg C} (\text{g ash-free DM})^{-1} \text{ hr}^{-1}$	0.07	0.21	0.84	5	[30]	Oscillatoriales	Davey 1989 [Broady 1991, trichome width 3-5 μm]	Ash/DM=0.60; AFDM/WM=0.04-0.067
43. <i>Phormidium autumnale</i> (Ag.) Gom.	D	$\text{mg C} (\text{g ash-free DM})^{-1} \text{ hr}^{-1}$	0.1	0.3	0.85	10	[30]	Oscillatoriales	Davey 1989 [Broady 1991, trichome width 3-5 μm]	Ash/DM=0.60; AFDM/WM=0.04-0.067
44. <i>Phormidium autumnale</i> (Ag.) Gom.	D	$\text{mg C} (\text{g ash-free DM})^{-1} \text{ hr}^{-1}$	0.17	0.5	1.00	15	[30]	Oscillatoriales	Davey 1989 [Broady 1991, trichome width 3-5 μm]	Ash/DM=0.60; AFDM/WM=0.04-0.067
45. <i>Phormidium autumnale</i> (Ag.) Gom.	D	$\text{mg C} (\text{g ash-free DM})^{-1} \text{ hr}^{-1}$	0.3	0.9	1.27	20	[30]	Oscillatoriales	Davey 1989 [Broady 1991, trichome width 3-5 μm]	Ash/DM=0.60; AFDM/WM=0.04-0.067
46. <i>Phormidium luridum</i>	D	$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	4.3-5.9	8.5	8.50	25	[4.4]	Oscillatoriales	Biggins 1969 [Pasteur Culture Collection, www.pasteur.fr, PCC 7602, width/diameter 1.8-2.3 μm , filamentous]	stable respiration during 8 hr in darkness
47. <i>Planktothrix agardhii</i>	D	$\text{mg O}_2 (\text{mg Chl a})^{-1} \text{ hr}^{-1}$ at Chl a/ODM = 0.0075	0.613	5.4	7.64	20	[19]	Oscillatoriales	Fietz & Nicklisch 2002 [Tonk et al. 2005, diam of filaments 3.3 μm]	ODM/V=0.42 C/N=4.0 C/ODM=0.42 Chl a/ODM=0.0075-0.0088; dark respiration measured for 20 min

48. <i>Plectonema boryanum</i>	Pr	nmol O ₂ (mg protein) ⁻¹ min ⁻¹	5	5.6	5.23	26	[4.4]	Oscillatoriales	Padan et al. 1971 [Pasteur Culture Collection, www.pasteur.fr , PCC 6306, width/diameter 1.8-2.3 μm, filamentous]	respiration of log-harvested cells after incubation for several hr in darkness; stable for several days
49. <i>Plectonema boryanum</i>	Pr	nmol O ₂ (mg protein) ⁻¹ min ⁻¹	15-20	20	18.66	26	[4.4]	Oscillatoriales	Padan et al. 1971 [Pasteur Culture Collection, www.pasteur.fr , PCC 6306, width/diameter 1.8-2.3 μm, filamentous]	respiration of log-harvested cells immediately after harvest
50. <i>Plectonema boryanum</i>	Pr	nmol O ₂ (mg protein) ⁻¹ min ⁻¹	55	62	57.85	26	[4.4]	Oscillatoriales	Padan et al. 1971 [Pasteur Culture Collection, www.pasteur.fr , PCC 6306, width/diameter 1.8-2.3 μm, filamentous]	maximum respiration of log-harvested after light incubation in optimal conditions for 8-10 hr
51. <i>Prochloron</i> sp. (isolated from <i>Lissoclinum patella</i>)	Chl	μmol O ₂ (mg Chl) ⁻¹ min ⁻¹	0.9	30	24.37	28	[5600]	Chroococcales	Alberte et al. 1986 [Cox 1986, Fig. 26, diam 22 μm]	Chl/cell=5.5 pg; low-light culture
52. <i>Prochloron</i> sp. (isolated from <i>Lissoclinum patella</i>)	Chl	μmol O ₂ (mg Chl) ⁻¹ min ⁻¹	3.8	127	103.16	28	[5600]	Chroococcales	Alberte et al. 1986 [Cox 1986, Fig. 26, diam 22 μm]	Chl/cell=2.7 pg; low-light culture
53. <i>Schizothrix calcicola</i>	Chl	mg O ₂ (mg Chl a) ⁻¹ hr ⁻¹	0.58	10	10.00	25		Oscillatoriales	Tang & Vincent 2000	Mat-forming Arctic species; daylength 8 hr
54. <i>Schizothrix calcicola</i>	Chl	mg O ₂ (mg Chl a) ⁻¹ hr ⁻¹	0.49	8.5	17.00	15		Oscillatoriales	Tang & Vincent 2000	Mat-forming Arctic species; daylength 8 hr
55. <i>Schizothrix calcicola</i>	Chl	mg O ₂ (mg Chl a) ⁻¹ hr ⁻¹	1.38	24	24.00	25		Oscillatoriales	Tang & Vincent 2000	Mat-forming Arctic species; daylength 24 hr
56. <i>Schizothrix calcicola</i>	Chl	mg O ₂ (mg Chl a) ⁻¹ hr ⁻¹	0.99	17	34.00	15		Oscillatoriales	Tang & Vincent 2000	Mat-forming Arctic species; daylength 24 hr
57. <i>Spirulina platensis</i> Compère 1968-3786	D	μmol O ₂ (mg Chl a) ⁻¹ hr ⁻¹	46	9	9.00	25	[180]	Chroococcales	Gordillo et al. 1999 [Ciferri 1983, cell diameter 6-8 μm, filamentous]	C/DM=0.697 N/DM=0.045 Chl a/DM=0.0053 soluble proteins/DM=0.083 nitrogen-limited, normal CO ₂
58. <i>Spirulina platensis</i> Compère 1968-3786	D	μmol O ₂ (mg Chl a) ⁻¹ hr ⁻¹	116	15	15.00	25	[180]	Chroococcales	Gordillo et al. 1999 [Ciferri 1983, cell diameter 6-8 μm, filamentous]	C/DM=0.583 N/DM=0.042 Chl a/DM=0.0035 soluble proteins/DM=0.070; nitrogen-unlimited, high CO ₂
59. <i>Spirulina platensis</i> Compère 1968-3786	D	μmol O ₂ (mg Chl a) ⁻¹ hr ⁻¹	33	19	19.00	25	[180]	Chroococcales	Gordillo et al. 1999 [Ciferri 1983, cell diameter 6-8 μm, filamentous]	C/DM=0.513 N/DM=0.108 Chl a/DM=0.0153 soluble proteins/DM=0.197; nitrogen-unlimited, high CO ₂
60. <i>Spirulina platensis</i> Compère 1968-3786	D	μmol O ₂ (mg Chl a) ⁻¹ hr ⁻¹	31	25	25.00	25	[180]	Chroococcales	Gordillo et al. 1999 [Ciferri 1983, cell diameter 6-8 μm, filamentous]	C/DM=0.574 N/DM=0.131 Chl a/DM=0.0215 soluble proteins/DM=0.285; nitrogen-unlimited, normal CO ₂
61. <i>Spirulina platensis</i> P 511	Chl	μmol O ₂ (mg Chl a) ⁻¹ hr ⁻¹	33	18	12.73	30	[180]	Chroococcales	Berry et al. 2003 [Ciferri 1983, cell diameter 6-8 μm, filamentous]	100 mM NaCl + 50 μM DBIMB

62. <i>Spirulina platensis</i> P 511	Chl	$\mu\text{mol O}_2 (\text{mg Chl a})^{-1} \text{ hr}^{-1}$	45	25	17.68	30	[180]	Chroococcales	Berry et al. 2003 [Ciferri 1983, cell diameter 6-8 μm , filamentous]	100 mM KCl
63. <i>Spirulina platensis</i> P 511	Chl	$\mu\text{mol O}_2 (\text{mg Chl a})^{-1} \text{ hr}^{-1}$	62	35	24.75	30	[180]	Chroococcales	Berry et al. 2003 [Ciferri 1983, cell diameter 6-8 μm , filamentous]	100 mM NaCl + 10 mM NaHCO ₃
64. <i>Spirulina platensis</i> P 511	Chl	$\mu\text{mol O}_2 (\text{mg Chl a})^{-1} \text{ hr}^{-1}$	64	36	25.46	30	[180]	Chroococcales	Berry et al. 2003 [Ciferri 1983, cell diameter 6-8 μm , filamentous]	100 mM NaCl
65. <i>Synechococcus</i> sp. RF-1 (PCC 8801)	Chl	$\text{nmol O}_2 (10^8 \text{ cells})^{-1} \text{ min}^{-1}$ at 15 $\mu\text{g Chl a cell}^{-1}$	3	2	1.74	27	[14]	Chroococcales	Chen et al. 1989 [Pasteur Culture Collection, www.pasteur.fr, PCC 8801, width/diameter 3 μm]	Chl a/cell=0.15 pg
66. <i>Synechocystis aquatilis</i>	Chl	$\mu\text{mol O}_2 (\text{mg Chl a})^{-1} \text{ hr}^{-1}$	61.8	34	24.04	30	8	Chroococcales	de Magalhães et al. 2004 [Fig. 6C, cells nearly spherical ~2.5 μm in diam]	0.11-0.23 $\mu\text{g Chl a}$ per 10^6 cells; control cells (not zinc-treated); isolated in Brasil
67. <i>Synechocystis</i> PCC 6803	Chl	$\text{mmol O}_2 (\text{g Chl a})^{-1} (30 \text{ s})^{-1}$	0.036	2.4	2.40	25	[8]	Chroococcales	Avendaño-Coletta & Schubert 2005 [Stanier et al. 1971, Fig. 5, sphere diam 2.5 μm]	Dark respiration during the first 30 s of 5min light:dark regime at 200 $\mu\text{mol photons m}^{-2} \text{ s}^{-1}$; dark respiration rates at 5:5 and 10:10 regimes and other photon densities (25, 35 and 100) are up to 7 W kg ⁻¹ [total 12 data entries]
68. <i>Synechocystis</i> PCC 6803	Chl	$\mu\text{mol O}_2 (\text{mg Chl})^{-1} \text{ hr}^{-1}$	43	24	20.89	27	[8]	Chroococcales	Hammouda & El-Sheekh 1994 [Stanier et al. 1971, Fig. 5, sphere diam 2.5 μm]	
69. <i>Trichodesmium</i> spp.	D	$\text{mg O}_2 (\text{mg Chl a})^{-1} \text{ hr}^{-1}$	2.41	7	5.27	29.1	[785]	Oscillatoriales	Carpenter & Roenneberg 1995 [Carpenter et al. 2004, Table 10]	Chl a/colony=50 ng C/colony=10 μg This means C/Chl a=200 and DM/Chl a=400, see also Trichodesmium Note ; September natural day/night light regime
70. <i>Trichodesmium</i> spp.	D	$\text{mg O}_2 (\text{mg Chl a})^{-1} \text{ hr}^{-1}$	4.93	14	10.76	28.8	[785]	Oscillatoriales	Carpenter & Roenneberg 1995 [Carpenter et al. 2004, Table 10]	Chl a/colony=50 ng C/colony=10 μg This means C/Chl a=200 and DM/Chl a=400, see also Trichodesmium Note
71. <i>Trichodesmium</i> spp.	D	$\text{mg O}_2 (\text{mg Chl a})^{-1} \text{ hr}^{-1}$	17.6	51	49.26	25.5	[785]	Oscillatoriales	Carpenter & Roenneberg 1995 [Carpenter et al. 2004, Table 10]	Chl a/colony=50 ng C/colony=10 μg This means C/Chl a=200 and DM/Chl a=400, see also Trichodesmium Note
72. <i>Trichodesmium</i> spp.	Chl	$\text{mg O}_2 (\text{mg Chl a})^{-1} \text{ hr}^{-1}$	12.5	218	189.78	27	[785]	Oscillatoriales	Roenneberg & Carpenter 1993 [Carpenter et al. 2004, Table 10]	calculated assumed DM/Chl a=67, but see Trichodesmium Note

73. <i>Trichodesmium</i> spp.	Chl	mg O ₂ (mg Chl a) ⁻¹ hr ⁻¹	13.4	233	202.84	27	[785]	Oscillatoriales	Roenneberg & Carpenter 1993 [Carpenter et al. 2004, Table 10]	calculated assumed DM/Chl a=67, but see <i>Trichodesmium</i> Note
74. <i>Trichodesmium</i> spp.	Chl	mg O ₂ (mg Chl a) ⁻¹ hr ⁻¹	18.0	313	272.48	27	[785]	Oscillatoriales	Roenneberg & Carpenter 1993 [Carpenter et al. 2004, Table 10]	calculated assumed DM/Chl a=67, but see <i>Trichodesmium</i> Note
75. <i>Trichodesmium</i> spp.	Chl	mg O ₂ (mg Chl a) ⁻¹ hr ⁻¹	26.9	468	407.42	27	[785]	Oscillatoriales	Roenneberg & Carpenter 1993 [Carpenter et al. 2004, Table 10]	calculated assumed DM/Chl a=67, but see <i>Trichodesmium</i> Note

Note on Chl a content in *Trichodesmium*

LaRoche and Breitbarth (2005) in their Table 1 give Chl a/C=96.5-320 µmol/mol. This corresponds to the C/Chl a mass ratio from 42 to 139 (assuming Chl a molar mass of 893.5 g/mol). LaRoche and Breitbarth (2005) refer to <http://www.nioz.nl/projects/ironages> for text and references. In Appendix 7, based on data of Berman-Frank et al. (2001), values of 0.018, 0.17, 0.19, 0.25 and 0.29 µg/µmol are listed for the Chl a/C ratio in *Trichodesmium*. This corresponds to C/Chl a mass ratio from 667 to 41.4.

However, in Appendix 2b of LaRoche and Breitbarth (2005), according to the data of Mague et al. (1977), carbon content per colony is 9.7×10³ ng, while Chl a content is 34 µg/colony. This gives a mass ratios C/Chl a=285. This is consistent with data obtained by Carpenter (1983) as cited by Carpenter and Roenneberg (1995): 10 µg C/colony at 50 ng Chl a/colony (C/Chl a=200). In Table 2 of LaRoche and Breitbarth (2005) it is said that C/colony=0.81-0.92 µmol=9.7-11 µg, while Chl a/colony = 89.5 fmol/colony = 79 pg Chl a/colony, too low a figure to be realistic.

Carpenter et al. (2004) in their Table 5 list seven measurements for C and Chl a content per colony in April (C/Chl a mass ratio ranges from 54 to 131 with a mean of 100), and five measurements for C and Chl a content per colony in October (C/Chl a mass ratio ranges from 160 to 343 with a mean of 240). These data, too, indicate that *Trichodesmium* possesses a lower percentage of chlorophyll than algae on average (Chl a/DM ratio of 0.015) (APHA 1992). (E.g. Stal & Moezelaar (1999) for cyanobacteria employed Pr/DM = 0.55 and Pr/Chl a = 27, which gives Chl a/DM = 0.02).

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Dataset S8. Dark respiration rates in eukaryotic microalgae

Notes to Table S8:

Data on dark respiration rates in unicellular, non-filamentous eukaryotic microalgae are presented. Taxonomic status (the “**Phylum: Class**” column) was determined for each genus following www.algaebase.org.

Abbreviations and universal conversions: DM – dry mass; WM – wet mass; N – nitrogen mass; Chl a – Chl a mass; C – carbon mass; Pr – protein mass; X/Y – X by Y mass ratio in the cell, e.g. DM/WM is the ratio of dry to wet cell mass; X/V – the ratio of variable X to volume, $\text{pg } \mu\text{m}^3$, e.g. N/V is how many pg nitrogen is contained in $1 \mu\text{m}^3$ of cell volume; X/cell is the amount of X in one cell, pg; $1 \text{ W} = 1 \text{ J s}^{-1}$; $1 \text{ mol O}_2 = 32 \text{ g O}_2$; $1 \text{ mol C} = 12 \text{ g C}$.

Column “**U**” (mass units of respiration rate measurements): D – dry mass or Chl a mass with known Chl a/DM ratio; W – wet mass without information on DM/WM ratio; Chl – chlorophyll mass without information on Chl a/DM ratio; Pr – protein mass; C – carbon mass.

“**Original units**” are the units of dark respiration rate measurements as given in the original publication (“**Source**”); **qou** is the numeric value of dark respiration rate in the original units. E.g., if it is “ $\text{mg O}_2 (\text{g DM})^{-1} \text{ hr}^{-1}$ ” in the column “**Original units**” and “1.1” in the column “**qou**”, this means that dark respiration rate of the corresponding species, as given in the original publication indicated in the column “**Source**”, is $1.1 \text{ mg O}_2 (\text{g DM})^{-1} \text{ hr}^{-1}$.

qWkg is dark respiration rate converted to W (kg WM)^{-1} (Watts per kg wet mass) using the following conversion factors: $\text{C/DM} = 0.5$ (Kratz & Myers 1955; Bratbak & Dundas 1984; Stal & Moezelaar 1997), $\text{Chl a/DM} = 0.015$ (APHA 1992), $\text{Pr/DM} = 0.5$ (Otte et al. 1999; Zubkov et al. 1999; Stal & Moezelaar 1997) and $\text{DM/WM} = 0.3$ as a crude mean for all taxa applied in the analysis (SI Methods, Table S12a). If the Chl a/DM ratio is known (shown in the “**Comments**” column), while **qou** is per unit Chl a mass, the dark respiration rate is first calculated per unit dry mass and then converted to **qWkg** using the reference $\text{DM/WM} = 0.3$. Energy conversion: $1 \text{ ml O}_2 = 20 \text{ J}$. Where **qou** is calculated on cell basis, and carbon content of the cell is known, **qWkg** is first expressed per unit carbon mass and then per unit wet mass using the conversion factors above, rather than obtained by dividing by the known cell mass.

TC is ambient temperature during measurements, degrees Celsius.

q25Wkg is dark respiration rate converted to 25°C using $Q_{10} = 2$, $\text{q25Wkg} = \text{qWkg} \times 2^{(25 - \text{TC})/10}$, dimension W (kg WM)^{-1} . For each species rows are arranged in the order of increasing **q25Wkg**.

MIN – indicates the minimum **q25Wkg** value for each species, that was used in the analyses in the paper.

Mpg: estimated cell mass, pg (1 pg = 10⁻¹² g). Square brackets around **Mpg** value indicate that this value was obtained from a different source than the source of dark respiration rate data. When converting cell volume to cell mass, cell density of 1 g ml⁻¹ was assumed.

Source: the first, unbracketed reference in this column is where the value of **qou** is taken from; references and data in square brackets refer to cell size determination.

Growth rate, day⁻¹: this column contains available growth rates as well as other information on culture conditions, including illumination (it is given in braces, where available, e.g. 150 μmol m⁻² s⁻¹); Log – logarithmic, exp – exponential, stat – stationary phase of cell cycle.

Comments: information on cell composition

qN: dark respiration rate per unit nitrogen mass, **100 W (kg N)⁻¹**. I.e., **qN** = 5 means that dark respiration rate is 500 W (kg N)⁻¹. Where available, values of **qN** corresponding to minimal **q25Wkg** values (those in MIN rows), were converted to 25 °C in the same manner as **qWkg** (using Q₁₀ = 2) Analysis of these **directly available** nitrogen-based dark respiration rates (17 temperature-transformed values for 17 species) yielded a geometric mean value of q_N = 4×10² W (kg N)⁻¹, in good agreement with the value of 3.7×10² W (kg N)⁻¹ in Table 1 of the paper, that was obtained by transforming mean data set **q25Wkg** value (8.8 W (kg WM)⁻¹, n = 47) with use of mean conversion coefficients as q_N = **q25Wkg**/0.3/0.08, where 0.3 = DM/WM and 0.08 = N/DM. The value of N/DM = 0.08 was calculated for the studied species with the known N/C ratio assuming C/DM = 0.5 (SI Methods, Table S12b).

Log₁₀-transformed values of **q25Wkg** (W (kg WM)⁻¹), minimum for each species, were used in the analyses shown in Figures 1-2 and Table 1 in the paper (a total of 47 values for n = 47 species). These values are in rows marked **MIN** and highlighted in blue.

Species	U	MIN	Original units	qou	qWkg	TC	q25Wkg	Mpg	Phylum: Class	Source	Growth rate, day ⁻¹	Comments	qN
1. Asterionella formosa	W	MIN	mg O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.17	1.613	10	4.562	410	Bacillariophyta: Fragilariophyceae	Talling 1957			
2. Asterionella formosa	W		mg O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.23	2.182	14	4.677	410	Bacillariophyta: Fragilariophyceae	Talling 1957			
3. Asterionella formosa	W		mg O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.21	1.992	11	5.257	410	Bacillariophyta: Fragilariophyceae	Talling 1957			
4. Asterionella formosa	W		mg O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.24	2.276	11	6.006	410	Bacillariophyta: Fragilariophyceae	Talling 1957			
5. Asterionella formosa	W		mg O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.35	3.320	16	6.195	410	Bacillariophyta: Fragilariophyceae	Talling 1957			

Table 1. Growth rates of algae in continuous culture													
Algal strain	Culture medium	Light intensity (μmol photons m ⁻² s ⁻¹)	Light limitation	Growth rate (d ⁻¹)	Yield (g dry wt m ⁻² d ⁻¹)	Specific growth rate (d ⁻¹)	Specific yield (g dry wt g ⁻¹ C d ⁻¹)	Algal group	Reference	Light limitation	Yield (g dry wt m ⁻² d ⁻¹)	Specific growth rate (d ⁻¹)	Specific yield (g dry wt g ⁻¹ C d ⁻¹)
6. <i>Asterionella formosa</i>	W		mg O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.21	2.094	7	7.292	390	Bacillariophyta: Fragilariophyceae	Talling 1957			
7. <i>Asterionella formosa</i>	W		mg O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.21	1.992	6	7.434	410	Bacillariophyta: Fragilariophyceae	Talling 1957			
8. <i>Asterionella formosa</i>	W		mg O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.20	1.897	5	7.588	410	Bacillariophyta: Fragilariophyceae	Talling 1957			
9. <i>Asterionella formosa</i>	W		mg O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.41	3.889	10	11.000	410	Bacillariophyta: Fragilariophyceae	Talling 1957			
10. <i>Asterionella formosa</i>	W		mg O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.85	8.063	16	15.046	410	Bacillariophyta: Fragilariophyceae	Talling 1957			
11. <i>Asterionella formosa</i>	W		mg O ₂ (10 ⁹ cells) ⁻¹ hr ⁻¹	0.45	4.268	5	17.072	410	Bacillariophyta: Fragilariophyceae	Talling 1957			
12. <i>Chaetoceros furcellatus</i>	C	MIN	g C (g cell C) ⁻¹ hr ⁻¹	0.00048	0.75	0.5	4.098	[150]	Bacillariophyta: Coscinodiscophyceae	Sakshaug et al. 1991 [calculated assuming C/V=0.15 pg/μm ³]	0.09	N/cell=3.8 pg C/cell=22 pg Chl a/cell=0.79 pg	0.2
13. <i>Chaetoceros furcellatus</i>	C		g C (g cell C) ⁻¹ hr ⁻¹	0.0017	2.6	0.5	14.207	[300]	Bacillariophyta: Coscinodiscophyceae	Sakshaug et al. 1991 [calculated assuming C/V=0.15 pg/μm ³]	0.12	N/cell=8.9 pg C/cell=47 pg Chl a/cell=0.94 pg	0.9
14. <i>Chaetoceros furcellatus</i>	C		g C (g cell C) ⁻¹ hr ⁻¹	0.0028	4.4	0.5	24.042	[130]	Bacillariophyta: Coscinodiscophyceae	Sakshaug et al. 1991 [calculated assuming C/V=0.15 pg/μm ³]	0.32	N/cell=2.9 pg C/cell=19 pg Chl a/cell=0.39 pg	1.9
15. <i>Chaetoceros furcellatus</i>	C		g C (g cell C) ⁻¹ hr ⁻¹	0.0036	5.6	0.5	30.599	[200]	Bacillariophyta: Coscinodiscophyceae	Sakshaug et al. 1991 [calculated assuming C/V=0.15 pg/μm ³]	0.30	N/cell=4.9 pg C/cell=32 pg Chl a/cell=0.23 pg	2.4
16. <i>Chlamydomonas reinhardtii</i>	Chl	MIN	mmol O ₂ (mol Chl) ⁻¹ s ⁻¹	4	9	22	11.080	[2000]	Chlorophyta: Chlorophyceae	Polle et al. 2001 [http://protist.i.hosei.ac.jp]		1 mol Chl=893.5 g Chl/cell=3.9 pg	
17. <i>Chlamydomonas reinhardtii</i>	Chl		mmol O ₂ (mol Chl) ⁻¹ s ⁻¹	26.7	15	25	15.000	[2000]	Chlorophyta: Chlorophyceae	Coleman & Colman 1980 [http://protist.i.hosei.ac.jp]			
18. <i>Chlamydomonas reinhardtii</i>	Chl		mmol O ₂ (mol Chl) ⁻¹ s ⁻¹	54.9	31	35	15.500	[2000]	Chlorophyta: Chlorophyceae	Coleman & Colman 1980 [http://protist.i.hosei.ac.jp]			
19. <i>Chlamydomonas reinhardtii</i>	Chl		mmol O ₂ (mol Chl) ⁻¹ s ⁻¹	17.3	9.7	15	19.400	[2000]	Chlorophyta: Chlorophyceae	Coleman & Colman 1980 [http://protist.i.hosei.ac.jp]			
20. <i>Chlorella kessleri</i>	Chl	MIN	μmol O ₂ (mg Chl) ⁻¹ hr ⁻¹	25	14	27	12.188	[90]	Chlorophyta: Trebouxiophyceae	Hammouda & El-Sheekh 1994 [Lee & Lee 2001, Fig. 1b; Krienitz et al. 2004, Fig. 8, diam 5-6 μm]			
21. <i>Chlorella pyrenoidosa</i>	Chl	MIN	μmol O ₂ (mg Chl a) ⁻¹ hr ⁻¹	9.9/9.0	0.6	24	0.643		Chlorophyta: Trebouxiophyceae	Burris 1977	late exp/ early stat		

22. Chlorella pyrenoidosa	W		$\mu\text{l O}_2 (0.2 \text{ g WM})^{-1} (30 \text{ min})^{-1}$	20	1	25	1.000		Chlorophyta: Trebouxiophyceae	Gest & Kamen 1948			
23. Chlorella pyrenoidosa	D		$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	4.6	7.7	25	7.700	30	Chlorophyta: Trebouxiophyceae	Myers & Graham 1971	0.35	DM/V=0.179 $\text{pg}/\mu\text{m}^3$ Chl/DM=5.19%	
24. Chlorella pyrenoidosa	D		$\mu\text{mol O}_2 (\text{g DM})^{-1} \text{ min}^{-1}$	4.14	9.3	25	9.300		Chlorophyta: Trebouxiophyceae	Pickett 1975	0.282		
25. Chlorella pyrenoidosa	D		$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	5.9	9.8	25	9.800	32	Chlorophyta: Trebouxiophyceae	Myers & Graham 1971	0.78	DM/V=0.197 $\text{pg}/\mu\text{m}^3$ Chl/DM=4.52%	
26. Chlorella pyrenoidosa	D		$\mu\text{mol O}_2 (\text{g DM})^{-1} \text{ min}^{-1}$	5.00	11.2	25	11.200		Chlorophyta: Trebouxiophyceae	Pickett 1975	0.361		
27. Chlorella pyrenoidosa	D		$\mu\text{mol O}_2 (\text{g DM})^{-1} \text{ min}^{-1}$	5.16	11.6	25	11.600		Chlorophyta: Trebouxiophyceae	Pickett 1975	0.439		
28. Chlorella pyrenoidosa	D		$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	8.7	15	25	15.000	34	Chlorophyta: Trebouxiophyceae	Myers & Graham 1971	1.3	DM/V=0.215 $\text{pg}/\mu\text{m}^3$ Chl/DM=3.92%	
29. Chlorella pyrenoidosa	D		$\mu\text{mol O}_2 (\text{g DM})^{-1} \text{ min}^{-1}$	7.43	16.6	25	16.600		Chlorophyta: Trebouxiophyceae	Pickett 1975	0.613		
30. Chlorella pyrenoidosa	D		$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	11.2	19	25	19.000	49	Chlorophyta: Trebouxiophyceae	Myers & Graham 1971	1.8	DM/V=0.252 $\text{pg}/\mu\text{m}^3$ Chl/DM=3.10%	
31. Chlorella pyrenoidosa	D		$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	11.9	20	25	20.000	92	Chlorophyta: Trebouxiophyceae	Myers & Graham 1971	2.3	DM/V=0.256 $\text{pg}/\mu\text{m}^3$ Chl/DM=1.53%	
32. Chlorella pyrenoidosa	D		$\mu\text{l O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	13.8	23	25	23.000	75	Chlorophyta: Trebouxiophyceae	Myers & Graham 1971	2.4	DM/V=0.242 $\text{pg}/\mu\text{m}^3$ Chl/DM=1.14%	
33. Coscinodiscus sp. C38B	C	MIN	$\text{g C (g cell C)}^{-1} \text{ hr}^{-1}$	0.0038	5.9	18	9.585	275000	Bacillariophyta: Coscinodiscophyceae	Blasco et al. 1982	0.62; Log	C/V=0.044 $\text{pg}/\mu\text{m}^3$ C/N=7.9 Chl a/C=0.93%	3.1
34. Coscinodiscus sp. CoA	C	MIN	$\text{g C (g cell C)}^{-1} \text{ hr}^{-1}$	0.0035	5.4	18	8.772	6200000	Bacillariophyta: Coscinodiscophyceae	Blasco et al. 1982	0.55; Log	C/V=0.026 $\text{pg}/\mu\text{m}^3$ C/N=6.7 Chl a/C=1.33%	2.4
35. Ditylum brightwellii	C	MIN	$\text{g C (g cell C)}^{-1} \text{ hr}^{-1}$	0.0019	3.0	18	4.874	118000	Bacillariophyta: Coscinodiscophyceae	Blasco et al. 1982	1.0; one division away from stationary phase	C/V=0.023 $\text{pg}/\mu\text{m}^3$ C/N=5.9 Chl a/C=1.41%	1.2
36. Dunaliella salina	W	MIN	$\text{ml O}_2 (10^8 \text{ cells})^{-1} \text{ hr}^{-1}$	1.08	4	20	5.657	1570	Chlorophyta: Chlorophyceae	Vladimirova & Zotin 1983, 1985 (data of Mironyuk & Einor 1968)			
37. Dunaliella salina	Chl		$\mu\text{mol O}_2 (\text{mg Chl})^{-1} \text{ hr}^{-1}$	33.5	18	24	19.292		Chlorophyta: Chlorophyceae	Liska 2004			
38. Dunaliella tertiolecta	C	MIN	$10^{-5} \text{ mol O}_2 (\text{mol C})^{-1} \text{ s}^{-1}$	0.17	9.5	18	15.433	[120]	Chlorophyta: Chlorophyceae	Quigg & Beardall 2003 [Sciandra et al. 1997, Fig. 1]	0.2	Chl a/C=6.9% Pr/cell=15.8 pg C/N=4.4	2.8
39. Dunaliella tertiolecta	C		$\mu\text{mol O}_2 \text{ cell}^{-1} \text{ min}^{-1} \times 10^{10} \text{ at } 40 \text{ pg C per cell}$	3.1	8.7	15	17.400	69	Chlorophyta: Chlorophyceae	Falkowski & Owens 1980	0; Log	C/V=0.580 $\text{pg}/\mu\text{m}^3$ C/N=3.1	1.8
40. Dunaliella tertiolecta	C		$\mu\text{mol O}_2 \text{ cell}^{-1} \text{ min}^{-1} \times 10^{10} \text{ at } 41 \text{ pg C per cell}$	4.0	11	15	22.000	73	Chlorophyta: Chlorophyceae	Falkowski & Owens 1980	0; Log	C/V=0.562 $\text{pg}/\mu\text{m}^3$ C/N=3.0	2.2

41. <i>Dunaliella tertiolecta</i>	C		$\mu\text{mol O}_2 \text{ cell}^{-1} \text{ min}^{-1} \times 10^{10}$ at 37 pg C per cell	5.2	16	15	32.000	84	Chlorophyta: Chlorophyceae	Falkowski & Owens 1980	0.09; Log	C/V=0.440 pg/ μm^3 C/N=3.2	3.4
42. <i>Dunaliella tertiolecta</i>	C		$10^{-5} \text{ mol O}_2 (\text{mol C})^{-1} \text{ s}^{-1}$	0.48	27	18	43.862	[120]	Chlorophyta: Chlorophyceae	Quigg & Beardall 2003 [Sciandra et al. 1997, Fig. 1]	0.7	Chl a/C=8.4% Pr/cell=8.24 pg C/N=7.6	14
43. <i>Dunaliella tertiolecta</i>	C		$\mu\text{mol O}_2 \text{ cell}^{-1} \text{ min}^{-1} \times 10^{10}$ at 31 pg C per cell	7.2	26	15	52.000	90	Chlorophyta: Chlorophyceae	Falkowski & Owens 1980	0.42; Log	C/V=0.341 pg/ μm^3 C/N=3.4	5.9
44. <i>Dunaliella tertiolecta</i>	C		$10^{-5} \text{ mol O}_2 (\text{mol C})^{-1} \text{ s}^{-1}$	0.61	34	18	55.233	[120]	Chlorophyta: Chlorophyceae	Quigg & Beardall 2003 [Sciandra et al. 1997, Fig. 1]	0.3	Chl a/C=6.6% Pr/cell=8.75 pg C/N=5.6	13
45. <i>Dunaliella tertiolecta</i>	C		$\mu\text{mol O}_2 \text{ cell}^{-1} \text{ min}^{-1} \times 10^{10}$ at 28 pg C per cell	7.3	29	15	58.000	104	Chlorophyta: Chlorophyceae	Falkowski & Owens 1980	0.66; Log	C/V=0.269 pg/ μm^3 C/N=3.8	7.4
46. <i>Dunaliella tertiolecta</i>	C		$\mu\text{mol O}_2 \text{ cell}^{-1} \text{ min}^{-1} \times 10^{10}$ at 30 pg C per cell	8.4	31	15	62.000	112	Chlorophyta: Chlorophyceae	Falkowski & Owens 1980	0.87; Log	C/V=0.269 pg/ μm^3 C/N=4.1	8.6
47. <i>Dunaliella tertiolecta</i>	C		$10^{-5} \text{ mol O}_2 (\text{mol C})^{-1} \text{ s}^{-1}$	0.72	40	18	64.980	[120]	Chlorophyta: Chlorophyceae	Quigg & Beardall 2003 [Sciandra et al. 1997, Fig. 1]	1.1	Chl a/C=7.2% Pr/cell=8.12 pg C/N=2.6	6.9
48. <i>Dunaliella tertiolecta</i>	C		$\mu\text{mol O}_2 \text{ cell}^{-1} \text{ min}^{-1} \times 10^{10}$ at 29 pg C per cell	8.9	34	15	68.000	115	Chlorophyta: Chlorophyceae	Falkowski & Owens 1980	1.25; Log	C/V=0.252 pg/ μm^3 C/N=5.3	12
49. <i>Dunaliella tertiolecta</i>	C		$10^{-5} \text{ mol O}_2 (\text{mol C})^{-1} \text{ s}^{-1}$	1.6	90	18	146.205	[120]	Chlorophyta: Chlorophyceae	Quigg & Beardall 2003 [Sciandra et al. 1997, Fig. 1]	1.0	Chl a/C=10.3% C/N=4.2	25
50. <i>Dunaliella tertiolecta</i>	C		$10^{-5} \text{ mol O}_2 (\text{mol C})^{-1} \text{ s}^{-1}$	1.6	90	18	146.205	[120]	Chlorophyta: Chlorophyceae	Quigg & Beardall 2003 [Sciandra et al. 1997, Fig. 1]	1.2	Chl a/C=4.1% C/N=3.7	22
51. <i>Dunaliella tertiolecta</i>	C		$10^{-5} \text{ mol O}_2 (\text{mol C})^{-1} \text{ s}^{-1}$	1.7	95	18	154.328	[120]	Chlorophyta: Chlorophyceae	Quigg & Beardall 2003 [Sciandra et al. 1997, Fig. 1]	1.4	Chl a/C=3.3% C/N=4.3	27
52. <i>Emiliana huxleyi</i>	C	MIN	$\mu\text{mol O}_2 (\text{mg Chl a})^{-1} \text{ hr}^{-1}$ at Chl a/Corg = 0.031	14	8	15	16.000	[64]	Haptophyta: Prymnesiophyceae	Nielsen 1997 [Verity et al. 1992]	0.27	Chl a/Corg=3.1% Corg/Ctot=0.62 Corg/N=4.8	2.6
53. <i>Emiliana huxleyi</i>	C		$\mu\text{mol O}_2 (\text{mg Chl a})^{-1} \text{ hr}^{-1}$ at Chl a/Corg = 0.028	16	8.4	15	16.800	[64]	Haptophyta: Prymnesiophyceae	Nielsen 1997 [Verity et al. 1992]	0.30	Chl a/Corg=2.8% Corg/Ctot=0.62 Corg/N=6.0	3.3
54. <i>Emiliana huxleyi</i>	Chl		$\text{mg O}_2 (\text{mg Chl})^{-1} \text{ hr}^{-1}$	0.60	10	17.5	16.818	[64]	Haptophyta: Prymnesiophyceae	Flameling & Kromkamp 1998 [Verity et al. 1992]	cells dark-adapted for 15 mins		
55. <i>Emiliana huxleyi</i>	C		$\mu\text{mol O}_2 (\text{mg Chl a})^{-1} \text{ hr}^{-1}$ at Chl a/Corg = 0.009	65	11	15	22.000	[64]	Haptophyta: Prymnesiophyceae	Nielsen 1997 [Verity et al. 1992]	0.87	Chl a/Corg=0.9% Corg/Ctot=0.62 Corg/N=5.1	3.7
56. <i>Emiliana huxleyi</i>	C		$\mu\text{mol O}_2 (\text{mg Chl a})^{-1} \text{ hr}^{-1}$ at Chl a/Corg = 0.029	22	12	15	24.000	[64]	Haptophyta: Prymnesiophyceae	Nielsen 1997 [Verity et al. 1992]	0.64	Chl a/Corg=2.9% Corg/Ctot=0.62 Corg/N=5.6	4.4

57. <i>Emiliana huxleyi</i>	C		$\mu\text{mol O}_2 \text{ (mg Chl a)}^{-1} \text{ hr}^{-1}$ at Chl a/Corg = 0.024	28	12	15	24.000	[64]	Haptophyta: Prymnesiophyceae	Nielsen 1997 [Verity et al. 1992]	0.68	Chl a/Corg=2.4% Corg/Ctot=0.62 Corg/N=5.7	4.8
58. <i>Emiliana huxleyi</i>	C		$\mu\text{mol O}_2 \text{ (mg Chl a)}^{-1} \text{ hr}^{-1}$ at Chl a/Corg = 0.020	32	12	15	24.000	[64]	Haptophyta: Prymnesiophyceae	Nielsen 1997 [Verity et al. 1992]	0.66	Chl a/Corg=2.0% Corg/Ctot=0.62 Corg/N=6.3	5.0
59. <i>Emiliana huxleyi</i>	C		$\mu\text{mol O}_2 \text{ (mg Chl a)}^{-1} \text{ hr}^{-1}$ at Chl a/Corg = 0.015	48	14	15	28.000	[64]	Haptophyta: Prymnesiophyceae	Nielsen 1997 [Verity et al. 1992]	0.85	Chl a/Corg=1.5% Corg/Ctot=0.62 Corg/N=4.9	4.4
60. <i>Emiliana huxleyi</i>	C		$\mu\text{mol O}_2 \text{ (mg Chl a)}^{-1} \text{ hr}^{-1}$ at Chl a/Corg = 0.019	42	15	15	30.000	[64]	Haptophyta: Prymnesiophyceae	Nielsen 1997 [Verity et al. 1992]	0.80	Chl a/Corg=1.9% Corg/Ctot=0.62 Corg/N=6.8	6.8
61. <i>Emiliana huxleyi</i>	C		$\mu\text{mol O}_2 \text{ (mg Chl a)}^{-1} \text{ hr}^{-1}$ at Chl a/Corg = 0.019	43	15	15	30.000	[64]	Haptophyta: Prymnesiophyceae	Nielsen 1997 [Verity et al. 1992]	0.72	Chl a/Corg=1.9% Corg/Ctot=0.62 Corg/N=5.3	5.4
62. <i>Emiliana huxleyi</i>	C		$\mu\text{mol O}_2 \text{ (mg Chl a)}^{-1} \text{ hr}^{-1}$ at Chl a/Corg = 0.018	46	16	15	32.000	[64]	Haptophyta: Prymnesiophyceae	Nielsen 1997 [Verity et al. 1992]	0.75	Chl a/Corg=1.8% Corg/Ctot=0.62 Corg/N=6.0	6.2
63. <i>Emiliana huxleyi</i>	C		$\mu\text{mol O}_2 \text{ (mg Chl a)}^{-1} \text{ hr}^{-1}$ at Chl a/Corg = 0.022	40	17	15	34.000	[64]	Haptophyta: Prymnesiophyceae	Nielsen 1997 [Verity et al. 1992]	0.73	Chl a/Corg=2.2% Corg/Ctot=0.62 Corg/N=6.3	6.9
64. <i>Emiliana huxleyi</i>	Chl		$\text{mg O}_2 \text{ (mg Chl)}^{-1} \text{ hr}^{-1}$	1.75	30	17.5	50.454	[64]	Haptophyta: Prymnesiophyceae	Flameling & Kromkamp 1998 [Verity et al. 1992]	cells dark-adapted for 15 mins		
65. <i>Emiliana huxleyi</i>	Chl		$\text{mg O}_2 \text{ (mg Chl)}^{-1} \text{ hr}^{-1}$	1.92	33	17.5	55.499	[64]	Haptophyta: Prymnesiophyceae	Flameling & Kromkamp 1998 [Verity et al. 1992]			
66. <i>Euglena gracilis</i>	W	MIN	$\text{ml O}_2 \text{ (10}^8 \text{ cells)}^{-1} \text{ hr}^{-1}$	12.8	7	20	9.899	10300	Euglenozoa: Euglenophyceae	Vladimirova & Zotin 1983, 1985 (data of Cook 1966)			
67. <i>Euglena gracilis</i>	W		$\text{ml O}_2 \text{ (10}^8 \text{ cells)}^{-1} \text{ hr}^{-1}$	16.0	9	20	12.728	9700	Euglenozoa: Euglenophyceae	Vladimirova & Zotin 1983, 1985 (data of Cook 1966)			
68. <i>Fragilaria crotonensis</i>	W	MIN	$\text{mg O}_2 \text{ (10}^9 \text{ cells)}^{-1} \text{ hr}^{-1}$	1.78	23	16	42.920	300	Bacillariophyta: Fragilariophyceae	Talling 1957			
69. <i>Glenodinium</i> sp.	Chl	MIN	$\mu\text{mol O}_2 \text{ (mg Chl a)}^{-1} \text{ hr}^{-1}$	61.7/1.3	26	24	27.866		Myzozoa: Dinophyceae	Burris 1977	late exp/early stat late log		
70. <i>Gonyaulax polyedra</i>	Pr	MIN	$\mu\text{mol O}_2 \text{ (mg Chl)}^{-1} \text{ hr}^{-1}$ at Chl a/Pr = 0.0048	64.1	5.7	20	8.061	21000	Myzozoa: Dinophyceae	Sweeney 1986			Pr/cell=7100 pg Chl a/cell=33.9 pg
71. <i>Gonyaulax tamarensis</i>	C	MIN	$10^{-3} \mu\text{mol O}_2 \text{ (}\mu\text{g C)}^{-1} \text{ hr}^{-1}$	0.64	12.0	15	24.000	15150	Myzozoa: Dinophyceae	Langdon 1987	0.520	C/V=0.277 pg/ μm^3 C/N=8.4 C/Chl a=87.0 C/cell=4190 pg	6.7
72. <i>Gonyaulax tamarensis</i>	C		$10^{-3} \mu\text{mol O}_2 \text{ (}\mu\text{g C)}^{-1} \text{ hr}^{-1}$	0.82	15.3	15	30.600	14137	Myzozoa: Dinophyceae	Langdon 1987	0.310	C/V=0.218 pg/ μm^3 C/N=7.5 C/Chl a=106.0 C/cell=3078 pg	7.7

73. <i>Gonyaulax tamarensis</i>	C		$10^{-3} \mu\text{mol O}_2 (\mu\text{g C})^{-1} \text{ hr}^{-1}$	1.13	21	15	42.000	13036	Myzozoa: Dinophyceae	Langdon 1987	0.023	C/V=0.195 pg/ μm^3 C/N=7.2 C/Chl a=57.0 C/cell=2539 pg	10
74. <i>Gonyaulax tamarensis</i>	C		$10^{-3} \mu\text{mol O}_2 (\mu\text{g C})^{-1} \text{ hr}^{-1}$	1.22	22.8	15	45.600	13306	Myzozoa: Dinophyceae	Langdon 1987	0.098	C/V=0.209 pg/ μm^3 C/N=8.2 C/Chl a=90.0 C/cell=2776 pg	12
75. <i>Gonyaulax tamarensis</i>	C		$10^{-3} \mu\text{mol O}_2 (\mu\text{g C})^{-1} \text{ hr}^{-1}$	1.52	28.4	15	56.800	16210	Myzozoa: Dinophyceae	Langdon 1987	0.600	C/V=0.249 pg/ μm^3 C/N=9.0 C/Chl a=172.0 C/cell=4035 pg	17
76. <i>Gonyaulax tamarensis</i>	C		$10^{-3} \mu\text{mol O}_2 (\mu\text{g C})^{-1} \text{ hr}^{-1}$	1.58	29.5	15	59.000	10653	Myzozoa: Dinophyceae	Langdon 1987	0.580	C/V=0.184 pg/ μm^3 C/N=8.2 C/Chl a=171.0 C/cell=1957 pg	16
77. <i>Gymnodinium nelsoni</i>	W	MIN	$\text{ml O}_2 (10^8 \text{ cells})^{-1} \text{ hr}^{-1}$	28.5	1.1	20	1.556	148000	Myzozoa: Dinophyceae	Vladimirova & Zotin 1983, 1985 (data of Hochachka & Teal 1964)			
78. <i>Isochrysis galbana</i>	C	MIN	$\text{g C (g cell C)}^{-1} \text{ hr}^{-1}$	0.09	5.8	18	9.422	30	Haptophyta: Prymnesiophyceae	Herzig & Falkowski 1989	growth rate 0.18 day ⁻¹	C/cell=18pg N/cell=1pg C/V=0.6 pg/ μm^3 Chl a/cell=0.07pg	7.0
79. <i>Isochrysis galbana</i>	C		$\text{g C (g cell C)}^{-1} \text{ hr}^{-1}$	0.09	5.8	18	9.422	30	Haptophyta: Prymnesiophyceae	Herzig & Falkowski 1989	growth rate 0.25 day ⁻¹	C/cell=16pg N/cell=1.2pg C/V=0.54 pg/ μm^3 Chl a/cell=0.08pg	5.2
80. <i>Isochrysis galbana</i>	C		$\text{g C (g cell C)}^{-1} \text{ hr}^{-1}$	0.10	6.5	18	10.559	30	Haptophyta: Prymnesiophyceae	Herzig & Falkowski 1989	growth rate 0.37 day ⁻¹	C/cell=15pg N/cell=1.3pg C/V=0.5pg/ μm^3 Chl a/cell=0.1pg	5.0
81. <i>Isochrysis galbana</i>	C		$\text{g C (g cell C)}^{-1} \text{ hr}^{-1}$	0.12	7.8	18	12.671	32	Haptophyta: Prymnesiophyceae	Herzig & Falkowski 1989	growth rate 0.48 day ⁻¹	C/cell=15pg N/cell=1.4pg C/V=0.44 pg/ μm^3 Chl a/cell=0.11pg	5.6
82. <i>Isochrysis galbana</i>	C		$\text{g C (g cell C)}^{-1} \text{ hr}^{-1}$	0.12	7.8	18	12.671	33	Haptophyta: Prymnesiophyceae	Herzig & Falkowski 1989	growth rate 0.53 day ⁻¹	C/cell=14pg N/cell=1.4pg C/V=0.43 pg/ μm^3 Chl a/cell=0.11pg	5.2
83. <i>Isochrysis galbana</i>	C		$\text{g C (g cell C)}^{-1} \text{ hr}^{-1}$	0.12	7.8	18	12.671	31	Haptophyta: Prymnesiophyceae	Herzig & Falkowski 1989	growth rate 0.59 day ⁻¹	C/cell=13pg N/cell=1.4pg C/V=0.42 pg/ μm^3 Chl a/cell=0.12pg	4.8
84. <i>Isochrysis galbana</i>	C		$\text{g C (g cell C)}^{-1} \text{ hr}^{-1}$	0.15	9.7	18	15.758	34	Haptophyta: Prymnesiophyceae	Herzig & Falkowski 1989	growth rate 0.72 day ⁻¹	C/cell=13pg N/cell=1.5pg C/V=0.38 pg/ μm^3 Chl a/cell=0.13pg	5.6

Table 1. Growth rates and nutrient requirements of Isochrysis galbana and Leptocylindrus danicus												
Strain	Medium	Light	Temperature	Salinity	pH	CO ₂	CO ₂ concentration	CO ₂ source	CO ₂ concentration	CO ₂ source	CO ₂ concentration	CO ₂ source
85. Isochrysis galbana	C	g C (g cell C) ⁻¹ hr ⁻¹	0.18	11.7	18	19.007	34	Haptophyta: Prymnesiophyceae	Herzig & Falkowski 1989	growth rate 0.87 day ⁻¹	C/cell=11pg N/cell=1.5pg C/V=0.32 pg/μm ³ Chl a/cell=0.13	5.7
86. Isochrysis galbana	C	μmol O ₂ cell ⁻¹ min ⁻¹ × 10 ¹⁰ at 10.3 pg C per cell	1.1	12	18	19.494	37	Haptophyta: Prymnesiophyceae	Falkowski et al. 1985	0.30 (30 μmol m ⁻² s ⁻¹); steady growth	C/cell=10.3 pg C/V=0.278 pg/μm ³ C/N=6.3 Chl/C=2.38%	5.0
87. Isochrysis galbana	C	μmol O ₂ cell ⁻¹ min ⁻¹ × 10 ¹⁰ at 12.4 pg C per cell	1.3	12	18	19.494	50	Haptophyta: Prymnesiophyceae	Falkowski et al. 1985	0.70 (70 μmol m ⁻² s ⁻¹); steady growth	C/cell=12.4 pg C/V=0.248 pg/μm ³ C/N=5.5 Chl/C=1.72%	4.3
88. Isochrysis galbana	C	μmol O ₂ cell ⁻¹ min ⁻¹ × 10 ¹⁰ at 20.0 pg C per cell	2.3	13	18	21.119	57	Haptophyta: Prymnesiophyceae	Falkowski et al. 1985	1.2 (600 μmol m ⁻² s ⁻¹); steady growth	C/cell=20.0 pg C/V=0.351 pg/μm ³ C/N=7.5 Chl/C=0.58%	6.4
89. Isochrysis galbana	C	g C (g cell C) ⁻¹ hr ⁻¹ at 10 pg C per cell	0.20	13	18	21.119	33	Haptophyta: Prymnesiophyceae	Herzig & Falkowski 1989	growth rate 0.96 day ⁻¹	C/cell=10pg N/cell=1.5pg C/V=0.3pg/μm ³ Chl a/cell=0.13pg	5.8
90. Isochrysis galbana	C	μmol O ₂ cell ⁻¹ min ⁻¹ × 10 ¹⁰ at 15 pg C per cell	2.0	15	18	24.368	55	Haptophyta: Prymnesiophyceae	Falkowski et al. 1985	1.10 (150 μmol m ⁻² s ⁻¹); steady growth	C/cell=15.0 pg C/V=0.273 pg/μm ³ C/N=5.7 Chl/C=1.37%	5.7
91. Isochrysis galbana	C	μmol O ₂ cell ⁻¹ min ⁻¹ × 10 ¹⁰ at 16.5 pg C per cell	2.2	20	18	32.490	52	Haptophyta: Prymnesiophyceae	Falkowski et al. 1985	1.2 (320 μmol m ⁻² s ⁻¹); steady growth	C/cell=16.5 pg C/V=0.317 pg/μm ³ C/N=3.7 Chl/C=0.88%	4.9
92. Leptocylindrus danicus	C	MIN pg C (pg cell C) ⁻¹ hr ⁻¹	0.0025	4	15	8.000	1158	Bacillariophyta: Coscinodiscophyceae	Verity 1981, 1982a,b	mean minimum value among 49 combinations of temperature, daylength and irradiance, growth rates from 0.1 to 3.0 day ⁻¹	log C(pg)=0.707 log V(μm ³) -0.225 log N(pg)=0.718 log V(μm ³) -1.024 C/Chl a=20-100 depending on temperature and day length C/N=5.8	1.5

93. <i>Monochrysis lutheri</i>	C	MIN	g C (g cell C) ⁻¹ hr ⁻¹	0.0052	8	20	11.314	[40]	Haptophyta: Prymnesiophyceae?	Laws & Caperon 1976 [estimated from C content using the formula of Verity et al. 1992 for non- diatomous algae]	0.19	C/cell=11.3 pg N/cell=0.885 pg [Verity et al. 1992: C/V (pg//μm ³)= =0.545×V ^{-0.181}]	6.9
94. <i>Monochrysis lutheri</i>	C		g C (g cell C) ⁻¹ hr ⁻¹	0.0054	8.4	20	11.879	[25]	Haptophyta: Prymnesiophyceae?	Laws & Caperon 1976 [estimated from C content using the formula of Verity et al. 1992 for non- diatomous algae]	0.38	C/cell=7.58 pg N/cell=0.854 pg [Verity et al. 1992: C/V (pg//μm ³)= =0.545×V ^{-0.181}]	5.0
95. <i>Monochrysis lutheri</i>	C		g C (g cell C) ⁻¹ hr ⁻¹	0.0082	13	20	18.385	[26]	Haptophyta: Prymnesiophyceae?	Laws & Caperon 1976 [estimated from C content using the formula of Verity et al. 1992 for non- diatomous algae]	0.60	C/cell=7.87 pg N/cell=1.058 pg [Verity et al. 1992: C/V (pg//μm ³)= =0.545×V ^{-0.181}]	6.4
96. <i>Monochrysis lutheri</i>	C		g C (g cell C) ⁻¹ hr ⁻¹	0.0091	14	20	19.799	[29]	Haptophyta: Prymnesiophyceae?	Laws & Caperon 1976 [estimated from C content using the formula of Verity et al. 1992 for non- diatomous algae]	0.77	C/cell=8.62 pg N/cell=1.289 pg [Verity et al. 1992: C/V (pg//μm ³)= =0.545×V ^{-0.181}]	6.2
97. <i>Nannochloris atomus</i>	C	MIN	pg C (pg cell C) ⁻¹ day ⁻¹	0.14	9	23	10.338	[6]	Chlorophyta: Chlorophyceae	Geider & Osborne 1989 [Sobrino et al. 2005, diam 2 μm, 0.025 pg Chl a/cell]			
98. <i>Navicula pelliculosa</i>	D	MIN	μmol O ₂ (10 ⁸ cells) ⁻¹ hr ⁻¹ at 28.6 pg DM (cell) ⁻¹	2	13	20	18.385	[95]	Bacillariophyta: Bacillariophyceae	Coombs et al. 1967a,b [calculated from dry mass]		Pr/DM=0.393 C/DM=0.412 Carbohydr./DM=0.1 54 Lipids/DM=0.338	
99. <i>Ochromonas malhamensis</i>	D	MIN	μl O ₂ (mg DM) ⁻¹ hr ⁻¹	15	25	28	20.306		Ochrophyta: Chrysophyceae	Weiss & Brown 1959	starved 24 hr in the dark		
100. <i>Ochromonas</i> sp.	W	MIN	nl O ₂ (cell) ⁻¹ hr ⁻¹	7.0×10 ⁻⁵	5.6	20	7.920	70	Ochrophyta: Chrysophyceae	Fenchel & Finlay 1983	starved		
101. <i>Olisthodiscus luteus</i>	C		10 ⁻³ μmol O ₂ (μg C) ⁻¹ hr ⁻¹	0.26	4.9	15	9.800	1023	Ochrophyta: Raphidophyceae	Langdon 1987	0.340	C/V=0.199 pg/μm ³ C/N=7.9 C/Chl a=46.0 C/cell=204 pg	2.6
102. <i>Olisthodiscus luteus</i>	C		10 ⁻³ μmol O ₂ (μg C) ⁻¹ hr ⁻¹	0.34	6.4	15	12.800	905	Ochrophyta: Raphidophyceae	Langdon 1987	0.610	C/V=0.229 pg/μm ³ C/N=8.3 C/Chl a=46.0 C/cell=207 pg	3.5
103. <i>Olisthodiscus luteus</i>	C		10 ⁻³ μmol O ₂ (μg C) ⁻¹ hr ⁻¹	0.38	7.1	15	14.200	945	Ochrophyta: Raphidophyceae	Langdon 1987	0.174	C/V=0.194 pg/μm ³ C/N=7.5 C/Chl a=33.0 C/cell=183 pg	3.6

Number	Species	Light	Temperature	Salinity	Medium	Chl a	Chl b	Chl c	Chl d	Phylogenetic Group	Reference	Chl a fluorescence	Chl a/C	Chl a/C/N	Chl a/C/V	Chl a/C/Pr
104.	<i>Olisthodiscus luteus</i>	C	10 ⁻³ μmol O ₂ (μg C) ⁻¹ hr ⁻¹	0.47	8.8	15	17.600	833		Ochrophyta: Raphidophyceae	Langdon 1987	0.056	C/V=0.202 pg/μm ³ C/N=7.5 C/Chl a=38.0 C/cell=168 pg			4.4
105.	<i>Olisthodiscus luteus</i>	C	10 ⁻³ μmol O ₂ (μg C) ⁻¹ hr ⁻¹	0.56	10.5	15	21.000	1124		Ochrophyta: Raphidophyceae	Langdon 1987	0.880	C/V=0.266 pg/μm ³ C/N=9.5 C/Chl a=73.5 C/cell=299 pg			6.7
106.	<i>Olisthodiscus luteus</i>	C	10 ⁻³ μmol O ₂ (μg C) ⁻¹ hr ⁻¹	0.70	13.1	15	26.200	1150		Ochrophyta: Raphidophyceae	Langdon 1987	0.810	C/V=0.273 pg/μm ³ C/N=9.7 C/Chl a=62.0 C/cell=314 pg			8.5
107.	<i>Peridinium gatunense</i>	Chl	MIN	μmol O ₂ (5 mg Chl) ⁻¹ min ⁻¹	5	33	22	40.628	[10000]	Myzozoa: Dinophyceae	Sukenik et al. 2002, Fig. 5 [Berman-Frank & Erez 1996]					
108.	<i>Phaeocystis globosa</i>	Chl	MIN	mg O ₂ (mg Chl) ⁻¹ hr ⁻¹	1.49	26	17.5	43.727	[106]	Haptophyta: Prymnesiophyceae	Flameling & Kromkamp 1998 [Olenina et al. 2006]	cells dark-adapted for 15 mins				
109.	<i>Phaeocystis globosa</i>	Chl		mg O ₂ (mg Chl) ⁻¹ hr ⁻¹	1.83	32	17.5	53.817	[106]	Haptophyta: Prymnesiophyceae	Flameling & Kromkamp 1998 [Olenina et al. 2006]	cells dark-adapted for 15 mins				
110.	<i>Phaeocystis globosa</i>	Chl		mg O ₂ (mg Chl) ⁻¹ hr ⁻¹	2.63	46	17.5	77.362	[106]	Haptophyta: Prymnesiophyceae	Flameling & Kromkamp 1998 [Olenina et al. 2006]	cells dark-adapted for 15 mins				
111.	<i>Phaeodactylum tricornutum</i>	C	MIN	10 ⁻⁵ mol O ₂ (mol C) ⁻¹ s ⁻¹	0.38	21	18	34.115	[120]	Bacillariophyta: Bacillariophyceae	Quigg & Beardall 2003 [Glover et al. 1987]	1.2				
112.	<i>Phaeodactylum tricornutum</i>	Chl		mg O ₂ (mg Chl) ⁻¹ hr ⁻¹	1.23	21	15	42.000	[120]	Bacillariophyta: Bacillariophyceae	Flameling & Kromkamp 1998 [Glover et al. 1987]	cells dark-adapted for 15 mins				
113.	<i>Phaeodactylum tricornutum</i>	Chl		mg O ₂ (mg Chl) ⁻¹ hr ⁻¹	1.33	23	15	46.000	[120]	Bacillariophyta: Bacillariophyceae	Flameling & Kromkamp 1998 [Glover et al. 1987]	cells dark-adapted for 15 mins				
114.	<i>Phaeodactylum tricornutum</i>	C		10 ⁻⁵ mol O ₂ (mol C) ⁻¹ s ⁻¹	0.51	29	18	47.111	[120]	Bacillariophyta: Bacillariophyceae	Quigg & Beardall 2003 [Glover et al. 1987]	0.55	Chl a/C=4.5% Pr/cell=3.49 pg C/N=4.9			9.3
115.	<i>Phaeodactylum tricornutum</i>	Chl		mg O ₂ (mg Chl) ⁻¹ hr ⁻¹	1.49	26	15	52.000	[120]	Bacillariophyta: Bacillariophyceae	Flameling & Kromkamp 1998 [Glover et al. 1987]	cells dark-adapted for 15 mins				
116.	<i>Phaeodactylum tricornutum</i>	C		10 ⁻⁵ mol O ₂ (mol C) ⁻¹ s ⁻¹	0.63	35	18	56.858	[120]	Bacillariophyta: Bacillariophyceae	Quigg & Beardall 2003 [Glover et al. 1987]	0.2	Chl a/C=4.6% Pr/cell=2.73 pg C/N=4.4			10
117.	<i>Phaeodactylum tricornutum</i>	C		μmol O ₂ cell ⁻¹ min ⁻¹ × 10 ¹⁰ at 13 pg C per cell	4.21	36	18	58.482	[60]	Bacillariophyta: Bacillariophyceae	Greene et al. 1991 [calculated from Chl a content]		Chl a/cell=0.250 pg Chl a/C=1.94% C/N=5.3			13

118. Phaeodactylum tricornutum	C		10^{-5} mol O ₂ (mol C) ⁻¹ s ⁻¹	0.67	38	18	61.731	[120]	Bacillariophyta: Bacillariophyceae	Quigg & Beardall 2003 [Glover et al. 1987]	0.95	Chl a/C=3.4% C/N=4.4	11.8
119. Phaeodactylum tricornutum	C		10^{-5} mol O ₂ (mol C) ⁻¹ s ⁻¹	0.76	43	18	69.854	[120]	Bacillariophyta: Bacillariophyceae	Quigg & Beardall 2003 [Glover et al. 1987]	1.15	Chl a/C=2.1% Pr/cell=2.34 pg C/N=5.3	15.2
120. Phaeodactylum tricornutum	C		10^{-5} mol O ₂ (mol C) ⁻¹ s ⁻¹	0.80	45	18	73.103	[120]	Bacillariophyta: Bacillariophyceae	Quigg & Beardall 2003 [Glover et al. 1987]	1	Chl a/C=5.6% C/N=4.6	14.5
121. Phaeodactylum tricornutum	C		10^{-5} mol O ₂ (mol C) ⁻¹ s ⁻¹	0.93	52	18	84.474	[120]	Bacillariophyta: Bacillariophyceae	Quigg & Beardall 2003 [Glover et al. 1987]	0.4	Chl a/C=7.5% Pr/cell=3.85 pg	12.1
122. Prorocentrum micans	C	MIN	$\mu\text{mol O}_2 \text{ cell}^{-1} \text{ min}^{-1} \times 10^{10}$ at 1068 pg C per cell	70	7.3	18	11.859	4340	Myzozoa: Dinophyceae	Falkowski et al. 1985	0.075 (70 $\mu\text{mol m}^{-2} \text{ s}^{-1}$) ; steady growth	C/cell=1068 pg C/V=0.246 pg/ μm^3 C/N=3.7 Chl/C=0.62%	0.5
123. Prorocentrum micans	C		$\mu\text{mol O}_2 \text{ cell}^{-1} \text{ min}^{-1} \times 10^{10}$ at 1096 pg C per cell	85	8.7	18	14.133	5096	Myzozoa: Dinophyceae	Falkowski et al. 1985	0.108 (150 $\mu\text{mol m}^{-2} \text{ s}^{-1}$) ; steady growth	C/cell=1096 pg C/V=0.215 pg/ μm^3 C/N=4.1 Chl/C=0.47%	2.0
124. Prorocentrum micans	C		$\mu\text{mol O}_2 \text{ cell}^{-1} \text{ min}^{-1} \times 10^{10}$ at 1117 pg C per cell	125	13	18	21.119	5122	Myzozoa: Dinophyceae	Falkowski et al. 1985	0.164 (320 $\mu\text{mol m}^{-2} \text{ s}^{-1}$) ; steady growth	C/cell=1117 pg C/V=0.218 pg/ μm^3 C/N=4.1 Chl/C=0.29%	3.4
125. Prorocentrum micans	C		$\mu\text{mol O}_2 \text{ cell}^{-1} \text{ min}^{-1} \times 10^{10}$ at 1178 pg C per cell	150	14	18	22.743	5350	Myzozoa: Dinophyceae	Falkowski et al. 1985	0.178 (600 $\mu\text{mol m}^{-2} \text{ s}^{-1}$) ; steady growth	C/cell=1178 pg C/V=0.220 pg/ μm^3 C/N=3.9 Chl/C=0.24%	3.7
126. Scenedesmus obliquus	W	MIN	ml O ₂ (10 ⁸ cells) ⁻¹ hr ⁻¹	0.04	1	20	1.414	220	Chlorophyta: Chlorophyceae	Vladimirova & Zotin 1983, 1985 (data of Margalef 1954)			
127. Scenedesmus protuberans	Chl	MIN	mg O ₂ (mg Chl) ⁻¹ hr ⁻¹	0.36	6.3	20	8.910		Chlorophyta: Chlorophyceae	Flameling & Kromkamp 1998	cells dark-adapted for 15 mins		
128. Scenedesmus protuberans	Chl		mg O ₂ (mg Chl) ⁻¹ hr ⁻¹	0.50	8.7	20	12.304		Chlorophyta: Chlorophyceae	Flameling & Kromkamp 1998	cells dark-adapted for 15 mins		
129. Scenedesmus protuberans	Chl		mg O ₂ (mg Chl) ⁻¹ hr ⁻¹	0.67	12	20	16.971		Chlorophyta: Chlorophyceae	Flameling & Kromkamp 1998	cells dark-adapted for 15 mins		

Table 1. Growth rate, photosynthesis, and nutrient content of <i>Scenedesmus quadricauda</i> and <i>Selenastrum capricornutum</i> under various nutrient conditions.													
Strain	Nutrient	Light	Photosynthesis	Chl a	Chl b	Chl c	Chl d	Chl e	Phytoplankton	Source	Deprivation	Chl a/C	Chl b/C
130. <i>Scenedesmus quadricauda</i>	C	MIN	$\mu\text{mol O}_2 (\text{mg C})^{-1} (6 \text{ hr})^{-1}$	1.3	4	20	5.657	[80]	Chlorophyta: Chlorophyceae	Healey 1979 [calculated from C content]	3 days of nitrate deprivation	C/cell=8 pg C/N=15 C/P=33 Chl a/C=0.8% Pr/C=0.45 Carbohydr/C=0.81	4.0
131. <i>Scenedesmus quadricauda</i>	C		$\mu\text{mol O}_2 (\text{mg C})^{-1} (6 \text{ hr})^{-1}$	1.4	4.4	20	6.223	[70]	Chlorophyta: Chlorophyceae	Healey 1979 [calculated from C content]	7 days of P deprivation	C/cell=10 pg C/N=6.2 C/P=204 Chl a/C=1.5% Pr/C=0.72 Carbohydr/C=0.82	1.8
132. <i>Scenedesmus quadricauda</i>	C		$\mu\text{mol O}_2 (\text{mg C})^{-1} (6 \text{ hr})^{-1}$	2.0	6.2	20	8.768		Chlorophyta: Chlorophyceae	Healey 1979	addition of N after 3 days of nitrate deprivation	24 hr after addition of N to 2 days' N-starved cells: C/cell=9pg C/N=5.1 C/P=19 Chl a/C=1.3% Pr/C=0.72 Carbohydr/C=0.72	
133. <i>Scenedesmus quadricauda</i>	C		$\mu\text{mol O}_2 (\text{mg C})^{-1} (6 \text{ hr})^{-1}$	2.2	6.8	20	9.617		Chlorophyta: Chlorophyceae	Healey 1979	addition of P after 7 days of P deprivation	36 hr after addition of P to 3 days' P-starved cells: C/cell=12pg C/N=4.7 C/P=26 Chl a/C=3.4% Pr/C=0.98 Carbohydr/C=0.54	
134. <i>Selenastrum capricornutum</i>	Chl	MIN	$\text{nmol O}_2 (10^6 \text{ cells})^{-1} \text{ min}^{-1}$ at 0.31 pg Chl a per cell	0.12	13	21	17.154	[50]	Chlorophyta: Chlorophyceae	Beardall et al. 1997 [Hall & Golding 1998]	(550 $\mu\text{mol m}^{-2} \text{ s}^{-1}$; 22 hr in darkness)		
135. <i>Selenastrum capricornutum</i>	Chl		$\text{nmol O}_2 (10^6 \text{ cells})^{-1} \text{ min}^{-1}$ at 0.35 pg Chl a per cell	0.15	14	21	18.473	[50]	Chlorophyta: Chlorophyceae	Beardall et al. 1997 [Hall & Golding 1998]	(100 $\mu\text{mol m}^{-2} \text{ s}^{-1}$; 0 hr in darkness)		
136. <i>Selenastrum capricornutum</i>	Chl		$\text{nmol O}_2 (10^6 \text{ cells})^{-1} \text{ min}^{-1}$ at 0.29 pg Chl a per cell	0.12	14	21	18.473	[50]	Chlorophyta: Chlorophyceae	Beardall et al. 1997 [Hall & Golding 1998]	(100 $\mu\text{mol m}^{-2} \text{ s}^{-1}$; 3.8 hr in darkness)		
137. <i>Selenastrum capricornutum</i>	Chl		$\text{nmol O}_2 (10^6 \text{ cells})^{-1} \text{ min}^{-1}$ at 0.38 pg Chl a per cell	0.16	14	21	18.473	[50]	Chlorophyta: Chlorophyceae	Beardall et al. 1997 [Hall & Golding 1998]	(100 $\mu\text{mol m}^{-2} \text{ s}^{-1}$; 22 hr in darkness)		
138. <i>Selenastrum capricornutum</i>	Chl		$\text{nmol O}_2 (10^6 \text{ cells})^{-1} \text{ min}^{-1}$ at 0.28 pg Chl a per cell	0.14	17	21	22.432	[50]	Chlorophyta: Chlorophyceae	Beardall et al. 1997 [Hall & Golding 1998]	(300 $\mu\text{mol m}^{-2} \text{ s}^{-1}$; 22 hr in darkness)		

Table 1. Growth characteristics of the algae strains used in the study													
Strain	Genus	Species	Strain	Genus	Species	Strain	Genus	Species	Strain	Genus	Species	Strain	Genus
139. Selenastrum capricornutum	Chl		nmol O ₂ (10 ⁶ cells) ⁻¹ min ⁻¹ at 0.22 pg Chl a per cell	0.14	21	21	27.710	[50]	Chlorophyta: Chlorophyceae	Beardall et al. 1997 [Hall & Golding 1998]	(300 μmol m ⁻² s ⁻¹ ; 3.8 hr in darkness)		
140. Selenastrum capricornutum	Chl		nmol O ₂ (10 ⁶ cells) ⁻¹ min ⁻¹ at 0.32 pg Chl a per cell	0.20	21	21	27.710	[50]	Chlorophyta: Chlorophyceae	Beardall et al. 1997 [Hall & Golding 1998]	(550 μmol m ⁻² s ⁻¹ ; 3.8 hr in darkness)		
141. Selenastrum capricornutum	Chl		nmol O ₂ (10 ⁶ cells) ⁻¹ min ⁻¹ at 0.30 pg Chl a per cell	0.23	26	21	34.307	[50]	Chlorophyta: Chlorophyceae	Beardall et al. 1997 [Hall & Golding 1998]	(300 μmol m ⁻² s ⁻¹ ; 0 hr in darkness)		
142. Selenastrum capricornutum	Chl		nmol O ₂ (10 ⁶ cells) ⁻¹ min ⁻¹ at 0.28 pg Chl a per cell	0.48	57	21	75.212	[50]	Chlorophyta: Chlorophyceae	Beardall et al. 1997 [Hall & Golding 1998]	(550 μmol m ⁻² s ⁻¹ ; 0 hr in darkness)		
143. Selenastrum minutum	Chl	MIN	fmol O ₂ cell ⁻¹ hr ⁻¹	4	6.4	20	9.051	77	Chlorophyta: Chlorophyceae	Theodorou et al. 1991	0.6	Chl/cell=0.235 pg	
144. Selenastrum minutum	Chl		fmol O ₂ cell ⁻¹ hr ⁻¹	24	43	20	60.811	69	Chlorophyta: Chlorophyceae	Theodorou et al. 1991	1.68	Chl/cell=0.717 pg	
145. Skeletonema costatum	C	MIN	10 ⁻³ μmol O ₂ (μg C) ⁻¹ hr ⁻¹	0.11	2.1	15	4.200	65	Bacillariophyta: Coscinodiscophyceae	Langdon 1987	0.056	C/cell=16 pg C/V=0.246 pg/μm ³ C/N=6.8 C/Chl a=22.9	0.9
146. Skeletonema costatum	C		μmol O ₂ cell ⁻¹ min ⁻¹ × 10 ¹⁰ at 14.6 pg C per cell	0.47	2.6	15	5.200	77	Bacillariophyta: Coscinodiscophyceae	Falkowski & Owens 1980	0.19; Log	C/V=0.260 pg/μm ³ C/N=4.5	0.7
147. Skeletonema costatum	C		μmol O ₂ cell ⁻¹ min ⁻¹ × 10 ¹⁰ at 20 pg C per cell	0.59	3.3	15	6.600	79	Bacillariophyta: Coscinodiscophyceae	Falkowski & Owens 1980	0.28; Log	C/V=0.253 pg/μm ³ C/N=3.8	0.8
148. Skeletonema costatum	C		μmol O ₂ cell ⁻¹ min ⁻¹ × 10 ¹⁰ at 19 pg C per cell	1.0	5.9	15	11.800	82	Bacillariophyta: Coscinodiscophyceae	Falkowski & Owens 1980	0.45; Log	C/V=0.232 pg/μm ³ C/N=4.3	1.7
149. Skeletonema costatum	C		g C (g cell C) ⁻¹ hr ⁻¹	0.0055	8.6	18	13.971	127	Bacillariophyta: Coscinodiscophyceae	Blasco et al. 1982	1.87; one division away from stationary phase	C/V=0.094 pg/μm ³ C/N=6.5 Chl a/C=1.56%	3.7
150. Skeletonema costatum	C		μmol O ₂ cell ⁻¹ min ⁻¹ × 10 ¹⁰ at 17.5 pg C per cell	1.5	9.3	15	18.600	84	Bacillariophyta: Coscinodiscophyceae	Falkowski & Owens 1980	0.62; Log	C/V=0.214 pg/μm ³ C/N=3.4	2.1
151. Skeletonema costatum	C		10 ⁻³ μmol O ₂ (μg C) ⁻¹ hr ⁻¹	0.60	11.2	15	22.400		Bacillariophyta: Coscinodiscophyceae	Langdon 1987	0.330	C/cell=15 pg C/N=8.2 C/Chl a=18.0	6.1
152. Skeletonema costatum	Chl		μmol O ₂ (mg Chl) ⁻¹ hr ⁻¹	20	11.2	15	22.400		Bacillariophyta: Coscinodiscophyceae	Smith et al. 1992			
153. Skeletonema costatum	Chl		μmol O ₂ (mg Chl) ⁻¹ hr ⁻¹	23	13	15	26.000		Bacillariophyta: Coscinodiscophyceae	Smith et al. 1992			
154. Skeletonema costatum	C		10 ⁻³ μmol O ₂ (μg C) ⁻¹ hr ⁻¹	0.80	14.9	15	29.800		Bacillariophyta: Coscinodiscophyceae	Langdon 1987	1.410	C/cell=14 pg C/N=7.5 C/Chl a=18.9	7.5

155. Skeletonema costatum	Chl		$\mu\text{mol O}_2 \text{ (mg Chl)}^{-1} \text{ hr}^{-1}$	27.5	15.4	15	30.800		Bacillariophyta: Coscinodiscophyceae	Smith et al. 1992			
156. Skeletonema costatum	C		$10^{-3} \mu\text{mol O}_2 \text{ (}\mu\text{g C)}^{-1} \text{ hr}^{-1}$	0.83	15.5	15	31.000		Bacillariophyta: Coscinodiscophyceae	Langdon 1987	2.410	C/cell=31 pg C/N=8.1	8.4
157. Skeletonema costatum	C		$10^{-3} \mu\text{mol O}_2 \text{ (}\mu\text{g C)}^{-1} \text{ hr}^{-1}$	0.84	15.7	15	31.400		Bacillariophyta: Coscinodiscophyceae	Langdon 1987	1.240	C/Chl a=82.4 C/cell=16 pg C/N=6.4 C/Chl a=29.4	6.7
158. Skeletonema costatum	C		$\mu\text{mol O}_2 \text{ cell}^{-1} \text{ min}^{-1} \times 10^{10} \text{ at 14 pg C per cell}$	2.3	16	15	32.000	91	Bacillariophyta: Coscinodiscophyceae	Falkowski & Owens 1980	0.95; Log	C/V=0.154 pg/ μm^3 C/N=5.6	6.9
159. Skeletonema costatum	C		$\mu\text{mol O}_2 \text{ cell}^{-1} \text{ min}^{-1} \times 10^{10} \text{ at 16 pg C per cell}$	2.3	16	15	32.000	88	Bacillariophyta: Coscinodiscophyceae	Falkowski & Owens 1980	0.77; Log	C/V=0.182 pg/ μm^3 C/N=3.7	4.0
160. Skeletonema costatum	Chl		$\mu\text{mol O}_2 \text{ (mg Chl)}^{-1} \text{ hr}^{-1}$	28	16	15	32.000		Bacillariophyta: Coscinodiscophyceae	Smith et al. 1992			
161. Skeletonema costatum	C		$10^{-3} \mu\text{mol O}_2 \text{ (}\mu\text{g C)}^{-1} \text{ hr}^{-1}$	0.90	16.8	15	33.600	124	Bacillariophyta: Coscinodiscophyceae	Langdon 1987	1.880	C/cell=16 pg C/V=0.129 pg/ μm^3 C/N=6.4 C/Chl a=51.3	7.2
162. Skeletonema costatum	C		$\mu\text{mol O}_2 \text{ cell}^{-1} \text{ min}^{-1} \times 10^{10} \text{ at 15 pg C per cell}$	2.2	18	15	36.000	92	Bacillariophyta: Coscinodiscophyceae	Falkowski & Owens 1980	0.88; Log	C/V=0.163 pg/ μm^3 C/N=5.0	6.1
163. Skeletonema costatum	C		$10^{-3} \mu\text{mol O}_2 \text{ (}\mu\text{g C)}^{-1} \text{ hr}^{-1}$	1.05	19.6	15	39.200		Bacillariophyta: Coscinodiscophyceae	Langdon 1987	2.340	C/cell=18 pg C/Chl a=54.4	
164. Skeletonema costatum	C		$10^{-3} \mu\text{mol O}_2 \text{ (}\mu\text{g C)}^{-1} \text{ hr}^{-1}$	1.20	22.4	15	44.800	124	Bacillariophyta: Coscinodiscophyceae	Langdon 1987	1.930	C/cell=22 pg C/V=0.177 pg/ μm^3 C/N=8.2	12
165. Skeletonema costatum	C		$10^{-3} \mu\text{mol O}_2 \text{ (}\mu\text{g C)}^{-1} \text{ hr}^{-1}$	1.20	22.4	15	44.800		Bacillariophyta: Coscinodiscophyceae	Langdon 1987	2.370	C/Chl a=29.0 C/cell=29 pg C/N=9.0 C/Chl a=38.7	13
166. Skeletonema costatum	Chl		$\mu\text{mol O}_2 \text{ (mg Chl)}^{-1} \text{ hr}^{-1}$	62.6	35	15	70.000		Bacillariophyta: Coscinodiscophyceae	Smith et al. 1992			
167. Skeletonema costatum	Chl		$\mu\text{mol O}_2 \text{ (mg Chl)}^{-1} \text{ hr}^{-1}$	96	54	15	108.000		Bacillariophyta: Coscinodiscophyceae	Smith et al. 1992			
168. Stephanodiscus neoastraea	C	MIN	$\text{mg O}_2 \text{ (mg Chl a)}^{-1} \text{ hr}^{-1} \text{ at Chl a/C} = 0.052$	0.3	8.4	20	11.879	[14000]	Bacillariophyta: Coscinodiscophyceae	Fietz & Nicklisch 2002 [Olenina et al. 2006]		Silicates/DM=38% ODM/V=0.27 C/N=4.6 C/ODM=0.46 Chl a/ODM=0.024	2.8
169. Strombidium capitatum	C	MIN	$\text{nI O}_2 \text{ cell}^{-1} \text{ hr}^{-1} \text{ at } 2.8 \times 10^3 \text{ pg C per cell}$	0.67	19	15	38.000	200000	Ologotrichia (ciliate)	Crawford & Stoecker 1996, Fig. 3	largest cells, little mortality during 24 hr starvation, active swimmer	The C/V ratio applied by the authors was 0.14 pg/ μm^3 based on the data of Putt & Stoecker 1989 for ciliates	

Table 1. Photosynthetic rates and biomass of symbiotic dinoflagellates and diatoms from the coral <i>Solenastrea bournoni</i> (Scleractinia: Scleractinia) from the Great Barrier Reef, Australia. The data were collected from the same coral colony, but from different depths (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 188, 189, 190, 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 210, 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249, 250, 251, 252, 253, 254, 255, 256, 257, 258, 259, 260, 261, 262, 263, 264, 265, 266, 267, 268, 269, 270, 271, 272, 273, 274, 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 291, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329, 330, 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 385, 386, 387, 388, 389, 390, 391, 392, 393, 394, 395, 396, 397, 398, 399, 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 411, 412, 413, 414, 415, 416, 417, 418, 419, 420, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 432, 433, 434, 435, 436, 437, 438, 439, 440, 441, 442, 443, 444, 445, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455, 456, 457, 458, 459, 460, 461, 462, 463, 464, 465, 466, 467, 468, 469, 470, 471, 472, 473, 474, 475, 476, 477, 478, 479, 480, 481, 482, 483, 484, 485, 486, 487, 488, 489, 490, 491, 492, 493, 494, 495, 496, 497, 498, 499, 500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 512, 513, 514, 515, 516, 517, 518, 519, 520, 521, 522, 523, 524, 525, 526, 527, 528, 529, 530, 531, 532, 533, 534, 535, 536, 537, 538, 539, 540, 541, 542, 543, 544, 545, 546, 547, 548, 549, 550, 551, 552, 553, 554, 555, 556, 557, 558, 559, 560, 561, 562, 563, 564, 565, 566, 567, 568, 569, 570, 571, 572, 573, 574, 575, 576, 577, 578, 579, 580, 581, 582, 583, 584, 585, 586, 587, 588, 589, 590, 591, 592, 593, 594, 595, 596, 597, 598, 599, 600, 601, 602, 603, 604, 605, 606, 607, 608, 609, 610, 611, 612, 613, 614, 615, 616, 617, 618, 619, 620, 621, 622, 623, 624, 625, 626, 627, 628, 629, 630, 631, 632, 633, 634, 635, 636, 637, 638, 639, 640, 641, 642, 643, 644, 645, 646, 647, 648, 649, 650, 651, 652, 653, 654, 655, 656, 657, 658, 659, 660, 661, 662, 663, 664, 665, 666, 667, 668, 669, 670, 671, 672, 673, 674, 675, 676, 677, 678, 679, 680, 681, 682, 683, 684, 685, 686, 687, 688, 689, 690, 691, 692, 693, 694, 695, 696, 697, 698, 699, 700, 701, 702, 703, 704, 705, 706, 707, 708, 709, 710, 711, 712, 713, 714, 715, 716, 717, 718, 719, 720, 721, 722, 723, 724, 725, 726, 727, 728, 729, 730, 731, 732, 733, 734, 735, 736, 737, 738, 739, 740, 741, 742, 743, 744, 745, 746, 747, 748, 749, 750, 751, 752, 753, 754, 755, 756, 757, 758, 759, 760, 761, 762, 763, 764, 765, 766, 767, 768, 769, 770, 771, 772, 773, 774, 775, 776, 777, 778, 779, 780, 781, 782, 783, 784, 785, 786, 787, 788, 789, 790, 791, 792, 793, 794, 795, 796, 797, 798, 799, 800, 801, 802, 803, 804, 805, 806, 807, 808, 809, 810, 811, 812, 813, 814, 815, 816, 817, 818, 819, 820, 821, 822, 823, 824, 825, 826, 827, 828, 829, 830, 831, 832, 833, 834, 835, 836, 837, 838, 839, 840, 841, 842, 843, 844, 845, 846, 847, 848, 849, 850, 851, 852, 853, 854, 855, 856, 857, 858, 859, 860, 861, 862, 863, 864, 865, 866, 867, 868, 869, 870, 871, 872, 873, 874, 875, 876, 877, 878, 879, 880, 881, 882, 883, 884, 885, 886, 887, 888, 889, 890, 891, 892, 893, 894, 895, 896, 897, 898, 899, 900, 901, 902, 903, 904, 905, 906, 907, 908, 909, 910, 911, 912, 913, 914, 915, 916, 917, 918, 919, 920, 921, 922, 923, 924, 925, 926, 927, 928, 929, 930, 931, 932, 933, 934, 935, 936, 937, 938, 939, 940, 941, 942, 943, 944, 945, 946, 947, 948, 949, 950, 951, 952, 953, 954, 955, 956, 957, 958, 959, 960, 961, 962, 963, 964, 965, 966, 967, 968, 969, 970, 971, 972, 973, 974, 975, 976, 977, 978, 979, 980, 981, 982, 983, 984, 985, 986, 987, 988, 989, 990, 991, 992, 993, 994, 995, 996, 997, 998, 999, 1000													
170. <i>Symbiodinium</i> sp. (zooxanthellae from <i>Aiptasia pallida</i>)	Chl	MIN	$\mu\text{g O}_2 (\mu\text{g Chl})^{-1} \text{ hr}^{-1}$	0.86	15	25	15.000	[500]	Myzozoa: Dinophyceae	Goulet et al. 2005 [calculated from mean Chl a content]		Chl a/cell=1.9 pg	
171. <i>Symbiodinium</i> sp. (zooxanthellae from <i>Aiptasia pallida</i>)	Chl		$\mu\text{g O}_2 (\mu\text{g Chl})^{-1} \text{ hr}^{-1}$	1.24	22	25	22.000	[500]	Myzozoa: Dinophyceae	Goulet et al. 2005 [calculated from mean Chl a content]		Chl a/cell=2.8 pg	
172. <i>Symbiodinium</i> sp. (zooxanthellae from <i>Aiptasia pallida</i>)	Chl		$\mu\text{g O}_2 (\mu\text{g Chl})^{-1} \text{ hr}^{-1}$	2.33	41	32	25.238	[500]	Myzozoa: Dinophyceae	Goulet et al. 2005 [calculated from mean Chl a content]		Chl a/cell=2.8 pg	
173. <i>Symbiodinium</i> sp. (zooxanthellae from <i>Aiptasia pallida</i>)	Chl		$\mu\text{g O}_2 (\mu\text{g Chl})^{-1} \text{ hr}^{-1}$	4.26	77	32	47.399	[500]	Myzozoa: Dinophyceae	Goulet et al. 2005 [calculated from mean Chl a content]		Chl a/cell=1.9 pg	
174. <i>Symbiodinium</i> sp. (zooxanthellae from <i>Aiptasia pallida</i>)	Chl		$\mu\text{g O}_2 (\mu\text{g Chl})^{-1} \text{ hr}^{-1}$	6.26	109	34	58.412	[500]	Myzozoa: Dinophyceae	Goulet et al. 2005 [calculated from mean Chl a content]		Chl a/cell=2.8 pg	
175. <i>Symbiodinium</i> sp. (zooxanthellae from <i>Aiptasia pallida</i>)	Chl		$\mu\text{g O}_2 (\mu\text{g Chl})^{-1} \text{ hr}^{-1}$	6.26	109	34	58.412	[500]	Myzozoa: Dinophyceae	Goulet et al. 2005 [calculated from mean Chl a content]		Chl a/cell=1.9 pg	
176. <i>Symbiodinium</i> sp. (zooxanthellae from <i>Pocillopora capitata</i>)	Chl	MIN	$\mu\text{mol O}_2 (\text{mg Chl a})^{-1} \text{ hr}^{-1}$	16.5/2.4	4	24	4.287		Myzozoa: Dinophyceae	Burris 1977			
177. <i>Terpsinoe musica</i>	D	MIN	$\text{mg O}_2 (\text{g DM})^{-1} \text{ hr}^{-1}$	2.5	2.9	25	2.900		Bacillariophyta: Coscinodiscophyceae	Necchi 2004			
178. <i>Thalassiosira allenii</i>	C	MIN	$\text{g C (g cell C)}^{-1} \text{ day}^{-1}$	0.04	2.6	20	3.677	[300]	Bacillariophyta: Coscinodiscophyceae	Geider & Osborne 1989 (data of Laws & Wong 1978) [Cózar & Echevarría 2005, Table 1]			
179. <i>Thalassiosira nordenskiöldii</i>	C	MIN	$\text{g C (g cell C)}^{-1} \text{ hr}^{-1}$	0.0013	2	0.5	10.928	[500]	Bacillariophyta: Coscinodiscophyceae	Sakshaug et al. 1991 [calculated from C content using the formula of Verity 1981 for the similar-sized diatom <i>Leptocyldrus danicus</i>]	0.12	N/cell=9.0 pg C/cell=48 pg Chl a/cell=3.1 pg	0.7
180. <i>Thalassiosira nordenskiöldii</i>	C		$\text{g C (g cell C)}^{-1} \text{ hr}^{-1}$	0.0015	2.3	0.5	12.568	[470]	Bacillariophyta: Coscinodiscophyceae	Sakshaug et al. 1991 [calculated from C content using the formula of Verity 1981 for the similar-sized diatom <i>Leptocyldrus danicus</i>]	0.33	N/cell=9.1 pg C/cell=46 pg Chl a/cell=1.9 pg	0.7
181. <i>Thalassiosira nordenskiöldii</i>	C		$\text{g C (g cell C)}^{-1} \text{ hr}^{-1}$	0.0024	3.7	0.5	20.217	[430]	Bacillariophyta: Coscinodiscophyceae	Sakshaug et al. 1991 [calculated from C content using the formula of Verity 1981 for the similar-sized diatom <i>Leptocyldrus danicus</i>]	0.10	N/cell=7.8 pg C/cell=43 pg Chl a/cell=2.7 pg	1.4

182. <i>Thalassiosira nordenskiöldii</i>	C		g C (g cell C) ⁻¹ hr ⁻¹	0.0029	4.5	0.5	24.589	[830]	Bacillariophyta: Coscinodiscophyceae	Sakshaug et al. 1991 [calculated from C content using the formula of Verity 1981 for the similar-sized diatom <i>Leptocylindrus danicus</i>]	0.33	N/cell=13 pg C/cell=69 pg Chl a/cell=1.2 pg	1.6
183. <i>Thalassiosira pseudonana</i>	Chl	MIN	μmol O ₂ (mg Chl a) ⁻¹ hr ⁻¹	74.6/8.3	5	24	5.359		Bacillariophyta: Coscinodiscophyceae	Burris 1977	late exp/ early stat		
184. <i>Thalassiosira pseudonana</i>	C		g C (g cell C) ⁻¹ hr ⁻¹	0.0080	12.4	18	20.144	77	Bacillariophyta: Coscinodiscophyceae	Blasco et al. 1982	1.92; Log	C/V=0.142 pg/μm ³ C/N=7.9 Chl a/C=1.33%	6.5
185. <i>Thalassiosira weissflogii</i>	C	MIN	μmol O ₂ cell ⁻¹ min ⁻¹ × 10 ¹⁰ at 275 pg C per cell	20	8.1	18	13.158	1172	Bacillariophyta: Coscinodiscophyceae	Falkowski et al. 1985	0.25 (30 μmol m ⁻² s ⁻¹); steady growth	C/cell=275 pg C/V=0.235 pg/μm ³ C/N=6.3 Chl/C=5.00%	3.4
186. <i>Thalassiosira weissflogii</i>	C		g C (g cell C) ⁻¹ hr ⁻¹	0.0097	15.1	18	24.530	529	Bacillariophyta: Coscinodiscophyceae	Blasco et al. 1982	1.25; Log	C/V=0.331 pg/μm ³ C/N=8.3 Chl a/C=1.43%	8.4
187. <i>Thalassiosira weissflogii</i>	C		μmol O ₂ cell ⁻¹ min ⁻¹ × 10 ¹⁰ at 277 pg C per cell	47	19	18	30.866	1460	Bacillariophyta: Coscinodiscophyceae	Falkowski et al. 1985	0.62 (70 μmol m ⁻² s ⁻¹); steady growth	C/cell=277 pg C/V=0.190 pg/μm ³ C/N=6.6 Chl/C=4.27%	8.4
188. <i>Thalassiosira weissflogii</i>	C		μmol O ₂ cell ⁻¹ min ⁻¹ × 10 ¹⁰ at 328 pg C per cell	75	26	18	42.237	1364	Bacillariophyta: Coscinodiscophyceae	Falkowski et al. 1985	1.73 (320 μmol m ⁻² s ⁻¹); steady growth	C/cell=328 pg C/V=0.240 pg/μm ³ C/N=8.3 Chl/C=2.11%	14
189. <i>Thalassiosira weissflogii</i>	C		μmol O ₂ cell ⁻¹ min ⁻¹ × 10 ¹⁰ at 291 pg C per cell	72	28	18	45.486	1675	Bacillariophyta: Coscinodiscophyceae	Falkowski et al. 1985	1.15 (150 μmol m ⁻² s ⁻¹); steady growth	C/cell=291 pg C/V=0.174 pg/μm ³ C/N=6.5 Chl/C=3.46%	12
190. <i>Thalassiosira weissflogii</i>	C		μmol O ₂ cell ⁻¹ min ⁻¹ × 10 ¹⁰ at 326 pg C per cell	82	28	18	45.486	1480	Bacillariophyta: Coscinodiscophyceae	Falkowski et al. 1985	1.80 (600 μmol m ⁻² s ⁻¹); steady growth	C/cell=326 pg C/V=0.220 pg/μm ³ C/N=8.1 Chl/C=1.55%	15
191. <i>Trebouxia</i> sp. (phycobiont of <i>Caloplaca holocarpa</i>)	D	MIN	μl O ₂ (mg DM) ⁻¹ hr ⁻¹	0.77	1.4	20	1.980	[1000]	Chlorophyta: Trebouxiophyceae	Showman 1972 [Hirose & Yamagishi 1977, genus, diam 10-15 μm]	overnight incubation in darkness	Chl a/WM=0.000625	
192. <i>Trebouxia</i> sp. (phycobiont of <i>Cladonia cristatella</i>)	D	MIN	μl O ₂ (mg DM) ⁻¹ hr ⁻¹	1.49	2.6	20	3.677	[1000]	Chlorophyta: Trebouxiophyceae	Showman 1972 [Hirose & Yamagishi 1977, genus, diam 10-15 μm]	overnight incubation in darkness	Chl a/WM=0.000358	

193. <i>Trebouxia</i> sp. (phycobiont of <i>Lecanora dispersa</i>)	D	MIN	$\mu\text{l O}_2 \text{ (mg DM)}^{-1} \text{ hr}^{-1}$	0.96	1.7	20	2.404	[1000]	Chlorophyta: Trebouxiophyceae	Showman 1972 [Hirose & Yamagishi 1977, genus, diam 10-15 μm]	overnight incubation in darkness	Chl a/WM=0.000354
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Dataset S9. Dark respiration rates in eukaryotic macroalgae

Notes to Table **S9**:

Data on dark respiration rates in eukaryotic macroalgae (including uniseriate filaments) are presented. Taxonomic status (the “**Phylum: Class**” column) was determined for each genus following www.algaebase.org.

Abbreviations and universal conversions: DM – dry mass; WM – wet mass; N – nitrogen mass; Chl a – Chl a mass; C – carbon mass; Pr – protein mass; X/Y – X by Y mass ratio in the cell, e.g. DM/WM is the ratio of dry to wet cell mass; $1\text{ W} = 1\text{ J s}^{-1}$; $1\text{ mol O}_2 = 32\text{ g O}_2$.

“**Original units**” are the units of dark respiration rate measurements as given in the original publication (“**Source**”); **qou** is the numeric value of dark respiration rate in the original units. E.g., if it is “ $\text{mg O}_2 (\text{g DM})^{-1} \text{ hr}^{-1}$ ” in the column “**Original units**” and “1.1” in the column “**qou**”, this means that dark respiration rate of the corresponding species, as given in the original publication indicated in the column “**Source**”, is $1.1\text{ mg O}_2 (\text{g DM})^{-1} \text{ hr}^{-1}$.

Column “**U**” (mass units of respiration rate measurements): D – dry mass or wet mass with known DM/WM ratio; W – wet mass without information on DM/WM ratio; Chl – chlorophyll mass.

qWkg is dark respiration rate converted to W $(\text{kg WM})^{-1}$ (Watts per kg wet mass) using the following conversion factors. If the DM/WM (dry mass to wet mass) ratio is unknown, while **qou** is reported per unit dry mass, the ratio $\text{DM/WM} = 0.3$ was used, as a crude mean for all taxa applied in the analysis (SI Methods, Table S12a). If the DM/WM ratio is known, while **qou** is reported per unit wet mass, the dark respiration rate is first calculated per unit dry mass and then converted to **qWkg** using the reference $\text{DM/WM} = 0.3$. This procedure was applied to make DM- and WM-based data comparable whenever possible. The respiratory quotient of unity was used (1 mol CO_2 released per 1 mol O_2 consumed). Energy conversion: $1\text{ ml O}_2 = 20\text{ J}$. In four cases **qou** was reported per unit chlorophyll a mass. In these cases mass ratio $\text{Chl a/DM} = 0.003$ was adopted to express dark respiration rate per unit dry mass, which was then converted to **qWkg** at $\text{DM/WM} = 0.3$. The ratio $\text{Chl a/DM} = 0.003$ was used as the mean for the studied species with known Chl a/DM ratio (range 0.00016-0.012, $N = 42$, Table **S9**). If **qou** was reported per unit wet mass with no information on the DM/WM ratio available, **qWkg** was obtained from **qou** without mass unit conversions applied.

TC is ambient temperature during measurements, degrees Celsius.

q25Wkg, temperature conversions: Regression of $\log_{10} \text{qWkg}$ on TC for DM-based measurements ($N = 77$) yielded the following results, $\log_{10} \text{qWkg} = a + b \text{ TC}$, where $a = 0.20 \pm 0.06$ ($\pm 1\text{ s.e.}$), $b = 0.015 \pm 0.004$ ($\pm 1\text{ s.e.}$), $N = 77$, $p = 0.0007$, $R^2 = 0.14$. This corresponds to $Q_{10} = 10^{10b} = 1.4$.

(Makarieva et al. 2006), with 95% C.I. for Q_{10} from 1.2 to 1.7. This Q_{10} was used to convert qWkg to 25 ° C, column “**q25Wkg**”, as follows: **q25Wkg** = **qWkg** $\times 1.4^{(25 - TC)/10}$, dimension W (kg WM)⁻¹. For each species rows are arranged in the order of increasing **q25Wkg**.

Log₁₀-transformed values of **q25Wkg** (W (kg WM)⁻¹), minimum for each species, were used in the analyses shown in Figures 1 and 2 and Table 1 in the paper (a total of 88 values for n = 88 species). The corresponding rows are highlighted in [blue](#).

References within Table **S9** to Tables, Figures etc. refer to the corresponding items in the original literature indicated in the **Source** column.

Table **S9**. Dark respiration rates in eukaryotic macroalgae.

Species	U	Original units	qou	qWkg	TC	q25Wkg	Phylum: Class	Source	Comments
1. <i>Acrosiphonia penicilliformis</i>	W	μmol O ₂ (g WM) ⁻¹ hr ⁻¹	1.56	0.19	1.5	0.42	Chlorophyta: Ultrapophyceae	Aguilera et al. 1999	Arctic; dark respiration measured for 3-4 hrs after 2 hr exposure to artificial photosynthetic active radiation (PAR) or PAR and ultraviolet radiation
2. <i>Adenocystis utricularis</i>	D	μmol CO ₂ (g WM) ⁻¹ hr ⁻¹ at DM/WM = 0.212	2.80	0.50	0	1.16	Ochrophyta: Phaeophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0030
3. <i>Antarctosaccion applanatum</i>	D	μmol CO ₂ (g WM) ⁻¹ hr ⁻¹ at DM/WM = 0.196	62.10	11.8	0	27.37	Ochrophyta: Phaeothamniophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0064
4. <i>Ascophyllum nodosum</i>	W	μmol O ₂ (g WM) ⁻¹ hr ⁻¹	2.9	0.36	10	0.60	Ochrophyta: Phaeophyceae	Skene 2004	Scotland; mean temperature of coastal North Sea; measurements were taken over a 10-min period, which was long enough to observe a constant rate of change of O ₂ concentration
5. <i>Ascoseira mirabilis</i>	D	μmol CO ₂ (g WM) ⁻¹ hr ⁻¹ at DM/WM = 0.132	11.75	3.3	0	7.65	Ochrophyta: Phaeophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0065
6. <i>Audouinella hermannii</i>	D	mg O ₂ (g DM) ⁻¹ hr ⁻¹	4	4.5	15	6.30	Rhodophyta: Florideophyceae	Necchi & Zucchi 2001	Brazil, freshwater; dark respiration measured for 45 minutes; temperature closest to the naturally encountered is chosen; studied temperatures 10, 15, 20, 25 °C; culture specimens
7. <i>Audouinella pygmaea</i>	D	mg O ₂ (g DM) ⁻¹ hr ⁻¹	7.5	8.7	20	10.29	Rhodophyta: Florideophyceae	Necchi & Zucchi 2001	Brazil, freshwater; dark respiration measured for 45 minutes; temperature closest to the naturally encountered is chosen; studied temperatures 10, 15, 20, 25 °C; culture specimens; at 25 °C respiration is less than at 20 °C.
8. <i>Ballia callitricha</i>	D	μmol O ₂ (g WM) ⁻¹ hr ⁻¹ at DM/WM = 0.212	2	0.35	0	0.81	Rhodophyta: Florideophyceae	Eggert & Wiencke 2000	Antarctic species; Chl a/DM = 0.0006-0.0011 depending on growth temperature
9. <i>Ballia callitricha</i>	D	μmol CO ₂ (g WM) ⁻¹ hr ⁻¹ at DM/WM = 0.547	33	7.5	0	17.39	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0012
10. <i>Bangia atropurpurea</i>	D	mg O ₂ (mg DM) ⁻¹ hr ⁻¹	1.34	1.5	15	2.10	Rhodophyta: Bangiophyceae	Graham & Graham 1987	Lake Ontario
11. <i>Batrachospermum ambiguum</i>	D	mg O ₂ (g DM) ⁻¹ hr ⁻¹	6	7.1	20	8.40	Rhodophyta: Florideophyceae	Necchi & Zucchi 2001	Brazil, freshwater; dark respiration measured for 45 minutes; temperature closest to the naturally encountered is chosen; studied temperatures 10, 15, 20, 25 °C

12. <i>Batrachospermum delicatulum</i>	D	mg O ₂ (g DM) ⁻¹ hr ⁻¹	1.1	1.4	20	1.66	Rhodophyta: Florideophyceae	Necchi & Alves 2005	Brazil; The lowest value from Table 2; 'Chantransia' stage; samples of field populations were collected or measured (noon ±2 h) at the end of the typical growth period in this region (September to October); respiration ranged from 1.1. to 10.3; in cultured algae from 0.6 to 13.6 original units
13. <i>Batrachospermum delicatulum</i>	D	mg O ₂ (g DM) ⁻¹ hr ⁻¹	4.5	5.3	25	5.30	Rhodophyta: Florideophyceae	Necchi 2004	Brazil; also studied temperatures 10, 15, 20 ° C
14. <i>Batrachospermum macrosporum</i>	D	mg O ₂ (g DM) ⁻¹ hr ⁻¹	1.2	1.5	20	1.77	Rhodophyta: Florideophyceae	Necchi & Zucchi 2001	Brazil, freshwater; dark respiration measured for 45 minutes; temperature closest to the naturally encountered is chosen; studied temperatures 10, 15, 20, 25 °C; culture specimens "Chantransia" stage
15. <i>Batrachospermum vogesiacum</i>	D	mg O ₂ (g DM) ⁻¹ hr ⁻¹	4.4	5.1	25	5.10	Rhodophyta: Florideophyceae	Necchi & Zucchi 2001	Brazil, freshwater; dark respiration measured for 45 minutes; temperature closest to the naturally encountered is chosen; studied temperatures 10, 15, 20, 25 °C
16. <i>Bostrychia moritziana</i>	W	μmol O ₂ (mg Chl a) hr ⁻¹ at Chl a/WM = 0.001	18	2.2	25	2.20	Rhodophyta: Florideophyceae	Karsten et al. 1993	Isolates from Venezuela; Australian samples respired at 62.6/7.6 orig. units (rate of photosynthesis divided by the ratio of photosynthesis to respiration)
17. <i>Callophyllis</i> sp.	D	μmol CO ₂ (g WM) ⁻¹ hr ⁻¹ at DM/WM = 0.272	6.71	0.93	0	2.16	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0023
18. <i>Callophyllis variegata</i>	D	μmol CO ₂ (g WM) ⁻¹ hr ⁻¹ at DM/WM = 0.150	6.15	1.54	0	3.57	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0015
19. <i>Caulerpa taxifolia</i>	D	μmol O ₂ (g DM) ⁻¹ min ⁻¹	0.27	0.6	22	0.66	Chlorophyta: Bryopsidophyceae	Chisholm et al. 2000	Subtropical Australia; 22 °C is within the natural temperature range of this species; respiration increases with decreasing temperature (negative Q ₁₀); Chl a/ DM = 0.0007-0.0031
20. <i>Chaetomorpha</i> sp.	Chl	μmol C (mg Chl a) ⁻¹ hr ⁻¹	3.0	0.3	24	0.31	Chlorophyta: Ulvophyceae	Burris 1977	The original value is the ratio of (net photosynthetic rate) in the air (Table 1) to the (steady-state dark respiration rate); temperature is close to the ambient temperature in the northern Gulf of California at the time of collection
21. <i>Chara braunii</i>	D	mg O ₂ (g DM) ⁻¹ hr ⁻¹	3	3.5	25	3.50	Charophyta: Charophyceae	Vieira & Necchi 2003	Brazil; also studied temperatures 10, 15, 20 ° C
22. <i>Chara guarinensis</i>	D	mg O ₂ (g DM) ⁻¹ hr ⁻¹	3.5	4.1	25	4.10	Charophyta: Charophyceae	Vieira & Necchi 2003	Brazil; also studied temperatures 10, 15, 20 ° C
23. <i>Chara hispida</i>	D	mg O ₂ (g ash-free DM) ⁻¹ hr ⁻¹	2	2.4	30	2.03	Charophyta: Charophyceae	Menendez & Sanchez 1998	Spain; dark respiration in May, period of maximum photosynthesis; also studied temperatures 10, 20 °C; Q ₁₀ between 20 and 30 °C is less than unity; Chl a/WM = 0.00005-0.0004 depending on the season
24. <i>Cladophora glomerata</i>	D	mg O ₂ (g DM) ⁻¹ hr ⁻¹	3.8	4.4	25	4.40	Chlorophyta: Ulvophyceae	Necchi 2004	Brazil; also studied temperatures 10, 15, 20, 30 ° C
25. <i>Cladophora glomerata</i>	D	mg O ₂ (g DM) ⁻¹ hr ⁻¹	4	4.7	25	4.70	Chlorophyta: Ulvophyceae	Necchi 2004	Brazil; also studied temperatures 10, 15, 20, 30 ° C
26. <i>Compsopogon coeruleus</i>	D	mg O ₂ (g DM) ⁻¹ hr ⁻¹	3.4	3.9	20	4.61	Rhodophyta: Compsopogonophyceae	Necchi & Zucchi 2001	Brazil, freshwater; dark respiration measured for 45 minutes; temperature closest to the naturally encountered is chosen; studied temperatures 10, 15, 20, 25 °C
27. <i>Compsopogon coeruleus</i>	D	mg O ₂ (g DM) ⁻¹ hr ⁻¹	7.2	12	25	12.00	Rhodophyta: Compsopogonophyceae	Necchi 2004	Brazil; also studied temperatures 10, 15, 20, 30 ° C

28. <i>Curdiea racovitzae</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.337	5.59	0.62	0	1.44	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0011
29. <i>Cystosphaera jacquinotii</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.151	10.07	2.5	0	5.80	Ochrophyta: Phaeophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0028
30. <i>Delesseria lancifolia</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.209	31.33	5.60	0	12.99	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0023
31. <i>Desmarestia aculeata</i>	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	2.75	0.34	1.5	0.75	Ochrophyta: Phaeophyceae	Aguilera et al. 1999	Arctic; dark respiration measured for 3-4 hrs after 2 hr exposure to artificial photosynthetic active radiation (PAR) or PAR and ultraviolet radiation
32. <i>Desmarestia anceps</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.187	21.26	4.25	0	9.86	Ochrophyta: Phaeophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0037
33. <i>Desmarestia antarctica</i> (adult)	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.287	8.39	1.09	0	2.53	Ochrophyta: Phaeophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0032; juveniles respire at $22.38 \mu\text{mol CO}_2 (\text{g FM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.166 ($\text{qWkg} = 5 \text{ W kg}^{-1}$) and have Chl a/DM = 0.0038
34. <i>Desmarestia menziesii</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.141	12.31	3.26	0	7.56	Ochrophyta: Phaeophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0075
35. <i>Devaleraea ramentacea</i>	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	0.90	0.11	1.5	0.24	Rhodophyta: Florideophyceae	Aguilera et al. 1999	Arctic; dark respiration measured for 3-4 hrs after 2 hr exposure to artificial photosynthetic active radiation (PAR) or PAR and ultraviolet radiation
36. <i>Enteromorpha bulbosa</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.193	11.19	2.2	0	5.10	Chlorophyta: Ulvophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0077
37. <i>Enteromorpha</i> sp.	Chl	$\mu\text{mol C (mg Chl a)}^{-1} \text{ hr}^{-1}$	7.7	0.9	20	1.06	Chlorophyta: Ulvophyceae	Burris 1977	The original value is the ratio of (net photosynthetic rate) in the air (Table 1) to the (steady-state dark respiration rate); species collected in California
38. <i>Fucus distichus</i>	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	0.88	0.11	1.5	0.24	Ochrophyta: Phaeophyceae	Aguilera et al. 1999	Arctic; dark respiration measured for 3-4 hrs after 2 hr exposure to artificial photosynthetic active radiation (PAR) or PAR and ultraviolet radiation
39. <i>Fucus serratus</i>	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	1.5	0.19	10	0.31	Ochrophyta: Phaeophyceae	Skene 2004	Scotland; mean temperature of coastal North Sea; measurements were taken over a 10-min period, which was long enough to observe a constant rate of change of O_2 concentration
40. <i>Fucus spiralis</i>	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	4.2	0.52	10	0.86	Ochrophyta: Phaeophyceae	Skene 2004	Scotland; mean temperature of coastal North Sea; measurements were taken over a 10-min period, which was long enough to observe a constant rate of change of O_2 concentration
41. <i>Fucus vesiculosus</i>	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	2.5	0.31	10	0.51	Ochrophyta: Phaeophyceae	Skene 2004	Scotland; mean temperature of coastal North Sea; measurements were taken over a 10-min period, which was long enough to observe a constant rate of change of O_2 concentration
42. <i>Geminocarpus geminatus</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.229	25.73	4.19	0	9.72	Ochrophyta: Phaeophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0095
43. <i>Georgiella confluens</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.201	15.66	2.91	0	6.75	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0040
44. <i>Gigartina skottsbergii</i>	D	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ h}^{-1}$ at DM/WM = 0.222	2.5	0.42	0	0.97	Rhodophyta: Florideophyceae	Eggert & Wiencke 2000	Antarctic species; Chl a/DM = 0.0009
45. <i>Gigartina skottsbergii</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.331	5	0.56	0	1.30	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0007
46. <i>Gymnogongrus antarcticus</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ h}^{-1}$ at DM/WM = 0.110	2.5	0.85	0	1.97	Rhodophyta: Florideophyceae	Eggert & Wiencke 2000	Antarctic species; Chl a/DM = 0.0022-0.0026 depending on growth temperature

47. <i>Gymnogongrus antarcticus</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.228	11.2	1.8	0	4.17	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0021
48. <i>Halopteris obovata</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.415	30.77	2.77	0	6.42	Ochrophyta: Phaeophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0028
49. <i>Himantothallus grandifolius</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.127	3.92	1.16	0	2.69	Ochrophyta: Phaeophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0025
50. <i>Hymenocladopsis crustigena</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.122	10.07	3.07	0	7.12	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0028
51. <i>Kallymenia antarctica</i>	D	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ h}^{-1}$ at DM/WM = 0.140	1.2	0.32	0	0.74	Rhodophyta: Florideophyceae	Eggert & Wiencke 2000	Antarctic species; Chl a/DM = 0.00059-0.00073 depending on growth temperature
52. <i>Kallymenia antarctica</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.183	5	1	0	2.32	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0003
53. <i>Laminaria digitata</i>	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	0.66	0.082	1.5	0.18	Ochrophyta: Phaeophyceae	Aguilera et al. 1999	Arctic; dark respiration measured for 3-4 hrs after 2 hr exposure to artificial photosynthetic active radiation (PAR) or PAR and ultraviolet radiation
54. <i>Laminaria saccharina</i>	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	1.00	0.12	1.5	0.26	Ochrophyta: Phaeophyceae	Aguilera et al. 1999	Arctic; dark respiration measured for 3-4 hrs after 2 hr exposure to artificial photosynthetic active radiation (PAR) or PAR and ultraviolet radiation
55. <i>Laminaria saccharina</i>	D	$\mu\text{mol O}_2 (\text{cm}^{-2} \text{ leaf area}) \text{ hr}^{-1}$ at 32.2 mg WM cm^{-2} and DM/WM = 0.131	0.12	1	13	1.50	Ochrophyta: Phaeophyceae	Gerard 1988	Three habitats in the vicinity of New York, roughly similar data for shallow, deep and turbid habitat; DM/WM = 0.11-0.179 (largest at high light regime); C/DM = 0.267-0.352; N/DM = 1.90-3.00%
56. <i>Laminaria solidungula</i>	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	0.78	0.097	1.5	0.21	Ochrophyta: Phaeophyceae	Aguilera et al. 1999	Arctic; dark respiration measured for 3-4 hrs after 2 hr exposure to artificial photosynthetic active radiation (PAR) or PAR and ultraviolet radiation
57. <i>Iridaea cordata</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.287	5.59	0.73	0	1.69	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0012
58. <i>Mastocarpus stellatus</i>	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	7.8	0.97	10	1.61	Rhodophyta: Florideophyceae	Skene 2004	Scotland; mean temperature of coastal North Sea; measurements were taken over a 10-min period, which was long enough to observe a constant rate of change of O_2 concentration
59. <i>Monostroma arcticum</i>	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	5.13	0.64	1.5	1.41	Chlorophyta: Ulvophyceae	Aguilera et al. 1999	Arctic; dark respiration measured for 3-4 hrs after 2 hr exposure to artificial photosynthetic active radiation (PAR) or PAR and ultraviolet radiation
60. <i>Monostroma harti</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.240	16.78	2.6	0	6.03	Chlorophyta: Ulvophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0057
61. <i>Mougeotia</i>	D	$\text{mg O}_2 (\text{mg DM})^{-1} \text{ hr}^{-1}$	2.67	3.1	15	4.34	Charophyta: Zygnematophyceae	Graham et al. 1996	Wisconsin, USA
62. <i>Myriogramme mangini</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.172	6.71	1.47	0	3.41	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0027
63. <i>Myriogramme smithii</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.255	34.69	5.08	0	11.78	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0029
64. <i>Nitella furcata</i> var. <i>sieberi</i>	D	$\text{mg O}_2 (\text{g DM})^{-1} \text{ hr}^{-1}$	1.2	2	25	2.00	Charophyta: Charophyceae	Necchi 2004	Brazil; also studied temperatures 10, 15, 20, 30 ° C
65. <i>Nitella</i> sp.	D	$\text{mg O}_2 (\text{g DM})^{-1} \text{ hr}^{-1}$	7.2	8.4	25	8.40	Charophyta: Charophyceae	Vieira & Necchi 2003	Brazil; also studied temperatures 10, 15, 20 ° C
66. <i>Nitella subglomerata</i>	D	$\text{mg O}_2 (\text{g DM})^{-1} \text{ hr}^{-1}$	6.7	7.8	25	7.80	Charophyta: Charophyceae	Vieira & Necchi 2003	Brazil; also studied temperatures 10, 15, 20 ° C

67. <i>Odonthalia dentata</i>	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	2.47	0.31	1.5	0.68	Rhodophyta: Florideophyceae	Aguilera et al. 1999	Arctic; dark respiration measured for 3-4 hrs after 2 hr exposure to artificial photosynthetic active radiation (PAR) or PAR and ultraviolet radiation
68. <i>Palmaria decipiens</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.107	5.59	1.96	0	4.55	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0027
69. <i>Palmaria palmata</i>	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	1.16	0.14	1.5	0.31	Rhodophyta: Florideophyceae	Aguilera et al. 1999	Arctic; dark respiration measured for 3-4 hrs after 2 hr exposure to artificial photosynthetic active radiation (PAR) or PAR and ultraviolet radiation
70. <i>Palmaria palmata</i>	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	6.5	0.81	10	1.34	Rhodophyta: Florideophyceae	Skene 2004	Scotland; mean temperature of coastal North Sea; measurements were taken over a 10-min period, which was long enough to observe a constant rate of change of O_2 concentration
71. <i>Pantoneura plocamioides</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.283	11.19	1.47	0	3.41	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0019
72. <i>Pelvetia canaliculata</i>	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	8.3	1	10	1.66	Ochrophyta: Phaeophyceae	Skene 2004	Scotland; mean temperature of coastal North Sea; measurements were taken over a 10-min period, which was long enough to observe a constant rate of change of O_2 concentration
73. <i>Phaeurus antarcticus</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.193	10.07	1.94	0	4.50	Ochrophyta: Phaeophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0098
74. <i>Phycodrys quercifolia</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.325	40.28	4.62	0	10.71	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0033
75. <i>Phycodrys rubens</i>	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	1.53	0.19	1.5	0.42	Rhodophyta: Florideophyceae	Aguilera et al. 1999	Arctic; dark respiration measured for 3-4 hrs after 2 hr exposure to artificial photosynthetic active radiation (PAR) or PAR and ultraviolet radiation
76. <i>Phyllophora ahnfeltioides</i>	D	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ h}^{-1}$ at DM/WM = 0.310	0.8	0.32	0	0.74	Rhodophyta: Florideophyceae	Eggert & Wiencke 2000	Antarctic species; Chl a/DM = 0.00012-0.00017 depending on growth temperature
77. <i>Phyllophora ahnfeltioides</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.272	28	3.8	0	8.81	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0006
78. <i>Phyllophora antarctica</i>	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	2.3	0.3	0	0.70	Rhodophyta: Florideophyceae	Schwarz et al. 2003	Antarctic species; minimum respiration at studied depths (10-25 m); Chl a/WM = 0.000048-0.000071
79. <i>Phyllophora appendiculata</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.230	10.07	1.63	0	3.78	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0027
80. <i>Phyllophora truncata</i>	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	0.63	0.078	1.5	0.17	Rhodophyta: Florideophyceae	Aguilera et al. 1999	Arctic; dark respiration measured for 3-4 hrs after 2 hr exposure to artificial photosynthetic active radiation (PAR) or PAR and ultraviolet radiation
81. <i>Picconiella plumosa</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.233	26.85	4.30	0	9.97	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0027
82. <i>Pithophora oedogonia</i>	D	$\text{mg O}_2 (\text{g DM})^{-1} \text{ hr}^{-1}$	0.3	0.35	5	0.69	Chlorophyta: Ulvophyceae	Spencer et al. 1985	USA; uniseriate filament
83. <i>Plocamium cartilagineum</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.203	26.85	4.94	0	11.46	Rhodophyta: Florideophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0028
84. <i>Porphyra endiviifolium</i>	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.322	24.62	2.85	0	6.61	Rhodophyta: Bangiophyceae	Weykam et al. 1996	Antarctic species; Chl a/DM = 0.0079
85. <i>Porphyra umbilicalis</i>	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	20	2.5	10	4.14	Rhodophyta: Bangiophyceae	Skene 2004	Scotland; mean temperature of coastal North Sea; measurements were taken over a 10-min period, which was long enough to observe a constant rate of change of O_2 concentration

86. <i>Prasiola crispa</i>	D	mg C (g ash-free DM) ⁻¹ hr ⁻¹	0.05	0.15	2	0.33	Chlorophyta: Trebouxiophyceae	Davey 1989	Antarctic species, terrestrial; at 2, 5, 10, 15 and 20 °C respired at 0.05, 0.14, 0.16, 0.25 and 0.5 original units, respectively; monostromatic blades or uniseriate filaments; 9% ash in dry mass; AFDM/WM = 0.17-0.20.
87. <i>Prasiola crispa</i>	D	mg C (g ash-free DM) ⁻¹ hr ⁻¹	0.14	0.42	5	0.82	Chlorophyta: Trebouxiophyceae	Davey 1989	Antarctic species, terrestrial; at 2, 5, 10, 15 and 20 °C respired at 0.05, 0.14, 0.16, 0.25 and 0.5 original units, respectively; monostromatic blades or uniseriate filaments; 9% ash in dry mass; AFDM/WM = 0.17-0.20.
88. <i>Prasiola crispa</i>	D	mg C (g ash-free DM) ⁻¹ hr ⁻¹	0.16	0.48	10	0.80	Chlorophyta: Trebouxiophyceae	Davey 1989	Antarctic species, terrestrial; at 2, 5, 10, 15 and 20 °C respired at 0.05, 0.14, 0.16, 0.25 and 0.5 original units, respectively; monostromatic blades or uniseriate filaments; 9% ash in dry mass; AFDM/WM = 0.17-0.20.
89. <i>Prasiola crispa</i>	D	mg C (g ash-free DM) ⁻¹ hr ⁻¹	0.27	0.81	15	1.13	Chlorophyta: Trebouxiophyceae	Davey 1989	Antarctic species, terrestrial; at 2, 5, 10, 15 and 20 °C respired at 0.05, 0.14, 0.16, 0.25 and 0.5 original units, respectively; monostromatic blades or uniseriate filaments; 9% ash in dry mass; AFDM/WM = 0.17-0.20.
90. <i>Prasiola crispa</i>	D	mg C (g ash-free DM) ⁻¹ hr ⁻¹	0.5	1.5	20	1.77	Chlorophyta: Trebouxiophyceae	Davey 1989	Antarctic species, terrestrial; at 2, 5, 10, 15 and 20 °C respired at 0.05, 0.14, 0.16, 0.25 and 0.5 original units, respectively; monostromatic blades or uniseriate filaments; 9% ash in dry mass; AFDM/WM = 0.17-0.20.
91. <i>Ptilota plumosa</i>	W	μmol O ₂ (g WM) ⁻¹ hr ⁻¹	0.75	0.093	1.5	0.21	Rhodophyta: Florideophyceae	Aguilera et al. 1999	Arctic; dark respiration measured for 3-4 hrs after 2 hr exposure to artificial photosynthetic active radiation (PAR) or PAR and ultraviolet radiation
92. <i>Saccorhiza dermatodea</i>	W	μmol O ₂ (g WM) ⁻¹ hr ⁻¹	0.63	0.078	1.5	0.17	Ochrophyta: Phaeophyceae	Aguilera et al. 1999	Arctic; dark respiration measured for 3-4 hrs after 2 hr exposure to artificial photosynthetic active radiation (PAR) or PAR and ultraviolet radiation
93. <i>Sargassum natans</i>	D	mg C (g DM) ⁻¹ hr ⁻¹	0.31	0.9	27	0.84	Ochrophyta: Phaeophyceae	Lapointe 1995	Mean value for oceanic populations (natural ambient temperature from 18 to 29 °C); neritic populations (temp from 24 to 30) respired at 0.51 orig. units
94. <i>Sargassum</i> sp.	Chl	μmol C (mg Chl a) ⁻¹ hr ⁻¹	7.9	0.9	24	0.93	Ochrophyta: Phaeophyceae	Burris 1977	The original value is the ratio of (net photosynthetic rate) in the air (Table 1) to the (steady-state dark respiration rate); temperature is close to the ambient temperature in the northern Gulf of California at the time of collection
95. <i>Spirogyra</i> sp.	D	mg O ₂ (g DM) ⁻¹ hr ⁻¹	7.5	12.5	25	12.50	Charophyta: Zygnematophyceae	Necchi 2004	Brazil; also studied temperatures 10, 15, 20, 30 °C
96. <i>Thorea hispida</i>	D	mg O ₂ (g DM) ⁻¹ hr ⁻¹	6.2	7.2	25	7.20	Rhodophyta: Florideophyceae	Necchi & Zucchi 2001	Brazil, freshwater; dark respiration measured for 45 minutes; temperature closest to the naturally encountered is chosen; studied temperatures 10, 15, 20, 25 °C; no change of respiration between 20 and 25 °C
97. <i>Thorea hispida</i>	D	mg O ₂ (g DM) ⁻¹ hr ⁻¹	6.2	10.4	25	10.40	Rhodophyta: Florideophyceae	Necchi 2004	Brazil; also studied temperatures 10, 15, 20, 30 °C
98. <i>Udotea flabellum</i>	Chl	μmol O ₂ (mg Chl a) ⁻¹ hr ⁻¹	2.8	0.3	23	0.32	Chlorophyta: Udoteaceae	Reiskind & Bowes 1991	Florida, Gulf of Mexico; measurements of dark respiration at predetermined optimum temperature
99. <i>Ulothrix zonata</i>	D	mg O ₂ (mg DM) ⁻¹ hr ⁻¹	2.00	2.3	15	3.22	Chlorophyta: Ulvophyceae	Graham et al. 1985	Lake Huron, USA; prolonged darkness
100. <i>Ulothrix zonata</i>	D	mg O ₂ (mg DM) ⁻¹ hr ⁻¹	2.68	3	10	4.97	Chlorophyta: Ulvophyceae	Graham et al. 1985	Lake Huron, USA
101. <i>Ulothrix zonata</i>	D	mg O ₂ (mg DM) ⁻¹ hr ⁻¹	3.06	3.6	15	5.04	Chlorophyta: Ulvophyceae	Graham et al. 1985	Lake Huron, USA
102. <i>Ulothrix zonata</i>	D	mg O ₂ (mg DM) ⁻¹ hr ⁻¹	3.01	3.5	20	4.14	Chlorophyta: Ulvophyceae	Graham et al. 1985	Lake Huron, USA

103.	Ulva lactuca	W	$\mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$	24	3.0	10	4.97	Phaeophyta	Skene 2004	Scotland; mean temperature of coastal North Sea; measurements were taken over a 10-min period, which was long enough to observe a constant rate of change of O_2 concentration; data requested from the author (author's reply to A.M. Makarieva of 07.08.2006) Antarctic species; Chl a/DM = 0.0029
104.	Unknown sp. (CW / MC 56)	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.218	10.07	1.72	0	3.99	Rhodophyta	Weykam et al. 1996	
105.	Urospora penicilliformis	D	$\mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ at DM/WM = 0.162	29.09	6.7	0	15.54	Chlorophyta: Ulvophyceae	Weykam et al. 1996	
106.	Vaucheria fontinalis	D	$\text{mg O}_2 (\text{g DM})^{-1} \text{ hr}^{-1}$	8.5	14	25	14.00	Ochrophyta: Xanthophyceae	Necchi 2004	

Examples of ***qou*** to ***qWkg*** conversions:

Acrosiphonia penicilliformis ***qou* = $1.56 \mu\text{mol O}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$** = $1.56 \times 22.4 \mu\text{l O}_2 (0.001 \text{ kg WM})^{-1} \text{ hr}^{-1}$ = $35 \text{ ml O}_2 (\text{kg WM})^{-1} (3600 \text{ s})^{-1}$ = $35 \times 20 \text{ J (kg WM)}^{-1} (3600 \text{ s})^{-1}$ = **$0.2 \text{ W (kg WM)}^{-1} = qWkg$**

Chara braunii ***qou* = $1.31 \mu\text{g O}_2 (\text{g DM})^{-1} \text{ hr}^{-1}$** = $1.31/32 \times 22.4 \times 0.001 \text{ ml O}_2 \times (20 \text{ J / ml O}_2) (0.001 \text{ kg DM})^{-1} (3600 \text{ s})^{-1}$ = $5.1 \text{ W (kg DM)}^{-1} = 5.1 \times 0.3 \text{ W (kg WM)}^{-1}$ = **$1.5 \text{ W (kg WM)}^{-1} = qWkg$**

Gymnogongrus antarcticus ***qou* = $2.5 \mu\text{mol CO}_2 (\text{g WM})^{-1} \text{ hr}^{-1}$ (at DM/WM = 0.11)** = $2.5 \times 22.4 \times 0.001 \text{ ml O}_2 \times (20 \text{ J / ml O}_2) / 0.11 (0.001 \text{ kg DM})^{-1} (3600 \text{ s})^{-1}$ = $2.8 \text{ W (kg DM)}^{-1} = 2.8 \times 0.3 \text{ W (kg WM)}^{-1}$ = **$qWkg$**

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Dataset S10. Dark respiration rates in green leaves

Notes to Table S10:

The basis for this data set is formed by the data of Wright et al. (2004). Supplementary data to the paper of Wright et al. (2004), available at www.nature.com, contain 274 dark respiration values for 246 species. For each species the minimal value in the data set was taken. To the resulting set of 246 values, 25 values for 25 species were added taken from the literature. These data are marked with a reference in the “Ref” column; references are given below the table. Data without a reference are from Wright et al. (2004).

LogRdmass is the original unit of dark respiration rate in the data set of Wright et al. (2004). It is equal to the decimal logarithm of dark respiration rate **Rdmass** measured in $\text{nmol CO}_2 (\text{g DM})^{-1} \text{ s}^{-1}$, where DM is dry leaf mass. Using the respiration coefficient of unity (1 mol CO_2 released per 1 mol O_2 consumed), energy conversion coefficient of 20 J $(\text{ml O}_2)^{-1}$ and dry mass/ wet mass ratio DM/WM = 0.3 (crude mean for all taxa applied in the analysis (SI Methods, Table S12a), **Rdmass** was converted to **qWkg** (W (kg WM)^{-1}) as follows: **qWkg** = **Rdmass** $\times 22.4 \text{ ml O}_2 \text{ mol}^{-1} \times 20 \text{ J (ml O}_2)^{-1} \times 0.3 \times 10^{-3} = 0.134 \text{ Rdmass W (kg WM)}^{-1}$. Or, for the decimal logarithms, **LogqWkg** = **LogRdmass** – 0.87. The resulting 271 **qWkg** values for 271 species were used in the analyses shown in Figures 1 and 2 and Table 1 in the paper. **Logqdd** is an axiliary variable equal to the decimal logarithm of dark respiration rate expressed in W (kg DM)^{-1} .

Main reference:

Wright I.J., Reich P.B., Westoby M., Ackerly D.D., Baruch Z., Bongers F., Cavender-Bares J., Chapin T., Cornelissen J.H.C., Diemer M., Flexas J., Garnier E., Groom P.K., Gulias J., Hikosaka K., Lamont B.B., Lee T., Lee W., Lusk C., Midgley J.J., Navas M.-L., Niinemets Ü., Oleksyn J., Osada N., Poorter H., Poot P., Prior L., Pyankov V.I., Roumet C., Thomas S.C., Tjoelker M.G., Veneklaas E.J., Villar R. (2004) The worldwide leaf economics spectrum. *Nature* 428: 821-827.

Table S10. Dark respiration in green leaves

Species	logRdmass	Logqdd	LogqWkg	Ref
1. <i>Abies lasiocarpa</i>	0.54	0.19	-0.33	
2. <i>Acacia colletioides</i>	0.76	0.41	-0.11	
3. <i>Acacia doratoxylon</i>	0.83	0.48	-0.04	
4. <i>Acacia floribunda</i>	1.03	0.68	0.16	
5. <i>Acacia havilandiorum</i>	0.95	0.60	0.08	
6. <i>Acacia oswaldii</i>	0.65	0.30	-0.22	
7. <i>Acacia suaveolens</i>	0.84	0.49	-0.03	
8. <i>Acacia willhelmiana</i>	1.06	0.71	0.19	
9. <i>Acer rubrum</i>	0.93	0.58	0.06	
10. <i>Acer saccharum</i>	0.85	0.50	-0.02	
11. <i>Achillea millefolium</i>	1.18	0.83	0.31	
12. <i>Acomastylis rosii</i>	1.32	0.97	0.45	
13. <i>Adenanthos cygnorum</i>	0.64	0.29	-0.23	
14. <i>Adinandra dumosa</i>	0.99	0.64	0.12	
15. <i>Aextoxicon punctatum</i>	0.57	0.22	-0.30	
16. <i>Agropyron repens</i>	1.13	0.78	0.26	
17. <i>Agrostis scabra</i>	1.51	1.16	0.64	
18. <i>Allocauarina</i> sp	0.89	0.54	0.02	
19. <i>Ambrosia artemisiifolia</i>	1.42	1.07	0.55	
20. <i>Amomyrtus luma</i>			-0.08	Lusk & del Poso 2002
21. <i>Amorpha canescens</i>	1.24	0.89	0.37	
22. <i>Anacardium excelsum</i>	1.41	1.06	0.54	
23. <i>Andersonia heterophylla</i>	1.08	0.73	0.21	
24. <i>Andropogon gerardi</i>	1.13	0.78	0.26	
25. <i>Anemone cylindrica</i>	1.11	0.76	0.24	

26.	<i>Anthocephalus chinensis</i>	1.37	1.02	0.50	Feng et al. 2004
27.	<i>Antirrhoea trichantha</i>	1.26	0.91	0.39	
28.	<i>Arctostaphylos uva-ursi</i>	0.72	0.37	-0.15	
29.	<i>Aristotelia chilensis</i>			0.14	Lusk & del Poso 2002
30.	<i>Asclepias syriaca</i>	1.45	1.10	0.58	
31.	<i>Asclepias tuberosa</i>	1.28	0.93	0.41	
32.	<i>Aster azureus</i>	1.24	0.89	0.37	
33.	<i>Aster ericoides</i>	1.30	0.95	0.43	
34.	<i>Astragalus candensis</i>	1.32	0.97	0.45	
35.	<i>Astroloma xerophyllum</i>	0.86	0.51	-0.01	
36.	<i>Astrotricha floccosa</i>	0.95	0.60	0.08	
37.	<i>Atriplex canescens</i>	1.09	0.74	0.22	
38.	<i>Atriplex stipitata</i>	1.59	1.24	0.72	
39.	<i>Austrocedrus chilensis</i>	0.79	0.44	-0.08	
40.	<i>Baccharis angustifolia</i>	1.24	0.89	0.37	
41.	<i>Banksia attenuata</i>	0.80	0.45	-0.07	
42.	<i>Banksia marginata</i>	0.68	0.33	-0.19	
43.	<i>Banksia menziesii</i>	0.75	0.40	-0.12	
44.	<i>Baptisia leucophaea</i>	1.56	1.21	0.69	
45.	<i>Barringtonia macrostachya</i>	0.98	0.63	0.11	Feng et al. 2004
46.	<i>Bellucia grossularioides</i>	0.90	0.55	0.03	
47.	<i>Bertya cunninghamii</i>	1.01	0.66	0.14	
48.	<i>Beyeria opaca</i>	0.97	0.62	0.10	
49.	<i>Bistorta bistortoides</i>	1.45	1.10	0.58	
50.	<i>Boronia ledifolia</i>	0.79	0.44	-0.08	
51.	<i>Bossiaea eriocarpa</i>	0.96	0.61	0.09	
52.	<i>Bossiaea walkeri</i>	0.83	0.48	-0.04	
53.	<i>Bouteloua curtipendula</i>	1.19	0.84	0.32	
54.	<i>Brachychiton populneus</i>	0.67	0.32	-0.20	
55.	<i>Bromus inermis</i>	1.13	0.78	0.26	
56.	<i>Calamovilfa longifolia</i>	1.10	0.75	0.23	
57.	<i>Caldecluvia paniculata</i>			0.04	Lusk & del Poso 2002
58.	<i>Callitris glaucophylla</i>	0.62	0.27	-0.25	
59.	<i>Calophyllum polyanthum</i>	0.93	0.58	0.06	Feng et al. 2004
60.	<i>Calytrix flavescens</i>	0.88	0.53	0.01	
61.	<i>Carya glabra</i>	1.01	0.66	0.14	
62.	<i>Cassinia laevis</i>	1.11	0.76	0.24	
63.	<i>Castanopsis sieboldii</i>	0.65	0.30	-0.22	
64.	<i>Castilla elastica</i>	1.43	1.08	0.56	
65.	<i>Cecropia ficifolia</i>	1.26	0.91	0.39	
66.	<i>Cecropia longipes</i>	1.34	0.99	0.47	
67.	<i>Chionochloa macra</i>	0.18	-0.17	-0.69	Mark 1975
68.	<i>Chionochloa oreophila</i>	0.31	-0.04	-0.56	Mark 1975
69.	<i>Chionochloa rigida</i>	-0.09	-0.44	-0.96	Mark 1975
70.	<i>Conostephium pendulum</i>	0.92	0.57	0.05	
71.	<i>Coreopsis palmata</i>	1.06	0.71	0.19	
72.	<i>Cornus florida</i>	1.04	0.69	0.17	
73.	<i>Correa reflexa</i>	0.66	0.31	-0.21	
74.	<i>Corylus americanus</i>	1.00	0.65	0.13	
75.	<i>Corymbia gummifera</i>	0.75	0.40	-0.12	
76.	<i>Cryptocarya alba</i>	0.97	0.62	0.10	
77.	<i>Dasyphyllum diacanthoides</i>	0.89	0.54	0.02	
78.	<i>Dasyphyllum diacanthoides</i>			0.15	Lusk & del Poso 2002
79.	<i>Desmodium canadense</i>	1.20	0.85	0.33	
80.	<i>Dillenia suffruticosa</i>	1.02	0.67	0.15	
81.	<i>Dodonaea triquetra</i>	1.00	0.65	0.13	
82.	<i>Dodonaea viscosa spatulata</i>	1.19	0.84	0.32	
83.	<i>Drimys winteri</i>	0.97	0.62	0.10	
84.	<i>Drimys winteri</i>			0.09	Lusk & del Poso 2002
85.	<i>Echinacea purpurea</i>	1.24	0.89	0.37	
86.	<i>Eleagnus angustifolia</i>	1.37	1.02	0.50	
87.	<i>Embothrium coccineum</i>	1.20	0.85	0.33	
88.	<i>Eremaea pauciflora</i>	0.96	0.61	0.09	
89.	<i>Eremophila deserti</i>	0.81	0.46	-0.06	
90.	<i>Eremophila glabra</i>	1.00	0.65	0.13	
91.	<i>Eremophila mitchelli</i>	1.04	0.69	0.17	
92.	<i>Eriostemon australasius</i>	0.84	0.49	-0.03	
93.	<i>Erythronium americanum</i>	1.72	1.37	0.85	
94.	<i>Eucalyptus dumosa</i>	0.57	0.22	-0.30	
95.	<i>Eucalyptus haemostoma</i>	0.78	0.43	-0.09	
96.	<i>Eucalyptus intertexta</i>	0.74	0.39	-0.13	
97.	<i>Eucalyptus paniculata</i>	0.79	0.44	-0.08	
98.	<i>Eucalyptus socialis</i>	0.68	0.33	-0.19	
99.	<i>Eucalyptus umbra</i>	0.80	0.45	-0.07	
100.	<i>Eucryphia cordifolia</i>	0.71	0.36	-0.16	
101.	<i>Eucryphia cordifolia</i>			0.08	Lusk & del Poso 2002

102. Eupatorium rugesum	1.65	1.30	0.78	
103. Eutaxia microphylla	1.17	0.82	0.30	
104. Fraxinus sp	1.07	0.72	0.20	
105. Galax aphylla	0.99	0.64	0.12	
106. Geijera parviflora	0.76	0.41	-0.11	
107. Gevuina avellana	0.81	0.46	-0.06	
108. Gompholobium grandiflorum	0.95	0.60	0.08	
109. Grevillea aneura	1.02	0.67	0.15	
110. Grevillea buxifolia	1.03	0.68	0.16	
111. Grevillea speciosa	0.87	0.52	0.00	
112. Gutierrezia sarothrae	1.16	0.81	0.29	
113. Hakea dactyloides	0.59	0.24	-0.28	
114. Hakea tephrosperma	0.38	0.03	-0.49	
115. Hakea teretifolia	0.61	0.26	-0.26	
116. Helianthus microcephalus	1.55	1.20	0.68	
117. Helichrysum apiculatum	1.55	1.20	0.68	
118. Hibbertia bracteata	0.87	0.52	0.00	
119. Hibbertia huegelii	0.77	0.42	-0.10	
120. Hibbertia subvaginata	0.87	0.52	0.00	
121. Jacksonia floribunda	0.92	0.57	0.05	
122. Juniperus monosperma	0.68	0.33	-0.19	
123. Kalmia latifolia	0.95	0.60	0.08	
124. Koeleria cristata	1.09	0.74	0.22	
125. Lambertia formosa	0.56	0.21	-0.31	
126. Larrea tridentata	0.89	0.54	0.02	
127. Lasiopetalum ferrugineum	0.63	0.28	-0.24	
128. Laurelia philippiana	0.79	0.44	-0.08	
129. Laurelia philippiana			0.12	Lusk & del Poso 2002
130. Lemna gibba			-0.60	Ullrich-Eberius et al. 1981
131. Leptospermum polygalifolium	1.15	0.80	0.28	
132. Leptospermum trinervium	1.00	0.65	0.13	
133. Lespedeza capitata	1.19	0.84	0.32	
134. Leucopogon conostephioides	1.04	0.69	0.17	
135. Liatris aspera	1.05	0.70	0.18	
136. Licania heteromorpha	0.72	0.37	-0.15	
137. Linociera insignis	0.94	0.59	0.07	Feng et al. 2004
138. Liquidambar styraciflua	0.53	0.19	-0.33	Hamilton et al. 2001
139. Liriodendron tulipifera	1.09	0.74	0.22	
140. Lomatia hirsuta	0.91	0.56	0.04	
141. Lomatia silaifolia	0.91	0.56	0.04	
142. Luehea seemannii	1.33	0.98	0.46	
143. Luma apiculata	0.97	0.62	0.10	
144. Lupinus perennis	1.32	0.97	0.45	
145. Lyginia barbata	0.60	0.25	-0.27	
146. Lyonia lucida	0.76	0.41	-0.11	
147. Macaranga heynei	0.87	0.52	0.00	
148. Machilus thunbergii	0.75	0.40	-0.12	
149. Macrozamia communis	0.60	0.25	-0.27	
150. Macrozamia riedlei	0.52	0.17	-0.35	
151. Magnolia fraseri	1.00	0.65	0.13	
152. Mallotus paniculatus	1.16	0.81	0.29	
153. Manihot esculenta	1.52	1.17	0.65	
154. Maytenus oleodes	0.59	0.24	-0.28	
155. Melaleuca acerosa	0.91	0.56	0.04	
156. Melaleuca uncinata	0.96	0.61	0.09	
157. Melastoma malabathricum	0.84	0.49	-0.03	
158. Miconia dispar	0.72	0.37	-0.15	
159. Micromyrtus sessilis	0.83	0.48	-0.04	
160. Myrceugenia planipes	0.72	0.37	-0.15	
161. Myrceugenia planipes			-0.08	Lusk & del Poso 2002
162. Neolitsea sericea	0.52	0.17	-0.35	
163. Nothofagus dombeyi	0.95	0.60	0.08	
164. Nothofagus dombeyi			0.32	Lusk & del Poso 2002
165. Nothofagus nitida			0.26	Lusk & del Poso 2002
166. Nuytsia floribunda	0.69	0.34	-0.18	
167. Nyssa sylvatica	1.04	0.69	0.17	
168. Ocotea costulata	0.70	0.35	-0.17	
169. Olearia decurrens	1.09	0.74	0.22	
170. Olearia pimelioides	1.14	0.79	0.27	
171. Oxydendron arboreum	1.10	0.75	0.23	
172. Panicum virgatum	1.06	0.71	0.19	
173. Patersonia occidentalis	0.70	0.35	-0.17	
174. Penstemon grandiflorus	1.20	0.85	0.33	
175. Persea borbonia	0.83	0.48	-0.04	
176. Persea lingue	0.64	0.29	-0.23	
177. Persoonia levis	0.40	0.05	-0.47	

178. <i>Persoonia linearis</i>	0.58	0.23	-0.29	
179. <i>Persoonia saccata</i>	0.72	0.37	-0.15	
180. <i>Petalostemum villosum</i>	1.25	0.90	0.38	
181. <i>Petrophile linearis</i>	0.79	0.44	-0.08	
182. <i>Philotheca difformis</i>	0.99	0.64	0.12	
183. <i>Phyllota phyllicoides</i>	1.03	0.68	0.16	
184. <i>Picea engelmannii</i>	0.51	0.16	-0.36	
185. <i>Picea glauca</i>	0.60	0.25	-0.27	
186. <i>Pimelea linifolia</i>	1.22	0.87	0.35	
187. <i>Pimelea microcephala</i>	1.42	1.07	0.55	
188. <i>Pinus banksiana</i>	0.78	0.43	-0.09	
189. <i>Pinus elliotii</i>	-0.10	-0.45	-0.97	Ryan et al. 1994
190. <i>Pinus flexilis</i>	0.60	0.25	-0.27	
191. <i>Pinus palustris</i>	0.56	0.21	-0.31	
192. <i>Pinus resinosa</i>	0.85	0.50	-0.02	Ryan et al. 1994
193. <i>Pinus rigida</i>	0.72	0.37	-0.15	
194. <i>Pinus serotina</i>	0.70	0.35	-0.17	
195. <i>Pinus strobus</i>	0.67	0.32	-0.20	
196. <i>Pinus sylvestris</i>	0.79	0.44	-0.08	
197. <i>Pinus taeda</i>	0.42	0.07	-0.45	Ryan et al. 1994
198. <i>Platanus occidentalis</i>	0.89	0.54	0.02	
199. <i>Poa pratensis</i>	1.28	0.93	0.41	
200. <i>Podocarpus nubigena</i>	0.63	0.28	-0.24	
201. <i>Podocarpus saligna</i>	0.67	0.32	-0.20	
202. <i>Podophyllum peltatum</i>	1.43	1.08	0.56	
203. <i>Pomaderris ferruginea</i>	0.75	0.40	-0.12	
204. <i>Populus deltoides</i>	1.15	0.80	0.28	
205. <i>Populus fremontii</i>	1.16	0.81	0.29	
206. <i>Populus tremuloides</i>	1.35	1.00	0.48	
207. <i>Potamogeton pectinatus</i>		0.67	0.15	Menendez & Sanchez 1998
208. <i>Potentilla arguta</i>	1.12	0.77	0.25	
209. <i>Prosopis glandulosa</i>	0.94	0.59	0.07	
210. <i>Protea acaulos</i>	0.66	0.31	-0.21	
211. <i>Protea neriifolia</i>	0.35	0.00	-0.52	
212. <i>Protea nitida</i>	0.50	0.15	-0.37	
213. <i>Protea repens</i>	0.63	0.28	-0.24	
214. <i>Protium sp1</i>	0.78	0.43	-0.09	
215. <i>Protium sp2</i>	0.76	0.41	-0.11	
216. <i>Prumnopitys andina</i>	0.61	0.26	-0.26	
217. <i>Psychrophila leptosepala</i>	1.18	0.83	0.31	
218. <i>Pterocaulon pycnostachyum</i>	1.19	0.84	0.32	
219. <i>Pultenea daphnoides</i>	0.89	0.54	0.02	
220. <i>Pultenea flexilis</i>	1.15	0.80	0.28	
221. <i>Quercus alba</i>	1.05	0.70	0.18	
222. <i>Quercus coccinea</i>	1.06	0.71	0.19	
223. <i>Quercus ellipsoidalis</i>	1.12	0.77	0.25	
224. <i>Quercus laevis</i>	0.80	0.45	-0.07	
225. <i>Quercus macrocarpa</i>	1.25	0.90	0.38	
226. <i>Quercus myrsinaefolia</i>	0.95	0.60	0.08	
227. <i>Quercus prinus</i>	1.08	0.73	0.21	
228. <i>Quercus rubra</i>	0.92	0.57	0.05	
229. <i>Quercus turbinella</i>	1.19	0.84	0.32	
230. <i>Quercus virginia</i> var <i>geminata</i>	0.76	0.41	-0.11	
231. <i>Regelia ciliata</i>	0.98	0.63	0.11	
232. <i>Rhododendron maximum</i>	0.58	0.23	-0.29	
233. <i>Robinia pseudoacacia</i>	1.05	0.70	0.18	
234. <i>Rudbeckia serotina</i>	1.17	0.82	0.30	
235. <i>Salix glauca</i>	1.25	0.90	0.38	
236. <i>Salix planifolia</i>	1.39	1.04	0.52	
237. <i>Sanguinaria canadensis</i>	1.81	1.46	0.94	
238. <i>Santalum acuminatum</i>	0.59	0.24	-0.28	
239. <i>Saxegothea conspicua</i>	0.68	0.33	-0.19	
240. <i>Schizachyrium scoparium</i>	1.09	0.74	0.22	
241. <i>Scholtzia involucrata</i>	1.00	0.65	0.13	
242. <i>Senna artemisioides</i> 3lft	1.09	0.74	0.22	
243. <i>Silphium integrifolium</i>	1.28	0.93	0.41	
244. <i>Silphium terebinthinaceum</i>	1.25	0.90	0.38	
245. <i>Solanum ferocissimum</i>	1.42	1.07	0.55	
246. <i>Solanum straminifolia</i>	1.53	1.18	0.66	
247. <i>Solidago nemoralis</i>	1.16	0.81	0.29	
248. <i>Solidago rigida</i>	1.08	0.73	0.21	
249. <i>Sorghastrum nutans</i>	1.10	0.75	0.23	
250. <i>Spartothamnella puberula</i>	1.52	1.17	0.65	
251. <i>Stipa spartea</i>	1.04	0.69	0.17	
252. <i>Stirlingia latifolia</i>	0.86	0.51	-0.01	
253. <i>Syncarpia glomulifera</i>	0.79	0.44	-0.08	

254. Synoum glandulosum	0.96	0.61	0.09	
255. Taxodium distichum	1.01	0.66	0.14	
256. Thalassodendron ciliatum			0.03	Titlyanov et al. 1992
257. Tilia americana	1.02	0.67	0.15	
258. Trema tomentosa	1.25	0.90	0.38	
259. Triodia scabra	0.91	0.56	0.04	
260. Tsuga canadensis	0.49	0.14	-0.38	
261. Urera caracasana	1.46	1.11	0.59	
262. Vaccinium arboreum	0.81	0.46	-0.06	
263. Vaccinium corymbosum	1.10	0.75	0.23	
264. Vaccinium myrtillus	1.03	0.68	0.16	
265. Veratrum parviflorum	1.27	0.92	0.40	
266. Verticordia nitens	0.78	0.43	-0.09	
267. Vismia japurensis	0.97	0.62	0.10	
268. Vismia lauriformis	1.04	0.69	0.17	
269. Weinmannia trichosperma			0.18	Lusk & del Pozo 2002
270. Xanthorrhoea preissii	0.49	0.14	-0.38	
271. Xylomelum pyriforme	0.59	0.24	-0.28	

Additional references to Table S10:

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Dataset S11. Dark respiration rates in seedlings and saplings of vascular plants

Notes to Table S11:

Dark respiration rates of whole vascular plants (seedlings and tree saplings) are taken from the work of Reich, P. B., Tjoelker, M. G., Machado, J.-L. & Oleksyn, J. Universal scaling of respiratory metabolism, size and nitrogen in plants. *Nature* 439, 457–461 (2006).

"Organism" — designation of the group to which the studied individual belongs as described in Reich et al. (2006);

"LogDMg" — decimal logarithm of whole plant dry mass DM, g;

"LogNMg" — decimal logarithm of whole plant nitrogen mass NMg, g;

"LogQd" — decimal logarithm of whole plant dark respiration Qd, nmol CO₂ s⁻¹, at 24 °C;

"LogqWkg" — decimal logarithm of whole plant dark respiration in W (kg wet mass)⁻¹, converted from Qd/DMg using respiratory quotient of unity (1 mol CO₂ released = 1 mol O₂ consumed), energy conversion 1 ml O₂ = 20 J and DM/WM (dry mass to wet mass ratio) = 0.3 (SI Methods, Table S12a);

"Logq25Wkg" — decimal logarithm of dark respiration rate converted to 25 °C using Q₁₀ = 2, $q_{25Wkg} = q_{Wkg} \times Q_{10}^{(25 - 24)/10}$.

See Reich et al. (2006) for details of this data base.

Table S11.

Organism	Lab/field	Seedling/ Sapling	LogDMg	LogqWkg	Logq25Wkg	LogQd	LogNMg
1. 9speciesGH	lab	Seedling	-2.11492162	0.654	0.685	-0.59127355	-3.41943994
2. 9speciesGH	lab	Seedling	-2.03151705	0.506	0.537	-0.65557661	-3.47041567
3. 9speciesGH	lab	Seedling	-1.83713701	0.665	0.696	-0.30223692	-3.464225
4. 9speciesGH	lab	Seedling	-1.79048499	0.392	0.423	-0.52858176	-3.14212497
5. 9speciesGH	lab	Seedling	-1.70333481	0.356	0.387	-0.47737077	-3.22476831
6. 9speciesGH	lab	Seedling	-1.68193667	0.061	0.092	-0.75095274	-3.25380187
7. 9speciesGH	lab	Seedling	-1.67468963	0.547	0.578	-0.25789094	-3.02632962
8. 9speciesGH	lab	Seedling	-1.65560773	0.548	0.579	-0.23774719	-3.00143118
9. 9speciesGH	lab	Seedling	-1.64397414	0.535	0.566	-0.23876391	-2.9897976
10. 9speciesGH	lab	Seedling	-1.64397414	0.552	0.583	-0.22181822	-2.94849247
11. 9speciesGH	lab	Seedling	-1.61439373	0.086	0.117	-0.6584135	-3.2044606
12. 9speciesGH	lab	Seedling	-1.60730305	0.640	0.671	-0.09771324	-2.98096268
13. 9speciesGH	lab	Seedling	-1.58252831	0.167	0.198	-0.54535481	-3.14477774
14. 9speciesGH	lab	Seedling	-1.54287537	0.735	0.766	0.06210409	-2.8473937
15. 9speciesGH	lab	Seedling	-1.4828041	0.236	0.267	-0.37728039	-3.07287098
16. 9speciesGH	lab	Seedling	-1.47886192	0.480	0.511	-0.12842409	-3.03638215
17. 9speciesGH	lab	Seedling	-1.47755577	0.219	0.250	-0.38869971	-3.02576933
18. 9speciesGH	lab	Seedling	-1.4710833	0.517	0.548	-0.08428971	-2.82272329
19. 9speciesGH	lab	Seedling	-1.46852108	0.510	0.541	-0.08895892	-3.079355
20. 9speciesGH	lab	Seedling	-1.44309473	0.172	0.203	-0.40121433	-3.01495994
21. 9speciesGH	lab	Seedling	-1.4424928	0.527	0.558	-0.04587786	-2.99070636
22. 9speciesGH	lab	Seedling	-1.38668684	0.347	0.378	-0.16993324	-2.81381524
23. 9speciesGH	lab	Seedling	-1.37830454	0.731	0.762	0.22306291	-2.724128
24. 9speciesGH	lab	Seedling	-1.3580304	0.490	0.521	0.00235317	-2.91555063
25. 9speciesGH	lab	Seedling	-1.33068312	0.179	0.210	-0.28122492	-2.90254833
26. 9speciesGH	lab	Seedling	-1.326058	0.427	0.458	-0.02890524	-2.88357823
27. 9speciesGH	lab	Seedling	-1.31740371	0.651	0.682	0.20342265	-2.69106334
28. 9speciesGH	lab	Seedling	-1.29670862	0.210	0.241	-0.21635533	-2.85895806

29.	9speciesGH	lab	Seedling	-1.27245874	-0.088	-0.057	-0.4905511	
30.	9speciesGH	lab	Seedling	-1.26841123	0.417	0.448	0.01850373	-2.8166248
31.	9speciesGH	lab	Seedling	-1.24795155	0.668	0.699	0.2897444	
32.	9speciesGH	lab	Seedling	-1.24641694	0.359	0.390	-0.01720775	-2.67354534
33.	9speciesGH	lab	Seedling	-1.23882419	0.365	0.396	-0.00361166	
34.	9speciesGH	lab	Seedling	-1.23844801	0.507	0.538	0.13898247	-2.81193675
35.	9speciesGH	lab	Seedling	-1.23582387	0.570	0.601	0.20381927	-2.78403743
36.	9speciesGH	lab	Seedling	-1.23545028	0.454	0.485	0.08901859	-2.79769972
37.	9speciesGH	lab	Seedling	-1.22914799	0.202	0.233	-0.15677624	-2.85075009
38.	9speciesGH	lab	Seedling	-1.22767829	0.415	0.446	0.05756739	-2.78992773
39.	9speciesGH	lab	Seedling	-1.22548303	0.187	0.218	-0.16823196	-2.65261143
40.	9speciesGH	lab	Seedling	-1.22184875	0.516	0.547	0.16405003	-2.59550838
41.	9speciesGH	lab	Seedling	-1.21681131	0.268	0.299	-0.07901018	-2.64393971
42.	9speciesGH	lab	Seedling	-1.21112488	0.518	0.549	0.17679226	-2.58478452
43.	9speciesGH	lab	Seedling	-1.17619985	0.361	0.392	0.05460618	-2.74968858
44.	9speciesGH	lab	Seedling	-1.17230771	0.184	0.215	-0.11846308	-2.81057988
45.	9speciesGH	lab	Seedling	-1.1697322	0.179	0.210	-0.1208239	-2.80800436
46.	9speciesGH	lab	Seedling	-1.12784373	0.500	0.531	0.24179694	
47.	9speciesGH	lab	Seedling	-1.11918641	0.394	0.425	0.1445964	-2.58389229
48.	9speciesGH	lab	Seedling	-1.11662251	0.324	0.355	0.07696062	-2.75489467
49.	9speciesGH	lab	Seedling	-1.11153968	0.263	0.294	0.02150317	-2.74981185
50.	9speciesGH	lab	Seedling	-1.09151498	0.696	0.727	0.47472278	-2.45402525
51.	9speciesGH	lab	Seedling	-1.08354605	0.512	0.543	0.29806678	-2.54825193
52.	9speciesGH	lab	Seedling	-1.079355	0.165	0.196	-0.04388578	-2.54406088
53.	9speciesGH	lab	Seedling	-1.0639892	0.602	0.633	0.40819904	-2.49696284
54.	9speciesGH	lab	Seedling	-1.05256628	0.706	0.737	0.52330929	-2.52881981
55.	9speciesGH	lab	Seedling	-1.0501223	0.346	0.377	0.16636893	
56.	9speciesGH	lab	Seedling	-1.03857891	0.388	0.419	0.21904618	-2.50328479
57.	9speciesGH	lab	Seedling	-1.03778856	0.201	0.232	0.03277438	-2.57092094
58.	9speciesGH	lab	Seedling	-1.03464149	0.158	0.189	-0.00686165	-2.56777387
59.	9speciesGH	lab	Seedling	-1.02159121	0.389	0.420	0.23718865	-2.67523823
60.	9speciesGH	lab	Seedling	-1.00833099	0.388	0.419	0.24942049	-2.58181973
61.	9speciesGH	lab	Seedling	-1.0015227	0.455	0.486	0.32387073	-2.65516972
62.	9speciesGH	lab	Seedling	-0.99934905	0.399	0.430	0.26969792	-2.56001635
63.	9speciesGH	lab	Seedling	-0.98927613	0.247	0.278	0.12750387	-2.52240851
64.	9speciesGH	lab	Seedling	-0.986952	0.430	0.461	0.31289544	-2.56044074
65.	9speciesGH	lab	Seedling	-0.97819072	0.549	0.580	0.44064929	-2.52640429
66.	9speciesGH	lab	Seedling	-0.97819072	0.597	0.628	0.48864194	
67.	9speciesGH	lab	Seedling	-0.97305837	0.512	0.543	0.40943524	
68.	9speciesGH	lab	Seedling	-0.96517108	0.607	0.638	0.51200214	-2.41372109
69.	9speciesGH	lab	Seedling	-0.96377046	0.452	0.483	0.35794523	
70.	9speciesGH	lab	Seedling	-0.95272513	0.726	0.757	0.64337597	-2.48290312
71.	9speciesGH	lab	Seedling	-0.93930216	0.542	0.573	0.47240992	-2.48293613
72.	9speciesGH	lab	Seedling	-0.93685436	0.436	0.467	0.36895867	-2.59050139
73.	9speciesGH	lab	Seedling	-0.92996213	0.573	0.604	0.51337431	-2.32465709
74.	9speciesGH	lab	Seedling	-0.92811799	0.378	0.409	0.31972063	
75.	9speciesGH	lab	Seedling	-0.91703521	0.345	0.376	0.29821519	
76.	9speciesGH	lab	Seedling	-0.90135627	0.110	0.141	0.0784276	-2.53028841
77.	9speciesGH	lab	Seedling	-0.89414933	0.204	0.235	0.1794889	
78.	9speciesGH	lab	Seedling	-0.89211197	0.689	0.720	0.66671147	
79.	9speciesGH	lab	Seedling	-0.88322627	0.438	0.469	0.4249101	-2.41340426
80.	9speciesGH	lab	Seedling	-0.87778412	0.189	0.220	0.1813797	-2.5639169
81.	9speciesGH	lab	Seedling	-0.87305728	0.600	0.631	0.5974059	-2.30603092
82.	9speciesGH	lab	Seedling	-0.8711163	0.334	0.365	0.33283072	-2.55724908
83.	9speciesGH	lab	Seedling	-0.85949196	0.296	0.327	0.30698231	-2.54562474
84.	9speciesGH	lab	Seedling	-0.85356186	0.647	0.678	0.663663	
85.	9speciesGH	lab	Seedling	-0.84924356	0.198	0.229	0.21920565	-2.38237594
86.	9speciesGH	lab	Seedling	-0.84103474	0.312	0.343	0.34049637	-2.37121272

87.	9speciesGH	lab	Seedling	-0.80548566	0.765	0.796	0.82959025	
88.	9speciesGH	lab	Seedling	-0.80465394	0.550	0.581	0.61581204	-2.35286751
89.	9speciesGH	lab	Seedling	-0.78648224	0.482	0.513	0.56519937	-2.20784303
90.	9speciesGH	lab	Seedling	-0.77910775	0.276	0.307	0.36664078	-2.46524053
91.	9speciesGH	lab	Seedling	-0.77481999	0.645	0.676	0.74057044	
92.	9speciesGH	lab	Seedling	-0.77237035	0.398	0.429	0.49573596	-2.30254833
93.	9speciesGH	lab	Seedling	-0.76095091	0.694	0.725	0.80324373	-2.23720444
94.	9speciesGH	lab	Seedling	-0.75808015	0.649	0.680	0.76062054	-2.20663015
95.	9speciesGH	lab	Seedling	-0.73992861	0.452	0.483	0.58246759	-2.28356258
96.	9speciesGH	lab	Seedling	-0.72170379	0.326	0.357	0.47465283	-2.3994845
97.	9speciesGH	lab	Seedling	-0.72010502	0.349	0.380	0.49884813	-2.4552872
98.	9speciesGH	lab	Seedling	-0.70929776	0.465	0.496	0.62582601	-2.33638575
99.	9speciesGH	lab	Seedling	-0.70071067	0.569	0.600	0.73821748	
100.	9speciesGH	lab	Seedling	-0.68486968	0.697	0.728	0.88177544	-2.04737995
101.	9speciesGH	lab	Seedling	-0.68455448	0.627	0.658	0.81276009	-2.31164248
102.	9speciesGH	lab	Seedling	-0.68402965	0.468	0.499	0.65404968	-2.22766362
103.	9speciesGH	lab	Seedling	-0.68235446	0.541	0.572	0.72894653	
104.	9speciesGH	lab	Seedling	-0.64781748	0.306	0.337	0.52769094	-2.27674962
105.	9speciesGH	lab	Seedling	-0.64653259	0.315	0.346	0.53846725	-2.38171476
106.	9speciesGH	lab	Seedling	-0.62333181	0.259	0.290	0.50568184	-2.30111252
107.	9speciesGH	lab	Seedling	-0.60730305	0.400	0.431	0.66293084	-2.34248522
108.	9speciesGH	lab	Seedling	-0.59808275	0.242	0.273	0.51363214	-2.27586345
109.	9speciesGH	lab	Seedling	-0.5866164	0.599	0.630	0.8824728	
110.	9speciesGH	lab	Seedling	-0.58269442	0.614	0.645	0.9016204	-2.05894795
111.	9speciesGH	lab	Seedling	-0.58004425	0.316	0.347	0.60574966	-2.31522643
112.	9speciesGH	lab	Seedling	-0.51074483	0.522	0.553	0.88167063	-2.12335501
113.	9speciesGH	lab	Seedling	-0.50675113	0.600	0.631	0.96361639	-2.13383913
114.	9speciesGH	lab	Seedling	-0.49825627	0.530	0.561	0.90198872	-2.11086644
115.	9speciesGH	lab	Seedling	-0.41566878	0.557	0.588	1.01114576	-2.02827895
116.	9speciesGH	lab	Seedling	-0.40285351	0.690	0.721	1.15670058	-1.93896052
117.	9speciesGH	lab	Seedling	-0.39577395	0.508	0.539	0.98175349	-2.03974809
118.	9speciesGH	lab	Seedling	-0.37960365	0.618	0.649	1.10825289	-2.00669165
119.	9speciesGH	lab	Seedling	-0.37263414	0.586	0.617	1.08294951	-1.79399493
120.	9speciesGH	lab	Seedling	-0.36151074	0.568	0.599	1.07671791	-1.97412092
121.	9speciesGH	lab	Seedling	-0.35418488	0.698	0.729	1.21425723	
122.	9speciesGH	lab	Seedling	-0.33254705	0.393	0.424	0.93078237	-1.727242
123.	9speciesGH	lab	Seedling	-0.29379446	0.599	0.630	1.17515806	
124.	9speciesGH	lab	Seedling	-0.27511	0.785	0.816	1.37994387	-1.86856982
125.	9speciesGH	lab	Seedling	-0.24949161	0.607	0.638	1.2275249	-1.64418656
126.	9speciesGH	lab	Seedling	-0.21631076	0.617	0.648	1.27053232	-1.66486077
127.	9speciesGH	lab	Seedling	-0.21253953	0.305	0.336	0.96202542	-1.85651367
128.	9speciesGH	lab	Seedling	-0.20276709	0.663	0.694	1.33044983	-1.79622691
129.	9speciesGH	lab	Seedling	-0.1877553	0.487	0.518	1.16913513	-1.83172945
130.	9speciesGH	lab	Seedling	-0.18508682	0.601	0.632	1.28561339	-1.60644761
131.	9speciesGH	lab	Seedling	-0.1150065	0.641	0.672	1.3955971	-1.70846632
132.	9speciesGH	lab	Seedling	-0.07696333	0.583	0.614	1.37598768	-1.61307034
133.	9speciesGH	lab	Seedling	0.0186381	0.651	0.682	1.53974614	-1.57482172
134.	9speciesGH	lab	Seedling	0.01996742	0.666	0.697	1.55607927	-1.55842865
135.	9speciesGH	lab	Seedling	0.04095817	0.548	0.579	1.4591865	-1.49514884
136.	9speciesGH	lab	Seedling	0.11085896	0.614	0.645	1.59496034	-1.46753712
137.	9speciesGH	lab	Seedling	0.1248953	0.469	0.500	1.46357681	-1.41121171
138.	9speciesGH	lab	Seedling	0.13700578	0.565	0.596	1.57202423	-1.4413903
139.	9speciesGH	lab	Seedling	0.16286299	0.304	0.335	1.33668443	-1.54488094
140.	9speciesGH	lab	Seedling	0.21789173	0.509	0.540	1.59726783	-1.36050435
141.	9speciesGH	lab	Seedling	0.26670197	0.470	0.501	1.60705217	
142.	9speciesGH	lab	Seedling	0.34271869	0.390	0.421	1.6031299	
143.	9speciesGH	lab	Seedling	0.37759763	0.525	0.556	1.77229729	
144.	9speciesGH	lab	Seedling	0.38118731	0.519	0.550	1.77041612	

145.	9speciesGH	lab	Seedling	0.42084654	0.673	0.704	1.9634765	
146.	9speciesGH	lab	Seedling	0.47026345	0.454	0.485	1.7940897	-1.23748048
147.	9speciesGH	lab	Seedling	0.53576235	0.430	0.461	1.8362488	
148.	9speciesGH	lab	Seedling	0.58629468	0.554	0.585	2.01037272	
149.	9speciesGH	lab	Seedling	0.6404218	0.447	0.478	1.95709161	
150.	9speciesGH	lab	Seedling	0.64819407	0.353	0.384	1.87108773	
151.	9speciesGH	lab	Seedling	0.65705585	0.394	0.425	1.92099253	
152.	9speciesGH	lab	Seedling	0.74264659	0.449	0.480	2.06209307	-0.96509734
153.	9speciesGH	lab	Seedling	0.76826802	0.357	0.388	1.99514949	-1.01424804
154.	9speciesGH	lab	Seedling	0.78653848	0.455	0.486	2.11197982	-0.93016029
155.	9speciesGH	lab	Seedling	0.79427887	0.423	0.454	2.08747872	-0.92241991
156.	9speciesGH	lab	Seedling	0.80848106	0.315	0.346	1.99336603	-0.974035
157.	9speciesGH	lab	Seedling	0.83289194	0.383	0.414	2.08556949	-0.88380683
158.	9speciesGH	lab	Seedling	0.83537345	0.281	0.312	1.98607161	-0.9471426
159.	9speciesGH	lab	Seedling	0.86607457	0.412	0.443	2.14783756	-0.8506242
160.	9speciesGH	lab	Seedling	0.87155542	0.379	0.410	2.12066637	-0.83618851
161.	9speciesGH	lab	Seedling	0.92298482	0.297	0.328	2.08958834	-0.85953124
162.	9speciesGH	lab	Seedling	0.96156346	0.280	0.311	2.11130565	-0.83161066
163.	9speciesGH	lab	Seedling	0.98154681	0.273	0.304	2.12437085	-0.81162732
164.	9speciesGH	lab	Seedling	1.03350417	0.378	0.409	2.2810345	-0.75966995
165.	9speciesGH	lab	Seedling	1.11571034	0.385	0.416	2.37111728	-0.67746379
166.	BorealGCB	lab	Seedling	-1.84403676	0.787	0.818	-0.1873045	
167.	BorealGCB	lab	Seedling	-1.78059915	0.761	0.792	-0.14950921	
168.	BorealGCB	lab	Seedling	-1.76848145	0.330	0.361	-0.56871751	
169.	BorealGCB	lab	Seedling	-1.72515831	1.007	1.038	0.1515138	
170.	BorealGCB	lab	Seedling	-1.67177268	0.586	0.617	-0.21562247	
171.	BorealGCB	lab	Seedling	-1.66672731	0.393	0.424	-0.40390884	
172.	BorealGCB	lab	Seedling	-1.66409863	0.342	0.373	-0.45222843	
173.	BorealGCB	lab	Seedling	-1.62836038	0.521	0.552	-0.2374705	
174.	BorealGCB	lab	Seedling	-1.60221293	0.759	0.790	0.02657622	
175.	BorealGCB	lab	Seedling	-1.6016569	0.950	0.981	0.21846933	
176.	BorealGCB	lab	Seedling	-1.58861068	0.642	0.673	-0.0764476	
177.	BorealGCB	lab	Seedling	-1.58719161	0.295	0.326	-0.42259547	
178.	BorealGCB	lab	Seedling	-1.57383607	0.711	0.742	0.007529	
179.	BorealGCB	lab	Seedling	-1.55887627	0.356	0.387	-0.33241796	
180.	BorealGCB	lab	Seedling	-1.54518674	0.373	0.404	-0.30245844	
181.	BorealGCB	lab	Seedling	-1.5439263	0.317	0.348	-0.35695663	
182.	BorealGCB	lab	Seedling	-1.53420337	0.642	0.673	-0.02186812	
183.	BorealGCB	lab	Seedling	-1.45198735	0.640	0.671	0.05769425	
184.	BorealGCB	lab	Seedling	-1.44018042	0.568	0.599	-0.00255182	
185.	BorealGCB	lab	Seedling	-1.43085616	0.650	0.681	0.08936966	
186.	BorealGCB	lab	Seedling	-1.4010724	0.700	0.731	0.16925447	
187.	BorealGCB	lab	Seedling	-1.3985619	0.642	0.673	0.11372711	
188.	BorealGCB	lab	Seedling	-1.3928738	0.570	0.601	0.04762384	
189.	BorealGCB	lab	Seedling	-1.3772967	0.727	0.758	0.21929648	
190.	BorealGCB	lab	Seedling	-1.36303744	0.490	0.521	-0.00255412	
191.	BorealGCB	lab	Seedling	-1.35756484	0.331	0.362	-0.15633302	
192.	BorealGCB	lab	Seedling	-1.34479543	0.402	0.433	-0.07295616	
193.	BorealGCB	lab	Seedling	-1.33776748	0.341	0.372	-0.12636596	
194.	BorealGCB	lab	Seedling	-1.33440507	0.548	0.579	0.08376197	
195.	BorealGCB	lab	Seedling	-1.304694	0.332	0.363	-0.10288943	
196.	BorealGCB	lab	Seedling	-1.29663085	0.316	0.347	-0.11057881	
197.	BorealGCB	lab	Seedling	-1.28504177	0.435	0.466	0.01976813	
198.	BorealGCB	lab	Seedling	-1.28362062	0.660	0.691	0.24649645	
199.	BorealGCB	lab	Seedling	-1.24202113	0.704	0.735	0.33241455	
200.	BorealGCB	lab	Seedling	-1.23913232	0.894	0.925	0.52473695	
201.	BorealGCB	lab	Seedling	-1.23337063	0.511	0.542	0.14788767	
202.	BorealGCB	lab	Seedling	-1.22151099	0.591	0.622	0.23965009	

203.	BorealGCB	lab	Seedling	-1.21886859	0.356	0.387	0.00752078
204.	BorealGCB	lab	Seedling	-1.2153766	0.365	0.396	0.0197505
205.	BorealGCB	lab	Seedling	-1.21483124	0.622	0.653	0.27728974
206.	BorealGCB	lab	Seedling	-1.20643051	0.495	0.526	0.1581039
207.	BorealGCB	lab	Seedling	-1.19675558	0.417	0.448	0.09020601
208.	BorealGCB	lab	Seedling	-1.19240145	0.631	0.662	0.30869797
209.	BorealGCB	lab	Seedling	-1.18267596	0.604	0.635	0.29153176
210.	BorealGCB	lab	Seedling	-1.1486706	0.551	0.582	0.27281959
211.	BorealGCB	lab	Seedling	-1.1451127	1.140	1.171	0.86473202
212.	BorealGCB	lab	Seedling	-1.11614819	0.478	0.509	0.23152874
213.	BorealGCB	lab	Seedling	-1.11132107	0.541	0.572	0.29968373
214.	BorealGCB	lab	Seedling	-1.09748161	0.459	0.490	0.23153083
215.	BorealGCB	lab	Seedling	-1.07562789	0.225	0.256	0.01976287
216.	BorealGCB	lab	Seedling	-1.06651571	0.276	0.307	0.07975732
217.	BorealGCB	lab	Seedling	-1.06314129	1.044	1.075	0.85048849
218.	BorealGCB	lab	Seedling	-1.06013766	0.608	0.639	0.41770058
219.	BorealGCB	lab	Seedling	-1.05559018	0.608	0.639	0.42258083
220.	BorealGCB	lab	Seedling	-1.05406257	0.527	0.558	0.34340627
221.	BorealGCB	lab	Seedling	-1.03779123	0.657	0.688	0.48951588
222.	BorealGCB	lab	Seedling	-1.02680859	0.662	0.693	0.504779
223.	BorealGCB	lab	Seedling	-1.01437529	0.602	0.633	0.45735259
224.	BorealGCB	lab	Seedling	-1.01287185	0.479	0.510	0.33658535
225.	BorealGCB	lab	Seedling	-1.00991495	0.445	0.476	0.30532163
226.	BorealGCB	lab	Seedling	-0.96258474	0.597	0.628	0.50477364
227.	BorealGCB	lab	Seedling	-0.9561018	0.415	0.446	0.32876669
228.	BorealGCB	lab	Seedling	-0.94918741	0.416	0.447	0.33657998
229.	BorealGCB	lab	Seedling	-0.94497292	0.796	0.827	0.72102624
230.	BorealGCB	lab	Seedling	-0.93468932	0.572	0.603	0.50773417
231.	BorealGCB	lab	Seedling	-0.93318242	0.898	0.929	0.83476141
232.	BorealGCB	lab	Seedling	-0.91832236	0.867	0.898	0.81913263
233.	BorealGCB	lab	Seedling	-0.8977889	0.519	0.550	0.49080056
234.	BorealGCB	lab	Seedling	-0.89276692	0.608	0.639	0.58549108
235.	BorealGCB	lab	Seedling	-0.87104183	0.353	0.384	0.351833
236.	BorealGCB	lab	Seedling	-0.86542793	0.416	0.447	0.42091628
237.	BorealGCB	lab	Seedling	-0.86154295	0.415	0.446	0.42341477
238.	BorealGCB	lab	Seedling	-0.8558831	0.485	0.516	0.49913315
239.	BorealGCB	lab	Seedling	-0.85563762	0.322	0.353	0.33658769
240.	BorealGCB	lab	Seedling	-0.85409369	0.570	0.601	0.58549608
241.	BorealGCB	lab	Seedling	-0.85119334	0.220	0.251	0.23882543
242.	BorealGCB	lab	Seedling	-0.77836583	0.219	0.250	0.31032546
243.	BorealGCB	lab	Seedling	-0.77731375	0.842	0.873	0.93445919
244.	BorealGCB	lab	Seedling	-0.75782616	0.357	0.388	0.46920928
245.	BorealGCB	lab	Seedling	-0.75560881	0.401	0.432	0.51576116
246.	BorealGCB	lab	Seedling	-0.74470611	0.688	0.719	0.81323205
247.	BorealGCB	lab	Seedling	-0.72853243	0.508	0.539	0.6490732
248.	BorealGCB	lab	Seedling	-0.72044603	0.558	0.589	0.70725857
249.	BorealGCB	lab	Seedling	-0.71149659	0.865	0.896	1.02325486
250.	BorealGCB	lab	Seedling	-0.69303245	0.566	0.597	0.74312403
251.	BorealGCB	lab	Seedling	-0.6876188	0.222	0.253	0.40434523
252.	BorealGCB	lab	Seedling	-0.68004783	0.266	0.297	0.45588991
253.	BorealGCB	lab	Seedling	-0.67295083	0.345	0.376	0.54173234
254.	BorealGCB	lab	Seedling	-0.66490317	0.623	0.654	0.82770751
255.	BorealGCB	lab	Seedling	-0.65598682	0.783	0.814	0.9966189
256.	BorealGCB	lab	Seedling	-0.62388197	0.737	0.768	0.98315951
257.	BorealGCB	lab	Seedling	-0.619392	0.779	0.810	1.03004013
258.	BorealGCB	lab	Seedling	-0.60506099	0.698	0.729	0.96340563
259.	BorealGCB	lab	Seedling	-0.59716811	0.508	0.539	0.7804654
260.	BorealGCB	lab	Seedling	-0.58342939	1.035	1.066	1.32197256

261.	BorealGCB	lab	Seedling	-0.56451332	0.990	1.021	1.2954292
262.	BorealGCB	lab	Seedling	-0.55450431	0.564	0.595	0.87935624
263.	BorealGCB	lab	Seedling	-0.53712313	0.595	0.626	0.92758629
264.	BorealGCB	lab	Seedling	-0.52148327	0.840	0.871	1.18806668
265.	BorealGCB	lab	Seedling	-0.51493472	0.591	0.622	0.94630375
266.	BorealGCB	lab	Seedling	-0.51180304	0.528	0.559	0.88654837
267.	BorealGCB	lab	Seedling	-0.50631129	0.348	0.379	0.71191717
268.	BorealGCB	lab	Seedling	-0.49231292	0.492	0.523	0.87005286
269.	BorealGCB	lab	Seedling	-0.48575539	0.455	0.486	0.8397261
270.	BorealGCB	lab	Seedling	-0.48178542	0.525	0.556	0.91349033
271.	BorealGCB	lab	Seedling	-0.47866463	0.410	0.441	0.80138664
272.	BorealGCB	lab	Seedling	-0.43980771	0.499	0.530	0.92905433
273.	BorealGCB	lab	Seedling	-0.43630005	0.714	0.745	1.14758391
274.	BorealGCB	lab	Seedling	-0.41226878	0.650	0.681	1.10736827
275.	BorealGCB	lab	Seedling	-0.40528514	0.781	0.812	1.24577655
276.	BorealGCB	lab	Seedling	-0.3781909	0.425	0.456	0.91678012
277.	BorealGCB	lab	Seedling	-0.37246081	0.497	0.528	0.99466071
278.	BorealGCB	lab	Seedling	-0.36241444	0.315	0.346	0.82232498
279.	BorealGCB	lab	Seedling	-0.35988862	0.732	0.763	1.24219381
280.	BorealGCB	lab	Seedling	-0.34929569	0.336	0.367	0.85631722
281.	BorealGCB	lab	Seedling	-0.33034277	0.871	0.902	1.41032408
282.	BorealGCB	lab	Seedling	-0.3290384	0.324	0.355	0.8651546
283.	BorealGCB	lab	Seedling	-0.328787	0.516	0.547	1.05701208
284.	BorealGCB	lab	Seedling	-0.32266413	0.421	0.452	0.9679327
285.	BorealGCB	lab	Seedling	-0.31850943	0.574	0.605	1.12520498
286.	BorealGCB	lab	Seedling	-0.31289815	0.657	0.688	1.21446213
287.	BorealGCB	lab	Seedling	-0.30066118	0.538	0.569	1.10782139
288.	BorealGCB	lab	Seedling	-0.27096555	0.184	0.215	0.78285222
289.	BorealGCB	lab	Seedling	-0.26489466	0.602	0.633	1.20676908
290.	BorealGCB	lab	Seedling	-0.26354347	0.666	0.697	1.27234629
291.	BorealGCB	lab	Seedling	-0.26318304	0.454	0.485	1.06082192
292.	BorealGCB	lab	Seedling	-0.25062986	0.608	0.639	1.22728418
293.	BorealGCB	lab	Seedling	-0.2018346	0.522	0.553	1.1904114
294.	BorealGCB	lab	Seedling	-0.19017988	0.953	0.984	1.63252706
295.	BorealGCB	lab	Seedling	-0.16604827	0.814	0.845	1.51843467
296.	BorealGCB	lab	Seedling	-0.16427608	0.592	0.623	1.29808612
297.	BorealGCB	lab	Seedling	-0.14345577	0.608	0.639	1.33495174
298.	BorealGCB	lab	Seedling	-0.13061282	0.327	0.358	1.06668509
299.	BorealGCB	lab	Seedling	-0.1285124	0.438	0.469	1.17922759
300.	BorealGCB	lab	Seedling	-0.10944367	0.670	0.701	1.43006878
301.	BorealGCB	lab	Seedling	-0.07090378	0.629	0.660	1.42814163
302.	BorealGCB	lab	Seedling	-0.06968546	0.514	0.545	1.3138272
303.	BorealGCB	lab	Seedling	-0.05515017	0.350	0.381	1.16471123
304.	BorealGCB	lab	Seedling	-0.05488394	0.577	0.608	1.39169537
305.	BorealGCB	lab	Seedling	-0.04734293	0.638	0.669	1.46020932
306.	BorealGCB	lab	Seedling	-0.04504638	0.334	0.365	1.15934764
307.	BorealGCB	lab	Seedling	-0.0381635	0.671	0.702	1.50260061
308.	BorealGCB	lab	Seedling	-0.03274643	0.641	0.672	1.47817011
309.	BorealGCB	lab	Seedling	-0.0205024	0.482	0.513	1.33163955
310.	BorealGCB	lab	Seedling	-0.00445054	0.696	0.727	1.56125122
311.	BorealGCB	lab	Seedling	-0.00404341	0.613	0.644	1.47911719
312.	BorealGCB	lab	Seedling	0.02884697	0.884	0.915	1.78246755
313.	BorealGCB	lab	Seedling	0.03675267	0.484	0.515	1.39059703
314.	BorealGCB	lab	Seedling	0.06018995	0.457	0.488	1.38679306
315.	BorealGCB	lab	Seedling	0.06452976	0.438	0.469	1.3722207
316.	BorealGCB	lab	Seedling	0.08102686	0.386	0.417	1.3374924
317.	BorealGCB	lab	Seedling	0.08214706	0.850	0.881	1.80188497
318.	BorealGCB	lab	Seedling	0.08235887	0.346	0.377	1.29788408

319.	BorealGCB	lab	Seedling	0.08995837	0.551	0.582	1.51057816	
320.	BorealGCB	lab	Seedling	0.09221892	0.751	0.782	1.71325214	
321.	BorealGCB	lab	Seedling	0.09887785	0.337	0.368	1.30540584	
322.	BorealGCB	lab	Seedling	0.10082294	0.593	0.624	1.56346004	
323.	BorealGCB	lab	Seedling	0.16304246	0.803	0.834	1.83636978	
324.	BorealGCB	lab	Seedling	0.17958787	0.508	0.539	1.55785895	
325.	BorealGCB	lab	Seedling	0.1816836	0.769	0.800	1.82020097	
326.	BorealGCB	lab	Seedling	0.2217474	0.450	0.481	1.54130522	
327.	BorealGCB	lab	Seedling	0.27432198	0.467	0.498	1.61131588	
328.	BorealGCB	lab	Seedling	0.29643481	0.464	0.495	1.6302515	
329.	BorealGCB	lab	Seedling	0.30078273	0.303	0.334	1.47366771	
330.	BorealGCB	lab	Seedling	0.32065762	0.367	0.398	1.55755997	
331.	BorealGCB	lab	Seedling	0.32200405	0.838	0.869	2.03037243	
332.	BorealGCB	lab	Seedling	0.3515702	0.351	0.382	1.57231685	
333.	BorealGCB	lab	Seedling	0.37020161	0.536	0.567	1.77640871	
334.	BorealGCB	lab	Seedling	0.42234141	0.491	0.522	1.78381874	
335.	BorealGCB	lab	Seedling	0.46713879	0.923	0.954	2.25993546	
336.	BorealGCB	lab	Seedling	0.47966737	0.795	0.826	2.14475283	
337.	BorealGCB	lab	Seedling	0.49567698	0.859	0.890	2.22475665	
338.	BorealGCB	lab	Seedling	0.5097555	0.578	0.609	1.95760313	
339.	BorealGCB	lab	Seedling	0.5276229	0.832	0.863	2.22982455	
340.	BorealGCB	lab	Seedling	0.53750975	0.400	0.431	1.80779008	
341.	BorealGCB	lab	Seedling	0.69488935	0.556	0.587	2.12113257	
342.	BorealGCB	lab	Seedling	0.73018523	0.599	0.630	2.19960229	
343.	BorealGCB	lab	Seedling	0.73380397	0.780	0.811	2.38426182	
344.	BorealGCB	lab	Seedling	0.75029967	0.830	0.861	2.44997641	
345.	BorealGCB	lab	Seedling	0.77139114	0.713	0.744	2.35391756	
346.	BorealGCB	lab	Seedling	0.78585503	0.862	0.893	2.5181878	
347.	BorealGCB	lab	Seedling	0.86344459	0.474	0.505	2.20744421	
348.	BorealGCB	lab	Seedling	0.89940405	0.546	0.577	2.31546143	
349.	BorealGCB	lab	Seedling	1.01911747	0.612	0.643	2.50155002	
350.	BorealGCB	lab	Seedling	1.05223447	0.416	0.447	2.33872079	
351.	BorealGCB	lab	Seedling	1.10244032	0.709	0.740	2.68142282	
352.	BorealGCB	lab	Seedling	1.16005351	0.779	0.810	2.80929901	
353.	BorealGCB	lab	Seedling	1.29503672	0.823	0.854	2.98770244	
354.	BorealGCB	lab	Seedling	1.31261932	0.584	0.615	2.76620319	
355.	BorealGCB	lab	Seedling	1.32923544	0.584	0.615	2.78281284	
356.	GHherbs	lab	Seedling	-1.89733766	0.681	0.712	-0.34626093	-3.35613434
357.	GHherbs	lab	Seedling	-1.70626924	0.704	0.735	-0.1318104	-3.14986344
358.	GHherbs	lab	Seedling	-1.68472957	0.629	0.660	-0.18534056	-3.0475583
359.	GHherbs	lab	Seedling	-1.5426228	0.475	0.506	-0.19795829	-3.17726163
360.	GHherbs	lab	Seedling	-1.52287875	0.622	0.653	-0.03077177	-3.14247591
361.	GHherbs	lab	Seedling	-1.5039934	0.675	0.706	0.04146487	-2.95301233
362.	GHherbs	lab	Seedling	-1.45181539	0.649	0.680	0.06718787	-3.06436504
363.	GHherbs	lab	Seedling	-1.41453927	0.485	0.516	-0.05992071	-2.8942994
364.	GHherbs	lab	Seedling	-1.33724217	0.735	0.766	0.26809868	-2.85273815
365.	GHherbs	lab	Seedling	-1.31875876	0.685	0.716	0.23657456	-2.9697586
366.	GHherbs	lab	Seedling	-1.29242982	0.426	0.457	0.00315031	-2.91766142
367.	GHherbs	lab	Seedling	-1.29242982	0.616	0.647	0.19371954	-2.74218525
368.	GHherbs	lab	Seedling	-1.28399666	0.747	0.778	0.33273033	-2.70833825
369.	GHherbs	lab	Seedling	-1.26760624	0.644	0.675	0.24654443	-2.69930525
370.	GHherbs	lab	Seedling	-1.26227741	0.504	0.535	0.11196978	-2.89870882
371.	GHherbs	lab	Seedling	-1.24667233	0.440	0.471	0.06304156	-2.96489981
372.	GHherbs	lab	Seedling	-1.16536739	0.684	0.715	0.38883856	-2.48184823
373.	GHherbs	lab	Seedling	-1.15697497	0.667	0.698	0.38036327	-2.50482117
374.	GHherbs	lab	Seedling	-1.14670748	0.729	0.760	0.45212556	-2.58665247
375.	GHherbs	lab	Seedling	-1.1426675	0.780	0.811	0.50734651	-2.56106514
376.	GHherbs	lab	Seedling	-1.11350927	0.698	0.729	0.45434044	-2.43574677

377.	GHherbs	lab	Seedling	-1.10513034	0.613	0.644	0.37744317	-2.62450481
378.	GHherbs	lab	Seedling	-1.08795517	0.467	0.498	0.24901408	-2.77222349
379.	GHherbs	lab	Seedling	-1.06173052	0.540	0.571	0.34814342	-2.53996549
380.	GHherbs	lab	Seedling	-1.03937069	0.726	0.757	0.55646463	-2.54425132
381.	GHherbs	lab	Seedling	-1.02687215	0.598	0.629	0.44090669	-2.45143844
382.	GHherbs	lab	Seedling	-0.98505965	0.582	0.613	0.46722996	-2.35641636
383.	GHherbs	lab	Seedling	-0.94884748	0.532	0.563	0.45287308	-2.45308922
384.	GHherbs	lab	Seedling	-0.93181414	0.585	0.616	0.52278741	-2.48471243
385.	GHherbs	lab	Seedling	-0.91009489	0.565	0.596	0.52475963	-2.50433135
386.	GHherbs	lab	Seedling	-0.90774735	0.800	0.831	0.762153	-2.35467478
387.	GHherbs	lab	Seedling	-0.87397688	0.637	0.668	0.6332009	-2.41912066
388.	GHherbs	lab	Seedling	-0.85490723	0.314	0.345	0.32906707	-2.40994391
389.	GHherbs	lab	Seedling	-0.82102305	0.659	0.690	0.70776786	-2.30460251
390.	GHherbs	lab	Seedling	-0.81530857	0.555	0.586	0.60974916	-2.50355884
391.	GHherbs	lab	Seedling	-0.77469072	0.763	0.794	0.85873546	-2.2941168
392.	GHherbs	lab	Seedling	-0.7721133	0.456	0.487	0.55389096	-2.3377522
393.	GHherbs	lab	Seedling	-0.64717456	0.664	0.695	0.88633125	-2.07308786
394.	GHherbs	lab	Seedling	-0.64589156	0.677	0.708	0.90077527	-2.22806128
395.	GHherbs	lab	Seedling	-0.63953282	0.450	0.481	0.68037114	-2.29929126
396.	GHherbs	lab	Seedling	-0.63638802	0.687	0.718	0.92067096	-1.97328461
397.	GHherbs	lab	Seedling	-0.61261017	0.534	0.565	0.79171284	-2.486155
398.	GHherbs	lab	Seedling	-0.59176003	0.571	0.602	0.84905272	-2.35292147
399.	GHherbs	lab	Seedling	-0.55232526	0.498	0.529	0.81524463	-1.99118964
400.	GHherbs	lab	Seedling	-0.537602	0.416	0.447	0.74834152	-2.22436477
401.	GHherbs	lab	Seedling	-0.48017201	0.580	0.611	0.97026605	-1.85786698
402.	GHherbs	lab	Seedling	-0.42365865	0.595	0.626	1.040991	-1.97700649
403.	GHherbs	lab	Seedling	-0.2670704	0.342	0.373	0.94455601	-2.03518811
404.	GHherbs	lab	Seedling	-0.25310644	0.582	0.613	1.19929655	-1.67204128
405.	GHherbs	lab	Seedling	-0.24565166	0.577	0.608	1.20134919	-1.67629463
406.	GHherbs	lab	Seedling	-0.21159592	0.548	0.579	1.20632452	-1.73775661
407.	GHherbs	lab	Seedling	-0.14752001	0.926	0.957	1.64799402	-1.66879187
408.	GHherbs	lab	Seedling	-0.14650459	0.575	0.606	1.29846634	-1.64775361
409.	GHherbs	lab	Seedling	-0.1352912	0.436	0.467	1.17107435	-1.92337997
410.	GHherbs	lab	Seedling	-0.10734897	0.596	0.627	1.35912029	-1.53979109
411.	GHherbs	lab	Seedling	-0.05469558	0.593	0.624	1.40877765	-1.47208243
412.	GHherbs	lab	Seedling	-0.04575749	0.512	0.543	1.33650842	-1.51670342
413.	GHherbs	lab	Seedling	-0.04415196	0.640	0.671	1.46629003	-1.53520334
414.	GHherbs	lab	Seedling	-0.00906946	0.512	0.543	1.37330382	-1.42457741
415.	GHherbs	lab	Seedling	-0.0070049	0.335	0.366	1.19801538	-1.71337809
416.	GHherbs	lab	Seedling	0.0150802	0.632	0.663	1.51733768	-1.48515853
417.	GHherbs	lab	Seedling	0.16066857	0.420	0.451	1.45114694	-1.39502301
418.	GHherbs	lab	Seedling	0.16583762	0.590	0.621	1.62595724	-1.26054702
419.	nMinnfield	field	Tree	-0.13846559	0.351	0.382	1.08301055	-2.12839788
			sapling					
420.	nMinnfield	field	Tree	0.07664044	-0.479	-0.448	0.46779468	-2.4481598
			sapling					
421.	nMinnfield	field	Tree	0.09025805	0.230	0.261	1.18990311	-1.89523983
			sapling					
422.	nMinnfield	field	Tree	0.22993769	0.169	0.200	1.26908521	-1.80894601
			sapling					
423.	nMinnfield	field	Tree	0.27760921	-0.714	-0.683	0.43400413	-1.98235686
			sapling					
424.	nMinnfield	field	Tree	0.3354579	-0.632	-0.601	0.57368775	-1.99749493
			sapling					
425.	nMinnfield	field	Tree	0.3494718	0.081	0.112	1.300798	-1.80152711
			sapling					
426.	nMinnfield	field	Tree	0.44138088	-0.265	-0.234	1.04658562	-1.8520085
			sapling					
427.	nMinnfield	field	Tree	0.45453998	-0.177	-0.146	1.14776508	-1.74881821
			sapling					

428.	nMinnfield	field	Tree	0.46893781	-0.014	0.017	1.32460269	-1.6302481
429.	nMinnfield	field	sapling					
429.	nMinnfield	field	Tree	0.48444221	-0.709	-0.678	0.64514178	-1.77128918
430.	nMinnfield	field	sapling					
430.	nMinnfield	field	Tree	0.49803472	-0.643	-0.612	0.72468137	-1.51655649
431.	nMinnfield	field	sapling					
431.	nMinnfield	field	Tree	0.50920252	-0.306	-0.275	1.07278712	-1.72510125
432.	nMinnfield	field	sapling					
432.	nMinnfield	field	Tree	0.53832233	-0.072	-0.041	1.33603242	-1.71113845
433.	nMinnfield	field	sapling					
433.	nMinnfield	field	Tree	0.55606116	-0.485	-0.454	0.94111776	-1.57088533
434.	nMinnfield	field	sapling					
434.	nMinnfield	field	Tree	0.62695595	-0.509	-0.478	0.98821929	-1.33178737
435.	nMinnfield	field	sapling					
435.	nMinnfield	field	Tree	0.65494623	-0.804	-0.773	0.72097286	-1.61314804
436.	nMinnfield	field	sapling					
436.	nMinnfield	field	Tree	0.66398345	-0.557	-0.526	0.97668253	-1.43201172
437.	nMinnfield	field	sapling					
437.	nMinnfield	field	Tree	0.66670514	-0.540	-0.509	0.99695878	-1.41413866
438.	nMinnfield	field	sapling					
438.	nMinnfield	field	Tree	0.71054045	-0.611	-0.580	0.9695777	-1.45489857
439.	nMinnfield	field	sapling					
439.	nMinnfield	field	Tree	0.72574833	0.092	0.123	1.68758593	-1.35641705
440.	nMinnfield	field	sapling					
440.	nMinnfield	field	Tree	0.7451529	-0.478	-0.447	1.13677602	-1.37328751
441.	nMinnfield	field	sapling					
441.	nMinnfield	field	Tree	0.78753132	-0.546	-0.515	1.11124796	-1.27229116
442.	nMinnfield	field	sapling					
442.	nMinnfield	field	Tree	0.82432112	-0.698	-0.667	0.996032	-1.48340395
443.	nMinnfield	field	sapling					
443.	nMinnfield	field	Tree	0.83142182	-0.526	-0.495	1.17571271	-1.46471031
444.	nMinnfield	field	sapling					
444.	nMinnfield	field	Tree	0.84016885	0.116	0.147	1.82576217	-1.17504284
445.	nMinnfield	field	sapling					
445.	nMinnfield	field	Tree	0.84348194	-0.698	-0.667	1.01502714	-1.2966725
446.	nMinnfield	field	sapling					
446.	nMinnfield	field	Tree	0.86219103	-0.449	-0.418	1.28338239	-1.28566865
447.	nMinnfield	field	sapling					
447.	nMinnfield	field	Tree	0.86940775	-0.729	-0.698	1.01004702	-1.23955429
448.	nMinnfield	field	sapling					
448.	nMinnfield	field	Tree	0.87128097	-0.316	-0.285	1.42505617	-1.28825463
449.	nMinnfield	field	sapling					
449.	nMinnfield	field	Tree	0.8986155	-0.301	-0.270	1.46788743	-1.4559605
450.	nMinnfield	field	sapling					
450.	nMinnfield	field	Tree	0.89927319	-0.097	-0.066	1.67225762	-1.31695386
451.	nMinnfield	field	sapling					
451.	nMinnfield	field	Tree	0.91645395	-0.546	-0.515	1.24052888	-1.35330109
452.	nMinnfield	field	sapling					
452.	nMinnfield	field	Tree	0.94384063	-0.480	-0.449	1.33366366	-1.24753974
453.	nMinnfield	field	sapling					
453.	nMinnfield	field	Tree	0.96857634	-0.586	-0.555	1.25233056	-1.208961
454.	nMinnfield	field	sapling					
454.	nMinnfield	field	Tree	0.99153618	-0.679	-0.648	1.18225876	-1.21041879
455.	nMinnfield	field	sapling					
455.	nMinnfield	field	Tree	1.01456254	-0.151	-0.120	1.73356328	-1.11970759
456.	nMinnfield	field	sapling					
456.	nMinnfield	field	Tree	1.02234588	-0.077	-0.046	1.81561456	-1.06166194
457.	nMinnfield	field	sapling					
457.	nMinnfield	field	Tree	1.02358166	-0.221	-0.190	1.67225589	-1.15135773
458.	nMinnfield	field	sapling					
458.	nMinnfield	field	Tree	1.02362279	-0.324	-0.293	1.5701185	-1.26519137
459.	nMinnfield	field	sapling					
459.	nMinnfield	field	Tree	1.03790436	-0.405	-0.374	1.50262181	-1.20677764

460.	nMinnfield	field	sapling Tree	1.09142073	-0.482	-0.451	1.47892316	-1.25856947
461.	nMinnfield	field	sapling Tree	1.10533984	-0.409	-0.378	1.56681568	-1.21648374
462.	nMinnfield	field	sapling Tree	1.12201917	-0.231	-0.200	1.76129579	-1.15036073
463.	nMinnfield	field	sapling Tree	1.13798673	-0.338	-0.307	1.67008173	-1.16363474
464.	nMinnfield	field	sapling Tree	1.15170686	-0.419	-0.388	1.60275779	
465.	nMinnfield	field	sapling Tree	1.16654851	-0.631	-0.600	1.40587803	-1.40008419
466.	nMinnfield	field	sapling Tree	1.21579613	-0.720	-0.689	1.36530463	-1.12474699
467.	nMinnfield	field	sapling Tree	1.21848299	-0.364	-0.333	1.72403876	-1.05251824
468.	nMinnfield	field	sapling Tree	1.23989982	-0.521	-0.490	1.58895585	-0.87883655
469.	nMinnfield	field	sapling Tree	1.24578402	-0.077	-0.046	2.03861582	-1.08675269
470.	nMinnfield	field	sapling Tree	1.26599637	-0.371	-0.340	1.76472393	-1.0644222
471.	nMinnfield	field	sapling Tree	1.30567375	-0.527	-0.496	1.64872379	-1.0111776
472.	nMinnfield	field	sapling Tree	1.32442651	-0.105	-0.074	2.08902784	-0.92145836
473.	nMinnfield	field	sapling Tree	1.32555671	-0.144	-0.113	2.05160138	-0.925128
474.	nMinnfield	field	sapling Tree	1.32565931	-0.710	-0.679	1.48574273	-0.86945898
475.	nMinnfield	field	sapling Tree	1.33328602	-0.365	-0.334	1.83782018	-1.02609711
476.	nMinnfield	field	sapling Tree	1.33391054	-0.287	-0.256	1.91739042	-0.99619708
477.	nMinnfield	field	sapling Tree	1.34068202	-0.498	-0.467	1.71242131	-1.20484116
478.	nMinnfield	field	sapling Tree	1.35351232	-0.458	-0.427	1.76588188	-0.82873066
479.	nMinnfield	field	sapling Tree	1.35545152	-0.117	-0.086	2.1087103	-0.83263599
480.	nMinnfield	field	sapling Tree	1.36163341	-0.362	-0.331	1.86973603	-1.00755717
481.	nMinnfield	field	sapling Tree	1.36225622	-0.376	-0.345	1.85588843	-0.96227796
482.	nMinnfield	field	sapling Tree	1.37421661	-0.165	-0.134	2.07939262	-0.81778342
483.	nMinnfield	field	sapling Tree	1.38098866	-0.309	-0.278	1.9416917	-0.81817756
484.	nMinnfield	field	sapling Tree	1.40052072	-0.359	-0.328	1.91107175	-0.90231902
485.	nMinnfield	field	sapling Tree	1.43018797	-0.625	-0.594	1.67475442	-0.85472181
486.	nMinnfield	field	sapling Tree	1.43396174	-0.460	-0.429	1.84360729	-1.19266187
487.	nMinnfield	field	sapling Tree	1.44218178	-0.379	-0.348	1.93281695	-0.9914867
488.	nMinnfield	field	sapling Tree	1.46874599	-0.326	-0.295	2.01243133	-0.87053086
489.	nMinnfield	field	sapling Tree	1.46982202	-0.294	-0.263	2.04563889	-0.93741162
490.	nMinnfield	field	sapling Tree	1.50749156	-0.358	-0.327	2.01931289	-0.93241589
			sapling					

491.	nMinnfield	field	Tree	1.51138865	-0.256	-0.225	2.12533851	-0.75662854
492.	nMinnfield	field	sapling	1.54709751	-0.362	-0.331	2.05505865	-0.82016739
493.	nMinnfield	field	Tree	1.55286281	-0.202	-0.171	2.22063918	-0.78089142
494.	nMinnfield	field	sapling	1.57047289	-0.118	-0.087	2.32278234	-0.71540844
495.	nMinnfield	field	Tree	1.57706649	-0.392	-0.361	2.05497158	-0.9004036
496.	nMinnfield	field	sapling	1.58344817	-0.240	-0.209	2.21356299	-0.64394985
497.	nMinnfield	field	Tree	1.58566373	-0.377	-0.346	2.07831735	-0.80851433
498.	nMinnfield	field	sapling	1.58868619	-0.361	-0.330	2.09760736	-0.68029259
499.	nMinnfield	field	Tree	1.59802407	-0.278	-0.247	2.18957808	-0.60291814
500.	nMinnfield	field	sapling	1.60829064	-0.402	-0.371	2.07606599	-0.81549624
501.	nMinnfield	field	Tree	1.61263554	-0.355	-0.324	2.12757022	-0.89936758
502.	nMinnfield	field	sapling	1.63157566	-0.245	-0.214	2.25628576	-0.70316376
503.	nMinnfield	field	Tree	1.66304097	-0.439	-0.408	2.09422438	-0.79105638
504.	nMinnfield	field	sapling	1.67717792	-0.210	-0.179	2.33673652	-0.74787994
505.	nMinnfield	field	Tree	1.68819728	-0.455	-0.424	2.1036367	-0.8177977
506.	nMinnfield	field	sapling	1.69642629	-0.341	-0.310	2.22539861	-0.67803495
507.	nMinnfield	field	Tree	1.70176652	-0.404	-0.373	2.16735069	-0.66660916
508.	nMinnfield	field	sapling	1.72309429	-0.194	-0.163	2.39860469	-0.52143091
509.	nMinnfield	field	Tree	1.72331608	-0.208	-0.177	2.38515128	-0.66134135
510.	nMinnfield	field	sapling	1.73602979	-0.464	-0.433	2.14153389	-0.82418142
511.	nMinnfield	field	Tree	1.7393587	-0.077	-0.046	2.53193406	-0.57054505
512.	nMinnfield	field	sapling	1.79049628	-0.332	-0.301	2.32883208	-0.53474671
513.	nMinnfield	field	Tree	1.81081701	-0.319	-0.288	2.36148938	-0.47885106
514.	nMinnfield	field	sapling	1.85912628	-0.121	-0.090	2.60787362	-0.39819654
515.	nMinnfield	field	Tree	1.88266112	-0.253	-0.222	2.49998549	-0.40013602
516.	nMinnfield	field	sapling	1.89420526	-0.140	-0.109	2.62430539	-0.37742097
517.	nMinnfield	field	Tree	1.89685677	-0.412	-0.381	2.35498493	-0.5182181
518.	nMinnfield	field	sapling	1.91324678	-0.203	-0.172	2.57979171	-0.33577852
519.	nMinnfield	field	Tree	1.93577421	-0.033	-0.002	2.77266111	-0.20793394
520.	nMinnfield	field	sapling	1.96056115	-0.315	-0.284	2.51509318	-0.29166027
521.	nMinnfield	field	Tree	1.97800238	-0.313	-0.282	2.53450929	-0.26178161
522.	nMinnfield	field	sapling	1.98298542	-0.156	-0.125	2.69663405	-0.22159382
			Tree					

523.	nMinnfield	field	sapling Tree	1.99044986	-0.312	-0.281	2.54818658	-0.44549424
524.	nMinnfield	field	sapling Tree	2.06256694	-0.267	-0.236	2.66546565	-0.37271184
525.	nMinnfield	field	sapling Tree	2.21065555	-0.257	-0.226	2.82378208	-0.06926812
526.	nMinnfield	field	sapling Tree	2.25967987	-0.147	-0.116	2.98309432	0.06698295
527.	nMinnfield	field	sapling Tree	2.31813432	-0.213	-0.182	2.97524473	0.10239417
528.	nMinnfield	field	sapling Tree	2.3205513	-0.257	-0.226	2.93353954	0.04923508
529.	nMinnfield	field	sapling Tree	2.34242702	-0.394	-0.363	2.81836986	0.02557341
530.	nMinnfield	field	sapling Tree	2.3582357	-0.078	-0.047	3.15016735	0.07073226
531.	nMinnfield	field	sapling Tree	2.38842804	-0.172	-0.141	3.08653488	0.21417073
532.	nMinnfield	field	sapling Tree	2.41218157	-0.037	-0.006	3.24563513	0.15472559
533.	nMinnfield	field	sapling Tree	2.41360819	-0.084	-0.053	3.1997201	0.09886431
534.	nMinnfield	field	sapling Tree	2.58421019	-0.262	-0.231	3.19178557	0.29602174
535.	nMinnfield	field	sapling Tree	2.61122821	-0.214	-0.183	3.26772626	0.33151245
536.	nMinnfield	field	sapling Tree	2.67995364	-0.102	-0.071	3.44751068	0.45422995
537.	nMinnfield	field	sapling Tree	2.74517242	-0.165	-0.134	3.44979971	0.54203771
			sapling					
